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MIL-HDBK-17-1C
Volume 1 of 3
28 February 1992
Superseding
MIL-HDBK-17-1B
29 FEBRUARY 1988

MILITARY HANDBOOK

POLYMER MATRIX COMPOSITES

VOLUME I. GUIDELINES



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FOREWORD

1. This Military Handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. Every effort has been made to reflect the latest information on polymeric composites. The Handbook is continually reviewed and revised to insure its completeness and currentness. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Department of the Army, Materials Technology Laboratory, ATTN: SLCMT-MEE, MIL-HDBK-17 Coordinator, Arsenal Street, Watertown, MA 02172-2719, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. Military Handbook 17 (Part 1) provides guidelines and material properties for polymer (organic) matrix composites materials. This handbook encompasses, but is not limited to, polymeric composites intended for aircraft and aerospace vehicles and military combat vehicle applications. Metal matrix composites (MMC), ceramic matrix composites (CMC), and carbon/carbon composites (C/C) will be covered in separate military handbooks as developments occur.

4. This standardization handbook has been developed and is being maintained as a joint effort of the Department of Defense and Federal Aviation Administration.

5. The information contained in this handbook was obtained from materials producers, industry, reports on Government sponsored research, the open literature, and by contact with research laboratories and those who participate in the MIL-HDBK-17 coordination activity.

6. All information and data contained in this handbook have been coordinated with industry and the U.S. Army, Navy, Air Force, NASA, and Federal Aviation Administration prior to publication.

7. Copies of this document and revisions thereto may be obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094

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VOLUME 1 - GUIDELINES

Page

Chapter 1. GENERAL INFORMATION 1-1
1.1 INTRODUCTION. 1-2
1.2 PURPOSE. 1-4
1.3 SCOPE. 1-4
1.4 USE OF THE DOCUMENT AND LIMITATIONS. 1-5
1.5 APPROVAL PROCEDURES. 1-6
1.6 SYMBOLS, ABBREVIATIONS, AND SYSTEMS OF UNITS. 1-6
1.6.1 Symbols and abbreviations. 1-6
1.6.2 System of units. 1-13
1.7 DEFINITIONS. 1-15
REFERENCES 1-36

Chapter 2. OBJECTIVES IN GENERATING PROPERTY DATA 2-1
2.1 INTRODUCTION 2-2
2.2 RECOMMENDATIONS FOR THE GENERATION OF PHYSICAL AND MECHANICAL PROPERTIES 2-2
2.2.1 General Guidelines 2-2
2.2.2 Moisture effects 2-2
2.2.3 Conditioning of samples 2-6
2.2.4 Statistical development of mechanical properties 2-8
2.2.5 Data pooling requirements 2-8
2.3 MATERIAL ACQUISITION AND PREPREG PHYSICAL PROPERTY CHARACTERIZATION 2-9
2.4 LAMINA PHYSICAL AND MECHANICAL PROPERTY TESTS 2-9
2.5 FILAMENT-WOUND MATERIALS PROPERTY TESTS 2-9
2.6 BRAIDED MATERIALS PROPERTY TESTS 2-9
2.7 THICK-SECTION COMPOSITES PROPERTY TESTS 2-9
2.8 OTHER USEFUL TEST MATRICES 2-13
2.8.1 Material system screening 2-13
2.8.2 Qualification guidelines and requirements for alternate composite materials 2-14
REFERENCES 2-22

Chapter 3. EVALUATION OF REINFORCEMENT FIBERS 3-1
3.1 INTRODUCTION. 3-3
3.2 CHEMICAL TECHNIQUES. 3-3
3.2.1 Elemental analysis. 3-3
3.2.2 Titration. 3-4
3.2.3 Fiber structure. 3-4
3.2.4 Fiber surface chemistry. 3-4
3.2.5 Sizing content and composition. 3-7
3.2.6 Moisture content. 3-7
3.2.7 Thermal stability and oxidative resistance. 3-7
3.2.8 Chemical resistance. 3-7

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Check the source to verify that this is the current version before use.

Page

3.3 PHYSICAL TECHNIQUES (INTRINSIC). 3-8
3.3.1 Filament diameter. 3-8
3.3.2 Density. 3-8
3.3.3 Electrical resistivity. 3-8
3.3.4 Coefficient of thermal expansion. 3-8
3.3.5 Thermal conductivity. 3-9
3.3.6 Specific heat. 3-9
3.3.7 Thermal transition temperatures. 3-9
3.4 PHYSICAL TECHNIQUES (EXTRINSIC). 3-9
3.4.1 Yield of yarn, strand, or roving. 3-10
3.4.2 Cross-sectional area of yarn or tow. 3-10
3.4.3 Twist of yarn. 3-10
3.4.4 Fabric construction. 3-10
3.4.5 Fabric areal density. 3-10
3.5 MECHANICAL PROPERTIES. 3-10
3.5.1 Tensile strength and Young's modulus. 3-11
3.5.2 Fiber strain and failure. 3-11
3.5.3 Transverse fiber modulus. 3-11
3.5.4 Poisson's ratio. 3-11
3.5.5 Shear moduli. 3-11
3.5.6 Interlaminar shear. 3-11
3.6 TEST METHODS. 3-11
3.6.1 Determination of pH. 3-12
3.6.2 Determination of amount of sizing on carbon fibers. 3-12
3.6.3 Determination of moisture content or moisture regain. 3-13
3.6.4 Determination of fiber diameter. 3-15
3.6.5 Determination of electrical resistivity. 3-17
REFERENCES 3-20
3-22

Chapter 4. MATRIX CHARACTERIZATION 4-1
4.1 INTRODUCTION. 4-2
4.2 CHEMICAL ANALYSIS TECHNIQUES. 4-2
4.2.1 Elemental analysis. 4-2
4.2.2 Functional group and wet chemical analysis. 4-2
4.2.3 Spectroscopic analysis. 4-2
4.2.4 Chromatographic analysis. 4-4
4.2.5 Molecular weight and molecular weight distribution analysis. 4-4
4.2.6 General scheme for resin material characterization. 4-5
4.3 PHYSICAL ANALYSIS TECHNIQUES. 4-12
4.3.1 Thermal analysis. 4-12
4.3.2 Rheological analysis. 4-14
4.3.3 Morphological Analysis. 4-15
4.3.4 Density. 4-15
4.4 VOLATILES DETERMINATION. 4-15
4.5 MOISTURE CONTENT. 4-15
4.6 MECHANICAL TEST METHODS. 4-15
4.6.1 Tensile Testing. 4-15
4.6.2 Compression Testing. 4-15
4.6.3 Flexural Testing. 4-15

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:30Z
Check the source to verify that this is the current version before use.

Page

4.6.4 Impact Strength Measurement. 4-15
4.6.5 Creep Testing. 4-15
4.6.6 Fatigue Resistance Testing. 4-15
4.6.7 Hardness Testing. 4-15
4.7 TOXICITY, HEALTH HAZARDS, AND SAFETY. 4-15
REFERENCES 4-16

Chapter 5. PREPREG MATERIALS CHARACTERIZATION 5-1
5.1 INTRODUCTION 5-2
5.2 CHARACTERIZATION TECHNIQUES - OVERVIEW. 5-2
5.3 SAMPLING. 5-3
5.4 GENERAL CHARACTERISTICS OF PREPREGS. 5-3
5.4.1 Physical description of reinforcement. 5-3
5.4.2 Resin content. 5-4
5.4.3 Fiber content. 5-4
5.4.4 Volatiles content. 5-4
5.4.5 Moisture content. 5-4
5.4.6 Inorganic fillers and additives content. 5-4
5.4.7 Areal weight. 5-5
5.4.8 Tack and drape. 5-5
5.4.9 Resin flow. 5-5
5.4.10 Gel time. 5-5
5.5 TEST METHODS. 5-6
5.5.1 Resin extraction procedure for epoxy resin prepregs. 5-6
5.5.2 Procedure for HPLC/HPSEC analysis of glass, aramid, and graphite fiber prepregs. 5-7
5.5.3 Procedure for Fourier transform infrared spectroscopy (FTIR). 5-9
5.5.4 Procedure for differential scanning calorimetry (DSC). 5-9
5.5.5 Procedure for dynamic mechanical analysis (DMA). 5-9
5.5.6 Procedure for rheological characterization. 5-10
REFERENCES 5-11

Chapter 6. LAMINA AND LAMINATE CHARACTERIZATION 6-1
6.1 INTRODUCTION. 6-2
6.2 NON-DESTRUCTIVE EVALUATION. 6-2
6.3 PHYSICAL ANALYSIS TECHNIQUES. 6-3
6.3.1 Thermal analysis. 6-3
6.3.2 Rheological analysis. 6-3
6.3.3 Fiber volume. 6-3
6.3.4 Void volume. 6-3
6.3.5 Density. 6-4
6.3.6 Dimensional stability. 6-4
6.3.7 Moisture weight gain. 6-4
6.4 ENVIRONMENTAL EXPOSURE TESTING. 6-4
6.5 FLAMMABILITY. 6-5
6.6 LAMINA, LAMINATE, AND FABRIC MECHANICAL PROPERTY TESTS. . . . 6-5
6.6.1 General. 6-5
6.6.3 Compression tests. 6-8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:30Z
Check the source to verify that this is the current version before use.

Page

6.6.4 Flexure tests. 6-8
6.6.5 Shear tests. 6-8
6.6.6 Fatigue. 6-10
6.6.7 Creep 6-10
6.6.8 Damage tolerance and laminate testing. 6-11
6.7 FILAMENT WINDING MECHANICAL PROPERTY TESTS. 6-12
6.8 BRAIDING MECHANICAL PROPERTY TESTS. 6-12
6.9 THICK-SECTION COMPOSITES MECHANICAL PROPERTY TESTS 6-14
6.9.1 Introduction. 6-14
6.10 THERMAL AND MOISTURE ABSORPTION PROPERTIES. 6-14
6.11 ELECTRICAL PROPERTIES. 6-14
REFERENCES 6-15

Chapter 7. STRUCTURAL ELEMENT CHARACTERIZATION 7-1
7.1 INTRODUCTION. 7-2
7.2 MECHANICALLY-FASTENED JOINTS 7-2
7.2.1 Definitions. 7-2
7.2.2 Failure modes. 7-2
7.2.3 Design requirements. 7-3
7.2.4 Material bearing strength. 7-5
7.2.5 Bearing strength of joints. 7-8
7.2.6 Notch tension/compression strength 7-17
7.2.7 Bearing/by-pass strength. 7-17
7.2.8 Shear-out strength. 7-17
7.2.9 Fastener pull-thru strength. 7-17
7.3 BONDED JOINTS. 7-17
7.4 OTHER TOPICS. 7-17
REFERENCES 7-18

Chapter 8. ANALYSIS AND PRESENTATION OF DATA 8-1
8.1 GENERAL. 8-5
8.1.1 Introduction. 8-5
8.1.2 Data documentation requirements. 8-5
8.1.3 Normalization. 8-5
8.1.4 Symbols. 8-8
8.2 MATERIAL AND SPECIMEN REQUIREMENTS. 8-10
8.2.1 Statistically-based material properties. 8-10
8.2.2 Typical properties. 8-10
8.3 DETERMINATION OF PROPERTIES. 8-11
8.3.1 Introduction. 8-11
8.3.2 Mechanical properties. 8-11
8.3.3 Chemical properties. 8-11
8.3.4 Physical properties. 8-11
8.3.5 Thermal properties. 8-12
8.3.6 Electrical properties. 8-12
8.3.7 Typical stress-strain curves. 8-12
8.4 PRESENTATION OF DATA. 8-17
8.4.1 Properties and definitions. 8-17
8.4.2 Organization of data in handbook. 8-17

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Check the source to verify that this is the current version before use.

Page

8.4.3 Sample tables. 8-18
8.4.4 Sample graphs. 8-22
8.5 CALCULATION OF STATISTICALLY-BASED MATERIAL PROPERTIES. . . 8-23
8.5.1 Introduction. 8-23
8.5.2 Definition of computational procedures. 8-27
8.5.3 Detecting outliers and characterizing the population. 8-29
8.5.4 Normal analysis procedure in the presence of batch-to-batch variation. 8-31
8.5.5 Normal analysis procedure for a single population. 8-33
8.5.6 Two-parameter Weibull analysis procedure for a single population. . 8-34
8.5.7 Nonparametric analysis procedure for a single population. 8-38
8.5.8 Linear statistical models. 8-39
8.6 STATISTICAL PROPERTIES. 8-43
8.6.1 Introduction. 8-43
8.6.2 Outlier detection procedure. 8-43
8.6.3 Subpopulation compatibility tests. 8-44
8.6.4 Goodness-of-fit tests. 8-45
8.6.5 Exploratory Data Analysis. 8-49
8.6.6 General linear statistical models. 8-53
8.6.7 Analysis of variance procedures. 8-59
8.6.8 Methods for developing A-basis values. 8-61
8.7 SAMPLES OF COMPUTATIONAL PROCEDURES. 8-64
8.7.1 Problem 1 - Outlier detection and multiple sample tests. 8-64
8.7.2 Problem 2 - Outlier detection and ANOVA method. 8-70
8.7.3 Problem 3 - Weibull distribution. 8-73
8.7.4 Problem 4 - Normal distribution. 8-76
8.7.5 Problem 5 - Lognormal distribution. 8-78
8.7.6 Problem 6 - Nonparametric method. 8-80
8.7.7 Problem 7 - Insufficient data. 8-81
8.7.8 Problem 8 - Regression analysis. 8-82
8.8 CALCULATIONS FOR STATISTICAL TABLES. 8-86
8.8.1 One-sided tolerance factors, k_B, for the normal distribution, 0.95 confidence. 8-86
8.8.2 One-sided tolerance factors, V, for the Weibull distribution, 0.95 confidence. 8-86
8.8.3 Ranks, r, of observations for determining B-values for an unknown distribution. 8-87
8.8.4 0.20 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. 8-87
8.8.5 0.95 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. 8-88
8.8.6 One-sided tolerance factors, T, for the one-way random effects ANOVA, 0.95 confidence. 8-89
8.8.7 Critical values for the k-sample Anderson-Darling test at the 0.05 significance level. 8-90
8.8.8 0.95 Quantiles of the chi-squared distribution associated with γ degrees of freedom. 8-90
8.8.9 Critical values for the MNR outlier detection procedure (α = 0.05). . 8-90
8.8.10 Nonparametric B-basis material properties for small samples. 8-91

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Check the source to verify that this is the current version before use.

Page

8.8.11 One-sided tolerance limit factors, V_A, for the Weibull distribution, 0.95 confidence. 8-91
8.8.12 One-sided tolerance limit factors, k_A, for the normal distribution, 0.95 confidence. 8-91
8.8.13 Ranks, r, of observations for determining A-basis values for an unknown distribution. 8-91
8.8.14 The Hanson-Koopmans method exponents for A-basis values. . . . 8-91
8.8.15 0.1225 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. 8-91
8.8.16 One-sided tolerance factors for A-basis calculations, T_A, for the one-way random effects ANOVA, 0.95 confidence. 8-92
STATISTICAL TABLES 8-93
REFERENCES 8-126

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Check the source to verify that this is the current version before use.

Chapter 1. GENERAL INFORMATION

Page

1.1 INTRODUCTION. 1-2

1.2 PURPOSE. 1-4

1.3 SCOPE. 1-4

1.4 USE OF THE DOCUMENT AND LIMITATIONS. 1-5

1.5 APPROVAL PROCEDURES. 1-6

1.6 SYMBOLS, ABBREVIATIONS, AND SYSTEMS OF UNITS. 1-6
1.6.1 Symbols and abbreviations. 1-6
1.6.1.1 Constituent properties. 1-10
1.6.1.2 Laminae and laminates. 1-11
1.6.1.3 Subscripts. 1-12
1.6.1.4 Superscripts. 1-13
1.6.2 System of units. 1-13

1.7 DEFINITIONS. 1-15

REFERENCES 1-36

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1.1 **INTRODUCTION.** The standardization of a statistically-based mechanical property data base, procedures, and overall material guidelines for characterization of composite material systems is recognized as being beneficial to both manufacturers and governmental agencies. It is also recognized that a complete characterization of the capabilities of any engineering material system is primarily dependent on the inherent material physical and chemical composition which precede, and are independent of, specific applications. Therefore, at the material system characterization level, the data and guidelines contained in this handbook are applicable to military and commercial products and provide the technical basis for establishing statistically valid design values acceptable to certifying or procuring agencies.

This handbook specifically provides statistically-based mechanical property data on current and emerging polymer matrix composite materials, provides guidelines for the analysis and presentation of data, and provides fabrication and characterization documentation to ensure repeatability of results or reliable detection of differences. The primary focus of MIL-HDBK-17 in the overall characterization/ design procedure as commonly applied to composites is shown in Figure 1.1.

The data contained herein, or appearing in approved items in the minutes of MIL-HDBK-17 coordination meetings are acceptable to the Army, the Navy, the Air Force, and the Federal Aviation Administration. Approval by the certifying or procuring agency must be obtained for the use of data or guidelines not contained herein.

This standardization handbook has been developed and is maintained as a joint effort of the Department of Defense and the Federal Aviation Administration. It is oriented toward the standardization of methods used to develop and analyze mechanical property data on current and emerging composite materials.

MIL-HDBK-17 is divided into three volumes. The first volume is oriented toward guidelines for data development and analyses. Chapter 2 provides guidelines for the generation of material properties, including the qualification of alternate materials and pooling of data from different sources. Chapters 3 through 5 define acceptable procedures for the evaluation of composite constituents including reinforcement fibers, resins, and prepreg materials. Chapter 6 addresses acceptable procedures for the evaluation of lamina and laminate materials. Procedures for analyzing composite structural details, and in particular, bolted joints are presented in Chapter 7. Chapter 8 of Volume 1 addresses the analysis and presentation of composite material property data. Important references are cited at the end of each chapter.

The second volume of MIL-HDBK-17 provides a compilation of statistically-based material properties for current and emerging composite materials. The first two chapters of this volume address matrix material and reinforcement fiber constituent properties. The remaining chapters address composite system properties and they are defined according to the primary reinforcement fiber types. Properties for composite materials reinforced with glass fibers, aramid fibers, carbon fibers, and so on are covered in separate chapters.

Volume 3 provides guidelines for the application of the data which are presented in Volume 2. Chapter 2 addresses typical composite materials and processes. The matrix materials and reinforcement fibers of interest in MIL-HDBK-17 are reviewed, as are the typical product forms. Fabrication methods are also covered for information purposes. Chapter 3 reviews important issues related to quality control in the production of composite materials. Recommended manufacturing inspection procedures are reviewed, along with techniques for material property verification and statistical quality control. Chapter 4 addresses the design and analysis of composite material systems. It provides an overview of the current techniques and describes how the various constituent properties

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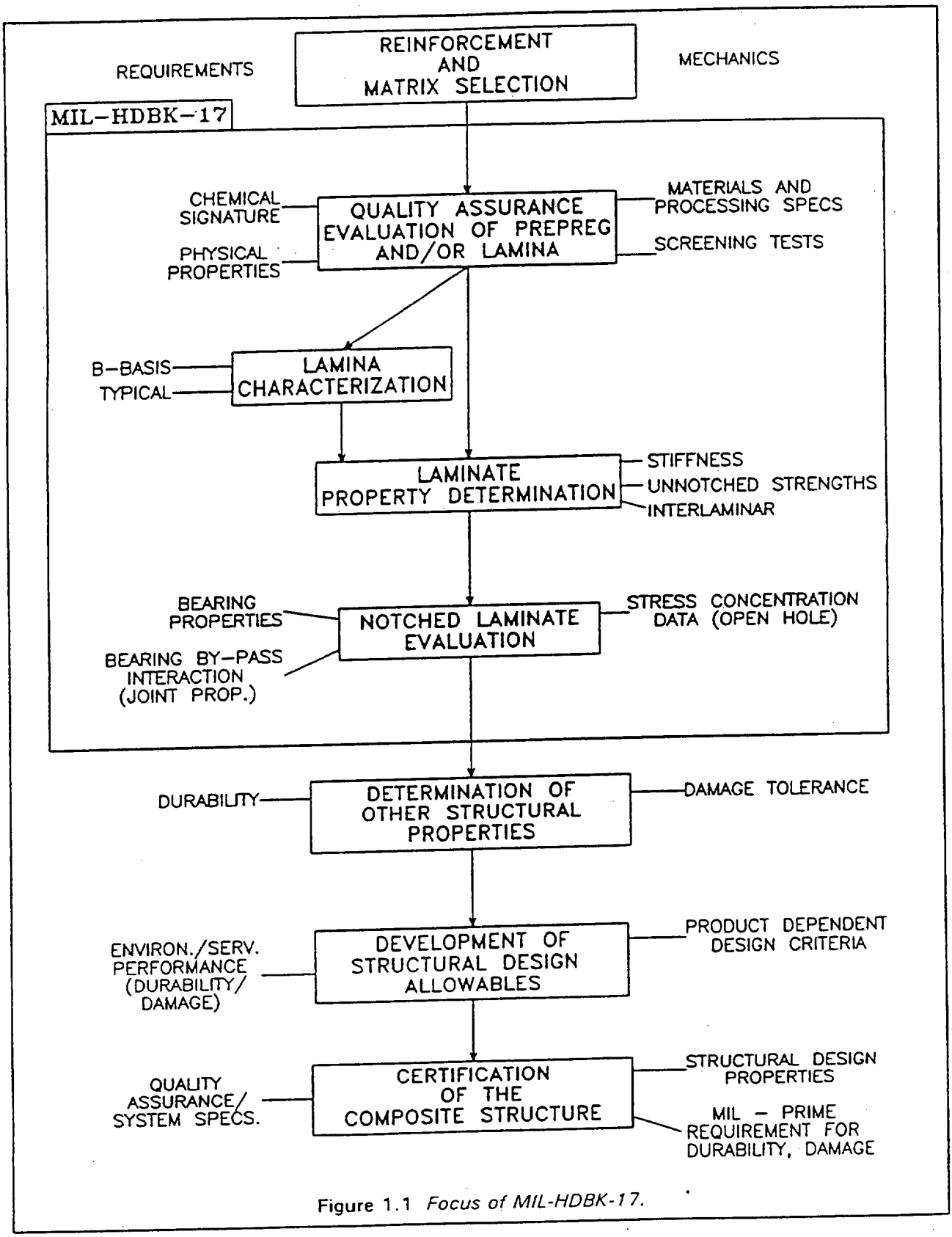


Figure 1.1 *Focus of MIL-HDBK-17.*

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reported in MIL-HDBK-17 are used in the design and analysis of a composite system.

The remainder of Chapter 1 of this volume provides additional information of importance to any user of MIL-HDBK-17. The purpose and scope of the document are defined and some comments are provided on its use and limitations. The accepted system of symbols, abbreviations, and units are also reviewed, and a large collection of terms pertinent to this handbook are defined.

1.2 PURPOSE. The purpose of this handbook is to provide a standard source of statistically-based mechanical property data for current and emerging composite materials. In order to serve this purpose the handbook must provide specific guidelines on how the necessary data should be developed and analyzed. Documentation requirements on the fabrication and characterization of these composites must also be clearly defined.

Twice yearly MIL-HDBK-17 coordination meetings are held for the specific purpose of reviewing and approving new guidelines and data proposals. These meetings consist of representatives from the DOD, FAA, and industry. Materials which are approved and included in this handbook or the minutes of the MIL-HDBK-17 coordination meetings are acceptable to the Army, Navy, Air Force, and Federal Aviation Administration and considered effective on the date approved by MIL-HDBK-17 coordination committee. The use of data or guidelines that are not approved and therefore not included in this document must be approved by the appropriate certification or procurement agency.

1.3 SCOPE. MIL-HDBK-17 is published in three volumes, and serves as a source for the following:

Volume 1 - Provides guidelines for the characterization of composite material systems to be used in vehicles and structures. Composite material systems must normally be evaluated in accordance with these, or equivalent guidelines, in order to be considered acceptable by government certification and procuring agencies.

Volume 2 - Provides a compilation of statistically-based mechanical property data for current and emerging composite material systems used in the aerospace industry. B-basis strength and strain-to-failure values will be presented along with related data.

Volume 3 - Provides information regarding materials and fabrication procedures, quality control, and design and analysis.

Specifically, Volume 1 provides guidelines for the physical, chemical, and mechanical characterization of composite materials. These guidelines address characterization of the fiber, matrix, and prepreg materials, which are the primary constituents in a typical composite system. The guidelines also address the characterization of the composite system, with particular emphasis on lamina properties, rather than laminate properties. Recommendations on the evaluation of bolted joints in composite materials are provided. Volume 1 also provides guidelines for the statistical analysis and presentation of data.

Volume 2 provides statistically-based mechanical property data for composite material systems used in the aerospace industry. Strength and strain-to-failure properties are reported either in terms of B-values or S-values (see Section 1.7). Stiffness properties are generally reported as typical values. The specific statistical significance of each of these quantities is defined in the first chapter of Volume 2. Physical, chemical, and mechanical properties of the composite constituents - the fibers, matrix material and prepreg - are reported where applicable. Later chapters include data summaries for the various composite systems. Individual chapters focus on particular reinforcement fibers.

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Volume 3 provides guidelines for the design and analysis of composite materials. The chapters in Volume 3 address related topics of 1) Materials and Processes, 2) Quality Control of Production Composites, and 3) Design and Analysis. This information is included primarily for background information and is not offered for regulatory purposes.

Statistically-based strength properties are defined for each composite material system over a range of potential usage conditions. The intent is to provide data at the upper and lower limits of the potential environmental conditions for a particular material, so that applications issues do not govern the mechanical property characterizations. If data are also available at intermediate environmental conditions, they are used to more exactly define the relationship between the mechanical properties and the effect of the environment on those properties. The statistically-based strength data which are available are tabulated in Volume 2. These data are useful as a starting point for establishing structural design allowables when stress and strength analysis capabilities permit lamina level margin of safety checks. Depending on the application, some structural design allowables will have to be determined empirically at the laminate and composite level, since MIL-HDBK-17 does not provide these data.

Additional information and properties are added as they become available and are demonstrated to meet the guideline criteria. Typical property values, as well as S-values (see definitions) are included if they meet the approval of the MIL-HDBK-17 Coordination Group.

When the guidelines or data requirements of MIL-HDBK-17 cannot be followed, the certifying or procuring government agency should be contacted to determine data requirements and other documentation which may be necessary to justify data values proposed or used by the manufacturer.

1.4 **USE OF THE DOCUMENT AND LIMITATIONS.** The information contained in MIL-HDBK-17 is obtained from materials producers and fabricators, the aerospace industry, reports on government-sponsored research, the open literature, and by contact with research laboratories and those who participate in the MIL-HDBK-17 coordination activity. All of the information and data contained in this document have been coordinated with representatives from industry and the Army, Navy, Air Force, and Federal Aviation Administration prior to publication. Every effort has been made to reflect the latest information on composite materials, vehicles, and structures. The handbook is continually reviewed and revised to insure its completeness and to keep it as current as possible.

All data included herein are based on test specimens only. Unless otherwise noted, test specimens dimensions conform with those specified for the particular test method which is used. Standard test methods are recommended where possible (ASTM standards are the primary source). The designer and all other users must be responsible for any translation of the data contained herein to other coupon dimensions, temperature, humidity, and other environmental conditions not covered in this document. Problems such as scale up effects and the influence of the test method selected on properties are also not addressed in this document. The manner in which S-basis values are used is also up to the discretion of the designer. In general, decisions concerning which properties to use for a specific application or design are the responsibility of the designer and are outside the scope of this handbook.

The data which are tabulated in this handbook are intended as an aid in assigning property values to a material. In specific cases where it is necessary or preferable to develop strength properties superior to those in this handbook acceptance of such values must be obtained from the appropriate procurement or certification agency. The applicability and interpretation of specific provisions of this handbook must also be defined by the appropriate procurement, regulatory, or certification agency.

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Reference information which is cited in this handbook may not comply in every respect with the guidelines or other criteria specified in this document. References are provided at the end of each chapter primarily as a source of additional information in a given subject area.

The use of tradenames and proprietary product names does not constitute an endorsement of those products by the Government.

1.5 **APPROVAL PROCEDURES.** The MIL-HDBK-17 Coordination Group is a joint government-industry activity that meets twice yearly. At each meeting, this group acts upon proposed changes or additions to the document that are submitted by any of the working groups. The agenda is usually mailed to attendees four weeks prior to the meeting date, and the minutes four weeks following the meeting. Requests for consideration of material for inclusion in the handbook should be submitted to the appropriate working group and the Secretariat well in advance of the mailing date.

Attachments containing proposed changes or additions to the document shall include specific notation of the changes or additions to be made and adequate documentation of supporting data and analytical procedures. Reproducible copies of drawings or photographs intended for inclusion in the document shall be furnished to the Secretariat.

Requests for inclusion of data in MIL-HDBK-17 should be submitted to the Secretariat with the documentation specified in Section 8.1.2. Following analysis and review of the data by the Secretariat, the data will be presented at the next meeting of the Coordination Group.

The choice of new materials to be included herein is governed by the MIL-HDBK-17 Coordination Group. Practical considerations preclude inclusion of all advanced composite materials that may be of interest. Reasonable attempts will be made to add new materials of interest in a timely manner.

1.6 **SYMBOLS, ABBREVIATIONS, AND SYSTEMS OF UNITS.** This section defines the symbols and abbreviations which are used within MIL-HDBK-17 and describes the system of units which is maintained. Common usage is maintained where possible. References 1.6(a), 1.6(b), and 1.6(c) served as primary sources for this information.

1.6.1 **Symbols and abbreviations.** The symbols and abbreviations used in this document are defined in this section with the exception of statistical symbols. These latter symbols are defined in Chapter 8.

- The symbols f and m, when used as either subscripts or superscripts, always denote fiber and matrix, respectively.
- The type of stress (e.g., cy - compression yield) is always used in the superscript position.
- Direction indicators (e.g., x, y, z, 1, 2, 3, etc.) are always used in the subscript position.
- Ordinal indicators of laminae sequence (e.g., 1, 2, 3, etc.) are used in the superscript position and must be parenthesized to distinguish them from mathematical exponents.
- Other indicators may be used in either subscript or superscript position, as appropriate for clarity.
- Compound symbols (i.e., basic symbol plus indicators) which deviate from these rules are shown in their specific form in the following list.

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The following general symbols and abbreviations are considered standard for use in MIL-HDBK-17. Where exceptions are made, they are noted in the text and tables.

A	- (1) area (m^2,in^2) - (2) ratio of alternating stress to mean stress - (3) A-basis for mechanical property values
ASTM	- American Society for Testing and Materials
a	- (1) length dimension (mm,in) - (2) acceleration (m/sec^2,ft/sec^2) - (3) amplitude - (4) crack or flaw dimension (mm,in)
B	- (1) B-basis for mechanical property values - (2) biaxial ratio
Btu	- British thermal unit(s)
b	- width dimension (mm,in), e.g. the width of a bearing or compression panel normal to load, or breadth of beam cross-section
C	- (1) specific heat (kJ/kg °C,BTU/lb °F) - (2) Celsius
CF	- centrifugal force (N,lbf)
CG	- (1) center of mass, "center of gravity" - (2) area or volume centroid
℄	- centerline
c	- column buckling end-fixity coefficient
$\bar{c}$	- honeycomb sandwich core depth (mm,in)
cpm	- cycles per minute
D	- (1) diameter (mm,in) - (2) hole or fastener diameter (mm,in) - (3) plate stiffness (N-m,lbf-in)
d	- mathematical operator denoting differential
E	- modulus of elasticity in tension, average ratio of stress to strain for stress below proportional limit (GPa,Msi)
E'	- storage modulus (GPa,Msi)
E"	- loss modulus (GPa,Msi)
E_c	- modulus of elasticity in compression, average ratio of stress to strain for stress below proportional limit (GPa,Msi)
$E_c^/$	- modulus of elasticity of honeycomb core normal to sandwich plane (GPa,Msi)
E^{sec}	- secant modulus (GPa,Msi)
E^{tan}	- tangent modulus (GPa,Msi)
e	- minimum distance from a hole center to the edge of the sheet (mm,in)
e/D	- ratio of edge distance to hole diameter (bearing strength)
F	- (1) stress (MPa,ksi) - (2) Fahrenheit
F^b	- bending stress (MPa,ksi)
F^{ccr}	- crushing or crippling stress (upper limit of column stress for failure) (MPa,ksi)
F^{su}	- ultimate stress in pure shear (this value represents the average shear stress over the cross-section) (MPa,ksi)
f	- (1) internal (or calculated) stress (MPa,ksi) - (2) stress applied to the gross flawed section (MPa,ksi) - (3) creep stress (MPa,ksi)
f^c	- internal (or calculated) compressive stress (MPa,ksi)

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f_c - (1) maximum stress at fracture (MPa,ksi)
- (2) gross stress limit (for screening elastic fracture data (MPa,ksi)

ft - foot, feet

G - modulus of rigidity (shear modulus) (GPa,Msi)

GPa - gigapascal(s)

g - (1) gram(s)
- (2) acceleration due to gravity (m/s^2,ft/s^2)

H/C - honeycomb (sandwich)

h - height dimension (mm,in) e.g. the height of a beam cross-section

hr - hour(s)

I - area moment of inertia (mm^4,in^4)

i - slope (due to bending) of neutral plane in a beam, in radians

in. - inch(es)

J - (1) torsion constant (= I_p for round tubes) (m^4,in^4)
- (2) Joule

K - (1) Kelvin
- (2) stress intensity factor (MPa$\sqrt{m}$,ksi$\sqrt{in}$)
- (3) coefficient of thermal conductivity (W/m °C, BTU/ft^2/hr/in/°F)
- (4) correction factor
- (5) dielectric constant

K_{app} - apparent plane strain fracture toughness or residual strength (MPa$\sqrt{m}$,ksi$\sqrt{in}$)

K_c - critical plane strain fracture toughness, a measure of fracture toughness at point of crack growth instability (MPa$\sqrt{m}$,ksi$\sqrt{in}$)

K_{Ic} - plane strain fracture toughness (MPa$\sqrt{m}$,ksi$\sqrt{in}$)

K_N - empirically calculated fatigue notch factor

K_s - plate or cylinder shear buckling coefficient

K_t - (1) theoretical elastic stress concentration factor
- (2) t_w/c ratio in H/C sandwich

K_x,K_y - plate or cylinder compression buckling coefficient

k - strain at unit stress (m/m,in/in)

L - cylinder, beam, or column length (mm,in)

L' - effective column length (mm,in)

lb - pound

M - applied moment or couple (N-m,in-lbf)

Mg - megagram(s)

MPa - megapascal(s)

MS - military standard

M.S. - margin of safety

MW - molecular weight

MWD - molecular weight distribution

m - (1) mass (kg,lb)
- (2) number of half wave lengths
- (3) metre
- (4) slope

N - (1) number of fatigue cycles to failure
- (2) number of laminae in a laminate
- (3) distributed in-plane forces on a panel (lbf/in)
- (4) Newton
- (5) normalized

NA - neutral axis

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NAS - National Aerospace Standard
n - (1) number of times in a set
- (2) number of half or total wavelengths
- (3) number of fatigue cycles endured
P - (1) applied load (N,lbf)
- (2) exposure parameter
- (3) probability
P^u - test ultimate load, (N,lb per fastener)
P^y - test yield load, (N,lb per fastener)
p - normal pressure (Pa,psi)
psi - pounds per square inch
Q - area static moment of a cross-section (mm^3,in^3)
q - shear flow (N/m,lbf/in)
R - (1) algebraic ratio of minimum load to maximum load in cyclic loading
- (2) reduced ratio
RA - reduction of area
R.H. - relative humidity
RMS - root-mean-square
RT - room temperature
r - (1) radius (mm,in)
- (2) root radius (mm,in)
- (3) reduced ratio (regression analysis)
S - (1) shear force (N,lbf)
- (2) nominal stress in fatigue (MPa,ksi)
- (3) S-basis for mechanical property values
S_a - stress amplitude in fatigue (MPa,ksi)
S_e - fatigue limit (MPa,ksi)
S_m - mean stress in fatigue (MPa,ksi)
S_{max} - highest algebraic value of stress in the stress cycle (MPa,ksi)
S_{min} - lowest algebraic value of stress in the stress cycle (MPa,ksi)
S_R - algebraic difference between the minimum and maximum stresses in one cycle (MPa,ksi)
S.F. - safety factor
s - (1) arc length (mm,in)
- (2) H/C sandwich cell size (mm,in)
T - (1) temperature (°C,°F)
- (2) applied torsional moment (N-m,in-lbf)
T_d - thermal decomposition temperature (°C,°F)
T_F - exposure temperature (°C,°F)
T_g - glass transition temperature (°C,°F)
T_m - melting temperature (°C,°F)
t - (1) thickness (mm,in)
- (2) exposure time (s)
- (3) elapsed time (s)
V - (1) volume (mm^3,in^3)
- (2) shear force (N,lbf)
W - (1) weight (N,lbf)
- (2) width (mm,in)
- (3) Watt
x - distance along a coordinate axis
Y - nondimensional factor relating component geometry and flaw size
y - (1) deflection (due to bending) of elastic curve of a beam (mm,in)

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- (2) distance from neutral axis to given point
- (3) distance along a coordinate axis

Z - section modulus, I/y (mm^3,in^3)
α - coefficient of thermal expansion (m/m/°C,in/in/°F)
γ - shear strain (m/m,in/in)
Δ - difference (used as prefix to quantitative symbols)
δ - elongation or deflection (mm,in)
ϵ - strain (m/m,in/in)
ϵ - elastic strain (m/m,in/in)
ϵ - plastic strain (m/m,in/in)
η - plasticity reduction factor
$[\eta]$ - intrinsic viscosity
η^* - dynamic complex viscosity
ν - Poisson's ratio
ρ - (1) density (kg/m3,lb/in3)
- (2) radius of gyration (mm,in)

ρ_c' - H/C sandwich core density (kg/m^3,lb/in^3)
Σ - total, summation
σ - standard deviation
σ_{ij}, τ_{ij} - stress in j direction on surface whose outer normal is in i direction (i, j = 1, 2, 3 or x, y, z) (MPa,ksi)
T - applied shear stress (MPa,ksi)
ω - angular velocity (radians/s)
∞ - infinity

1.6.1.1 *Constituent properties.* The following symbols apply specifically to the constituent properties of a typical composite material.

E^f - Young's modulus of filament material (MPa,ksi)
E^m - Young's modulus of matrix material (MPa,ksi)
E_x^g - Young's modulus of impregnated glass scrim cloth in the filament direction or in the warp direction of a fabric (MPa,ksi)
E_y^g - Young's modulus of impregnated glass scrim cloth transverse to the filament direction or to the warp direction in a fabric (MPa,ksi)
G^f - shear modulus of filament material (MPa,ksi)
G^m - shear modulus of matrix (MPa,ksi)
G_{xy}^g - shear modulus of impregnated glass scrim cloth (MPa,ksi)
G'_{cx} - shear modulus of sandwich core along X-axis (MPa,ksi)
G'_{cy} - shear modulus of sandwich core along Y axis (MPa,ksi)
ℓ - filament length (mm,in)
α^f - coefficient of thermal expansion for filament material (m/m/°C,in/in/°F)
α^m - coefficient of thermal expansion for matrix material (m/m/°C,in/in/°F)
α_x^g - coefficient of thermal expansion of impregnated glass scrim cloth in the filament direction or in the warp direction of a fabric (m/m/°C,in/in/°F)
α_y^g - coefficient of thermal expansion of impregnated glass scrim cloth transverse to the filament direction or to the warp direction in a fabric (m/m/°C,in/in/°F)
ν^f - Poisson's ratio of filament material

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ν^m - Poisson's ratio of matrix material

ν^g_{xy} - glass scrim cloth Poisson's ratio relating to contraction in the transverse (or fill) direction as a result of extension in the longitudinal (or warp) direction

ν^g_{yx} - glass scrim cloth Poisson's ratio relating to contraction in the longitudinal (or warp) direction as a result of extension in the transverse (or fill) direction

σ - applied axial stress at a point, as used in micromechanics analysis (MPa,ksi)

τ - applied shear stress at a point, as used in micromechanics analysis (MPa,ksi)

1.6.1.2 *Laminae and laminates.* The following symbols, abbreviations, and notations apply to composite laminae and laminates. At the present time the focus in MIL-HDBK-17 is on laminae properties. However, commonly used nomenclature for both laminae and laminates are included here to avoid potential confusion.

A_{ij} (i,j = 1,2,6) - extensional rigidities (N/m,lbf/in)

B_{ij} (i,j = 1,2,6) - coupling matrix (N,lbf)

C_{ij} (i,j = 1,2,6) - elements of stiffness matrix (Pa,psi)

D_x, D_y - flexural rigidities (N-m,lbf-in)

D_{xy} - twisting rigidity (N-m,lbf-in)

D_{ij} (i,j = 1,2,6) - flexural rigidities (N-m,lbf-in)

E_1 - Young's modulus of lamina parallel to filament or warp direction (GPa,Msi)

E_2 - Young's modulus of lamina transverse to filament or warp direction (GPa,Msi)

E_x - Young's modulus of laminate along x reference axis (GPa,Msi)

E_y - Young's modulus of laminate along y reference axis (GPa,Msi)

G_{12} - shear modulus of lamina in 12 plane (GPa,Msi)

G_{xy} - shear modulus of laminate in xy reference plane (GPa,Msi)

h_i - thickness of ith ply or lamina (mm,in)

M_x, M_y, M_{xy} - bending and twisting moment components (N-m/m, in-lbf/in in plate and shell analysis)

n_f - number of filaments per unit length per lamina

Q_x, Q_y - shear force parallel to z axis of sections of a plate perpendicular to x and y axes, respectively (N/m,lbf/in)

Q_{ij} (i,j = 1,2,6) - reduced stiffness matrix (Pa,psi)

u_x, u_y, u_z - components of the displacement vector (mm,in)

u^o_x, u^o_y, u^o_z - components of the displacement vector at the laminate's midsurface (mm,in)

V_v - void content (% by volume)

V_f - filament content or fiber volume (% by volume)

V_g - glass scrim cloth content (% by volume)

V_m - matrix content (% by volume)

V_x, V_y - edge or support shear force (N/m,lbf/in)

W_f - filament content (% by weight)

W_g - glass scrim cloth content (% by weight)

W_m - matrix content (% by weight)

W_s - weight of laminate per unit surface area (N/m^2,lbf/in^2)

α_1 - lamina coefficient of thermal expansion along 1 axis (m/m/°C,in/in/°F)

α_2 - lamina coefficient of thermal expansion along 2 axis (m/m/°C,in/in/°F)

α_x - laminate coefficient of thermal expansion along general reference x axis (m/m/°C,in/in/°F)

α_y - laminate coefficient of thermal expansion along general reference y axis (m/m/°C,in/in/°F)

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α_{xy} - laminate shear distortion coefficient of thermal expansion (m/m/°C,in/in/°F)
θ - angular orientation of a lamina in a laminate, i.e., angle between 1 and x axes (°)
λ_{xy} - product of ν_{xy} and ν_{yx}
ν_{12} - Poisson's ratio relating contraction in the 2 direction as a result of extension in the 1 direction.
ν_{21} - Poisson's ratio relating contraction in the 1 direction as a result of extension in the 2 direction.
ν_{xy} - Poisson's ratio relating contraction in the y direction as a result of extension in the x direction
ν_{yx} - Poisson's ratio relating contraction in the x direction as a result of extension in the y direction
ρ_c - density of a single lamina (kg/m^3,lb/in^3)
$\bar{\rho}_c$ - density of a laminate (kg/m^3,lb/in^3)
ϕ - (1) general angular coordinate, (°)
- (2) angle between x and load axes in off-axis loading (°)

1.6.1.3 *Subscripts.* The following subscript notations are considered standard in MIL-HDBK-17.

1, 2, 3 - laminae natural orthogonal coordinates (1 is filament or warp direction)
A - axial
a - (1) adhesive
- (2) alternating
app - apparent
c - composite system, specific filament/matrix composition. Composite as a whole, contrasted to individual constituents. Also, sandwich core when used in conjunction with prime (')
- (4) critical
cf - centrifugal force
e - fatigue or endurance
eff - effective
eq - equivalent
f - filament
g - glass scrim cloth
H - hoop
i - i^{th} position in a sequence
L - lateral
m - (1) matrix
- (2) mean
max - maximum
min - minimum
n - (1) n^{th} (last) position in a sequence
- (2) normal
p - polar
s - symmetric
st - stiffener
T - transverse
t - value of parameter at time t
x, y, z - general coordinate system
Σ - total, or summation
o - initial or reference datum

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( ) - format for indicating specific, temperature associated with term in parentheses. RT - room temperature (21°C,70°F); all other temperatures in °F unless specified.

1.6.1.4 *Superscripts.* The following superscript notations are considered standard in MIL-HDBK-17.

b	- bending
br	- bearing
c	- (1) compression
	- (2) creep
cc	- compression crippling
cr	- compression buckling
e	- elastic
f	- filament
g	- glass scrim cloth
is	- interlaminar shear
(i)	- i^{th} ply or lamina
lim	- limit, used to indicate limit loading
m	- matrix
p	- plastic
pl	- proportional limit
rup	- rupture
s	- shear
scr	- shear buckling
sec	- secant (modulus)
T	- temperature or thermal
t	- tension
tan	- tangent (modulus)
u	- ultimate
y	- yield
'	- secondary (modulus), or denotes properties of H/C core when used with subscript c

1.6.2 System of units. To comply with Department of Defense Instructive 5000.2, Part 6, Section M, "Use of the Metric System," dated February 23, 1991, the data in MIL-HDBK-17 are generally presented in both the International System of Units (SI units) and the U. S. Customary (English) system of units. ASTM E-380, Standard for Metric Practice, provides guidance for the application for SI units which are intended as a basis for worldwide standardization of measurement units (Reference 1.6.2(a)). Further guidelines on the use of the SI system of units and conversion factors are contained in the following publications (References 1.6.2(b) - (e)):

(1) DARCOM P 706-470, *Engineering Design Handbook: Metric Conversion Guide*, July 1976.

(2) NBS Special Publication 330, "The International System of Units (SI)," National Bureau of Standards, 1986 edition.

(3) NBS Letter Circular LC 1035, "Units and Systems of Weights and Measures, Their Origin, Development, and Present Status," National Bureau of Standards, November 1985.

(4) NASA Special Publication 7012, "The International System of Units Physical Constants and Conversion Factors", 1964.

English to SI conversion factors pertinent to MIL-HDBK-17 data are contained in Table 1.6.2.

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TABLE 1.6.2 *English to SI conversion factors.*

To convert from	to	Multiply by
Btu (thermochemical)/in^2-s	watt/meter^2 (W/m^2)	1.634 246 E+06
Btu-in/(s-ft^2-°F)	W/(m K)	5.192 204 E+02
degree Fahrenheit	degree Celsius (°C)	T = (T - 32)/1.8
degree Fahrenheit	kelvin (K)	T = (T + 459.67)/1.8
foot	meter (m)	3.048 000 E-01
ft^2	m^2	9.290 304 E-02
foot/second	meter/second (m/s)	3.048 000 E-01
ft/s^2	m/s^2	3.048 000 E-01
inch	meter (m)	2.540 000 E-02
in.^{2}	meter^2 (m^2)	6.451 600 E-04
in.^{3}	m^3	1.638 706 E-05
kilogram-force (kgf)	newton (N)	9.806 650 E+00
kgf/m^2	pascal (Pa)	9.806 650 E+00
kip (1000 lbf)	newton (N)	4.448 222 E+03
ksi (kip/in^2)	MPa	6.894 757 E+00
lbf-in	N-m	1.129 848 E-01
lbf-ft	N-m	1.355 818 E+00
lbf/in^2 (psi)	pascal (Pa)	6.894 757 E+03
lb/in^2	gm/m^2	7.030 696 E+05
lb/in^3	kg/m^3	2.767 990 E+04
Msi-(10^6 psi)	GPa	6.894 757 E+00
pound-force (lbf)	newton (N)	4.488 222 E+00
pound-mass (lb avoirdupois)	kilogram (kg)	4.535 924 E-01
torr	pascal (Pa)	1.333 22 E+02

*The letter "E" following the conversion factor stands for exponent and the two digits after the letter "E" indicate the power of 10 by which the number is to be multiplied.

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1.7 DEFINITIONS. The following definitions are used within MIL-HDBK-17. This glossary of terms is not totally comprehensive but it does represent nearly all commonly used terms. Where exceptions are made, they are noted in the text and tables. For ease of identification the definitions have been organized alphabetically.

A-Basis (or A-Value) -- The mechanical property value above which 99 percent of the population of values is expected to fall, with a confidence of 95 percent. Note - A-values are not currently presented in MIL-HDBK-17. This is a 95% lower confidence limit on the first percentile.

A-Stage -- An early stage in the reaction of thermosetting resins in which the material is still soluble in certain liquids and may be liquid or capable of becoming liquid upon heating. (Sometimes referred to as resol.)

Absorption -- A process in which one material (the absorbent) takes in or absorbs another (the absorbate).

Accelerator -- A material which, when mixed with a catalyzed resin, will speed up the chemical reaction between the catalyst and the resin.

Accuracy -- The degree of conformity of a measured or calculated value to some recognized standard or specified value. Accuracy involves the systematic error of an operation.

Addition Polymerization -- Polymerization in which monomers are linked together without the splitting off of water or other simple molecules.

Adhesion -- The state in which two surfaces are held together at an interface by forces or interlocking action or both.

Adhesive -- A substance capable of holding two materials together by surface attachment. In the handbook, the term is used specifically to designate structural adhesives, those which produce attachments capable of transmitting significant structural loads.

ADK -- Notation used for the k-sample Anderson-Darling statistic, which is used to test the hypothesis that k batches have the same distribution.

Aliquot -- A small, representative portion of a larger sample.

Aging -- The effect, on materials, of exposure to an environment for a period of time; the process of exposing materials to an environment for an interval of time.

Ambient -- The surrounding environmental conditions such as pressure or temperature.

Anelasticity -- A characteristic exhibited by certain materials in which strain is a function of both stress and time, such that, while no permanent deformations are involved, a finite time is required to establish equilibrium between stress and strain in both the loading and unloading directions.

Angleply -- Same as Crossply.

Anisotropic -- Not isotropic; having mechanical and/or physical properties which vary with direction relative to natural reference axes inherent in the material.

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Aramid -- A manufactured fiber in which the fiber-forming substance consisting of a long-chain synthetic aromatic polyamide in which at least 85% of the amide (-CONH-) linkages are attached directly to two aromatic rings.

Areal Weight of Fiber -- The weight of fiber per unit area of prepreg. This is often expressed as grams per square meter. See Table 1.6.2 for conversion factors.

Artificial Weathering -- Exposure to laboratory conditions which may be cyclic, involving changes in temperature, relative humidity, radiant energy and any other elements found in the atmosphere in various geographical areas.

Aspect Ratio -- In an essentially two-dimensional rectangular structure (e.g., a panel), the ratio of the long dimension to the short dimension. However, in compression loading, it is sometimes considered to be the ratio of the load direction dimension to the transverse dimension. Also, in fiber micro-mechanics, it is referred to as the ratio of length to diameter.

Autoclave -- A closed vessel for producing an environment of fluid pressure, with or without heat, to an enclosed object which is undergoing a chemical reaction or other operation.

Autoclave Molding -- A process similar to the pressure bag technique. The lay-up is covered by a pressure bag, and the entire assembly is placed in an autoclave capable of providing heat and pressure for curing the part. The pressure bag is normally vented to the outside.

B-Basis (or B-Value) -- The mechanical property value above which at least 90 percent of the population of values is expected to fall, with a confidence of 95 percent, specifically the same as A-Basis except on the tenth percentile. (See Volume 1, Section 8.5.1.1)

B-Stage -- An intermediate stage in the reaction of a thermosetting resin in which the material softens when heated and swells when in contact with certain liquids but does not entirely fuse or dissolve. Materials are usually precured to this stage to facilitate handling and processing prior to final cure. (Sometimes referred to as resitol.)

Bag Molding -- A method of molding or laminating which involves the application of fluid pressure to a flexible material which transmits the pressure to the material being molded or bonded. Fluid pressure usually is applied by means of air, steam, water or vacuum.

Balanced Laminate -- A composite laminate in which all laminae at angles other than 0 degrees and 90 degrees occur only in ± pairs (not necessarily adjacent).

Batch (or Lot) -- For fibers and resins, a quantity of material formed during the same process and having identical characteristics throughout. For prepregs, laminae, and laminates, material made from one batch of fiber and one batch of resin.

Bearing Area -- The product of the pin diameter and the specimen thickness.

Bearing Load -- A compressive load on an interface.

Bearing Yield Strength -- The bearing stress at which a material exhibits a specified limiting deviation from the proportionality of bearing stress to bearing strain.

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Bend Test -- A test of ductility by bending or folding, usually with steadily applied forces. In some instances the test may involve blows to a specimen having a cross section that is essentially uniform over a length several times as great as the largest dimension of the cross section.

Binder -- A bonding resin used to hold strands together in a mat or preform during manufacture of a molded object.

Binomial Random Variable -- The number of successes in independent trials where the probability of success is the same for each trial.

Birefringence -- The difference between the two principal refractive indices (of a fiber) or the ratio between the retardation and thickness of a material at a given point.

Bleeder Cloth -- A nonstructural layer of material used in the manufacture of composite parts to allow the escape of excess gas and resin during cure. The bleeder cloth is removed after the curing process and is not part of the final composite.

Bond -- The adhesion of one surface to another, with or without the use of an adhesive as a bonding agent.

Braiding -- A textile process where two or more strands, yarns or tapes are intertwined in the bias direction to form an integrated structure.

Broadgoods -- A term loosely applied to prepreg material greater than about 12 inches in width, usually furnished by suppliers in continuous rolls. The term is currently used to designate both collimated uniaxial tape and woven fabric prepregs.

Buckling (Composite) -- A mode of structural response characterized by an out-of-plane material deflection due to compressive action on the structural element involved. In advanced composites, buckling may take the form not only of conventional general instability and local instability but also a micro-instability of individual fibers.

Bundle -- A general term for a collection of essentially parallel filaments or fibers.

C-Stage -- The final stage of the curing reaction of a thermosetting resin in which the material has become practically infusable and insoluble. (Normally considered fully cured and sometimes referred to as resite.)

Carbon Fibers -- Fibers produced by the pyrolysis of organic precursor fibers such as rayon, polyacrylonitrile (PAN), and pitch in an inert atmosphere. The term is often used interchangeably with "graphite"; however, carbon fibers and graphite fibers differ in the temperature at which the fibers are made and heat-treated, and the amount of carbon produced. Carbon fibers typically are carbonized at about 2400°F (1300°C) and assay at 93 to 95% carbon, while graphite fibers are graphitized at 3450 to 5450°F (1900 to 3000°C) and assay at more than 99% elemental carbon.

Caul Plates -- Smooth metal plates, free of surface defects, the same size and shape as a composite lay-up, used immediately in contact with the lay-up during the curing process to transmit normal pressure and to provide a smooth surface on the finished laminate.

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Censoring -- Data is right (left) censored at M, if, whenever an observation is less than or equal to M (greater than or equal to M), the actual value of the observation is recorded. If the observation exceeds (is less than) M, the observation is recorded as M.

Chain-Growth Polymerization -- A chemical reaction in which polymer formation is initiated by a reactive species R^* produced from some compound I termed an initiator. The reactive species may be a free radical, cation or anion. The reactive center, once produced, adds monomer units in a chain reaction and grows rapidly to a large size. High molecular weight polymer forms immediately with the molecular weight changing slightly, if at all, as the monomer concentration decreases steadily during the reaction.

Chromatogram -- A plot of detector response against peak volume of solution (eluate) emerging from the system for each of the constituents which have been separated.

Circuit -- One complete traverse of the fiber feed mechanism of a winding machine; one complete traverse of a winding band from one arbitrary point along the winding path to another point on a plane through the starting point and perpendicular to the axis.

Cocuring -- The act of curing a composite laminate and simultaneously bonding it to some other prepared surface during the same cure cycle (see Secondary Bonding).

Coefficient of Linear Thermal Expansion -- The change in length per unit length resulting from a one-degree rise in temperature.

Coefficient of Variation -- The ratio of the population (or sample) standard deviation to the population (or sample) mean.

Collimated -- Rendered parallel.

Compatible -- The ability of different resin systems to be processed in contact with each other without degradation of end product properties. (See Compatible, Volume 1, Section 8.5.1.1)

Composite Class -- As used in the handbook, a major subdivision of composite construction in which the class is defined by the fiber system and the matrix class, e.g., organic-matrix filamentary laminate.

Composite Material -- Composites are considered to be combinations of materials differing in composition or form on a macroscale. The constituents retain their identities in the composite; that is, they do not dissolve or otherwise merge completely into each other although they act in concert. Normally, the components can be physically identified and exhibit an interface between one another.

Compound -- An intimate mixture of polymer or polymers with all the materials necessary for the finished product.

Condensation Polymerization -- This is a special type of step-growth polymerization characterized by the formation of water or other simple molecules during the stepwise addition of reactive groups.

Confidence Coefficient -- See Confidence Interval.

Confidence Interval -- A confidence interval is defined by a statement of one of the following forms:

$$(1)\ P\{a < \theta\} \leq 1-\alpha$$

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(2) $P\{\theta < b\} \leq 1-\alpha$
(3) $P\{a < \theta < b\} \leq 1-\alpha$

where $1-\alpha$ is called the confidence coefficient. A statement of type (1) or (2) is called a one-sided confidence interval and a statement of type (3) is called a two-sided confidence interval. In (1) a is a lower confidence limit and in (2) b is an upper confidence limit. With probability at least $1-\alpha$, the confidence interval will contain the parameter θ.

Constituent -- In general, an element of a larger grouping. In advanced composites, the principal constituents are the fibers and the matrix.

Continuous Filament -- A yarn or strand in which the individual filaments are substantially the same length as the strand.

Coupling Agent -- Any chemical substance designed to react with both the reinforcement and matrix phases of a composite material to form or promote a stronger bond at the interface. Coupling agents are applied to the reinforcement phase from an aqueous or organic solution or from a gas phase, or added to the matrix as an integral blend.

Crazing -- Apparent fine cracks at or under the surface of an organic matrix.

Creep -- The time dependent part of strain resulting from an applied stress.

Creep, Rate Of -- The slope of the creep-time curve at a given time.

Critical Value(s) -- When testing a one-sided statistical hypothesis, a critical value is the value such that, if the test statistic is greater than (less than) the critical value, the hypothesis is rejected. When testing a two-sided statistical hypothesis, two critical values are determined. If the test statistic is either less than the smaller critical value or greater than the larger critical value, then the hypothesis is rejected. In both cases, the critical value chosen depends on the desired risk (often 0.05) of rejecting the hypothesis when it is true.

Crossply -- Any filamentary laminate which is not uniaxial. Same as Angleply. In some references, the term crossply is used to designate only those laminates in which the laminae are at right angles to one another, while the term angleply is used for all others. In the handbook, the two terms are used synonymously. The reservation of a separate terminology for only one of several basic orientations is unwarranted because a laminate orientation code is used.

Cumulative Distribution Function -- See Volume 1, Section 8.5.1.1.

Cure -- To change the properties of a thermosetting resin irreversibly by chemical reaction, i.e., condensation, ring closure, or addition. Cure may be accomplished by addition of curing (cross-linking) agents, with or without catalyst, and with or without heat. Cure may occur also by addition, such as occurs with anhydride cures for epoxy resin systems.

Cure Cycle -- The schedule of time periods at specified conditions to which a reacting thermosetting material is subjected in order to reach a specified property level.

Cure Stress -- A residual internal stress produced during the curing cycle of composite structures. Normally, these stresses originate when different components of a lay-up have different thermal coefficients of expansion.

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Debond -- A deliberate separation of a bonded joint or interface, usually for repair or rework purposes. (See Disbond, Unbond).

Deformation -- The change in shape of a specimen caused by the application of a load or force.

Degradation -- A deleterious change in chemical structure, physical properties or appearance.

Delamination -- The separation of the layers of material in a laminate. This may be local or may cover a large area of the laminate. It may occur at any time in the cure or subsequent life of the laminate and may arise from a wide variety of causes.

Denier -- A direct numbering system for expressing linear density, equal to the mass in grams per 9000 meters of yarn, filament, fiber, or other textile strand.

Density -- The mass per unit volume.

Desorption -- A process in which an absorbed or adsorbed material is released from another material. Desorption is the reverse of absorption, adsorption, or both.

Deviation -- Variation from a specified dimension or requirement, usually defining the upper and lower limits.

Dielectric Constant -- The ratio of the capacity of a condenser having a dielectric constant between the plates to that of the same condenser when the dielectric is replaced by a vacuum; a measure of the electrical charge stored per unit volume at unit potential.

Dielectric Strength -- The average potential per unit thickness at which failure of the dielectric material occurs.

Disbond -- An area within a bonded interface between two adherends in which an adhesion failure or separation has occurred. It may occur at any time during the life of the structure and may arise from a wide variety of causes. Also, colloquially, an area of separation between two laminae in the finished laminate (in this case the term "delamination" is normally preferred.) (See Debond, Unbond, Delamination.)

Distribution -- A formula which gives the probability that a value will fall within prescribed limits. (See Normal, Weibull, and Lognormal Distributions, also Volume 1, Section 8.5.1.1).

Dry Fiber Area -- Area of fiber not totally encapsulated by resin.

Ductility -- The ability of a material to deform plastically before fracturing.

Elasticity -- The property of a material which allows it to recover its original size and shape immediately after removal of the force causing deformation.

Elongation -- The increase in gage length or extension of a specimen during a tension test, usually expressed as a percentage of the original gage length.

Eluate -- The liquid emerging from a column (in liquid chromatography).

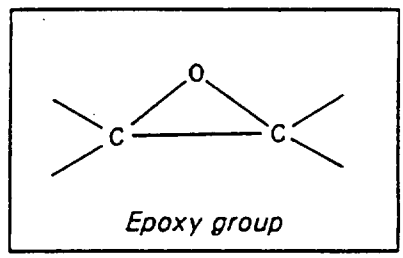
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Eluent -- The mobile phase used to sweep or elute the sample (solute) components into, through, and out of the column.

End -- A single fiber, strand, roving or yarn being or already incorporated into a product. An end may be an individual warp yarn or cord in a woven fabric. In referring to aramid and glass fibers, an end is usually an untwisted bundle of continuous filaments.

Epoxy Equivalent Weight -- The number of grams of resin which contain one chemical equivalent of the epoxy group.

Epoxy Resin -- Resins which may be of widely different structures but are characterized by the presence of the epoxy group. (The epoxy or epoxide group is usually present as a glycidyl ether, glycidyl amine, or as part of an aliphatic ring system. The aromatic type epoxy resins are normally used in composites.)



Epoxy group

Extensometer -- A device for measuring linear strain.

F-Distribution -- See Volume 1, Section 8.5.1.1.

Fabric, Nonwoven -- A textile structure produced by bonding or interlocking of fibers, or both, accomplished by mechanical, chemical, thermal, or solvent means, and combinations thereof.

Fabric, Woven -- A generic material construction consisting of interlaced yarns or fibers, usually a planar structure. Specifically, as used in this handbook, a cloth woven in an established weave pattern from advanced fiber yarns and used as the fibrous constituent in an advanced composite lamina. In a fabric lamina, the warp direction is considered the longitudinal direction, analogous to the filament direction in a filamentary lamina.

Fiber -- A general term used to refer to filamentary materials. Often, fiber is used synonymously with filament. It is a general term for a filament of finite length.

Fiber Content -- The amount of fiber present in a composite. This is usually expressed as a percentage volume fraction or weight fraction of the composite.

Fiber Count -- The number of fibers per unit width of ply present in a specified section of a composite.

Fiber Direction -- The orientation or alignment of the longitudinal axis of the fiber with respect to a stated reference axis.

Fiber System -- The type and arrangement of fibrous material which comprises the fiber constituent of an advanced composite. Examples of fiber systems are collimated filaments or filament yarns, woven fabric, randomly oriented short-fiber ribbons, random fiber mats, whiskers, etc.

Filament -- The smallest unit of a fibrous material. The basic units formed during spinning and which are gathered into strands of fiber, (for use in composites). Filaments usually are of extreme length and of very small diameter. Filaments normally are not used individually. Some textile filaments can function as a yarn when they are of sufficient strength and flexibility.

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Filamentary Composites -- A major form of advanced composites in which the fiber constituent consists of continuous filaments. Specifically, a filamentary composite is a laminate comprised of a number of laminae, each of which consists of a nonwoven, parallel, uniaxial, planar array of filaments (or filament yarns) embedded in the selected matrix material. Individual laminae are directionally oriented and combined into specific multiaxial laminates for application to specific envelopes of strength and stiffness requirements.

Filament Winding -- A reinforced-plastics process that employs a series of continuous, resin-impregnated fibers applied to a mandrel in a predetermined geometrical relationship under controlled tension.

Filament Wound -- Pertaining to an object created by the filament winding method of fabrication.

Fill -- Yarn oriented at right angles to the warp in a woven fabric.

Filler -- A relatively inert substance added to a material to alter its physical, mechanical, thermal, electrical, and other properties or to lower cost. Sometimes the term is used specifically to mean particulate additives.

Finish (or Size System) -- A material, with which filaments are treated, which contains a coupling agent to improve the bond between the filament surface and the resin matrix in a composite material. In addition, finishes often contain ingredients which provide lubricity to the filament surface, preventing abrasive damage during handling, and a binder which promotes strand integrity and facilitates packing of the filaments.

Fixed Effect -- A common systematic (nonrandom) shift in a group of measurements which is due to a controlled level change of a treatment or condition. (See Volume 1, Section 8.5.1.1.)

Flash -- Excess material which forms at the parting line of a mold or die, or which is extruded from a closed mold.

Fracture Ductility -- The true plastic strain at fracture.

Gage Length -- the original length of that portion of the specimen over which strain or change of length is determined.

Gel -- The initial jelly-like solid phase that develops during formation of a resin from a liquid. Also, a semi-solid system consisting of a network of solid aggregates in which liquid is held.

Gel Coat -- A quick-setting resin used in molding processes to provide an improved surface for the composite; it is the first resin applied to the mold after the mold-release agent.

Gel Point -- The stage at which a liquid begins to exhibit pseudo-elastic properties. (This can be seen from the inflection point on a viscosity-time plot.)

Gel Time -- The period of time from a pre-determined starting point to the onset of gelation (gel point) as defined by a specific test method.

Glass -- An inorganic product of fusion which has cooled to a rigid condition without crystallizing. In the handbook, all reference to glass will be to the fibrous form as used in filaments, woven fabric, yarns, mats, chopped fibers, etc.

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Glass Cloth -- Conventionally-woven glass fiber material (see Scrim).

Glass Fibers -- A fiber spun from an inorganic product of fusion which has cooled to a rigid condition without crystallizing.

Glass Transition -- The reversible change in an amorphous polymer or in amorphous regions of a partially crystalline polymer from (or to) a viscous or rubbery condition to (or from) a hard and relatively brittle one.

Glass Transition Temperature -- The approximate midpoint of the temperature range over which the glass transition takes place.

Graphite Fibers -- See Carbon Fibers.

Greige -- Fabric that has received no finish.

Hand Lay-up -- A process in which components are applied either to a mold or a working surface, and the successive plies are built up and worked by hand.

Hardness -- Resistance to deformation; usually measured by indention. Types of standard tests include Brinell, Rockwell, Knoop, and Vickers.

Heat Cleaned -- Glass or other fibers which have been exposed to elevated temperatures to remove preliminary sizings or binders which are not compatible with the resin system to be applied.

Heterogeneous -- Descriptive term for a material consisting of dissimilar constituents separately identifiable; a medium consisting of regions of unlike properties separated by internal boundaries. (Note that all nonhomogeneous materials are not necessarily heterogeneous).

Homogeneous -- Descriptive term for a material of uniform composition throughout; a medium which has no internal physical boundaries; a material whose properties are constant at every point, i.e., constant with respect to spatial coordinates (but not necessarily with respect to directional coordinates).

Horizontal Shear -- Sometimes used to indicate interlaminar shear. This is not an approved term for use in this handbook.

Humidity, Relative -- The ratio of the pressure of water vapor present to the pressure of saturated water vapor at the same temperature.

Hybrid -- A composite laminate comprised of laminae of two or more composite material systems. Or, a combination of two or more different fibers such as carbon and glass or carbon and aramid into a structure (tapes, fabrics and other forms may be combined).

Hygroscopic -- Capable of absorbing and retaining atmospheric moisture.

Hysteresis -- The energy absorbed in a complete cycle of loading and unloading.

Inclusion -- A physical and mechanical discontinuity occurring within a material or part, usually consisting of solid, encapsulated foreign material. Inclusions are often capable of transmitting some structural stresses and energy fields, but in a noticeably different manner from the parent material.

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Integral Composite Structure -- Composite structure in which several structural elements, which would conventionally be assembled by bonding or with mechanical fasteners after separate fabrication, are instead laid up and cured as a single, complex, continuous structure; e.g., spars, ribs, and one stiffened cover of a wing box fabricated as a single integral part. The term is sometimes applied more loosely to any composite structure not assembled by mechanical fasteners.

Interface -- The boundary between the individual, physically distinguishable constituents of a composite.

Interlaminar -- Descriptive term pertaining to some object (e.g., voids), event (e.g., fracture), or potential field (e.g., shear stress) referenced as existing or occurring between two or more adjacent laminae.

Interlaminar Shear -- Shearing force tending to produce a relative displacement between two laminae in a laminate along the plane of their interface.

Intermediate Bearing Stress -- The bearing stress at the point on the bearing load-deformation curve where the tangent is equal to the bearing stress divided by a designated percentage (usually 4%) of the original hole diameter.

Intralaminar -- Descriptive term pertaining to some object (e.g., voids), event (e.g., fracture), or potential field (e.g., temperature gradient) existing entirely within a single lamina without reference to any adjacent laminae.

Isotropic -- Having uniform properties in all directions. The measured properties of an isotropic material are independent of the axis of testing.

Knuckle Area -- The area of transition between sections of different geometry in a filament wound part.

k-Sample Data -- A collection of data consisting of values observed when sampling from k batches.

Lamina -- A single ply or layer in a laminate made up of a series of layers.

Laminae -- Plural of lamina.

Laminate -- A product made by bonding together two or more layers or laminae of material or materials.

Laminate Orientation -- The configuration of a crossplied composite laminate with regard to the angles of crossplying, the number of laminae at each angle, and the exact sequence of the lamina lay-up.

Lattice Pattern -- A pattern of filament winding with a fixed arrangement of open voids.

Lay-up -- A process of fabrication involving the assembly of successive layers of resin-impregnated material.

Lognormal Distribution -- A probability distribution for which the probability that an observation selected at random from this population falls between a and b ($0 < a < b < B$) is given by the area

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under the normal distribution between log a and log b. The common (base 10) or the natural (base e) logarithm may be used. (See Volume 1, Section 8.5.1.1.)

Lower Confidence Bound -- See Confidence Interval.

Macro -- In relation to composites, denotes the gross properties of a composite as a structural element but does not consider the individual properties or identity of the constituents.

Macrostrain -- The mean strain over any finite gage length of measurement which is large in comparison to the material's interatomic distance.

Mandrel -- A form fixture or male mold used for the base in the production of a part by lay-up or filament winding.

Mat -- A fibrous material consisting of randomly oriented chopped or swirled filaments loosely held together with a binder.

Material Acceptance -- The testing of incoming material to ensure that it meets requirements.

Material Qualification -- The procedures used to accept a material by a company or organization for production use.

Material System -- A specific composite material made from specifically identified constituents in specific geometric proportions and arrangements and possessed of numerically defined properties.

Material System Class -- As used in this handbook, a group consisting of material systems categorized by the same generic constituent materials, but without defining the constituents uniquely; e.g., the carbon/epoxy class.

Material Variability -- A source of variability due to variations in the spatial, consistency, and processing aspects of the material. (See Volume 1, Section 8.5.1.1.)

Matrix -- The essentially homogeneous material in which the fiber system of a composite is embedded.

Mean -- See Sample Mean and Population Mean.

Mechanical Properties -- The properties of a material that are associated with elastic and inelastic reaction when force is applied, or the properties involving the relationship between stress and strain.

Median -- See Sample Median and Population Median.

Micro -- In relation to composites, denotes the properties of the constituents, i.e., matrix and reinforcement and interface only, as well as their effects on the composite properties.

Microstrain -- The strain over a gage length comparable to the material's interatomic distance.

Modulus, Initial -- The slope of the initial straight portion of a stress-strain or load-elongation curve.

Modulus, Secant -- The ratio of change in stress to change in strain between two points on a stress-strain curve, particularly the points of zero stress and stress at a particular strain.

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Modulus, Tangent -- The ratio of change in stress to change in strain derived from the tangent to any point on a stress-strain curve.

Modulus, Young's -- The ratio of change in stress to change in strain below the elastic limit of a material. (Applicable to tension and compression).

Modulus of Rigidity (also Shear Modulus or Torsional Modulus) -- The ratio of stress to strain below the proportional limit for shear or torsional stress.

Modulus of Rupture, in Bending -- The maximum tensile or compressive stress (whichever causes failure) value in the extreme fiber of a beam loaded to failure in bending. The value is computed from the flexure equation:

$$F^b = \frac{Mc}{I} \qquad 1.7(a)$$

where M = maximum bending moment computed from the maximum load and the original moment arm,
c = initial distance from the neutral axis to the extreme fiber where failure occurs,
I = the initial moment of inertia of the cross section about its neutral axis.

Modulus of Rupture, in Torsion -- The maximum shear stress in the extreme fiber of a member of circular cross section loaded to failure in torsion calculated from the equation:

$$F^s = \frac{Tr}{J} \qquad 1.7(b)$$

where T = maximum twisting moment,
r = original outer radius,
J = polar moment of inertia of the original cross section.

Moisture Content -- The amount of moisture in a material determined under prescribed condition and expressed as a percentage of the mass of the moist specimen, i.e., the mass of the dry substance plus the moisture present.

Moisture Equilibrium -- The condition reached by a sample when it no longer takes up moisture from, or gives up moisture to, the surrounding environment.

Mold Release Agent -- A lubricant applied to mold surfaces to facilitate release of the molded article.

Molded Edge -- An edge which is not physically altered after molding for use in final form and particularly one which does not have fiber ends along its length.

Molding -- The forming of a polymer or composite into a solid mass of prescribed shape and size by the application of pressure and heat.

Monolayer -- The basic laminate unit from which crossplied or other laminates are constructed.

Monomer -- A compound consisting of molecules each of which can provide one or more constitutional units.

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NDE -- Nondestructive evaluation. Broadly considered synonymous with NDI.

NDI -- Nondestructive Inspection. A process or procedure for determining the quality or characteristics of a material, part, or assembly without permanently altering the subject or its properties.

NDT -- Nondestructive Testing. Broadly considered synonymous with NDI.

Necking -- A localized reduction in cross-sectional area which may occur in a material under tensile stress.

Negatively Skewed -- A distribution is said to be negatively skewed if the distribution is not symmetric and the longest tail is on the left.

Nominal Specimen Thickness -- The nominal ply thickness multiplied by the number of plies.

Nominal Value -- A value assigned for the purpose of a convenient designation. A nominal value exists in name only.

Normal Distribution -- A two parameter (μ,σ) family of probability distributions for which the probability that an observation will fall between a and b is given by the area under the curve

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \qquad 1.7(c)$$

between a and b. (See Volume 1, Section 8.5.1.1.)

Normalized Stress -- Stress calculated by multiplying the raw stress value by the ratio of measured fiber volume to the nominal fiber volume. This ratio is often approximated by the ratio of the measured specimen thickness to the nominal specimen thickness. Stresses for fiber-dominated failure modes are often normalized.

Observed Significance Level (OSL) -- The probability of observing a more extreme value of the test statistic when the null hypotheses is true.

Oligomer -- A polymer consisting of only a few monomer units such as a dimer, trimer, etc. or their mixtures.

One-Sided Tolerance Limit Factor -- See **Tolerance Limit Factor**.

Orthotropic -- Having three mutually perpendicular planes of elastic symmetry.

Oven Dry -- The condition of a material that has been heated under prescribed conditions of temperature and humidity until there is no further significant change in its mass.

PAN Fibers -- Reinforcement fiber derived from the controlled pyrolysis of poly(acrylonitrile) fiber.

Parallel Laminate -- A laminate of woven fabric in which the plies are aligned in the same position as originally aligned in the fabric roll.

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Peel Ply -- A layer of resin free material used to protect a laminate for later secondary bonding.

pH -- A measure of acidity or alkalinity of a solution, with neutrality represented by a value of 7, with increasing acidity corresponding to progressively smaller values, and increasing alkalinity corresponding to progressively higher values.

Pick Count -- The number of filling yarns per inch of woven fabric.

Pitch Fibers -- Reinforcement fiber derived from petroleum or coal tar pitch.

Plastic -- A material that contains one or more organic polymers of large molecular weight, is solid in its finished state, and, at some state in its manufacture or processing into finished articles, can be shaped by flow.

Plasticizer -- A material of lower molecular weight added to a polymer to separate the molecular chains. This results in a depression of the glass transition temperature, reduced stiffness and brittleness, and improved processability. (Note, many polymeric materials do not need a plasticizer.)

Poisson's Ratio -- The absolute value of the ratio of transverse strain to the corresponding axial strain resulting from uniformly distributed axial stress below the proportional limit of the material.

Polymer -- An organic material composed of molecules characterized by the repetition of one or more types of monomeric units.

Polymerization -- A chemical reaction in which the molecules of monomers are linked together to form polymers.

Population -- The set of measurements about which inferences are to be made or the totality of possible measurements which might be obtained in a given testing situation. For example, "all possible ultimate tensile strength measurements for carbon/epoxy system A, conditioned at 95% relative humidity and room temperature". In order to make inferences about a population, it is often necessary to make assumptions about its distributional form. The assumed distributional form may also be referred to as the population. (See Volume 1, Section 8.5.1.1.)

Population Mean -- The average of all potential measurements in a given population weighted by their relative frequencies in the population. (See Volume 1, Section 8.5.1.1.)

Population Median -- That value in the population such that the probability of exceeding it is 0.5 and the probability of being less than it is 0.5. (See Volume 1, Section 8.5.1.1.)

Population Variance -- A measure of dispersion in the population.

Porosity -- A condition of trapped pockets of air, gas, or vacuum within a solid material, usually expressed as a percentage of the total nonsolid volume to the total volume (solid plus nonsolid) of a unit quantity of material.

Positively Skewed -- A distribution is said to be positively skewed if the distribution is not symmetric and the longest tail is on the right.

Postcure -- Additional elevated temperature cure, usually without pressure, to improve final properties or complete the cure or both.

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Pot Life -- The period of time during which a reacting thermosetting composition remains suitable for its intended processing after mixing with a reaction initiating agent.

Precision -- The degree of agreement within a set of observations or test results obtained. Precision involves repeatability and reproducibility.

Precursor (for Carbon or Graphite Fiber) -- Either the PAN or pitch fibers from which carbon and graphite fibers are derived.

Preply -- A composite material lamina in the raw material stage ready to be fabricated into a finished laminate. The lamina is usually combined with other raw laminae prior to fabrication. A preply includes all of the fiber system placed in position relative to all or part of the required matrix material that together will comprise the finished lamina. An organic matrix preply is called a prepreg.

Prepreg -- Ready to mold or cure material in sheet form which may be fiber, cloth, or mat impregnated with resin and stored for use. The resin is partially cured to a B-stage and supplied to the fabricator for lay-up and cure.

Pressure -- The force or load per unit area.

Probability Density Function -- See Volume 1, Section 8.5.1.1.

Proportional Limit -- The maximum stress that a material is capable of sustaining without any deviation from the proportionality of stress to strain (also known as Hooke's law).

Quasi-Isotropic Laminate -- A laminate approximating isotropy by orientation of plies in several or more directions.

Random Effect -- A common shift in a group of measurements due to a random level change of a usually uncontrollable factor. (See Volume 1, Section 8.5.1.1.)

Random Error -- That part of data variation due to level changes in uncontrolled factors which affect each observation separately and independently. (See Volume 1, Section 8.5.1.1.)

Reduction of Area -- The difference between the original cross sectional area of a tension test specimen and the area of its smallest cross section, usually expressed as a percentage of the original area.

Refractive Index - The ratio of the velocity of light (of specified wavelength) in air to its velocity in the substance under examination. Also defined as the sine of the angle of incidence divided by the sine of the angle of refraction as light passes from air into the substance.

Reinforced Plastic -- A plastic with relatively high stiffness or very high strength fibers embedded in the composition. This improves some mechanical properties over that of the base resin.

Release Agent -- See **Mold Release Agent.**

Resilience -- A property of a material which is able to do work against restraining forces during return from a deformed condition.

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Resin -- A solid or pseudo-solid organic material usually of high molecular weight which exhibits a tendency to flow when subjected to stress, usually has a softening or melting range, and fractures conchoidally.

Resin Content -- The amount of matrix present in a composite either by percent weight or percent volume.

Resin Starved Area -- Area of composite part where the resin has a non-continuous smooth coverage of the fiber.

Resin System -- A mixture of resin, with ingredients such as catalyst, initiator, diluents, etc. required for the intended processing and final product.

Roving -- A number of strands, tows, or ends collected into a parallel bundle with little or no twist.

S-Basis (or S-Value) -- The mechanical property value which is usually the specified minimum value of the appropriate government specification or SAE Aerospace Material Specification for this material.

Sample -- A small portion of a material or product intended to be representative of the whole. Statistically, a sample is the collection of measurements taken from a specified population. (See Volume 1, Section 8.5.1.1.)

Sample Mean -- The arithmetic average of the measurements in a sample. The sample mean is an estimator of the population mean. (See Volume 1, Section 8.5.1.1.)

Sample Median -- Order the observation from smallest to largest. Then the sample median is the value of the middle observation if the sample size is odd; the average of the two central observations if n is even. If the population is symmetric about its mean, the sample median is also an estimator of the population mean. (See Volume 1, Section 8.5.1.1.)

Sample Standard Deviation -- The square root of the sample variance. (See Volume 1, Section 8.5.1.1.)

Sample Variance -- The sum of the squared deviations from the sample mean, divided by n-1. (See Volume 1, Section 8.5.1.1.)

Sandwich Construction -- A structural panel concept consisting in its simplest form of two relatively thin, parallel sheets of structural material bonded to, and separated by, a relatively thick, light-weight core.

Saturation -- An equilibrium condition in which the net rate of absorption under prescribed conditions falls essentially to zero.

Scrim (also called Glass Cloth, Carrier) -- A low cost fabric woven into an open mesh construction, used in the processing of tape or other B-stage material to facilitate handling.

Secondary Bonding -- The joining together, by the process of adhesive bonding, of two or more already-cured composite parts, during which the only chemical or thermal reaction occurring is the curing of the adhesive itself.

Selvedge -- The woven edge portion of a fabric parallel to the warp.

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Set -- The strain remaining after complete release of the force producing the deformation.

Shear Fracture (for crystalline type materials) -- A mode of fracture resulting from translation along slip planes which are preferentially oriented in the direction of the shearing stress.

Shelf Life -- The length of time a material, substance, product, or reagent can be stored under specified environmental conditions and continue to meet all applicable specification requirements and/or remain suitable for its intended function.

Significant -- Statistically, the value of a test statistic is significant if the probability of a value at least as extreme is less than or equal to a predetermined number called the significance level of the test.

Significant Digit -- Any digit that is necessary to define a value or quantity.

Size System -- See Finish.

Skewness -- See Positively Skewed, Negatively Skewed.

Slenderness Ratio -- The unsupported effective length of a uniform column divided by the least radius of gyration of the cross-sectional area.

Sliver -- A continuous strand of loosely assembled fiber that is approximately uniform in cross-sectional area and has no twist.

Solute -- The dissolved material.

Specific Gravity -- The ratio of the weight of any volume of a substance to the weight of an equal volume of another substance taken as standard at a constant or stated temperature. Solids and liquids are usually compared with water at 4°C (39°F).

Specific Heat -- The quantity of heat required to raise the temperature of a unit mass of a substance one degree under specified conditions.

Specimen -- A piece or portion of a sample or other material taken to be tested. Specimens normally are prepared to conform with the applicable test method.

Standard Deviation -- See Sample Standard Deviation.

Staple -- Either naturally occurring fibers or lengths cut from filaments.

Step-Growth Polymerization -- A chemical reaction in which polymers are formed by the stepwise intermolecular addition of molecules through reactive groups. Any two molecular species present can react. Monomers disappear early in the reaction and polymer molecular weight rises steadily throughout the reaction.

Strain -- the per unit change, due to force, in the size or shape of a body referred to its original size or shape. Strain is a nondimensional quantity, but it is frequently expressed in inches per inch, meters per meter, or percent.

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Strand -- Normally an untwisted bundle or assembly of continuous filaments used as a unit, including slivers, tow, ends, yarn, etc. Sometimes a single fiber or filament is called a strand.

Strength -- the maximum stress which a material is capable of sustaining.

Stress -- The intensity at a point in a body of the forces or components of forces that act on a given plane through the point. Stress is expressed in force per unit area (pounds-force per square inch, megapascals, etc.).

Stress Relaxation -- The time dependent decrease in stress in a solid under given constraint conditions.

Stress-Strain Curve (Diagram) -- A graphical representation showing the relationship between the change in dimension of the specimen in the direction of the externally applied stress and the magnitude of the applied stress. Values of stress usually are plotted as ordinates (vertically) and strain values as abscissa (horizontally).

Surfacing Mat -- A thin mat of fine fibers used primarily to produce a smooth surface on an organic matrix composite.

Symmetrical Laminate -- A composite laminate in which the sequence of plies below the laminate midplane is a mirror image of the stacking sequence above the midplane.

Tack -- Stickiness of the prepreg.

Tape -- Prepreg fabricated in widths up to 12 inches wide for carbon and 3 inches for boron. Cross stitched carbon tapes up to 60 inches wide are available commercially in some cases.

Tenacity -- The tensile stress expressed as force per unit linear density of the unstrained specimen i.e., grams-force per denier or grams-force per tex.

Tex -- A unit for expressing linear density equal to the mass or weight in grams of 1000 meters of filament, fiber, yarn or other textile strand.

Thermal Conductivity -- Ability of a material to conduct heat. The physical constant for quantity of heat that passes through unit cube of a substance in unit time when the difference in temperature of two faces is one degree.

Thermoplastic -- A plastic that repeatedly can be softened by heating and hardened by cooling through a temperature range characteristic of the plastic, and when in the softened stage, can be shaped by flow into articles by molding or extrusion.

Thermoset -- A plastic that is substantially infusible and insoluble after having been cured by heat or other means.

Tolerance -- The total amount by which a quantity is allowed to vary.

Tolerance Limit -- A lower (upper) confidence limit on a specified percentile of a distribution. For example, the B-basis value is a 95% lower confidence limit on the tenth percentile of a distribution.

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Tolerance Limit Factor -- The factor which is multiplied by the estimate of variability in computing the tolerance limit.

Toughness -- A measure of a material's ability to absorb work, or the actual work per unit volume or unit mass of material that is required to rupture it. Toughness is proportional to the area under the load-elongation curve from the origin to the breaking point.

Tow -- An untwisted bundle of continuous filaments. Commonly used in referring to man-made fibers, particularly carbon and graphite fibers, in the composites industry.

Transformation -- A transformation of data values is a change in the units of measurement accomplished by applying a mathematical function to all data values. For example, if the data is given by x, then $y = x + 1$, x, $1/x$, $\log x$, and $\cos x$ are transformations.

Transition, First Order -- A change of state associated with crystallization or melting in a polymer.

Transversely Isotropic -- Descriptive term for a material exhibiting a special case of orthotropy in which properties are identical in two orthotropic dimensions, but not the third; having identical properties in both transverse directions but not the longitudinal direction.

Twist -- The number of turns about its axis per unit of length in a yarn or other textile strand. It may be expressed as turns per inch (tpi).

Twist, Direction of -- The direction of twist in yarns and other textile strands is indicated by the capital letters S and Z. Yarn has S twist if, when held in a vertical position, the visible spirals or helices around its central axis are in the direction of slope of the central portion of the letter S, and Z twist is in the other direction.

Typical Basis -- A typical property value is a sample mean. Note that the typical value is defined as the simple arithmetic mean which has a statistical connotation of 50% reliability with a 50% confidence.

Unbond -- An area within a bonded interface between two adherends in which the intended bonding action failed to take place. Also used to denote specific areas deliberately prevented from bonding in order to simulate a defective bond, such as in the generation of quality standards specimens. (See Disbond, Debond).

Unidirectional Laminate -- A laminate with nonwoven reinforcements and all layers laid up in the same direction.

Upper Confidence Limit -- See Confidence Interval.

Vacuum Bag Molding -- A process in which the lay-up is cured under pressure generated by drawing a vacuum in the space between the lay-up and a flexible sheet placed over it and sealed at the edges.

Variance -- See Sample Variance.

Viscosity -- The property of resistance to flow exhibited within the body of a material.

Void -- A physical and mechanical discontinuity occurring within a material or part which may be two-dimensional (e.g., disbonds, delaminations) or three-dimensional (e.g., vacuum-, air-, or gas-filled

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pockets). Porosity is an aggregation of micro-voids. Voids are essentially incapable of transmitting structural stresses or nonradiative energy fields. (See Inclusion.)

Warp -- The longitudinally oriented yarn in a woven fabric (see Fill); a group of yarns in long lengths and approximately parallel.

Weibull Distribution (Two - Parameter) -- A probability distribution for which the probability that a randomly selected observation from this population lies between a and b ($0 < a < b < B$) is given by Equation 1.7(d) where a is called the scale parameter and b is called the shape parameter. (See Volume 1, Section 8.5.1.1.)

$$\exp\left[-\left(\frac{a}{\alpha}\right)^{\beta}\right] - \exp\left[-\left(\frac{b}{\alpha}\right)^{\beta}\right] \qquad 1.7(d)$$

Wet Lay-up -- A method of making a reinforced product by applying a liquid resin system while the reinforcement is put in place.

Wet Strength -- The strength of an organic matrix composite when the matrix resin is saturated with absorbed moisture. (See Saturation).

Wet Winding -- A method of filament winding in which the fiber reinforcement is coated with the resin system as a liquid just prior to wrapping on a mandrel.

Whisker -- A short single crystal fiber or filament. Whisker diameters range from 1 to 25 microns, with aspect ratios between 100 and 15,000.

Work Life -- The period during which a compound, after mixing with a catalyst, solvent, or other compounding ingredient, remains suitable for its intended use.

Woven Fabric Composite -- A major form of advanced composites in which the fiber constituent consists of woven fabric. A woven fabric composite normally is a laminate comprised of a number of laminae, each of which consists of one layer of fabric embedded in the selected matrix material. Individual fabric laminae are directionally oriented and combined into specific multiaxial laminates for application to specific envelopes of strength and stiffness requirements.

Yarn -- A generic term for strands or bundles of continuous filaments or fibers, usually twisted and suitable for making textile fabric.

Yarn, Plied -- Yarns made by collecting two or more single yarns together. Normally, the yarns are twisted together though sometimes they are collected without twist.

Yield Point -- The first stress in a material at which an increase in strain occurs without an increase in stress. (The stress is less than the maximum attainable.) It should be noted that only materials that exhibit the unique phenomenon of yielding have a "yield point".

Yield Strength -- The stress at which a material exhibits a specified limiting deviation from the proportionality of stress to strain. (The deviation is expressed in terms of strain such as 0.2 percent for the Offset Method or 0.5 percent for the Total Extension Under Load Method.)

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X-Axis -- In composite laminates, an axis in the plane of the laminate which is used as the 0 degree reference for designating the angle of a lamina.

X-Y Plane -- In composite laminates, the reference plane parallel to the plane of the laminate.

Y-Axis -- In composite laminates, the axis in the plane of the laminate which is perpendicular to the x-axis.

Z-Axis -- In composite laminates, the reference axis normal to the plane of the laminate.

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Chapter 2. OBJECTIVES IN GENERATING PROPERTY DATA

Page

2.1 INTRODUCTION 2-2

2.2 RECOMMENDATIONS FOR THE GENERATION OF PHYSICAL AND MECHANICAL PROPERTIES 2-2
2.2.1 General Guidelines 2-2
2.2.2 Moisture effects 2-2
2.2.3 Conditioning of samples 2-6
2.2.4 Statistical development of mechanical properties 2-8
2.2.5 Data pooling requirements 2-8

2.3 MATERIAL ACQUISITION AND PREPREG PHYSICAL PROPERTY CHARACTERIZATION 2-9

2.4 LAMINA PHYSICAL AND MECHANICAL PROPERTY TESTS 2-9

2.5 FILAMENT-WOUND MATERIALS PROPERTY TESTS 2-9

2.6 BRAIDED MATERIALS PROPERTY TESTS 2-9

2.7 THICK-SECTION COMPOSITES PROPERTY TESTS 2-9

2.8 OTHER USEFUL TEST MATRICES 2-13
2.8.1 Material system screening 2-13
2.8.2 Qualification guidelines and requirements for alternate composite materials 2-14
2.8.2.1 Introduction 2-14
2.8.2.2 Goal. 2-14
2.8.2.3 Material compatibility. 2-15
2.8.2.4 Key material or structural performance parameters. 2-15
2.8.2.5 Success criteria. 2-15
2.8.2.6 General approach. 2-15
2.8.2.7 Specific requirements for lamina mechanical property allowables. 2-17
2.8.2.8 Suggested requirements for laminate mechanical properties. . 2-17
2.8.2.9 Summary. 2-20
2.8.2.10 Statistical methods 2-20

REFERENCES 2-22

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2.1 INTRODUCTION. The purpose of Chapter 2 is to provide guidelines for the characterization of composite material properties. While current procedures emphasize the development of a lamina-level database, this does not preclude higher level testing. Further, it is possible that qualification can be achieved at the application level of testing.

This chapter also provides guidelines for the screening of new material systems. Mechanical properties testing is proposed which provides a preliminary, typical value, evaluation of key stiffness and strength properties. The intent of this screening test program is to provide an initial evaluation of new material systems under worst-case environmental and loading test conditions. The screening includes both lamina (ply) and laminate (application) level testing and should be sufficient to eliminate deficient material systems and to reveal promising new material systems for planning of in-depth evaluations.

2.2 RECOMMENDATIONS FOR THE GENERATION OF PHYSICAL AND MECHANICAL PROPERTIES.

2.2.1 General Guidelines. Generation of design allowables for advanced composite materials requires physical and chemical property characterization of the prepreg material as well as physical and mechanical property tests on cured lamina and laminates. These guidelines contain recommendations for the number and type of tests sufficient to establish B-basis mechanical properties for advanced composite unidirectional tape and woven fabric materials. Recommendations are also provided for vendor acceptance tests by the manufacturer of the prepreg materials.

These guidelines present the methods recommended by MIL-HDBK-17 for generating statistically-based material properties for lamina coupons. Other methods involving statistical analysis of mechanical property tests on oriented laminates are equally acceptable when agreed to by the contractor and certifying agency.

2.2.2 Moisture effects. Organic matrix composites absorb small amounts of moisture. The absorbed water causes a dimensional change (swelling), lowers the glass transition temperature of the resin, and results in a reduction of the matrix and matrix/fiber interface dependent mechanical properties of the composite, particularly at elevated temperatures. Except for the aramid fiber composites, it is usually assumed that the moisture absorption is limited to the organic matrix.

The degradation of composite properties as a result of exposure to high temperatures is nonlinear. A drastic reduction in properties occurs at a characteristic temperature level, defined here as the material operational limit (MOL). For composites, this behavior is more complicated because the introduction of moisture will result in more than one critical temperature, as illustrated in Figure 2.2.2(a). The various moisture contents are represented schematically by saturation at different relative humidity levels. There are no standard criteria for the determination of the MOL. One method (References 2.2.2(a) - (c)), illustrated in Figure 2.2.2(b), utilizes the glass transition temperature (T_g) with some safety factor K as the basis for the MOL. For epoxy composites, 50 F° (28 C°) has been proposed for the value of the safety factor. If service usage at high temperatures is limited to occasional excursions,it can be argued that lower safety factors are acceptable.

The elevated temperature wet testing required by MIL-HDBK-17 should be at the MOL temperature, corresponding to the highest practical relative humidity. For aircraft, the worst-case relative humidity is 85%. As can be seen from Figure 2.2.2(a), the effect of moisture is small for matrix-dependent properties at temperatures below room temperature (RT). However, if Figure 2.2.2(a) were produced for the fiber-dependent properties, degradation in properties would be observed below room temperature. Therefore, MIL-HDBK-17 requires testing at cold temperatures. Since the changes in

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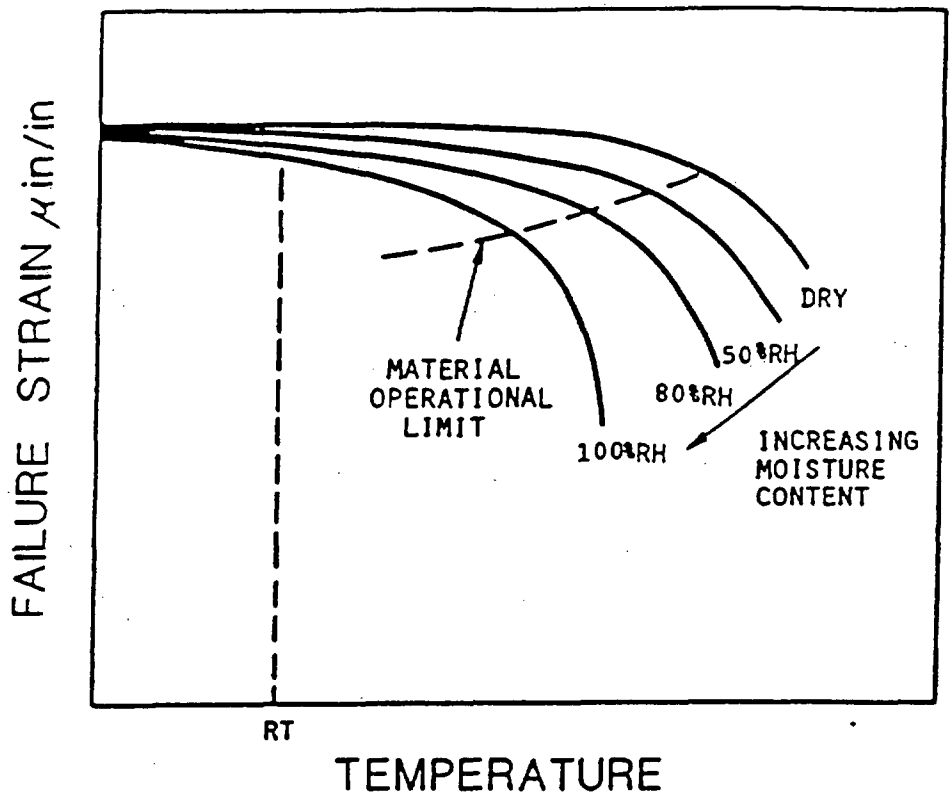


FIGURE 2.2.2(a) Influence of temperature and moisture on matrix-dependent failure strain.

properties at these temperatures are not drastic, the lowest service temperature will suffice and there is no need to determine a MOL.

The equilibrium level of weight gain due to moisture absorption is generally considered to be a function of relative humidity (RH) only. The relationship is not necessarily linear as shown by the test data fitted curves for epoxy composites in Figure 2.2.2(c). The rate of moisture absorption is very strongly dependent on temperature as shown in Figure 2.2.2(d); a decrease of 60 F° (33 C°) increases the time to absorb 1% moisture by five times. The relationship between the diffusion coefficient and absolute temperature is of the Arrhenius form

$$D = D_0 \exp\left[\frac{-E_d}{RT}\right] \qquad 2.2.2(a)$$

where D_0 and E_d are material constants and R is the universal gas constant. The moisture absorption and desorption is mathematically modeled using a one-dimensional Fickian equation.

$$\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2} \qquad 2.2.2(b)$$

where c is moisture concentration. The boundary conditions are:

$$c = c_1 \text{ at } t = 0$$

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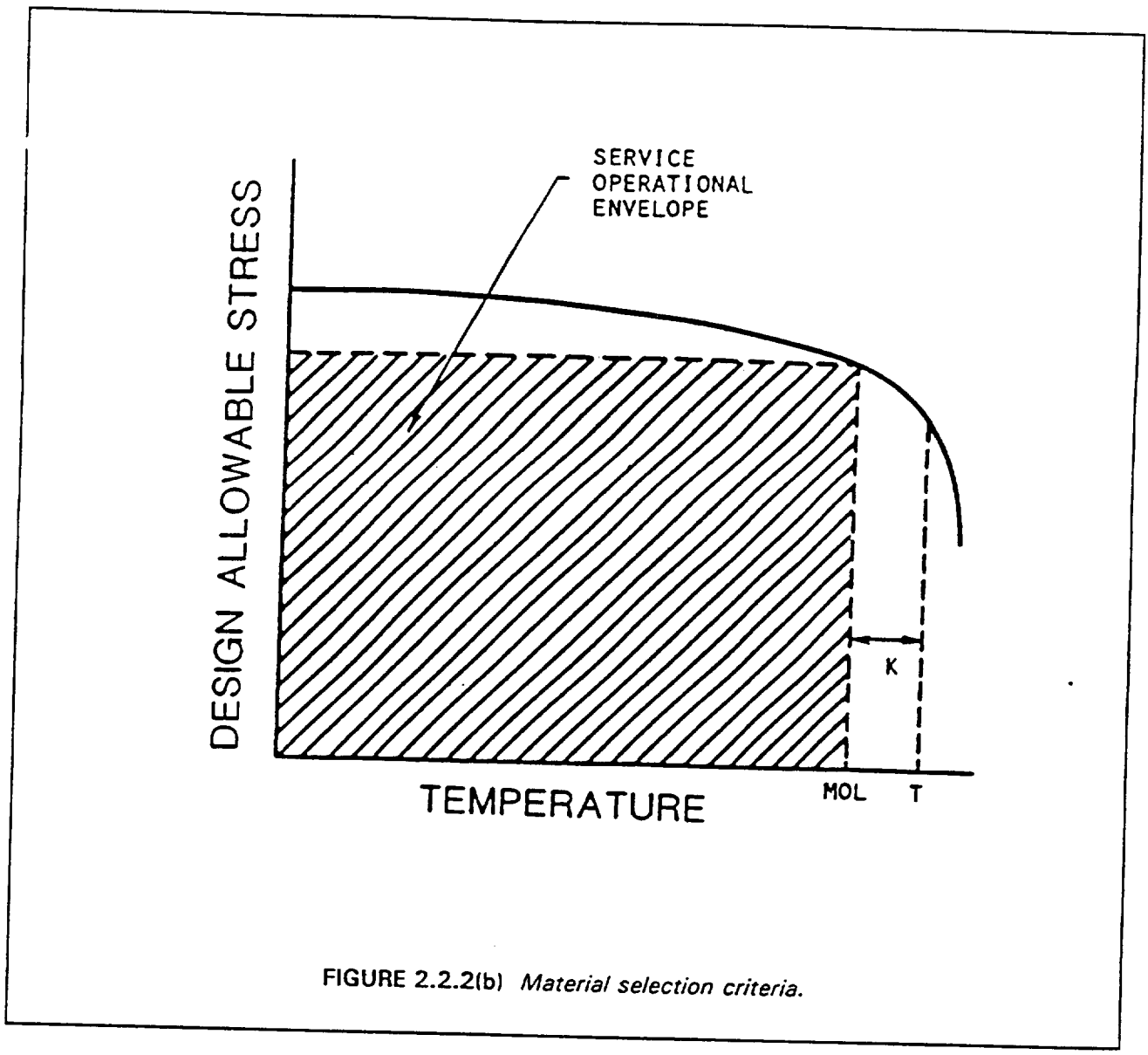


FIGURE 2.2.2(b) *Material selection criteria.*

$c = c_a$ at $t >> 0$

and x is the thickness parameter varying between 0 and h. This equation assumes that the moisture diffusion and temperature are constant inside the material. Because the rate of temperature diffusion is about 10^6 times faster than the moisture diffusion, and because diffusivity changes very little with moisture content, these assumptions are quite valid. If a further simplification of constant boundary conditions with time is made, solution to Equation 2.2.2(b) can be approximated (Reference 2.2.2(d)) as

$$M_{avg} = G(M_m - M_i) + M_i \qquad 2.2.2(c)$$

where

M_{avg} = moisture content of slab in percent by weight
M_m = moisture content at equilibrium for the boundary relative humidity
M_i = initial uniform moisture content (assumed to be zero)
G = time-dependent parameter defined below

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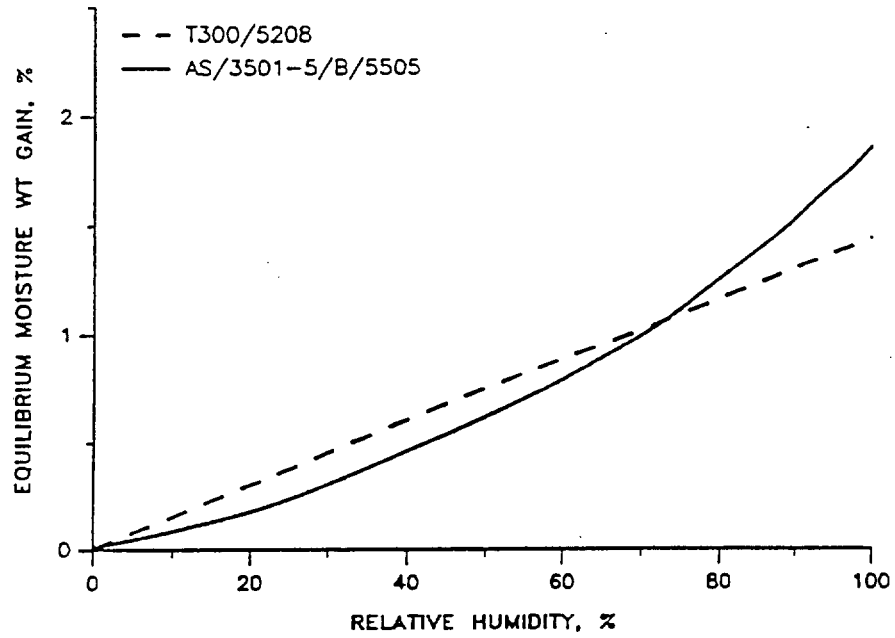


FIGURE 2.2.2(c) *Maximum moisture content as a function of relative humidity*

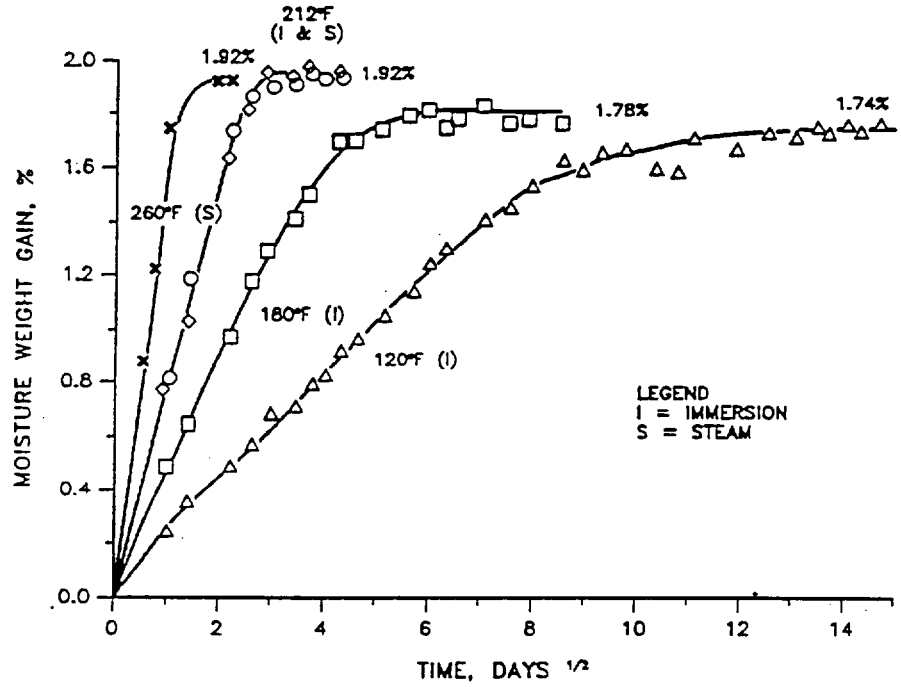


FIGURE 2.2.2(d) *Effect of temperature on moisture absorption rate in hybrid boron-graphite/epoxy (5505-AS/3501) laminate (3.0 x 0.5 x 12 in.; 76 x 13 x 3.0 mm) (Reference 2.2.2(b)).*

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The term G is a series solution parameter which can be adequately evaluated over the range of interest by the following approximation given in Reference 2.2.2(d).

$$G = 1 - \exp\left[-7.300\left(\frac{D_0 t}{h^2}\right)^{0.75}\right] \qquad 2.2.2(d)$$

where

h = slab thickness exposed on both sides to the same environment
t = exposure time

M can be obtained from Figure 2.2.2(c).

For time variable boundary conditions, the finite difference method (Reference 2.2.2(c)) can be used to numerically integrate Equation 2.2.2(b) with respect to time.

In the design of a composite structure, the effects of moisture are accounted for in the design allowables. The end-of-life moisture content (usually defined as the design service moisture content) is either calculated using Equation 2.2.2(b) or established by previous experimental or flight data. After the end-of-life moisture content is determined, mechanical property tests are performed with specimens preconditioned to that moisture level. The conditioning procedure to be performed is described in the next section.

Before end-of-life moisture contents can be calculated, environmental definition of component usage must be determined. An example is illustrated in Reference 2.2.2(d), where worldwide weather data and USAF aircraft basing data were gathered to define runway storage environmental spectra for each of the three classes of Air Force vehicles: fighters, bombers, and cargo/tankers. A ranking procedure was developed and applied to select baseline and worst-case locations with respect to the absorption of moisture by carbon/epoxy composite structures. A guide to environmental exposure parameters is provided in MIL-STD-810E (Reference 2.2.2(e)).

2.2.3 Conditioning of samples. To obtain test data for mechanical properties of wet material, specimens must be environmentally conditioned prior to mechanical testing. The goal for this procedure is a uniform moisture content throughout the thickness of the specimen. Uniformity in moisture content is achieved when the specimens reach an equilibrium moisture content. This moisture content is different at different humidity levels but is not, in general, a function of temperature, if the conditioning temperature is below the glass transition temperature of the material. The humidity level that is chosen depends on the application, the extreme being water immersion (100% RH). An 85% exposure would be more representative of worst usage conditions for aircraft based in the tropics. Because the moisture absorption rate is so strongly dependent on temperature, there is a tendency to increase conditioning temperatures to save time. However, long exposures to high temperatures combined with moisture are deleterious. It is essential that realistic exposure conditions be employed in conditioning samples such that moisture diffusion does not deviate significantly from Fickian behavior. Conditioning temperature and humidity specifications are material dependent. For epoxy matrices, the conditioning temperature should be kept below 170°F (77°C) for a 350°F (177°C) cure material and below 155°F (68°C) for a 250°F (121°C) cure material.

In addition to specifying conditioning temperature and RH, time of exposure has to be specified. The best way to specify conditioning is to require specimens to reach equilibrium, i.e. the maximum

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moisture content (MC) by weight that can be achieved at the given humidity level. A shortcut prevalent in the industry is the specification of a time period, e.g. thirty days. The drawbacks of this approach are illustrated in Figure 2.2.3 where, using the absorption rate and equilibrium level for IM6/3501-6 carbon/epoxy, the maximum moisture content has been calculated for various composite laminate thicknesses exposed for 30 days at 95% relative humidity and 140°F (60°C). The required 1.2% MC (equilibrium at 78% RH) by weight can be obtained only if the laminate thickness is approximately 0.07 in. (1.8 mm). For thicknesses larger than 0.07 in. (1.8 mm), the MC will be lower than desired and, for thicknesses smaller than 0.07 in. (1.8 mm), the MC will be higher than required. Furthermore, the moisture distribution across the thickness will be non-uniform for all thicknesses greater than 0.035 in. (0.9 mm). Because a typical test matrix includes specimens with different thicknesses, different exposure times should be specified. Figure 2.2.3 was generated for a known material; such information may not be readily available for a new material system. Specifying final MC instead of time of exposure implies knowledge of the material properties at the resin and void content of the laminate.

Two-step conditioning schemes consisting of 95% RH exposure followed by a lower RH exposure can also be used. These save time and are perfectly acceptable if the last step is exposure at the required RH until equilibrium.

Moisture content measurements are taken by weighing either the actual specimens or travellers that have been cut from the same panel and conditioned at the same time as the specimens. To minimize ingress of moisture through the edges, the thickness of the travellers must be small in comparison to the length and width dimensions; a ratio of 1:10 is usually adequate. For greater thicknesses, the edges may have to be sealed. Because the weight gain is approximately 1%, measurements must be taken on a fine balance to an accuracy of 0.0001 g. If the specimens have fiberglass tabs, adhesive materials, etc., travellers must be used, unless the equilibrium moisture content at a given temperature and humidity is specified, and even in this case the specimen may be too large or heavy to be conveniently weighed.

Testing, whether screening or qualification, is performed on specimens either in an unconditioned ("dry") or moisture-conditioned state. "Dry" is defined as the as-manufactured condition. Moisture conditioning is accomplished by 1) inserting the specimens in the chamber at the specified humidity level until equilibrium, or 2) in cases where absolute moisture content is required to be known, the drying procedure below should be used. The absolute moisture content can also be obtained using the first method by drying the specimens after moisture conditioning.

Both specimens and travellers should be dried in an oven prior to any additional conditioning. Care should be taken to avoid excessively high drying temperatures and high thermal excursions which may induce thermal cracking in specimens. For some but not all epoxy matrices, oven drying at temperatures up to 250°F (121°C) for 350°F (177°C) cure materials and 200°F (93°C) for 250°F (121°C) cure materials are often employed. Since the time from cure to completion of specimen fabrication may be long, a required time of drying cannot be specified. Periodic readings should be taken. In general, the time period between readings starts at twenty-four hours. Readings may be further apart towards the end of conditioning. For more information, see ASTM Standard Test Method for Moisture Absorption Properties and Equuilibrium Conditioning of Polymer-Matrix Composite Materials (Reference 2.2.3(a)). When three consecutive readings are within 0.01% weight change, the travellers and specimens are considered to be dry. The dry weight should be obtained at this time. The specimens and travellers should then be stored in plastic bags with a desiccator. From this time on, the travellers must accompany the specimens throughout the conditioning history. For travellers and specimens that are conditioned until equilibrium, equilibrium is defined as the point when three consecutive weighings give weight changes that average less than 1% of the total increase in weight.

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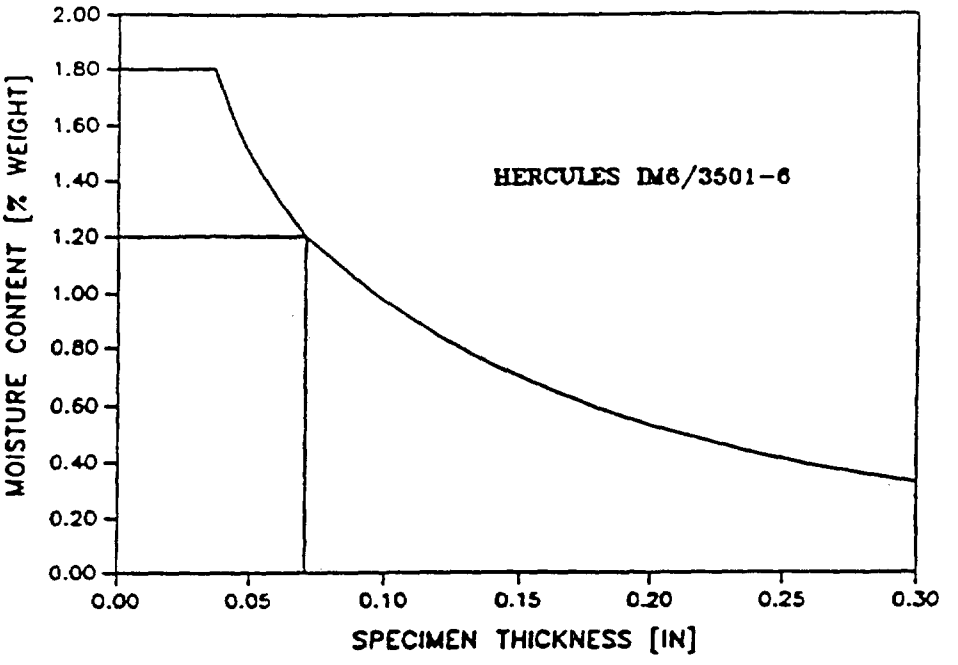


FIGURE 2.2.3 *Moisture absorption after 30 days at 95% RH and 140°F (60°C) for carbon/epoxy laminates with both sides exposed.*

ASTM Standard Test Method D 570-81 "Water Absorption of Plastics" and ASTM D 618-61 "Standard Methods of Conditioning Plastics and Electrical Insulating Materials for Testing" provide guidance for test procedures (References 2.2.3(b) and (c)).

The specimens or travellers should also be weighed just before and after mechanical testing, especially for elevated temperatures. At these temperatures, it is also important not to soak the specimens longer than needed so as not to drive-out moisture. A typical soak time after the thermocouples on the specimens reach the desired temperature is half a minute for carbon/epoxy for temperatures up to 350°F (177°C). Because thick laminates would require unacceptably long times to reach equilibrium moisture levels, thin laminates are usually used in moisture conditioning testing. The diffusion properties and equilibrium moisture content obtained from thin laminates are assumed to be applicable to thick laminates unless test data or calculations show otherwise.

2.2.4 Statistical development of mechanical properties. Data interpretation and statistical analysis procedures are defined in Chapter 8.

2.2.5 Data pooling requirements. The capability of pooling data may be desirable in order to obtain sufficient data to calculate B-basis values for material properties. Data to be pooled may be available for materials from different composite component manufacturers, different locations of one manufacturer, or slightly different processes from the same manufacturer. All properties should be examined for batch-to-batch variability (Section 8.5.3.2). A decision regarding the pooling of the data will be made in discussions with the certifying agency or, for MIL-HDBK-17 data, by the Data Review working group.

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2.3 MATERIAL ACQUISITION AND PREPREG PHYSICAL PROPERTY CHARACTERIZATION. For material property characterization suitable for inclusion in MIL-HDBK-17, five prepreg batches of each material type shall be prepared by the material supplier using prepreg production facilities. The five prepreg batches shall be made with five different fiber lots impregnated with five different production resin batches. The contractor shall prepare and coordinate a preliminary material specification with the supplier prior to manufacture of the prepreg batches. The preliminary material specification shall define agreed-to laminate and specimen fabrication procedures, test procedures, and physical and mechanical property requirements.

The five-batch requirement only applies to material properties that are to be incorporated in MIL-HDBK-17. An alternate number of tests and batches may be employed upon approval of the certifying agency. However, mechanical property test data should be examined by the recommended statistical methods in this document (MIL-HDBK-17) to ascertain that they are statistically acceptable.

Recommended tests to be performed on the prepreg materials are shown in Table 2.3. When applicable, the tests should be standard ASTM procedures.

2.4 LAMINA PHYSICAL AND MECHANICAL PROPERTY TESTS. A minimum of two test panels for each prepreg batch shall be cured in two separate autoclave runs to be used in fabricating test coupons. The prepreg material shall be laid up, bagged, and cured in accordance with a process specification prepared by the contractor. Cured test panels shall be non-destructively evaluated (NDE) using ultrasonic inspection. Imperfections in the test panels shall be reported with questionable areas marked on the panels. Test coupons shall not be extracted from these areas.

Suggested physical and mechanical property tests are shown in Tables 2.4(a) and 2.4(b). A test procedures document should be prepared by the contractor prior to the start of testing. It is recommended that, whenever possible, the tests should be ASTM standard methods.

The test matrix shown in Table 2.4(b) is based on thirty tests per data point (six tests per batch) to provide for parametric/nonparametric analysis in determining B-basis properties. Fewer tests per data point may be acceptable for non-critical properties as agreed to by the prime contractor and the certifying agency. For generating B-basis values, 0° and 90° tension, 0° and 90° compression, and in-plane shear tests are required. Short beam shear tests are required for determining process control test minimum values.

Extensometers or strain gages should be used to obtain strain measurements for each specimen. Two of each group of six compression specimens shall be strain gaged back-to-back with axial gages to assess possible premature buckling failure.

2.5 FILAMENT-WOUND MATERIALS PROPERTY TESTS.

2.6 BRAIDED MATERIALS PROPERTY TESTS.

2.7 THICK-SECTION COMPOSITES PROPERTY TESTS.

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TABLE 2.3 *Recommended physical and chemical property tests to be performed by material supplier and prime contractor.*

Test Property	Suggested Test Procedure[1]	Number of Tests per Batch[2]	Total Number of Tests
Resin Content, %	D 3529	3	15
Volatile Content, %	D 3530	3	15
Gel Time, min.	D 3532	3	15
Resin Flow, %	D 3531	3	15
Fiber Areal Wt., gm/M	†	3	15
Moisture Content, %	†	3	15
Tack	†	3	15
HPLC (High Performance Liquid Chromatography)	†	3	15
IR (Infrared Spectroscopy)	†	3	15
DMA (Dynamic Mechanical Analysis, neat resin only)	†	3	15
DSC (Differential Scanning Calorimetry)	†	3	15
RDS (Rheological Dynamic Spectroscopy)	†	3	15

[1]Test procedures shall be coordinated and agreed to prior to manufacture of prepreg material.
[2]Tests shall be performed on each of the five batches of prepreg material.
†Test procedures to be described at a later date.

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TABLE 2.4(a) *Cured lamina physical property tests.*

Physical Property	Suggested Test Procedure	Number of Tests Per Prepreg Batch[1]	Total Number of Tests
Fiber Volume, %	D 3171	3	15
Resin Volume, %	D 3171	3	15
Density, gm/cc	D 792	3	15
Measurements of per ply thickness	-	10	50
Glass Transition Temperature, °F (dry)[2]	-	3	15
Glass Transition Temperature, °F (wet)[2]	-	3	15

[1] Tests shall be performed on each of the five batches.

[2] Dry specimens are "as fabricated" specimens which have been maintained at ambient conditions in an environmentally-controlled test laboratory. Wet specimens are environmentally conditioned by exposing them in an elevated temperature humidity chamber until they attain an equilibrium moisture content agreed to by the contractor and customer, then packaged in a heat-sealed aluminized polyethylene bag until required for test. Tests shall be performed in a manner which maintains the moisture content in specimens at the levels agreed to by the contractor and certifying agency.

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TABLE 2.4(b) *Cured lamina mechanical property tests.*

Mechanical Property	Suggested Test Procedure[1]	Test Condition and Number of Tests Per Batch[2]			Number of Tests
Condition[3]		Min. Temp Dry	RT Dry	Max. Temp Wet	
0° Tension (warp)	D 3039	6	6	6	90
90° Tension (fill)	D 3039	6	6	6	90
0° Compression (warp)	D 3410	6	6	6	90
90° Compression (fill)	D 3410	6	6	6	90
In-plane Shear	D 3518-Tape	6	6	6	90
0° Short Beam Shear	D 2344[4]	-	6	-	30
					480

[1] Reader is referred to Section 6.6 Mechanical Property Tests for more information on these ASTM test methods.

[2] Tests shall be performed on each of the five batches.

[3] Minimum and maximum temperature tests shall be performed within ±5°F (±2.8°C) of the nominal test temperature. Nominal test temperatures will be as agreed to by contractor and certifying agency. Dry specimens are "as- fabricated" specimens which have been maintained at ambient conditions in an environmentally-controlled test laboratory. Wet specimens are environmentally-conditioned by exposing them in a humidity chamber until they attain an equilibrium moisture content agreed to by the contractor and certifying agency, and then packaging them in a heat-sealed aluminized polyethylene bag until required for test. Tests shall be performed in a manner which maintains the moisture content in specimens at the levels agreed to by the contractor and certifying agency.

[4] ASTM D 2344 is for quality control purposes only.

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2.8 OTHER USEFUL TEST MATRICES.

2.8.1 Material system screening. The objective of this screening process is to reveal key mechanical property attributes and/or inadequacies in new material system candidates through use of a reduced test matrix. The screening process will identify the critical test and environmental conditions, and any special considerations for a particular composite material system. Proper test selection and environmental conditioning will enable comparisons with current productions systems.

Table 2.8.1 shows a typical mechanical property screening test matrix primarily intended for epoxy resin systems. The test matrix will vary depending on the purpose of the investigation. The 0° axial tensile tests examine fiber dominant properties, and 0° axial compression tests monitor matrix/fiber interactions. The 0° axial tensile tests may reveal matrix-fiber interaction in some materials at high strain rates. These tests will yield static strength and stiffness properties. To test matrix characteristics, ±45° tensile tests will be used. The ±45° tensile tests permit actual shear modulus and effective shear strength values to be obtained. Finally, damage tolerance will be evaluated using impact testing.

Testing will be conducted under three environmental conditions: cold temperature dry (CTD), room temperature ambient (RTA), and elevated temperature wet (ETW). These conditions are selected based on results for the epoxy resin systems currently in use which show that the CTD condition is a fiber critical environmental state and the ETW condition is the most severe test for the matrix. Moisture conditioning may be accomplished by placing the dried test specimens in an environmental chamber until an equilibrium moisture weight gain at the desired relative humidity is obtained.

TABLE 2.8.1 *Composite material static strength screening test matrix.*

Test	Number of Specimens			Evaluation Emphasis
	CTD	RTA	ETW	
Lamina:				
0° Tension	3	3	3	fiber
0° Compression		3	3	fiber/matrix
±45° Tension		3		fiber/matrix (0°/90° shear - lamina) (±45° - laminate)
Laminate:				
Open Hole Compression[1]		3		stress riser
Open Hole Tension[1]		3		stress riser
Bolt-Bearing[1]		3		bearing
Compression after Impact[2]		3		impact damage

[1]Fastener hole effects
[2]per NASA Reference Publication 1092.

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The screening process presented allows key mechanical properties and environmental conditions to be evaluated. Dependent upon application, sensitivity to exposures to operational fluids and other special issues, additional tests may also be included in the screening evaluation. However, a reduced test matrix for mean property evaluations should permit preliminary efficient and economical assessment of new composite material systems.

The approach to screening is the performance of key static tests to provide sufficient data for mean property evaluations of stiffness and strength. The recommended specimens provide data at the lamina (ply) and laminate (application) levels. The lamina level tests provide key generic material stiffness and strength properties commonly used in classical lamination plate point stress analysis including tension, compression, and shear loadings. The laminate level tests provide screening strength data on application issues relating to effects of stress concentrations, manufactured stress risers (fastener holes), bolt bearing, or impact damage. Additional laminate level tests provide screening stiffness data to verify the use of lamina data with classical lamination plate theory for laminate stiffness predictions. The damage tolerance screening is patterned after NASA Reference Publication 1092 (Reference 2.8.1).

All laminate level tests suggested are performed at the RTD test condition. For screening purposes, it is assumed the lamina (ply) level unnotched specimens will conservatively provide mean data to assess environmental effects on strength and stiffness. The laminate level tests will provide mean data on the effects of stress concentrations.

2.8.2 Qualification guidelines and requirements for alternate composite materials.

2.8.2.1 *Introduction.* These guidelines are intended to apply to the situation where one composite material system, from a single supplier, has been qualified and it is necessary or desirable to qualify an alternate system and/or supplier. The approach assumes the existence of a body of data and experience developed with the original material (none exists for the alternate system) from which the mechanical property allowables have been developed. It also assumes higher level tests have been performed to qualify a product and verify its performance.

A drastic change, such as switching fiber from E-Glass to aramid, is not covered by this guideline. The focus is on materials that will meet the original material specification. A fiber class change, or comparable substitution, is considered a major revision or redesign. Processing and tooling changes are also considered beyond the scope of this section.

2.8.2.2 *Goal.* The ultimate goal in qualifying an alternate material is to be able to exchange this material with the original system without compromising manufacturing or structural performance. To accomplish this goal it is necessary to define the key material parameters that govern performance during specific phases, such as processing, manufacturing, and service. The ideal is to perform this evaluation at the material constituent level, or composite lamina level; i.e., measure and compare the chemical composition, fiber strength, matrix strength, and composite strength. This may be possible in the future, but is not adequate with current technology.

Successful qualification of an alternate material will not, in itself, be sufficient to permit mixing of this system with the original material in a given part. Intermixing is prohibited unless appropriate evaluations are carried out to demonstrate compatibility.

The focus of MIL-HDBK-17 is "B" basis lamina properties. Adequate alternate material qualification may require going beyond this level of evaluation into more complex demonstrations involving analysis

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and tests. These may include laminate, coupon, element and subcomponent tests such as open hole, filled hole, bolt bearing, low velocity impact, fatigue, and panel buckling.

2.8.2.3 *Material compatibility.* The extent to which an alternate material used in hardware applications must be evaluated to demonstrate equivalence, or superiority, with the original system is first, a function of its material compatibility and second, a function of hardware structural complexity and loading. Material compatibility is defined by the criteria shown in Table 2.8.2.3. The baseline system is a material from a single prepregger and a specific production line at that prepregger. For example, AS4/3501-6 produced from line 3 at Hercules, Inc. The most compatible alternate material, and the one requiring the minimum to demonstrate equivalence would be AS4/3501-6 produced from line 4 at Hercules, Inc. The least compatible material system would one from a different prepregger with a different resin and fiber. Thus, Fiberite C12K/934 is the least compatible system and requires the most effort to demonstrate equivalence. Situations not included in Table 2.8.2.3 must be evaluated with respect to their appropriate compatibility scale.

2.8.2.4 *Key material or structural performance parameters.* Key material or structural performance parameters are those measurable quantities which, if compared to the original values, can be used to quantify any difference in manufacturing or structural performance parameters, are material and hardware dependent, and may change with design, tooling, manufacturing, and usage factors. However, five categories of parameters have been defined in Table 2.8.2.4. This table lists examples of typical performance parameters appropriate for each category.

2.8.2.5 *Success criteria.* The relative importance and completeness of performance parameters varies with the part design, loading, and application. In some cases it is sufficient merely to report a measured value. In other cases the value must meet or exceed the original measurement. And in some cases the value must not vary significantly either higher or lower than the original value. As an example, this is generally true for modulus, fiber areal weight, resin content, and per-ply thickness.

Success criteria for each parameter must be defined at the beginning of the qualification program. Justification for each success criteria imposed must be provided. A tolerance on a given measurement should be part of the success criteria.

2.8.2.6 *General approach.* The general approach to be followed for qualification of an alternate material is as follows:

1. Identify the key material performance parameters and why they are crucial.
2. Define appropriate tests, measurements, or evaluations for each of the parameters. These must correspond exactly to the tests, measurements, or evaluations performed on the original material (i.e., same specimen type, same conditions, etc).
3. Define pass/fail (success) criteria for the tests, measurements, and evaluations.
4. Prepare test plan and obtain necessary approvals.
5. Perform tests, etc., and document results.
6. Accept or reject.

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Table 2.8.2.3 *Material compatibility criteria.*

MOST COMPATIBLE							LEAST COMPATIBLE	
MATERIAL FACTOR	1	2	3	4a	4b	5a	5b	6
Fiber Type	Y	Y	Y	N	Y	N	Y	N
Fiber Tow Size	Y	N	Y	Y/N	Y	Y/N	Y	N
Resin	Y	Y	Y	Y	N	Y	N	N
Prepregger	Y	Y	N	Y	Y	N	N	N
Production Line	N	Y	N	Y	Y	N	N	N

Y - remains the same in alternate material
N - changes in alternate material

a) Column 1 is a change in supplier production line. Often this is required for improved production rates to meet customer order requirements on schedule. This is the most compatible situation and would require the least effort to demonstrate acceptability.

b) Column 2 represents a change in fiber tow size; for example, from 6000 filament to 12000 filament tow. This is justified for economic reasons as the larger tow sizes are typically less expensive.

c) Column 3 is a change in prepreg supplier and production line. This situation is becoming more common today as resin systems are licensed between prepreg manufacturers; for example, the Navy's A-6 re-wing and V-22 Osprey programs where Hercules 3501-6 is licensed to ICI Fiberite. This cooperative licensing allows competitive bidding for prepreg supplies and provides the customer with nearly identical prepreg for production usage.

d) Column 4(a) represents a change in fiber type based on a new fiber with properties similar to the originally qualified fiber. This situation may occur for economic reasons or in the event of discontinued fiber supply.

e) Column 4(b) is a change in resin. This would be justified by development of new resin systems by the prepregger that would offer improved pricing and/or properties, such as damage tolerance, for the customer's program.

f) Column 5 represents a change in prepreg supplier, production line, and fiber or resin. This situation would occur when a customer needs and additional supplier but wishes to use the same fiber or resin due to second-source qualification budget constraints (assumes existing data base on the resin and/or fiber). Again, economic reasons justify this situation.

g) Column 6 involves qualifying a new source prepregger using a different fiber and resin system. An example of this situation is qualifying Fiberite C12K/934 to replace Hercules AS4/3501-6. This is the least compatible situation and would require the greatest effort to demonstrate acceptability.

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TABLE 2.8.2.4 *Examples of key material or structural performance parameters.*

PHYSICAL	PROCESSING	MECHANICAL	MANUFACTURING	HARDWARE SCALE-UP
TACK	PER-PLY-THICKNESS	LAMINA PROPERTIES	DRILLING	STATIC STRENGTH
RESIN CONTENT	CURE CYCLE	ENVIRONMENTAL EFFECTS	TOOLING	FATIGUE STRENGTH
AREAL WEIGHT	SENSITIVITY	DAMAGE TOLERANCE	NONDESTRUCTIVE INSPECTION	STIFFNESS
FLOW	FIBER VOLUME	INTERLAMINAR SHEAR	COST	FAILURE MODES
GLASS TRANSITION TEMPERATURE	THERMAL CYCLING	FLATWISE TENSION	LEAD TIME	QUALITY
FORM	DENSITY	FLAW GROWTH	AVAILABILITY	BEARING
OUT TIME	EXOTHERM	EFFECT OF DEFECTS	REPEATABILITY	CRIPPLING
SHELF LIFE	TOXICITY	PRESSURE BOTTLE TESTS	MACHINABILITY	OPEN HOLE TENSION
STORAGE REQUIREMENTS			UNIFORMITY	OPEN HOLE COMPRESSION
MOISTURE ABSORPTION				PANEL TESTS
SOLVENT RESISTANCE				FATIGUE TESTS

2.8.2.7 *Specific requirements for lamina mechanical property allowables.* MIL-HDBK-17 (Section 2.4) has defined minimum requirements to produce B-basis property values. These are six replicates of five batches of material for a total of thirty specimens at each point of interest. Three environments are recommended and five types of lamina property values. This produces a total of 450 tests. Since the alternate material qualification program is not intended to establish the design allowables, but rather to show compliance with the allowables, a reduced number of lamina tests is permitted. The total number depends on compatibility of the two material systems. The requirement for tape and fabric are shown in Tables 2.8.2.7(a) and (b) respectively. Appropriate statistical analysis must be performed to evaluate the test results. Tests shown in Tables 2.8.2.7(a) and 2.8.2.7(b) must be performed in the same way as tests performed on the original material.

2.8.2.8 *Suggested requirements for laminate mechanical properties.* The next higher level of testing that should be considered for qualification of an alternate material system is laminate mechanical properties. This level of testing confirms allowable stress (strain) values for strategic design should be the same as the laminate tested for the original material. The suggested tests are shown in Table 2.8.2.8(a). The extent to which the Table 2.8.2.8(a) tests are performed is governed by the material compatibility factor. The number of tests required is given in Table 2.8.2.8(b).

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Table 2.8.2.7(a) *Alternate material lamina test requirements - tape.*

No. of Batches								Replicates								Environments[3]								TOTAL							
COMPATIBILITY[1]								COMPATIBILITY[1]								COMPATIBILITY[1]								COMPATIBILITY[1]							
1[2]	2	3	4a	4b	5a	5b	6	1	2	3	4a	4b	5a	5b	6	1	2	3	4a	4b	5a	5b	6	1	2	3	4a	4b	5a	5b	6
1	2	2	3	3	3	3	3	4	4	4	4	4	5	5	6	2	2	2	2	2	2	2	2	8	16	16	24	24	30	30	36
-	-	2	3	3	3	3	3	-	-	4	4	4	5	5	6	-	-	2	2	2	2	2	2	-	-	16	24	24	30	30	36
1	2	2	3	3	3	3	3	4	4	4	4	4	5	5	6	2	2	2	2	2	2	2	2	8	16	16	24	24	30	30	36
-	-	2	3	3	3	3	3	4	4	4	4	4	5	5	6	-	-	2	2	2	2	2	2	-	-	16	24	24	30	30	36
1	2	2	3	3	3	3	3	4	4	4	4	4	5	5	6	2	2	2	2	2	2	2	2	8	16	16	24	24	30	30	36
																								24	48	80	120	120	150	150	180

[1] Compatibility is defined in Table 2.8.2.8

[2] If original material constituents are available (fiber lot, resin mix) do one batch on new line. If they are not available, produce two batches from a new fiber lot/resin mix, one on each line.

[3] The environments should be RTD and the worst case.

Quality assurance tests must be performed per individual specification.

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Table 2.8.2.7(b) *Alternate material lamina test requirements - fabric.*

No. of Batches								Replicates								Environments[3]								TOTAL							
COMPATIBILITY[1]								COMPATIBILITY[1]								COMPATIBILITY[1]								COMPATIBILITY[1]							
1[2]	2	3	4a	4b	5a	5b	6	1	2	3	4a	4b	5a	5b	6	1	2	3	4a	4b	5a	5b	6	1	2	3	4a	4b	5a	5b	6
1	1	2	3	3	3	3	3	4	4	4	4	4	5	5	6	2	2	2	2	2	2	2	2	8	8	16	24	24	30	30	36
-	1	-	3	3	3	3	3	-	4	-	4	4	5	5	6	-	2	-	2	2	2	2	2	-	8	-	24	24	30	30	36
1	1	2	3	3	3	3	3	4	4	4	4	4	5	5	6	2	2	2	2	2	2	2	2	8	8	16	24	24	30	30	36
-	1	-	3	3	3	3	3	-	4	-	4	4	5	5	6	-	2	-	2	2	2	2	2	-	8	16	24	24	30	30	36
1	1	2	3	3	3	3	3	4	4	4	4	4	5	5	6	2	2	2	2	2	2	2	2	8	8	16	24	24	30	30	36
																								24	32	48	120	120	150	150	180

[1] Compatibility is defined in Table 2.8.2.3.

[2] If original material constituents are available (fiber lot, resin mix) do one batch on new line. If they are not available, produce two batches from a new fiber lot/resin mix, one on each line.

[3] The environments should be RTD and the worst case.

Quality assurance tests must be performed per individual specification.

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TABLE 2.8.2.8(a) *Extent of laminate testing.*

Material Compatibility Factor	Laminate Tests	Total	
		Tape	Fabric
1	None Required	0	0
2	Unnotched Laminates	12	12
3	Unnotched Laminates	12	12
4	All Static Test, Two Environments	36	36
5	All Static Test, Two Environments	36	36
6	All Required	42	42

2.8.2.9 *Summary.* Many of the requirements defined in this guidelines document, such as physical and processing characteristics, are included in materials and processes specifications. Other requirements are more design related and may be complex to demonstrate. Yet, they could be of prime importance. All key performance parameters for a specific project or product must be considered.

Guidelines were given for satisfying lamina and laminate material property requirements. Higher level mechanical element/subcomponent tests may also be required depending on the extent of the material change and the key parameter definition. The approach to accomplishing these evaluations must be determined from the basic program.

The test plan document defines performance factors, methods of evaluation, and success criteria. It is prepared by the contractor and approved by the certifying agency. It is the focal point for understanding between the contractor and certifying agency.

2.8.2.10 *Statistical methods.*

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TABLE 2.8.2.8(b) *Number of suggested laminate tests.*

DESIGN PROPERTY	LOADING		NO. OF LAMINATE TYPES		NO. OF ENVIRONMENTS[1]	REPLICATES[2]	TOTAL NUMBER OF SPECIMENS	
	TENSION	COMPRESSION	TAPE	FABRIC			TAPE	FABRIC
Static								
Unnotched Laminate, strength and stiffness	X	X	1	1	2	3	12	12
Open Hole		X	1	1	2	3	6	6
Filled Hole	X		1	1	2	3	6	6
Impact Damage	X	X	1	1	1	3	6	6
Double Shear Bearing	X		1	1	1	3	3	3
Single Shear Bearing	X		1	1	1	3	3	3
							36	36
Fatigue[3]								
Open Hole	-	-	1	1	1	3	3	3
Impact Damage	-	-	1	1	1	3	3	3
							6	6
							42	42

[1] Where two environments are required, they should be RTD and worst case. Where one is required, it should be RTD.
[2] One batch of material is sufficient.
[3] Repeated load and residual strength: constant amplitude, R = -1, n = 1×10^6 cycles.

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Chapter 3. EVALUATION OF REINFORCEMENT FIBERS

Page

3.1 INTRODUCTION. 3-3

3.2 CHEMICAL TECHNIQUES. 3-3
3.2.1 Elemental analysis. 3-3
3.2.2 Titration. 3-4
3.2.3 Fiber structure. 3-4
3.2.4 Fiber surface chemistry. 3-4
3.2.5 Sizing content and composition. 3-7
3.2.6 Moisture content. 3-7
3.2.7 Thermal stability and oxidative resistance. 3-7
3.2.8 Chemical resistance. 3-7

3.3 PHYSICAL TECHNIQUES (INTRINSIC). 3-8
3.3.1 Filament diameter. 3-8
3.3.2 Density. 3-8
3.3.3 Electrical resistivity. 3-8
3.3.4 Coefficient of thermal expansion. 3-9
3.3.5 Thermal conductivity. 3-9
3.3.6 Specific heat. 3-9
3.3.7 Thermal transition temperatures. 3-9

3.4 PHYSICAL TECHNIQUES (EXTRINSIC). 3-10
3.4.1 Yield of yarn, strand, or roving. 3-10
3.4.2 Cross-sectional area of yarn or tow. 3-10
3.4.3 Twist of yarn. 3-10
3.4.4 Fabric construction. 3-10
3.4.5 Fabric areal density. 3-10

3.5 MECHANICAL PROPERTIES. 3-11
3.5.1 Tensile strength and Young's modulus. 3-11
3.5.2 Fiber strain and failure. 3-11
3.5.3 Transverse fiber modulus. 3-11
3.5.4 Poisson's ratio. 3-11
3.5.5 Shear moduli. 3-11
3.5.6 Interlaminar shear. 3-11

3.6 TEST METHODS. 3-12
3.6.1 Determination of pH. 3-12
3.6.1.1 Scope. 3-12
3.6.1.2 Apparatus. 3-12
3.6.1.3 Procedure. 3-12
3.6.2 Determination of amount of sizing on carbon fibers. 3-13
3.6.2.1 Scope. 3-13
3.6.2.2 Apparatus. 3-13

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:30Z
Check the source to verify that this is the current version before use.

3.6.2.3 Materials. 3-13
3.6.2.4 Procedure. 3-13
3.6.2.5 Calculation. 3-14
3.6.2.6 Preparation of crucibles for reuse. 3-14
3.6.3 Determination of moisture content or moisture regain. 3-15
3.6.3.1 Scope. 3-15
3.6.3.2 Apparatus. 3-15
3.6.3.3 Sample preparation. 3-15
3.6.3.4 Procedure. 3-15
3.6.3.5 Calculations. 3-16
3.6.4 Determination of fiber diameter. 3-17
3.6.4.1 Description and application. 3-17
3.6.4.2 Apparatus. 3-18
3.6.4.3 Calibration. 3-18
3.6.4.4 Prepare slide. 3-18
3.6.4.5 Measuring procedure. 3-18
3.6.4.6 Calculation. 3-19
3.6.5 Determination of electrical resistivity. 3-20
3.6.5.1 Scope. 3-20
3.6.5.2 Apparatus. 3-20
3.6.5.3 Sample preparation. 3-20
3.6.5.4 Procedure. 3-20
3.6.5.5 Calculation. 3-20
3.6.5.6 Calibration and maintenance. 3-20
3.6.5.7 Definition of units of measurement. 3-20
REFERENCES 3-22

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3.1 **INTRODUCTION.** This chapter describes techniques and test methods that are generally used to characterize the chemical, physical, and mechanical properties of reinforcement fibers for application in organic matrix composite materials. Reinforcements in the form of unidirectional yarns, strands, or tows, and bidirectional fabrics are covered. Sophisticated experimental techniques generally are required for fiber characterization, and test laboratories must be well-equipped and experienced for measuring fiber properties. It is also recognized that in many cases the measurement of a fiber property that manifests itself in the reinforced composite can best be accomplished with the composite. Sections 3.2 - 3.5 recommend general techniques and test methods for evaluating carbon, glass, organic (polymeric), and other specialty reinforcement fibers. Section 6 contains examples of test methods that can be used for evaluating fibers.

Most reinforcement fibers are surface treated or have a surface treatment (e.g., sizing) applied during their production to improve handleability and/or promote fiber-resin bonding. Surface treatments affect wettability of the fiber during impregnation as well as the dry strength and hydrolytic stability of the fiber-matrix bond during use. Because of the direct relation to composite properties, the effectiveness of any treatments to modify surface chemistry is generally measured on the composite itself by means of mechanical tests. The amount of sizing and its compositional consistency are significant in quality control of the fiber and measurement of these parameters is part of the fiber evaluation.

3.2 **CHEMICAL TECHNIQUES.** A wide variety of chemical and spectroscopic techniques and test methods are available to characterize the chemical structures and compositions of reinforcement fibers. Carbon fibers are found to range from 90-100% carbon. Typically, standard and intermediate modulus PAN carbon fibers are 90-95% carbon, with most of the remaining material being nitrogen. Minor constituents and trace elements can be extremely important when composites containing these fibers are considered for use at elevated temperatures (above 500°F or 260°C). Organic fibers usually contain significant amounts of hydrogen and one or more additional elements (e.g., oxygen, nitrogen, and sulfur) which can be identified by spectroscopic analysis. Glass fibers contain sulfur dioxide and usually aluminum and iron oxide. Depending upon the type of glass, calcium oxide, sodium oxide, and oxides of potassium, boron, barium, titanium, zirconium, sulfur, and arsenic may be found.

3.2.1 **Elemental analysis.** A variety of quantitative wet gravimetric and spectroscopic chemical analysis techniques may be applied to analyze the compositions and trace elements in fibers. ASTM Test Method C 169 may be used to determine the chemical compositions of borosilicate glass fibers (Reference 3.2.1(a)).

A suitable standardized method for carbon and hydrogen analysis, modified to handle carbon and polymeric fibers is provided by ASTM D 3178 (Reference 3.2.1(b)). Carbon and hydrogen concentrations are determined by burning a weighed quantity of sample in a closed system and fixing the products of combustion in an absorption train after complete oxidation and purification from interfering substances. Carbon and hydrogen concentrations are expressed as percentages of the total dry weight of the fiber. ASTM Method D 3174 (Reference 3.2.1(c) describes a related test in which metallic impurities may be determined by the analysis of ash residue.

Alternatively, a variety of commercial analytical instruments are available which can quickly analyze carbon, hydrogen, nitrogen, silicon, sodium, aluminum, calcium, magnesium and other elements in reinforcement fibers. X-ray fluorescence, atomic absorption (AA), flame emission, and inductively coupled plasma emission (ICAP) spectroscopic techniques may be employed for elemental analysis. Operating instructions and method details are available from the instrument manufacturers.

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Trace metallic constituents are significant in carbon and polymeric fibers because of their possible effect on the rate of fiber oxidation. The presence of metals is usually expressed as parts per million in the original dry fiber and can be determined by analyzing the ash residue. Semi-quantitative determinations are generally made using flame emission spectroscopy. When quantitative values are desired, atomic absorption methods are used. With respect to oxidation of carbon fibers, sodium is usually of most concern because of its tendency to catalyze the oxidation of carbon.

3.2.2 **Titration.** The potential chemical activity of surface groups on fibers may be determined by titration techniques. For example, the relative concentration of hydrolyzable groups introduced during the manufacture or post treatment of carbon fibers may be determined by measuring the pH (section 3.6.1). However, titration techniques are typically not used on commercial carbon fibers due to the low levels of surface functionality.

3.2.3 **Fiber structure.** X-Ray diffraction spectroscopy may be used to characterize the overall structure of crystalline or semi-crystalline fibers. The degree of crystallinity and orientation of crystallites have a direct effect on the modulus and other critical properties of carbon and polymeric fibers.

X-ray powder diffraction using commercial power supplies and diffractometer units is used to characterize the structure of carbon fibers. The fiber is ground into a fine powder and then the X-ray powder diffraction pattern is taken using CuK radiation. The patterns generally undergo computer analysis to determine the following parameters:

(a) Average graphite layer spacing: from the 002 peak position.
(b) Average crystal size L_c: from the 002 peak width
(c) Average crystal size L_a: from the 100 peak width.
(d) Average lattice dimension a-axis: from the 100 peak position.
(e) The ratio of peak area to the diffused area.
(f) The 002 peak area to the total diffraction area.
(g) The 100 peak area to the total diffraction area.
(h) The ratio of the 100 to 002 peak areas.
(i) Crystallinity index: from a comparison of the X-ray diffraction of known crystallized and amorphous carbons.

X-ray scattering of crystalline fibrous materials shows the presence of sharp and diffuse diffraction patterns which are indicative of crystal phases interdispersed with amorphous regions. The concept of the crystallinity index is derived from the fact that a portion of the scattering from a fiber is diffuse and thereby contributes to the so-called amorphous background. Thus, a simple method of estimating crystallinity is obtained by separating the diffraction pattern into crystalline (sharp) and amorphous (diffuse) components. The crystallinity index is a relative measure of crystallinity, and not an absolute numerical result, useful for correlating with physical properties of fibers.

Wide angle X-ray spectroscopy and infrared spectroscopy techniques have also been developed to determine the crystallinity and orientation of molecules in polymeric fibers. Testing and interpretation of results requires specialized equipment, sophisticated computer models, and a high level of technical expertise.

3.2.4 **Fiber surface chemistry.** Fibers generally are given a surface treatment to improve the adhesion between the fibers and resin matrix materials. Gases, plasmas, liquid chemical or electrolytic treatments are employed to modify fiber surfaces. Introducing surface oxidation is perhaps the most common approach to modifying fiber surfaces.

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Fiber surface structure, the modifications which surfaces undergo as a result of the different fiber surface treatments, and the relative importance of these modifications for composite properties are not well understood. This arises because of the small surface areas involved (0.5 to 1.5 m^2/g) and the very low concentrations of functional groups. If 20% of the surface was covered by one particular species, this would only amount to 1 μmole of chemical groups per gram of fiber. Surface characterization should be carried out on fibers which have not been sized. Residual size from solvent desized fiber can interfere with most techniques, while pyrolysis techniques may alter the fiber surface due to oxidation and char products.

The following techniques have been used for characterizing fiber surfaces:

(a) X-ray diffraction - provides information relating to crystallite size and orientation, degree of graphitization, and micropore characteristics.

(b) Electron diffraction - gives crystallite orientation, three-dimensional order, and degree of graphitization. (better for surfaces since penetration is only 1000Å).

(c) Transmission Electron Microscopy (TEM) - provides the highest resolution of any of the microscopic techniques routinely available. Ultramicrotomy can be used to prepare specimens, typically about 50 nanometers thick, for direct TEM analysis of the fiber surfaces. TEM provides information about surface fine structure and show fibrils and needle-like pores.

(d) Scanning Electron Microscopy (SEM) - Gives structural and surface features. SEM is a useful technique for determining fiber diameters and identifying morphological characteristics (scales, chips, deposits, pits) on fiber surfaces.

(e) Electron Spin Resonance (ESR) Spectroscopy - gives crystallite orientation.

(f) X-ray Photoelectron Spectroscopy (XPS) or Electron Spectroscopy for Chemical Analysis (ESCA) - measures the binding energy of core electrons in atoms excited by low energy X-rays. Changes in the chemical environment of a surface region 10-15 nanometers thick (the first few atomic layers) are revealed by slight shifts in the energy of these core electrons giving information on functional group types and concentrations. The surface sensitivity arises because the depth of the electrons is between 1 and 2 nanometers.

The ratios of total oxygen to total carbon and of oxidized carbon (including hydroxyl, ether, ester, carbonyl and carboxy functional groups) to total carbon may be determined in carbon fibers using XPS or ESCA.

(g) Auger Electron Spectroscopy (AES) - directs high energy electrons (1-5 KeV) onto surfaces to create vacancies in the core levels of atoms. These vacancies represent excited ions which may undergo de-excitation and thereby create Auger electrons. By analyzing the characteristic energies of all the back-scattered Auger electrons in the energy range 0-1 KeV, the elemental composition of the first 30 or 40 atomic layers (about 30 nanometers) is possible and in some cases molecular information can be obtained from analysis of data.

(h) Ion Scattering Spectroscopy (ISS) - uses an ion as a molecular probe to identify elements on the outermost surface layer. Only atomic information can be obtained and sensitivity depends upon the atomic element.

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(i) Secondary Ion Mass Spectroscopy (SIMS) - uses a controlled sputtering process with accelerated ions to remove surface atomic layers for direct analysis by mass spectroscopy. SIMS can be used to identify surface molecules and determine their concentrations.

(j) Infrared Spectroscopy (IRS) or Fourier Transform IRS (FTIRS) - absorption vibrational spectroscopy technique to obtain molecular information about surface composition. IRS yields both qualitative and quantitative information relating to the chemical composition of surface molecules. The quality of the IR analysis depends on the fiber composition and is directly related to the care taken during sample preparation.

For fibers with diameters between 0.015 and 0.03 mm, no sample preparation is required if an IR microscope is available to examine fibers directly. Organic fibers may be pressed (up to $1000/m^2$) into a film of fiber grids.

(k) Laser Raman spectroscopy - absorption/vibrational spectroscopic technique which complements IR and is relatively simple to apply. Little or no sample preparation is necessary. Fibers can be oriented in the path of the incident beam for direct analysis. Fiber sample must be stable to the high intensity incident light and should not contain species that fluoresce.

(l) Contact angle and wetting measurements - provide an indirect measurement of fiber surface free energy for use in predicting interfacial compatibility and thermodynamic equilibrium with matrix materials. Contact angle and wetting measurement information can be obtained by direct measurement of contact angle, mass pick-up, or surface velocity. Measurement of contact angles on small diameter fibers (< 10 microns) is difficult if done optically. If a fiber's dimensions are known, a simple force balance may be used to determine the contact angle by measuring the force induced by immersing the fiber into a liquid of known surface free energy. The apparatus usually employed for this test is the Wilhelmy balance (Reference 3.2.4(a)).

Contact angles θ also may be measured indirectly by the micro-Wilhelmy technique (References 3.2.4(b-e)). A single fiber is partially immersed in a liquid and the force exerted on the fiber due to the surface tension of the liquid is measured. The contact angle is determined from the relationship $F = C\gamma_{LV}\cos\theta$ where F is the force measured corrected for buoyancy, C is the circumference of the fiber, and γ_{LV} is the surface tension of the liquid. The results may be used to determine the fiber surface free energy and the contributions of polar and dispersive components to the free energy (References 3.2.4(c) and (d)).

(m) Physisorption and chemisorption measurements - adsorption of inert gas or organic molecules can be used to measure fiber surface area. To obtain accurate estimates of surface area, it is important that there is complete monolayer coverage of the surface, that the area occupied by the adsorbed gas is known and that significant amounts of the gas are not taken up in micropores. Additional complications arise when the adsorption of organic molecules is used in place of gas adsorption, since it may be necessary to know the orientation of the adsorbed molecules to calculate surface area. Adsorption may also occur only at specific active sites and, if solutions are used, solvent molecules may be adsorbed as well.

The chemical reactivity of fiber surfaces can be determined by oxygen chemisorption and desorption measurements. Topographical changes (e.g., pores, cracks and fissures) caused by surface treatments often can be readily detected by adsorption measurements. Flow microcalorimetry is a useful technique for directly measuring heats of adsorption (Reference 3.2.4(f)).

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(n) Thermal desorption measurements - desorption of volatile products from fibers by heat treatment in vacuo. Thermogravimetric analysis (TGA), gas chromatography (GC), mass spectroscopy (MS), infrared spectroscopy (IRS) analysis or combinations of pyrolysis GC/MS or TGA/IRS may be used to identify components desorbed from fiber surfaces. Below 150°C, CO_2, NH_3, CH and various organic molecules are observed depending upon the fiber type.

(o) Chemical identification of functional groups by titrimetric, coulometric and radiographic techniques.

3.2.5 Sizing content and composition. The amount of sizing contained on fibers is expressed as a percentage of the dry sized fiber weight. It is generally determined by extracting the fibers with a heated solvent; then the cleaned fibers are washed, dried, and weighed. ASTM Test Method C 613 (Reference 3.2.5) describes a suitable method utilizing Soxhlet extraction equipment; however, similar extractions using a laboratory hot plate and beaker are also common. The selection of a solvent which quantitatively removes all the sizing but not does dissolve the fiber is essential for accuracy in this determination.

Thermal removal techniques are also utilized and are most practical for the more difficult soluble sizings. Time, temperature, and atmosphere conditions must be predetermined to ensure the sizing is removed with out seriously affecting the fiber. The precise amounts of residue from decomposition of the sizing and weight loss of the fibers due to oxidation must also be known from control tests for greatest accuracy. SACMA recommended test method SRM 14-90 "Determination of Sizing Content on Carbon Fibers" describes a pyrolysis technique for carbon fibers.

Sizing compositions and lot-to-lot chemical consistency may be determined by spectroscopic and chromatographic analysis of materials isolated by extracting the fibers with a suitable solvent. Acetone, tetrahydrofuran and methylene chloride are commonly used solvents for extraction. Liquid and gas chromatography and diffuse infrared spectroscopy are used to analyze or "fingerprint" the chemical compositions of extracts.

3.2.6 Moisture content. The moisture content or moisture regain of fibers or textiles may be determined using the procedure shown in Section 3.6.3. Care must be taken when applying the procedure since volatile materials in addition to moisture may be removed. If possible, tests should be performed on fibers that have not been sized. Moisture content is expressed as weight percentage moisture based upon the dry weight of the specimen.

3.2.7 Thermal stability and oxidative resistance. The susceptibility of fibers and fiber surface to oxidation is measured as weight loss under given conditions of time, temperature, and atmosphere. This is especially important in the evaluation of carbon and organic fibers considered for use in plastics exposed to elevated temperatures since it contributes to the long term degradation of composite properties. Thermogravimetric analysis (TGA) may be used to determine the thermal decomposition temperature T_d of carbon and organic fibers and estimate the relative amounts of volatile, organic additives and inorganic residues.

A standard method for determining the weight loss of carbon fibers is given in ASTM Test Method D 4102 (Reference 3.2.7(a)). Variations in this test method regarding exposure of fibers have been studied and give similar results (Reference 3.2.7(b)). In order to minimize variability in test results, proper control of gas flow rates and currents is critical when performing TGA analyses.

3.2.8 Chemical resistance.

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3.3 PHYSICAL TECHNIQUES (INTRINSIC). The physical properties of fibers of importance in their applications in polymer matrix composites fall into two categories - those inherent in the filament itself (intrinsic), and those derived from the construction of filaments into yarns, tows, or fabrics (extrinsic). The former includes density, diameter, and electrical resistivity; the latter includes yield, cross-sectional area, twist, fabric construction and areal weight. Density and the derived properties are used in the calculations required for the construction and analysis of the composite products. Density and yield are useful measures of quality assurance. Filament diameter and electrical resistivity are important for the nonstructural aspects of aerospace and aircraft applications.

3.3.1 Filament diameter. The average diameter of fibers may be determined by using an indexing microscope fitted with an image splitting eyepiece or from a photomicrograph of the cross-sectional view of a group of mounted fibers. Since fibers are not always true cylinders, effective diameters may be calculated from the total cross-sectional area of the yarn or tow and dividing by the number of filaments in the bundle. The cross-sectional area may also be estimated from the ratio of mass per unit length to density. For irregular, but characteristically-shaped, fibers an area factor may be required in calculating the average fiber diameter.

Optical microscopy can provide information about fiber diameter and variation in diameter with length. The upper limit of resolution of the optical microscope is about one-tenth of a micron; hence features less than one micron can not be well-characterized by optical microscopy. A detailed procedure for the determination of fiber diameter is described in Section 3.6.4.

Other techniques, such as scanning electron microscopy (SEM), provide much higher resolution than optical microscopy for determining fiber diameter and cross-sectional characteristics. Features of fiber surfaces down to the 5 nanometer level can be observed. In addition, the large depth of field provided by SEM helps defined three-dimensional characteristics on fiber surfaces and define fiber topography.

3.3.2 Density. ASTM Standard Test Method D3800 is recommended for determining the density of high modulus fibers (Reference 3.3.2). The sample is weighed in air and then reweighed immersed in a liquid. The liquid used must thoroughly wet the sample and be of lower density than the sample. The difference in weight is the buoyant force which is converted to sample volume by dividing by the liquid density. The density of the yarn is calculated by dividing the sample weight in air by the sample volume.

Alternatively, the density of fibers, yarns and fabrics may be determined by means of an air comparison pycnometer (e.g., the Model 930 Purgeable Air Comparison Pycnometer, Beckman Instr. Inc.).

3.3.3 Electrical resistivity. The determination of electrical resistivity is recommended as a control measure for checking processing temperature and to determine compliance with specific resistance specifications, where required. Electrical resistivity is one of the properties dramatically affected by the structural anisotropy of carbon fibers. Measurements can be made on either a single filament or a yarn. The measured value is resistance per given length of fiber as read on an ohm meter or similar device. The contact resistance can be eliminated by obtaining the resistance for two different lengths of fiber and calculating the difference due to the longer length. This difference is then converted to resistance per unit length and then multiplied by the area of the fiber or yarn bundle expressed in consistent units. Resistivity is expressed as ohm-centimeter, ohm-meter, or ohm-inches and refers to the value in the axial direction. Transverse resistivity is seldom reported. A procedure for determining the electrical resistance of carbon cloth or felt is described in Section 3.6.5.

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3.3.4 **Coefficient of thermal expansion.** Standardized methods for measuring the coefficient of thermal expansion (CTE) of the fibers are not generally available although good correlations between laboratories making these measurements do exist. CTE's are directionally dependent, highly influenced by the anisotropy of fibers. Carbon fibers typically have a negative axial CTE and a slightly positive transverse CTE. Commercial instruments (e.g., DuPont Model 943 Thermomechanical Analyzer, or equivalent) can be used directly or modified to measure axial CTE.

The CTE of the fiber can also be derived from measurements made on composites with unidirectional reinforcement. Laser interferometry and dilatometry are the techniques most frequently used. Other techniques, including some applied to the unimpregnated fiber, have also been found satisfactory. When testing the composite, the unidirectional fibers may be oriented parallel or perpendicular to the direction of measurement to obtain the axial or transverse CTE. To perform the analysis, the modulus of the fiber, the modulus and CTE of the matrix, and the fiber loading must be known. It may be desirable to perform the measurements on composites with different fiber loadings in order to check the results.

3.3.5 **Thermal conductivity.** The thermal conductivity of fibers is generally determined analytically from measurements of axial thermal conductivity on unidirectional reinforced composites. However, some measurements have been made on both fiber bundles and single filaments. These have agreed quite well with values determined from composite measurements (Reference 3.3.5(a)). Both types of measurements require considerable operator skill and sophisticated equipment, and are perhaps best left to the thermophysics laboratory. A well defined relationship between axial thermal conductivity and axial electrical conductivity (or resistivity) has been developed for a wide range of carbon fibers. Since electrical resistivity is relatively easy to measure, reasonable estimates of thermal conductivity can be made for electrical resistivity measurements (Reference 3.3.5(b)). Transverse thermal conductivity can be determined for thin composites using a pulsed laser technique to measure thermal diffusivity. The thermal conductivity can then be calculated if the specific heat of the fiber is known.

3.3.6 Specific heat. This property is measured in a calorimeter such as described in ASTM D 2766 (Reference 3.3.6). This also is not a simple measurement and is best left to the experienced laboratory.

3.3.7 **Thermal transition temperatures.** Differential scanning calorimetry (DSC), differential thermal analysis (DTA) or thermomechanical analysis instrumentation may be applied to measure the glass transition temperature T_g and, if the fiber is semi-crystalline, its crystalline melting temperature T_m. General procedures for measuring T_g and T_m of organic fibers are given in ASTM standards D 3417 and D 3418 (References 3.3.7(a) and (b)).

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3.4 PHYSICAL TECHNIQUES (EXTRINSIC).

3.4.1 **Yield of yarn, strand, or roving.** Yield is generally expressed as length per unit weight, such as yards per lb, or as its reciprocal, linear density, expressed as weight per unit length. The latter is normally the measured value and is determined by accurately weighing in air a precise length of yarn, strand, and roving.

3.4.2 **Cross-sectional area of yarn or tow.** This property is calculated rather than measured. However, it is very useful in subsequent calculations of fiber loadings in prepregs and composites as well as in calculations for other physical and thermophysical properties. Often it is considered a quality assurance criterion for fiber manufacture. The cross-sectional area is determined by dividing the linear density, weight per unit length, by the volumetric density, weight per unit volume, using consistent units. It should be noted that this value includes only the cumulative total of the cross-sectional areas of all the individual filaments within the bundle. The cross-sectional area is not affected by any space between filaments nor related to any calculations based on yarn or bundle "diameter".

3.4.3 **Twist of yarn.** Twist is defined as the number of turns about its axis per unit length in a yarn or other textile strand. Twist is sometimes desirable to improve handleability and, at other times, undesirable because it restricts spreading of the yarn or tow. It can be measured according to the direct procedure described in ASTM D 1423 (Reference 3.4.3).

3.4.4 **Fabric construction.** Properties of fabrics such as handleability, drapability, physical stability, thickness, and the effectiveness of the translation of fiber properties to the fabric are all dependent on fabric construction. For the purpose of this document, fabric construction is defined according to the fiber used (by type and filament count), the weave style such as "plain" or "satin", and the number of yarns per inch of fabric in both warp and fill directions. The most common weave styles employed for carbon fabrics used in aircraft and aerospace applications are plain weave, crowfoot satin, five harness satin, and eight harness satin. For a given yarn, fabric physical stability decreases and drapability increases progressively from the plain weave to the eight harness satin weave. In order to maintain a satisfactory level of stability, more yarns per inch must be added progressively toward the 8-harness satin weave fabric, thus the lightest weight fabrics are of plain weave style. There are many construction-related tests applied in the textile industry which are beyond the scope of this document. Essential standards for measure of construction are Determination of Yarn Count (ASTM D 3775), Length (ASTM D 3773), Width (ASTM D 3774) and Weight (ASTM D 3776) (References 3.4.4(a) - (d)). Additional information on weaves is provided in Volume 3, Section 2.4.2.

3.4.5 **Fabric areal density.** This property, although related to the yarn count previously described, is itself useful in calculations for composite construction and analysis. Expressed as weight per unit area of fabric, fabric areal density along with the fiber density governs the thickness of a cured ply of impregnated fabric at a given fiber volume loading. It is measured according to the method described in ASTM D 3776 (Reference 3.4.4(d)).

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3.5 MECHANICAL PROPERTIES.

3.5.1 Tensile strength and Young's modulus.

3.5.2 Fiber strain and failure.

3.5.3 Transverse fiber modulus.

3.5.4 Poisson's ratio.

3.5.5 Shear moduli.

3.5.6 Interlaminar shear.

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3.6 TEST METHODS.

3.6.1 Determination of pH. (Reference 3.6.1)

3.6.1.1 *Scope.* This method describes a procedure for determining the pH of carbon and graphite fibers and fabrics by means of a pH meter. Measurements should be made on fibers that have not been sized. Due to the small amount of surface functionality on commercial fibers, these measurements require extreme care.

3.6.1.2 *Apparatus.* The apparatus needed for this procedure is as follows:

1. A pH meter equipped with glass and calomel electrodes or a single combination electrode which is preferable. It should have an accuracy of ±005 pH and conform to the requirements in the Method for Determination of pH of aqueous solutions with the Glass Electrode ASTM E-70.

2. Lipless beakers 100-ml. capacity with cover glasses.

3. Hot plate.

4. Shears for cutting samples.

5. Large pyrex flask one to two liters capacity for boiled distilled water. The pH of this water should be between 6.9 and 7.1 at 25°C. If it is impossible to meet these limits by boiling, the pH may be adjusted with extremely weak NaOH or HCl.

3.6.1.3 *Procedure.*

1. Prepare cloth samples by shearing into small (1/2" to 3/4") squares sufficient quantity to make 3.0 grams. Prepare yarn by cutting the sample into pieces 1/2" to 3/4" in length.

2. To 3 grams of sample add 30 ml. of the boiled distilled water, cover with a watch glass and boil very gently for 15 minutes. The use of Berzelius or lipless beakers prevents excessive loss of water. At the end of 15 minutes approximately 4 or 5 ml. of slurry should remain.

3. Set the covered beakers in a tray of cold water and cool to room temperature. Keep the beakers covered to prevent the absorption of chemical fumes that may be present in the room. After cooling remove the cover glasses but do not wash down.

4. When all is in readiness for the test, standardize the pH meter by use of a reliable buffer. Place buffer in a beaker, immerse the electrodes and adjust the meter to exactly the same value. A buffer should be chosen with a pH value in the same range as the sample to be tested. The temperature of the buffer and the sample should be the same temperature within ±1°.

5. After the meter is adjusted, remove the electrodes from the buffer, rinse thoroughly with distilled water and wipe dry with clean absorbent tissue.

6. Place the electrodes in the slurry and rotate the beaker gently in alternate directions until a constant pH value is obtained.

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3.6.2 Determination of amount of sizing on carbon fibers.

3.6.2.1 *Scope.*

This method describes the procedures for determining the sizing on carbon fibers, expressed as a percent of yarn weight.

3.6.2.2 *Apparatus.* The following equipment is needed for this procedure:

1. Balance - Analytical, Mettler Model B5-H26. Scientific Products Catalog No. B1253, or equivalent.

2. Desiccator - Scheibler including Coors desiccator plate, 250 mm I.D. Scientific Products Catalog No. D1450-5, or equivalent.

3. Crucible - Coors, 40 ml Cap, 47mm rim dia., 40 ml h. Scientific Products Catalog No. C8450-8, or equivalent.

4. Muffle Furnace - Thermolyne, Model No. F-A1730. Maximum temp. 2000°F. Chamber Dimensions: 9 1/2"W x 8 1/2"H x 13 1/2"D, or equivalent.

5. Sagger - Stainless steel (16 gage). Dimensions: 9"W x 3"H x 10"D, with a snug fitting stainless steel cover having a 1/2" hangover on all sides. Sagger must have a 1/8" stainless steel tube connected into it for purposes of nitrogen purging. In house fabricated.

3.6.2.3 *Materials.* The following materials are needed for this procedure:

1. Drierite or equivalent - Both indicating and non-indicating crystals.

2. Nitrogen Gas - Standard purity.

3.6.2.4 *Procedure.*

1. Wind approximately 1.5g of sample yarn into a small (25-40 mm dia.) coil and place it in a desiccator for 2 hours.

2. Using clean dry forceps, remove the sample coiled from the desiccator and weigh to nearest 0.1 mg. Record as W_1.

3. Obtain a clean dry crucible from the desiccator and place the sample coil in it. Weigh crucible plus yarn and record as W_2. NOTE: Wear clean dry cotton gloves to prevent any moisture from being picked up by the crucible.

4. Place the crucible plus sample in the sagger and cover it with lid. A stainless steel shelf with twelve 1-3/4" Dia. holes located 3/4" from the bottom of the sagger is suggested to prevent crucibles from overturning.

5. Purge the sagger with nitrogen at a rate of 7.5 S.C.F.H. for a minimum of 45 minutes.

6. While system is purging, set the control on the muffle furnace to 450°C and turn on.

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7. When purging time is complete and the furnace is at temperature, place the sagger in the furnace and heat for 1 hour. Note, for safety, wear protective asbestos gloves or equivalent when inserting or removing the sagger from the hot muffle furnace. The N_2 purge is continued throughout the heating and cool-down phases of this test.

8. After heating for 1 hour, remove the sagger from the furnace and place it in protected cool-down area.

9. Remove the crucible from the sagger and place it in the desiccator to cool to room temperature.

10. When cool, weigh crucible plus sample and record as W_3.

3.6.2.5 *Calculation.* Calculate the amount of sizing by determining the percent weight loss as follows:

$$\text{Sizing Content} = \frac{W_2 - W_3}{W_1} \times 100 \qquad 3.6.2.5$$

3.6.2.6 *Preparation of crucibles for reuse.* Before reusing crucibles, burn off any residue by placing them in the muffle furnace and heating them at 700°C in air for at least 2 hours. After the crucibles are removed from the furnace and cool to room temperature, blow them out with clean compressed air. Store clean crucibles in the desiccator.

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3.6.3 Determination of moisture content or moisture regain.

3.6.3.1 *Scope.* This method may be used to determine the amount of moisture in a textile either as it is received or when it is in moisture equilibrium for testing in the standard atmosphere as defined in the definitions of Terms Relating to Textile Materials (ASTM Designation: D 123, Reference 3.6.3.1).

3.6.3.2 *Apparatus.* The following equipment is needed for this procedure:

1. Weighing Bottle, glass, approximately 100-ml. capacity, fitted with a ground-glass cover, or an aluminum weighing can, approximately 100-ml. capacity, and having a tight-fitting cover.

2. Desiccator, containing anhydrous calcium chloride ($CaCl_2$) or other suitable dehydrating agent.

3. Chemical Balance, capable of weighing to 0.5 mg.

4. Oven, maintained at 105° to 110°C. Note that special equipment for drying specimens to constant weight, which is generally available in textile laboratories (conditioning ovens, etc.) may also be used. The apparatus here and the procedure described in Section 3.6.3.4 are provided for laboratories without such special equipment.

3.6.3.3 *Sample preparation.*

1. Cut samples for test. Approximately 2 grams are required. When sampling cloth use the Alfred Sutler Company sample cutter or equivalent, which cuts a circular disc slightly over two inches in diameter. Four discs will usually weigh about two grams. When sampling yarn, form sample into small coil. Hold in place by tying with one of the ends.

3.6.3.4 *Procedure.*

1. Dry the glass weighing bottle at 105° to 110°C. to constant weight. Place the weighing bottle and cover separately in the oven. After heating for 1 hour, replace the cover, transfer the weighing bottle to the desiccator and allow it to cool to room temperature. Remove the cover momentarily to equalize the pressure, and with the cover in place, weigh the container. Repeat the heating, cooling, and weighing until the weight of the empty weighing bottle is constant to within ±0.001 g.

2. Place the specimen to be tested in the container, cover, and weigh. Subtract the weight of the empty container (1.) from this weight to obtain the air-dry weight of the specimen, weight A.

3. Place the uncovered weighing bottle and specimen in the oven for 1½ hours at a temperature of 105° to 110°C. Cover and transfer the container to a desiccator. When the container has cooled to room temperature, remove the cover momentarily to adjust the pressure, replace the cover, and weigh. Repeat the heating for periods of not less than 20 minutes, cooling and weighing until the weight is constant to within ±0.001 g. Subtract the weight of the empty container from this weight to obtain the oven-dry weight of the specimen, weight B. When textiles are heated under the conditions described, volatile materials, in addition to moisture, may be removed. If this possibility is known or suspected, it should be reported that the percentage loss in weight of the textile does or may include volatile substances as well as moisture.

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3.6.3.5 *Calculations.* Calculate the moisture content of the specimen as follows:

$$\text{Moisture content, percent} = \frac{A - B}{A} \times 100 \qquad 3.6.3.5(a)$$

Calculate the moisture regain of the specimen as follows:

$$\text{Moisture regain, percent} = \frac{A - B}{B} \times 100 \qquad 3.6.3.5(b)$$

where:

A = air-dry weight of the specimen
B = Oven-dry weight of the specimen

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3.6.4 Determination of fiber diameter.

3.6.4.1 *Description and application.*

This method describes a procedure for determining the average diameter of fibers by means of a microscope fitted with an image splitting eyepiece. This instrument measures the distance across an object as it lays upon a glass slide. Therefore, this is a valid diameter measurement only if the fibers are essentially true cylinders. Since different types of fibers have characteristic shapes it is possible to use this procedure for irregularly shaped fibers by determining an area factor for the particular type of fiber being measured by means of microphotography.

Figure 3.6.4.1 is a sketch showing the optical scheme incorporated in the instrument. In the Image Splitter a prism system is interposed between microscope objective and eyepiece to produce a double image of the microscope field of view. This prism system is precisely rotatable by a micrometer screw. Upon rotation of the prisms double images of objects in the field of view transverse one another. Measurement is accomplished by reading off the micrometer the amount of prism rotation required to place on object's double images exactly edge to edge in the axis of desired measurement. Measurement is accomplished in the plane of the object.

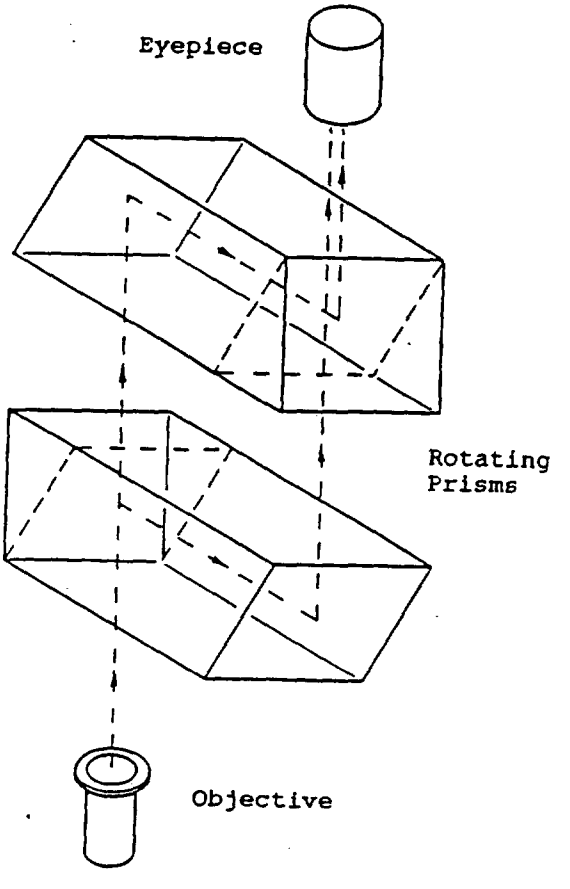


Figure 3.6.4.1 *Schematic diagram of the optics of an image-splitting microscope.*

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The accuracy which can be obtained and the size limits for various conventional microscope objectives are shown in Table 3.6.4.1.

3.6.4.2 *Apparatus.* The following equipment is needed for this procedure:

1. Microscope, Unitron monocular model MLU with 5X, 10X, 20X and 40X objectives, fitted with a Vickers AEI 10X image splitting eyepiece, or equivalent.

2. Microscope lamp, A.O. Spencer, Fisher Catalog #12-394, or equivalent.

3. Glass microscope slides, Fisher Catalog No. 12-550.

4. Sharp Knife or razor blade for cutting sample.

3.6.4.3 *Calibration.* Assemble and calibrate the microscope following the directions supplied with the instrument. If for any reason a different eyepiece or objective is used, the instrument must be recalibrated for the new part.

3.6.4.4 *Prepare slide.* Select representative strands from the sample and align in an even bundle. Place the bundle on a spare glass slide and hold it in place with the top of the finger. Cut the strands with a sharp instrument to secure smooth even ends. Place the strands on a clean slide so that the even ends are near one edge of the slide. (See Figure 3.6.4.4.) Cut the strands so that the pieces approximately 0.5 mm will be produced. The pieces will dust in a fine spray over the surface of the slide in an even pattern.

3.6.4.5 *Measuring procedure.*

1. Place the slide under the microscope and select at random a single fiber. The fiber will appear as parallel red and green lines.

2. Rotate the image splitter until the fiber is parallel with the axis of the micrometer barrel and the ends of the two image are even.

TABLE 3.6.4.1 *Accuracy and size limits.*

OBJECTIVE POWER		READING ACCURACY		MAXIMUM SIZE OBJECT WHICH CAN BE COMPLETELY SHEARED	
5X	(N.A. 0.15)	0.00008"	2.0μm	0.04"	1.0 mm
10X	(N.A. 0.28)	0.00004"	1.0μm	0.02"	0.5 mm
20X	(N.A. 0.50)	0.000026"	0.6μm	0.01"	0.25 mm
40X	(N.A. 0.65)	0.0000128"	0.325μm	0.005"	0.12 mm

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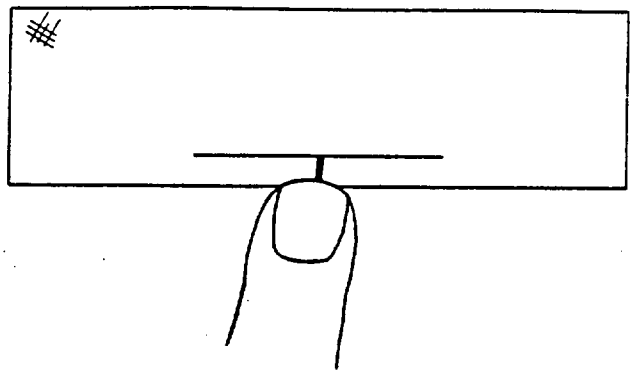


Figure 3.6.4.4 *Place the strands on a clean slide so that the even ends are near one edge of the slide.*

3. Bring into sharp focus. Rotate the micrometer until the red and green images overlap forming a single black line. Continue the rotation until the red and green lines again separate and a band of light shows between them. Reverse the rotation and slowly bring the colored images back together until they just touch. Neither a thin light nor black line should be visible. Record the micrometer reading. If the micrometer should be turned too far so that the black line appears back off and start over. This will eliminate the effect of any backlash in the mechanism.

4. Continue to rotate the micrometer until the images have completely overlapped and are just ready to separate on the opposite side. Turn the micrometer very slowly until the thick black line completely disappears. At this point the red and green images should be just touching with no light showing between them. Record the micrometer reading. The difference between the two readings is equal to twice the width of the object being measured.

5. Select other fibers and continue the measurements until twenty pieces have been measured. Move the slide in a uniform pattern to eliminate the possibility of measuring the same piece twice.

3.6.4.6 *Calculation.* List the difference in micrometer readings obtained in Paragraph 5 for each of the twenty measurements. Divide this number by two then multiply the result by the calibration factor for the microscope to obtain diameter measurement in microns. Calculate the average for the twenty measurements.

If the diameter measurement calculated above is to be used to calculate cross sectional area for strength determinations see Section 3.6.4.1. For example, the area of Thornel 25™ fibers may be calculated as follows: calculate the area using the average diameter determined in this procedure. Multiply this area by the area factor to obtain the actual area. (An area factor of 0.66 has been determined for Thornel 25™.)

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3.6.5 Determination of electrical resistivity.

3.6.5.1 *Scope.* This method describes a procedure for determining the electrical resistance of carbon and graphite cloth and felt. It is used as a control measure for checking temperature of processing and to determine compliance of materials with specific resistance specifications.

3.6.5.2 *Apparatus.* The following equipment is needed for this procedure:

1. Jig for clamping cloth.
2. Vacuum tube volt-ohmmeter. Triplet #850 or equivalent.
3. Scissors or other implement for cutting samples.

3.6.5.3 *Sample preparation.* For cloth, obtain a 1/2 yard long, full width piece of each roll of cloth to be tested. Cut five warp and five fill direction strips 1-1/4" wide and 11" to 12" long. Distribute the location of the strips over the entire area of the cloth sample. Ravel each strip to the nearest thread to one inch width. For felt samples, obtain a 1/2 yard long, full width piece of each roll of felt to be tested. Cut five strips in "warp" direction and five strips in "fill" direction using a 1" x 12" metal template. Distribute the location of strips over the entire area of the felt sample.

3.6.5.4 *Procedure.*

1. Adjust the silver jaws of the resistance jig to provide a test length of 10".

2. Clamp sample in jig and measure resistance.

3.6.5.5 *Calculation.* Divide observed resistance by 10 to obtain ohms/square value. (See Section 3.6.5.7.) Determine the average resistance for five strips and record on data sheet as ohms per inch per inch width.

3.6.5.6 *Calibration and maintenance.* Vacuum tube volt-ohmmeters are used for fabric and felt measurement. The meter and jig should be calibrated every six months with a standard resistance box, certified to NBS standards. Any incident requiring meter maintenance (tube replacement etc.) should be followed by recalibration regardless of the six month routine check.

The zero and full scale adjustment should be checked each shift the meter is used. If meter fails to adjust properly to zero and full scale, it must be checked by the electrical maintenance department. The zero adjustment compensates for lead wire and meter resistance errors.

3.6.5.7 *Definition of units of measurement.* The electrical resistance measurement on cloth and felt is expressed in ohms per square (unit area). This is not the same value as the specific resistance measured on bulk carbon. The fabric resistance value is proportional to specific resistance when a given grade is considered. For example, carbon cloth has higher resistance than graphite cloth, the same material after graphitization.

The size of the square (unit area) does not influence the resistance value of the cloth.or felt. This can be demonstrated with the standard equation for the relation between sample resistance and specific resistance for a solid rectangular shape.

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$$P = \frac{RTW}{L} \quad \text{or} \quad R = \frac{PL}{TW} \qquad 3.6.5.7(a)$$

where

P = specific resistance
R = sample resistance
L = sample length
T = sample thickness
W = sample width

For materials with identical specific resistance (P) such as copper, the resistance of one square inch of a given thickness would be:

$$R = \frac{PL}{TW} = \frac{P}{T} \times \frac{1}{1} \qquad 3.6.5.7(b)$$

Since T and P are constant, the resistance of the one inch square can be written:

$$R = K\frac{L}{W} = K\frac{1}{1} = K \qquad 3.6.5.7(c)$$

If the square is made twice as large (2 x 2), the resistance of the square will remain the same.

$$R = K\frac{L}{W} = K\frac{2}{2} = K \qquad 3.6.5.7(d)$$

Therefore, the resistance of fabrics is reported as ohms/square. The user can use any unit for the square he chooses, inch, centimeter, foot or yard.

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Chapter 4. MATRIX CHARACTERIZATION

Page

4.1 INTRODUCTION. 4-2

4.2 CHEMICAL ANALYSIS TECHNIQUES. 4-2
4.2.1 Elemental analysis. 4-2
4.2.2 Functional group and wet chemical analysis. 4-2
4.2.3 Spectroscopic analysis. 4-2
4.2.4 Chromatographic analysis. 4-4
4.2.5 Molecular weight and molecular weight distribution analysis. 4-4
4.2.6 General scheme for resin material characterization. 4-5

4.3 PHYSICAL ANALYSIS TECHNIQUES. 4-12
4.3.1 Thermal analysis. 4-12
4.3.2 Rheological analysis. 4-14
4.3.3 Morphological Analysis. 4-15
4.3.4 Density. 4-15

4.4 VOLATILES DETERMINATION. 4-15

4.5 MOISTURE CONTENT. 4-15

4.6 MECHANICAL TEST METHODS. 4-15
4.6.1 Tensile Testing. 4-15
4.6.2 Compression Testing. 4-15
4.6.3 Flexural Testing. 4-15
4.6.4 Impact Strength Measurement. 4-15
4.6.5 Creep Testing. 4-15
4.6.6 Fatigue Resistance Testing. 4-15
4.6.7 Hardness Testing. 4-15

4.7 TOXICITY, HEALTH HAZARDS, AND SAFETY. 4-15

REFERENCES 4-16

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4.1 INTRODUCTION. A wide range of polymeric resins and reactive resin precursor materials with additives are used as matrix materials for reinforcement fibers in prepregs and composite materials. Prior to the prepregging process, the resin or formulated resin components may exist in a variety of forms (liquid, solid, film, powder, pellets, etc.). Although polymeric resins, resin precursors, and additives are primarily organic materials, other components (e.g., catalysts, fillers, and processing aids) may be inorganic or contain metals. Generally a resin will be designed as either a thermoset or thermoplastic polymer. However, thermoset resin formulations may also contain thermoplastic or rubber additives.

This chapter describes techniques and test methods that are generally used to analyze and evaluate the chemical, physical, mechanical, and related properties of polymeric resin and resin precursor materials. The characterization of individual resin components, resin formulations, and "cured" or polymerized resin materials is addressed. Where possible, standard test methods and state-of-the-art techniques for characterizing resin materials are referenced. Selected test methods and practices that are not published in open literature or are difficult to access are listed in Section 4.18.

4.2 CHEMICAL ANALYSIS TECHNIQUES. Chemical characterization techniques are listed in Table 4.2. Elemental analysis and functional group analysis provide basic and quantitative information relating to chemical composition. Spectroscopic analysis provides detailed information about molecular structure, conformation, morphology, and physical-chemical characteristics of polymers. Chromatographic techniques separate sample components from one another, and thereby simplify compositional characterization and make a more accurate analysis possible. Employing spectroscopic techniques to monitor components separated by gas or liquid chromatography greatly enhances characterization, providing a means to identify and quantitatively analyze even the most minor components.

4.2.1 Elemental analysis. Elemental analysis techniques such as ion chromatography, atomic absorption (AA), X-ray fluorescence, or emission spectroscopy can be applied to analyze specific elements, such as boron or fluorine. When necessary, X-ray diffraction may also be used to identify crystalline components, such as fillers, and to determine the relative percent crystallinity for certain resins.

4.2.2 Functional group and wet chemical analysis. The analysis of reactive functional groups is particularly important in determining equivalent weights of prepolymers. Titration and wet chemical analysis for specific functional groups are useful techniques for characterizing individual epoxy components but have limited application and may provide misleading results when complex resin formulations are analyzed.

4.2.3 Spectroscopic analysis. Infrared spectroscopy (IRS) provides more useful information for identifying polymers and polymer precursors than any other absorption or vibrational spectroscopy technique and is generally available in most laboratories. IR yields both qualitative and quantitative information concerning a polymer sample's chemical nature, i.e., structural repeat units, end groups and branch units, additives and impurities (Reference 4.2.3(a)). Computerized libraries of spectra for common polymeric materials exist for direct comparison and identification of unknowns. Computer software allows the spectrum of a standard polymer to be subtracted from an unknown to estimate its concentration and perhaps to determine whether another type of polymer is also present in the sample.

Infrared (IR) spectroscopy is sensitive to changes in the dipole moments of vibrating groups in molecules and, accordingly, yields useful information for the identification of resin components. IR spectroscopy provides a fingerprint of the resin composition and is not limited by the solubility of resin

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TABLE 4.2 *Techniques for chemical characterization.*

Elemental Analysis -	Conventional Analytical Techniques X-Ray Fluorescence Atomic Absorption (AA) ICAP EDAX Neutron Activation Analysis
Functional Group Analysis -	Conventional Wet Chemical Techniques Potentiometric Titration Coulometry Radiography
Spectroscopic Analysis -	Infrared (Pellet, Film, Dispersion, Reflectance), Fourier Transform IR (FTIR), Photoacoustic FTIR, Internal Reflection IR, IR Microscopy, Dichroism Laser Raman Nuclear Magnetic Resonance (NMR) 13C, 1H, 15N; Conventional (Soluble Sample), Solid State (Machined or Molded Sample) Fluorescence, Chemiluminescence, Phosphorescence Ultraviolet-Visible (UV-VIS) Mass Spectroscopy (MS), Election Impact MS, Field Desorption MS, Laser Desorption MS, Secondary Ion Mass Spectroscopy (SIMS), Chemical Ionization MS Electron Spin Resonance (ESR) ESCA (Electron Spectroscopy for Chemical Analysis) X-Ray Photoelectron X-Ray Emission X-Ray Scattering (Small Angle-Saxs) Small-Angle Neutron Scattering (SANS) Dynamic Light Scattering
Chromatographic Analysis -	Gas Chromatography (GC) or GC/MS (Low MW Compounds) Pyrolysis-GC and GC/MS (Pyrolysis Products) Headspace GC/MS (Volatiles) Inverse GC (Thermodynamic Interaction Parameters) Size-Exclusion Chromatography (SEC), SEC-IR Liquid Chromatography (LC or HPLC), HPLC-MS, Multi-Dimensional/Orthogonal LC, Microbore LC Supercritical Fluid Chromatography (SFC) Thin-Layer Chromatography (TLC), 2-D TLC

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components (References 4.2.3(b) - 4.2.3(d)). Indeed, gases, liquids and solids may be analyzed by IR spectroscopy. Advances in technology have led to the development of Fourier transform infrared spectroscopy (FTIR), a computer-supported IR technique for rapidly scanning and storing infrared spectra. Multiple scans and Fourier transformation of the infrared spectra enhance the signal-to-noise ratio and provide improved spectra for interpretation. In addition, the FTIR attenuated total reflection (ATR) and diffuse reflectance techniques may be applied for quality assurance of thermoset composite materials to assess their state of cure; i.e., residual epoxide concentration. (See Section 5.6.2.)

Although not as popular as IR, laser Raman spectroscopy complements IR as an identification technique and is relatively simple to apply (Reference 4.2.3(a)). As long as the specimen is stable to the high intensity incident light and does not contain species that fluoresce, little or no sample preparation is necessary. Solid specimens need only be cut to fit into the sample holder. Transmission spectra are obtained directly with transparent specimens. For translucent specimens, a hole may be drilled into the specimen for passage of the incident light and a transmission spectra obtained by analyzing light scattered perpendicular to the incident beam. The spectrum of a turbid or highly scattering specimen is obtained by analyzing the light reflected from its front surface. Powdered samples are simply tamped into a transparent glass tube and fibers can be oriented in the path of the incident beam for direct analysis.

4.2.4 Chromatographic analysis. High performance liquid chromatography (HPLC) is the more versatile and economically viable quality assurance technique for soluble resin materials (References 4.2.4(a) - 4.2.4(g)). HPLC involves the liquid-phase separation and monitoring of separated resin components. Dilute solutions of resin samples are prepared and injected into a liquid mobile phase which is pumped through column(s) packed with a stationary phase to facilitate separation and then into a detector. The detector monitors concentrations of the separated components, and its signal response, recorded as a function of time after injection, provides a "fingerprint" of the sample's chemical composition. Quantitative information may be obtained if the sample components are known and sufficiently well-resolved, and if standards for the components are available. Size exclusion chromatography (SEC), an HPLC technique, is particularly useful in determining the average molecular weights and molecular weight distributions of thermoplastic resins (Reference 4.2.4(g)). Recent advances have resulted in improved and automated HPLC instrumentation that is relatively low cost and simple to operate and maintain.

A powerful, but technically more demanding, technique for directly analyzing polymers is pyrolysis GC/MS (gas chromatography/mass spectroscopy). In this case, the sample only needs to be rendered sufficiently small to fit onto the pyrolysis probe. Not only can the polymer type be identified by comparing the resulting spectrum with standards, but volatiles and additives can be identified rapidly and quantitatively, and polymer branching and crosslink density can sometimes be measured.

Other chromatographic and spectroscopic techniques have also been considered (References 4.2.3(a), 4.2.4(h) - 4.2.4(l)). Gas chromatography (GC), GC head-space analysis, and GC-mass spectroscopy are useful for analyzing residual solvents and some of the more volatile resin components. Combined thermal analysis - GC-mass spectroscopy can be used to identify volatile reaction products during cure (References 4.2.4(m) and 4.2.4(n)).

4.2.5 Molecular weight and molecular weight distribution analysis. Techniques for evaluating polymer molecular weight (MW), molecular weight distribution (MWD), and chain structure are listed in Table 4.2.5. Size-exclusion chromatography (SEC) is the most versatile and widely used method for analyzing polymer MW and MWD. Once the solubility characteristics of a polymer are known, a suitable solvent can be selected for dilute solution characterization. THF is most often the solvent of choice for SEC, however, toluene, chloroform, TCB, DMF (or DMP) and m-cresol are also used. If the

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polymer's Mark-Houwink constants, K and a, in the solvent are known, size-exclusion chromatography (SEC) can be applied to determine the polymer's average MW and MWD (Reference 4.2.5(a)). If the constants are unknown or the polymer has a complex structure (e.g., branched, a copolymer, or mixture of polymers), SEC still may be used to estimate the MWD and other parameters relating to the structure and composition of the polymer. Although SEC indicates the presence of soluble non-polymeric components, high performance liquid chromatography (HPLC) is the better technique for characterizing residual monomers, oligomers, and other soluble, low MW sample components.

Light scattering, osmometry, and viscometry are also used to analyze polymer MW. Although seldom applied to synthetic polymers, sedimentation is an excellent technique for characterizing the MW of polymers having very large MW. The "special" techniques tend to be somewhat empirical or have limited utility and therefore are used less often.

New techniques which show great promise for characterizing polymer chain structure also are listed in Table 4.2.5. One of the most promising new techniques is dynamic laser light scattering. Unlike SEC, dynamic light scattering can be applied to any soluble polymer, regardless of temperature or solvent, and does not require polymer standards for calibration. Figure 4.2.5 illustrates the MWD of poly(1,4-phenyleneterephthalamide) (i.e., Kevlar™) measured by the laser light scattering (Reference 4.2.5(b)). As indicated, the polymer's MWD can be fully characterized using very little sample and a single solution with concentrated sulfuric acid as the solvent.

Dilute solution viscometry is a simple technique for determining the limiting viscosity number or intrinsic viscosity $[\eta]$ of soluble polymers (Reference 4.2.5(a)). The apparatus is inexpensive and simple to assemble and operate. The $[\eta]$ of a polymer depends upon its hydrodynamic volume in the solvent and is related to the MW of the polymer.

4.2.6 General scheme for resin material characterization. The following questions deserve careful consideration when developing procedures for preparing and characterizing polymer and polymer precursor (thermosetting resins and resin formulations) samples -

What are the inherent characteristics of the polymer or prepolymer?

Will certain operations cause irreversible changes in the sample?

What requirements does the characterization technique impose upon the sample?

Is it necessary to isolate the polymer or prepolymer from other sample components?

It should be recognized that the properties of polymer compounds and prepolymer formulations are often quite different from those of the pure polymers and polymer precursors. Polymer properties are greatly influenced by the presence of other components, e.g. fillers, additives, processing aids, dyes, residual catalysts, impurities, solvents and other polymers, low MW oligomers and monomers.

One must decide whether the specimen needs to be modified or specially treated for a particular analysis. Chemical structure, thermal transition behavior and solubility determine what can be done with a specimen. Operations, such as heating or extraction, may alter morphology or change the chemical composition of a specimen and thereby affect its properties and compromise the validity of certain tests. Many characterization techniques require polymer specimens to be modified or have a particular shape or form. If a specimen does not conform precisely to test criteria, the test may be invalid. On the other hand, in order to apply certain techniques (e.g., light scattering and membrane osmometry for MW analysis), it is essential that the polymer be totally isolated from nonpolymeric

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TABLE 4.2.5 *Polymer molecular weights, molecular*

Standard Techniques	Parameters Measured
Size-Exclusion Chromatography	Mol. wgt. averages and MWD, also provides (SEC) information relating to polymer chain branching, copolymer composition, and polymer shape.
Light Scattering (Rayleigh)	Weight-average mol. wgt. M_w (g/mol), virial coefficient A_2 (mol. cc/g^2), radius of gyration $<R_g>_z$(A), polymer structure, anisotropy, polydispersity.
Membrane Osmometry	Number-average mol. wgt. M_n (g/mol), virial coefficient A_2 (mol cc/g^2). Good for polymers with MW's in the range $5000 < MW < 10^6$, lower MW species must be removed.
Vapor Phase Osmometry	Same as membrane osmometry except that the technique is best suited for polymers with MW < 20,000 g/mol.
Viscometry (dilute solution)	Viscosity-average mol. wgt. M_η (g/mol) as determined by intrinsic viscosity $[\eta]$ (ml/g) relationship $[\eta] = KM_v$ where K and a are constants.
Ultracentrifugation or Sedimentation	Sedimentation-diffusion average mol. wgt. M_{sd} as defined by the relationship $M_{sd} = S_w/D_w$. Number- and z-average mol. wgt., M_n and M_z. MWD determined by the relation $S = kM^a$ where k and a are constants. Also provides information on the size and shape of polymer molecules.

Special Techniques	Parameters Measured
Ebulliometry	Number-average mol. wgt. M_n (g/mol) for $M_n < 20{,}000$ g/mol.
Cryoscopy	Number-average mol. wgt. M_n (g/mol) for $M_n < 20{,}000$ g/mol.
End Group Analysis	Number-average mol. wgt. M_n (g/mol generally for $M_n <$ 10,000. Upper limit depends on the sensitivity of the analytical method used.
Turbidimetry	Weight-average mol. wgt. M_w (g/mol) and MWD based upon solubility considerations and fractional precipitation of polymers in very dilute solutions

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weight distribution and chain structure.

	Principle
	Liquid chromatography technique. Separates molecules according to their size in solution and employs various detectors to monitor concentrations and identify sample components. Requires calibration with standard polymers.
	Measurement of scattered light intensities from dilute polymer solutions dependent upon solute concentration and scattering angle. Requires solubility, isolation, and in some cases fractionation of polymer molecules.
	Measurement of pressure differential between dilute polymer solution and solvent separated by a semi-permeable membrane. Colligative property method based upon thermodynamic chemical potential for polymer mixing.
	Involves isothermal transfer of solvent from a saturated vapor phase to a polymer solution and measurement of energy required to maintain thermal equilibrium. A colligative property.
	Employs capillary or rotational viscometer to measure increase in viscosity of solvent caused by the presence of polymer molecules. Not an absolute method, requires standards.
	Strong centrifugal field is employed with optical detection to measure sedimentation velocity and diffusion equilibrium coefficients S_w and D_w. Sedimentation transport measurements of dilute polymer solutions corrected for pressure and diffusion provides the sedimentation coefficient S. Permits analysis of gel containing solutions.

	Principle
	Measures boiling point elevation by polymer in dilute solution. A colligative property.
	Measures freezing point depression by polymer in dilute solution. A colligative property.
	The number or concentration of polymer chain end groups per weight or concentration of polymer are determined by specific chemical or instrumental techniques.
	Optical techniques are applied to measure the extent of precipitation as polymer solution is titrated with a non-solvent under isothermal conditions or as the solution prepared with a poor solvent is slowly cooled.

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TABLE 4.2.5 *Polymer molecular weights, molecular*

Special Techniques	Parameters Measured
Chromatographic Fractionation	Molecular weight distribution. An absolute MW technique is needed to analyze fractions.
Melt Rheometry	Weight-average mol. wgt. M_w (g/mol) and weight-fraction differential molecular weight distribution semi-empirical method.
Gel-Sol Analysis of Crosslinked Polymers	Gel fraction, Crosslink density
Swelling Equilibrium	Network structure, crosslink density, number- average mol. wgt. of chains between crosslinks M_c.

Promising Techniques	Parameters Measured
Laser Light Scattering (quasi-elastic, line-broadening or dynamic)	Same as Rayleigh light scattering plus trans-diffusion coefficient, molecular weight distribution, and information relating to gel structure.
Field Flow Fractionation (FFF)	Mol. wgt. averages and MWD. Requires calibration.
Non-Aqueous Reverse-Phase High Performance Liquid Chromatography HPLC and Thin-Layer Chromatography TLC	Mol. wgt. averages and MWD. Requires calibration
Supercritical Fluid Chromatography (SFC)	Mol. wgt. averages and MWD. Requires calibration.
Neutron Scattering Small Angle (SANS)	Weight-average mol. wgt. M_w (g/mol), Virial coefficient A_2 (mol-cc/g^2), Radius of gyration $<R_g>_z$ (A)

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weight distribution and chain structure.

	Principle
	Polymer is coated onto silica particles packed in thermostated column and separated according using solvent gradient elution. Polymer solubility decreases with increasing MW.
	Dynamic melt rheological method involving measurement of spectrum of diffusional relaxation times for polymer during oscillatory deformation.
	Extraction, filtration, and centrifugation are employed to isolate soluble polymer from gel. MW of soluble polymer is determined separately.
	Molar volume of crosslinked polymer immersed in swelling liquid and density of the swollen polymer are determined. Theory of partial molar free energy of mixing is applied.

	Principle
	Same as above but also involves measurement of the low-frequency line broadening of the central Rayleigh line of the scattered light. The structure of polymers in both dilute and concentrated solutions can be analyzed.
	Separates polymers according to their size and shape in solution. An elution technique, like chromatography, except that a field/gradient (thermal, gravitational, flow, electrical, etc.) is applied perpendicular to the axis of solution flow through a capillary or ribbon-shaped channel and a single phase is employed.
	Liquid chromatography technique based upon equilibrium distribution of polymer molecules between a non-aqueous binary solvent mobile phase and a nonpolar stationary (packing) phase.
	Liquid chromatography technique involving the use of a mobile phase under supercritical conditions (100 bars, 250°C).
	Measurement of amplitude of neutron scattering momentum vector for polymer in dilute solution or blend with another polymer. Scattering angle and polymer concentration are varied. Deuterated solvents are used. Dilute solid solutions and polymer blends have been studied.

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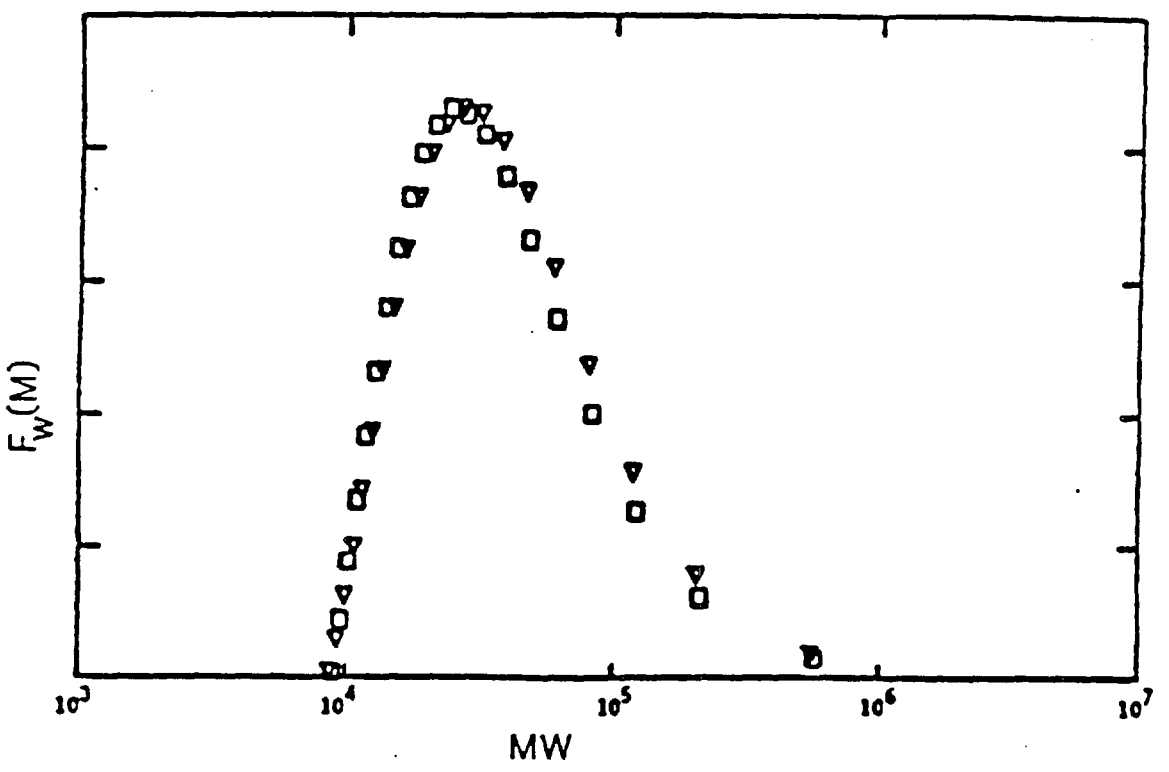


FIGURE 4.2.5 *Molecular weight distribution (MWD) of Kevlar™ in concentrated sulfuric acid, using dynamic laser light scattering.*

components.

Knowledge of the type of polymer or prepolymer is important in developing characterization procedures. If the material is unidentified, a simple series of tests (Level I in Figure 4.2.6(a)) may be applied, first to answer the question of whether the sample actually contains polymer, and then to determine its characteristics and identify the polymer or prepolymer.

Specimen modification for Level I merely involves breaking or cutting a small section from the sample and, if possible, further reducing the specimen size by grinding. To facilitate thermal and

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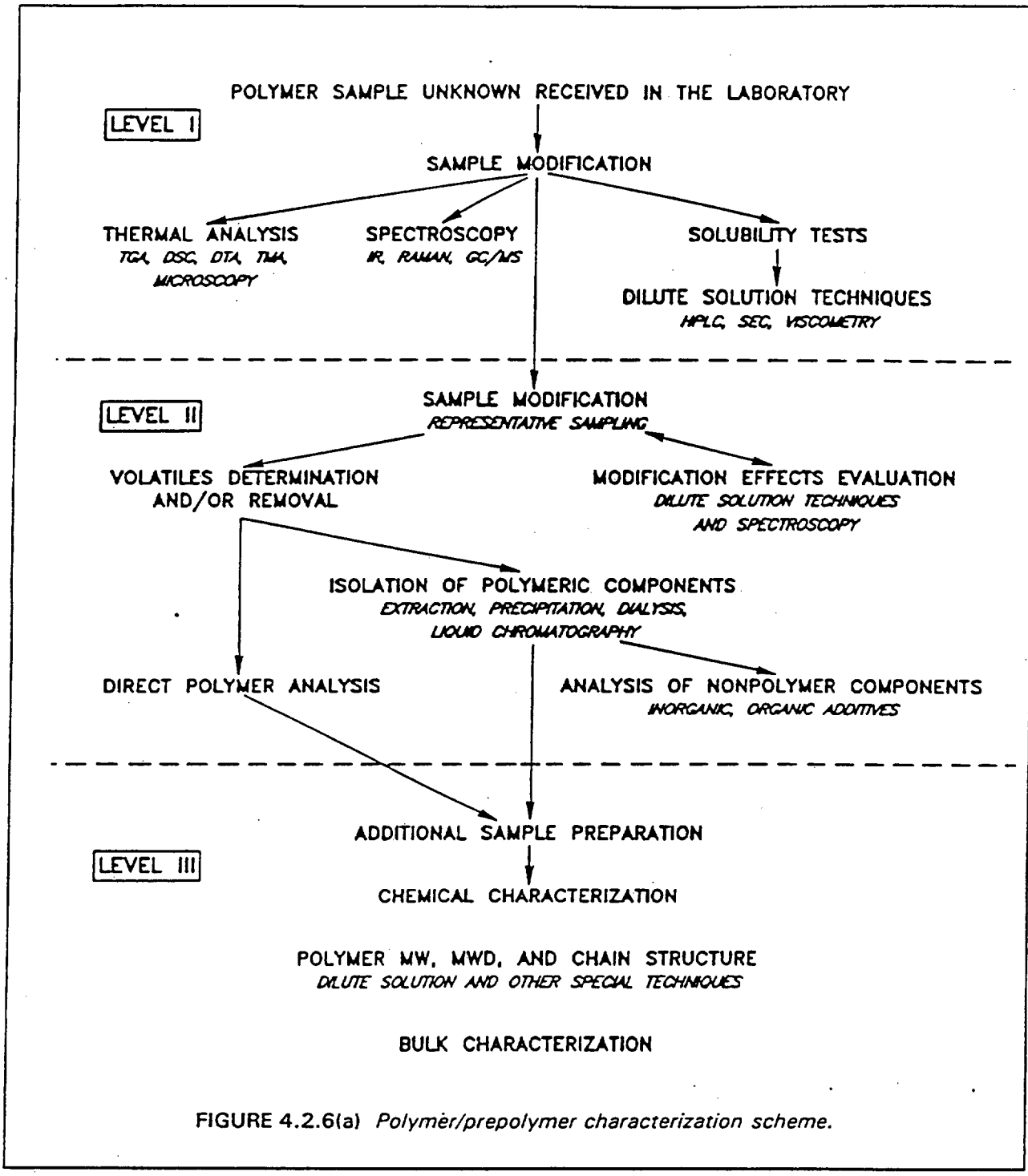


FIGURE 4.2.6(a) *Polymer/prepolymer characterization scheme.*

spectroscopic analysis and solubility testing, the specimen should have a large surface area. Liquid and heterogeneous specimens should be thoroughly mixed before removing an aliquot for analysis. Each test can be run using as little as 10 mg sample.

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Structural and compositional information obtained by the tests in Level I is used to help develop more sophisticated specimen preparation schemes and support the application of more detailed or specialized characterization techniques. The major concern of Level II is representative sampling and insuring that specimen modification procedures (cutting, grinding, molding, etc.) do not compromise polymer characteristics to be evaluated. Level II also addresses the "quantitative" aspects of sample composition (percent polymer, additives, volatiles, and inorganic and other organic residues) and, if necessary, deals with the identification of nonpolymeric components.

A general scheme for polymer analysis is illustrated in Figure 4.2.6(b). The polymer sample should be uniform and have a large surface area. Once volatile components are removed, the polymer can be directly analyzed, or a variety of techniques (e.g., extraction, precipitation, filtration, liquid chromatography) may be applied to isolate the polymer. If required, special procedures are applied to prepare the polymer sample for chemical characterization - molecular weight, molecular weight distribution, and chain structure evaluation, and bulk characterization (Level III in Figure 4.2.6(a)).

Whenever possible, complementary techniques should be used for the chemical quality assurance of resin materials. Techniques, such as HPLC and IR spectroscopy, are fundamentally different from one another and provide direct, but different, information about a resin's composition. If appropriate test methods are applied, HPLC and IR spectroscopy are usually powerful enough to detect differences or changes in the chemical compositions of resins. DTA and DSC complement HPLC and IR spectroscopy by providing information relating to the handleability (i.e.,the T_g and extent of reaction of the resin)and the processability of the prepreg. TGA and GC head-space analysis techniques for volatile components are secondary, but important, techniques. Special techniques for analyzing specific components or elements should be used if knowledge of the concentrations of the components is critical for processing the resin or if their presence could adversely effect the performance and durability of the cured composite. The information provided by mechanical, rheological, and dielectric analysis techniques is related to the chemical composition of the prepreg resin and thereby complements the more direct chemical techniques. However, caution is recommended in applying non-chemical techniques since the information obtained is complex and frequently ambiguous when attempts are made to relate measured parameters to chemical composition. (See Section 5.6.6.)

4.3 PHYSICAL ANALYSIS TECHNIQUES.

4.3.1 Thermal analysis. Thermal analytical techniques, such as thermal gravimetrical analysis (TGA), differential thermal analysis (DTA), differential scanning calorimetry (DSC), thermal mechanical analysis (TMA), dynamic mechanical analysis (DMA), and torsional braid analysis (TBA) provide useful information relating to the composition and processability of resins.

Thermal gravimetric analysis (TGA) monitors the weight changes in a sample as a function of temperature. Although primarily used for studying the degradation processes, TGA can also be applied as a quality assurance technique to provide information about the volatiles, resin, fiber, and inorganic residue content of prepreg materials (Reference 4.3.1(a)). Since dissimilar materials often degrade and volatilize at different temperature and rates, compositional differences may be reflected by differences in their TGA thermograms. Thermal oxidative degradation rates determined by TGA are useful for estimating the life cycles of resin materials (Reference 4.3.1(b)).

Differential scanning calorimetry (DSC) and differential thermal analysis (DTA) techniques are frequently employed for characterizing resins and composite materials (References 4.2.4(g), 4.2.3(b), 4.3.1(c), and 4.3.1(d)). Both DSC and DTA monitor enthalpy changes in materials as a function of temperature (DSC directly and DTA indirectly) and thereby provide similar information useful for quality assurance of prepreg materials. DTA measures the temperature difference (WT) between the epoxy

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Polymer Sample (fine powder or thin film)

Volatiles Removal and/or Determination

- Weight loss on drying
- TGA (Thermal Gravimetric Analysis)
- Head-Space analysis (GC/MS)
- Moisture analyzer

Isolation of Polymeric Component(s)

- Extraction
- Dissolution
- Filtration
- Precipitation
- Centrifugation

Chemical Characterization Techniques

- Elemental analysis
- Functional group analysis
- Spectroscopic analysis
- Chromatographic analysis

Polymer Molecular Weight, Molecular Weight Distribution, and Chain Structure

- Dilute solution techniques
- Other special techniques

Bulk Characterization Techniques

- Thermal analysis
- Microscopy
- Rheological analysis
- Mechanical testing
- Miscellaneous

FIGURE 4.2.6(b) *General scheme for polymer analysis.*

resin specimen and a reference material; whereas DSC measures the rate of heat evolution (dH/dt) or enthalpy absorption of the specimen relative to a reference. DTA and DSC measure thermal changes (1) as a function of time with both the specimen and reference material held at the same temperature (isothermal), or (2) as a function of temperature with both the specimen and reference material heated at the same heating rate (dynamic).

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For quality assurance applications, DTA and DSC are usually run in the dynamic mode with the weighed specimen in an aluminum specimen holder and an empty holder used as the reference. Dynamic DTA and DSC measure the glass transition temperature T and heat of reaction ΔH of the prepreg resin but do not provide information about chemical composition directly. By monitoring the fraction of heat evolved as a function of temperature or time, information relating to the extent of cure and curing kinetics can be obtained. DSC and DTA may also be applied to evaluate the melting temperature T and to estimate the degree of crystallinity of thermoplastic resins and composites. Since the average specimen size used in DSC is only about 10 mg (0.00002 lb), special care must be taken in obtaining representative materials. Multiple specimens runs are advisable.

Thermal mechanical analysis (TMA) is used in conjunction with DTA and DSC to study the thermal transition behavior (e.g., T_g) of prepreg resins and cured laminates. TMA simulates a linear dilatometer to measure the thermal expansion and contraction of specimens under dynamic or isothermal heating conditions. Adjustable loads are applied via a specially designed probe resting upon the specimen surface. Sensitive displacement devices are employed to monitor the "nominal" thermal response of a material. Since thermal transition behavior is related to the chemical composition and extent of cure of a prepreg resin, TMA can be applied as a quality assurance technique.

As previously discussed, TGA provides an indication of a sample's thermal decomposition temperature T_d and is used to estimate the relative amounts of volatiles, polymer, nonpolymeric additives, and inorganic residues. DSC or DTA is applied to evaluate the extent of cure and curing characteristics of thermoset resins, to determine a polymer's T , and, if the polymer is semi-crystalline, to determine its crystalline melting temperature T_m. Suggested procedures for measuring T_g and T_m are given in ASTM Standards D 3417 and D 3418 (References 4.3.1(e) and 4.3.1(f)). TMA can also be used to determine the T_g and to obtain further information about a polymer's heat distortion temperature and thermal expansion coefficient. For pelletized or molded samples, a razor blade or microtome can be used to cut samples to approximately fit the dimensions (thickness and diameter) of the sample holder. If the sample has been cut or is already in film or sheet form with a thickness no greater than 0.015 in (0.04mm), a punch or cork borer may be used to cut disks of an appropriate size.

Alternatively, a hot stage microscope may be used to observe the heat distortion temperature and onset of flow of powdered samples. Initially the powder particles have sharp, rough edges. As the sample is heated and the heat distortion temperature is approached, the edges first become blurred and then the particles start to agglomerate. Finally, at T_m, for semi-crystalline polymers, or T_g, in the case of glassy polymers, flow occurs and a clear melt or liquid forms. Microscopes equipped with cross polarizers are useful for defining crystal-crystal transitions and the onset of melting of semi-crystalline polymers.

4.3.2 Rheological analysis. Dynamic mechanical analysis (DMA), torsional braid analysis (TBA), and various mechanical spectrometers measure the rheological response of resins and composites as a function of frequency, temperature and/or state of cure. Both DMA and TBA provide information relating to the storage modulus (E') and the loss modulus (E") of the resin during cure and to the T of the cured resin (References 4.3.1(c), 4.3.2(a) - 4.3.2(c)). The dynamic complex viscosity (h*) and tan delta parameters can also be determined from rheological measurements of thermoset resins during cure. Rheological techniques are most often used to optimize processing parameters and evaluate the onset of gelation and vitrification during cure. However, since rheological properties are related to resin composition and morphology, rheological techniques may also be applied for the quality assurance of prepregs.

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Dynamic dielectric analysis (DDA) techniques involve the use of electrical measurements to monitor changes in the dielectric constant, the dissipation factor, capacitance, and/or conductance of the prepreg resin during processing as a function of frequency, time, and temperature. Measured electrical parameters are highly responsive to changes in resin viscosity and, similar to the rheological techniques, are often employed to investigate and optimize prepreg processing parameters such as resin flow and gelation time/temperature. Since chemical composition affects the electrical properties and curing behavior of prepregs, DDA techniques may also be applied for their quality assurance (References 4.2.3(c) - 4.2.3(j)). (See Section 5.6.5.)

4.3.3 Morphological Analysis.

4.3.4 Density.

4.4 VOLATILES DETERMINATION.

4.5 MOISTURE CONTENT.

4.6 MECHANICAL TEST METHODS.

4.6.1 Tensile Testing.

4.6.2 Compression Testing.

4.6.3 Flexural Testing.

4.6.4 Impact Strength Measurement.

4.6.5 Creep Testing.

4.6.6 Fatigue Resistance Testing.

4.6.7 Hardness Testing.

4.7 TOXICITY, HEALTH HAZARDS, AND SAFETY.

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Chapter 5. PREPREG MATERIALS CHARACTERIZATION

Page

5.1 INTRODUCTION 5-2

5.2 CHARACTERIZATION TECHNIQUES - OVERVIEW. 5-2

5.3 SAMPLING. 5-3

5.4 GENERAL CHARACTERISTICS OF PREPREGS. 5-3
5.4.1 Physical description of reinforcement. 5-3
5.4.1.1 Alignment. 5-3
5.4.1.2 Gaps. 5-3
5.4.1.3 Width. 5-4
5.4.1.4 Length. 5-4
5.4.1.5 Edges. 5-4
5.4.1.6 Splices. 5-4
5.4.2 Resin content. 5-4
5.4.3 Fiber content. 5-4
5.4.4 Volatiles content. 5-4
5.4.5 Moisture content. 5-4
5.4.6 Inorganic fillers and additives content. 5-4
5.4.7 Areal weight. 5-5
5.4.8 Tack and drape. 5-5
5.4.9 Resin flow. 5-5
5.4.10 Gel time. 5-5

5.5 TEST METHODS. 5-6
5.5.1 Resin extraction procedure for epoxy resin prepregs. 5-6
5.5.2 Procedure for HPLC/HPSEC analysis of glass, aramid, and graphite fiber prepregs. 5-7
5.5.2.1 Reverse phase HPLC analysis. 5-8
5.5.2.2 Size Exclusion Chromatography (SEC) analysis. 5-8
5.5.3 Procedure for Fourier transform infrared spectroscopy (FTIR). 5-9
5.5.4 Procedure for differential scanning calorimetry (DSC). 5-9
5.5.5 Procedure for dynamic mechanical analysis (DMA). 5-9
5.5.6 Procedure for rheological characterization. 5-10

REFERENCES 5-11

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5.1 **INTRODUCTION.** The processability and properties of high performance composites depend upon the composition of the fiber/resin preimpregnated materials (prepregs) from which they are manufactured. In general, prepregs consist of "modified" or surface-treated glass, graphite, or aramid fibers impregnated with 28-60 weight-percent of a reactive and chemically-complex thermoset resin formulation or a thermoplastic resin. A typical thermoset resin formulation may contain, for example, several different types of epoxy resins, curing agents, diluents, rubber modifiers, thermoplastic additives, accelerators or catalysts, residual solvents, and inorganic materials, plus various impurities and synthetic by-products. Furthermore, such resins are often "staged" or partially reacted during the prepregging process and may undergo compositional changes during transport, handling, and storage. Although less likely to undergo compositional changes, polymer molecular weight (MW), molecular weight distribution (MWD), and crystalline morphology have major effects on the processability and properties of thermoplastic prepregs and composites. Inadvertent or minor changes in resin composition may cause problems in processing and have deleterious effects on the performance and long-term properties of composites.

Modern analytical techniques and detailed knowledge relating to fibers, fiber surface treatments, and resin types and formulations are needed to characterize prepregs and composite materials. Characterization involves the identification and quantification of the fiber, fiber surface, and major resin components and should include information about the presence of impurities or contaminants. For thermoset resins and composites, characterization should include a description of the nature and extent of the prepreg resin reaction and the thermal/rheological and thermal/mechanical behavior. In the case of thermoplastics, the polymer molecular weight distribution, crystallinity, and time/temperature viscosity profile should also be analyzed. However, few laboratories are equipped or have the knowledgeable technical personnel to characterize prepregs and composites completely, and few studies have been published describing how variations in fiber type and resin chemistry/morphology affect the physical properties and long-term performance of composites. Also, until recently, prepreg compositions were considered proprietary, processing conditions were only recommended, and acceptance was based primarily upon mechanical testing of fabricated specimens. The purpose of this chapter is to provide an overview of characterization techniques and, more specifically, to address the application of state-of-the-art techniques for the chemical and physical characterization of resins and prepreg materials used in the manufacture of high performance organic matrix composites.

5.2 **CHARACTERIZATION TECHNIQUES - OVERVIEW.** According to a recent survey (Reference 5.2(a)), the most widely utilized techniques for the characterization and quality assurance of composite material precursors are -

1. High Performance Liquid Chromatography (HPLC)
2. Infrared (IR) Spectroscopy
3. Thermal Analysis
4. Rheological Analysis

HPLC and IR spectroscopy provide the capability for rapid screening and quality control fingerprinting of individual resin constituents as well as of the prepreg resin and, therefore, may be used advantageously by both the prepregger and composite manufacturer (References 5.2(b) - (f)). Thermal analytical techniques, such as thermal gravimetrical analysis (TGA), differential thermal analysis (DTA), differential scanning calorimetry (DSC), thermal mechanical analysis (TMA), dynamic mechanical analysis (DMA), and torsional braid analysis (TBA) are not strictly chemical analysis techniques; however, they provide useful information relating to the composition and processability of resins (Reference 5.2(g)). Similarly, rheological and dielectric techniques are used frequently to evaluate the chemoviscosity properties of thermoset resins during cure (Reference 5.2(h)), and there

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is increasing interest in applying such techniques for process monitoring and process control of both thermoset and thermoplastic resins.

Many of the chemical and physical analytical techniques described in Chapter 4 are also applicable to the characterization of prepreg materials. HPLC, IR spectroscopy, thermal analytical techniques and rheological methods are described in Section 5.5.

5.3 SAMPLING. Prepregs are commonly described by the purchaser's requirements which may include the manufacturer's trade name, resin type (e.g., 250°F) and lot number, fiber type and form (tape, fabric, roving, etc.), prepreg lot and roll numbers, and date of manufacture. The shipping date and expected shelf life are also usually designated along with recommended processing conditions. Generally, prepregs are shipped as rolls of impregnated woven fabric or unidirectional tape. (Typical widths are 38" (~97 cm) for woven aramid and glass, 42" (~107 cm) for woven graphite, and 12" (~30 cm) for tapes.) The prepreg layers are separated by thin, removable plastic or coated paper films which are removed when the prepregs are analyzed.

To check uniformity, it is recommended that sections be cut from the center and each side of the front-end (first off) of designated prepreg rolls. The amount and number of samples required for a particular analysis or test depends upon the homogeneity of the resin and the uniformity of the prepreg fibers. For some techniques, such as HPLC, 0.5 to 2.0 grams of prepreg may be needed to provide a representative sample. Other techniques (e.g., DSC) which utilize relatively small specimens (10 to 20 milligrams) may demand multiple specimens to provide an "average" value or test result.

Care must be taken not to contaminate or in any way alter samples during handling and storage. The samples should be placed in clean, dry, sealable containers and be carefully labeled. The containers must not react with the samples and precautions must be taken not to expose the prepregs to moisture nor allow them to stand unrefrigerated for long periods of time. For reactive prepreg resins such as epoxies, it is recommended that the prepregs be stored in hermetically sealed packaging at -0°F (-18°C). Upon removing the containers from cold storage, they should be allowed to achieve room temperature before opening to prevent condensation of moisture on the samples.

5.4 GENERAL CHARACTERISTICS OF PREPREGS.

5.4.1 Physical description of reinforcement. The physical description of the reinforcement used in a composite shall be described using the standard definitions of ASTM D 3878 (Reference 5.4.1). The filaments in the prepreg should be uniformly wetted by the resin. No particulates should be observable upon visual examination.

5.4.1.1 *Alignment.* In unidirectional prepregs, the filament bundles must be parallel to the longitudinal direction of the prepreg within an angle of 0.5° when examined visually using appropriate aids to measure angular alignment.

5.4.1.2 *Gaps.* Any gap within or between filament bundles in unidirectional prepregs generally should comply with the specifications. General guidelines are provided as follows:

a. No gap shall exceed 0.030 inch (0.76 mm) in width.

b. The length of any portion of the gap with an average width of 0.030 inch (0.76) shall not exceed 24 inches (0.61 m).

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c. Gaps in line with each other and no more than one inch (25 mm) apart shall be considered as one gap, regardless of number.

d. Gaps with excessive width or length shall be considered defective and will be the basis for flagging or replacing the prepreg.

5.4.1.3 *Width.* Width tolerance for unidirectional prepreg tape should be as specified, typically ±0.030 inch (0.76 mm).

5.4.1.4 *Length.* The length of each roll of prepreg shall be provided. Limits on the length of prepreg on any single roll shall be specified. Alternatively, suppliers and users may agree on limits for the prepreg weight or area per roll.

5.4.1.5 *Edges.* Acceptable waviness of edges should be specified. A typical acceptance value for any 12-inch (30-cm) length of tape shall be 0.030 inch (0.76 mm) from the edge when measured with an appropriate straight edge.

5.4.1.6 *Splices.* Prepreg splices may be permitted on any roll of tape where processing is continuous without change in fiber or resin batch. Such splices shall be appropriately marked as a nonconforming area.

5.4.2 Resin content. The resin content of prepregs may be determined by extracting the resin from the prepreg fibers using a solvent in which the resin material is fully soluble and the fibers are not dissolved. Soxhlet extraction procedures are described in ASTM C 613 (Reference 5.4.2(a)). Procedures for determining the resin content of carbon fiber-epoxy prepregs are provided in ASTM D 3529. Special procedures and solvents may be required to extract high molecular weight or thermoplastic resins. An alternate procedure for determining the resin content in epoxy resin prepregs is outlined in Section 5.5.1.

5.4.3 Fiber content. Procedures used to determine resin content often provide information about the fiber content of prepregs. Alternatively, acid digestion methods (ASTM D 3171) may be applied to remove the matrix resin from the fibers as long as the fibers are not degraded (Reference 5.4.3). Digestion methods are not preferred for graphite and aramid fiber prepregs since such fibers are subject to oxidative degradation. Section 5.5.1 describes a procedure for determining the fiber and resin contents of glass, carbon, and aramid fiber/epoxy resin prepregs.

5.4.4 Volatiles content. The volatiles content of prepregs may be determined by ASTM D 3530 (Reference 5.4.4). Thermogravimetric analysis (TGA) may also be applied to estimate weight percent volatiles in a prepreg.

5.4.5 Moisture content. The moisture content of prepregs may be determined by coulometry in accordance with ASTM D 4019 (Reference 5.4.5) or by automated moisture meters based on the Karl Fischer titration method.

5.4.6 Inorganic fillers and additives content. The quantitative determination of inorganic fillers and additives in the prepreg resin requires considerable care. For example, the weight percent inorganic fillers and additives in a prepreg resin may be determined according to the procedure described in Section 5.5.1. Assuming that the organic resin material is fully soluble in tetrahydrofuran (THF) and that the inorganic fillers and additives are insoluble, the solution prepared in step 6 can be centrifuged to precipitate the insoluble components. The precipitate is washed at least three times with the solvent, dried, and then weighed.

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5.4.7 **Areal weight.** The areal weight (mass per unit area) of a prepreg material may be determined using ASTM D 3776 (Reference 5.4.7).

5.4.8 **Tack and drape.** Tack refers to the ability of a prepreg to adhere to itself or to other material surfaces and is a key factor in determining prepreg suitability for component/part fabrication. There is no quantitative method for measuring tack. Subjective terms such as high, medium, and, low are often used in describing tack. Although there is no generally accepted method for measuring tack, some composite manufacturer's use a Monsanto Tack Tester™ to obtain a relative index for prepreg tack. Drape is also a subjective term which relates to the ease of handling and conforming prepregs to complex surfaces.

5.4.9 **Resin flow.** Resin flow under specified test conditions relates to the composition, extent of reaction, and/or morphology of the prepreg resin, as well as the resin content. Prepreg processability and resin content in the processed laminate are affected by resin flow. Test conditions (temperature, pressure, layers of prepreg, number of bleeder plies) depend upon the type of resin. The resin flow of prepreg materials may be determined by ASTM D 3531 (Reference 5.4.9).

5.4.10 **Gel time.** Gel time relates to the chemical composition and extent of reaction of thermosetting prepreg resins. Prepreg processability is affected by the resin gel time. The test temperature depends upon the type of resin. The gel times of prepregs may be determined by ASTM D 3532 (Reference 5.4.10).

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5.5 TEST METHODS. The following methods are examples of analytical techniques that can be used for prepreg characterization.

5.5.1 Resin extraction procedure for epoxy resin prepregs. This procedure is applicable for determining the fiber and resin contents of glass, carbon, and aramid fiber/epoxy resin prepregs. Solutions prepared according to this procedure can be used directly for HPLC analysis provided appropriate grade solvents are used. Recommended sampling, specimen handling procedures, and standard laboratory safety procedures should be followed.

1. Cut a rectangular specimen (approx. 1 g) from prepreg section and weigh on analytical balance. (±0.001g or better). Record weight as W_o (grams).

2. Place specimen in 25 mL Erlenmeyer flask (fitted with a ground-glass stopper) and add about 20 mL THF (tetrahydrofuran, fresh, HPLC grade, distilled-in-glass, with no inhibitor added).

3. Stopper the flask and allow the specimen to soak in the THF for at least 4 hours.

4. Place flask on vortex mixer and agitate for 1 minute.

5. Carefully decant the THF solution into a 50 mL volumetric flask. The fibers should remain bunched together in the 25 mL flask.

6. Add about 10 mL THF to rinse the fibers in the 25 mL flask, mix on the vortex mixer, and decant the THF into the 50 mL volumetric flask containing the primary solution (step 5).

7. Repeat step 6.

8. Add THF to fill the volumetric flask to the 50 mL mark.

9. Carefully remove the graphite fibers from the 25 mL Erlenmeyer flask (using forceps), wrap fibers in Kimwipes™ or equivalent, place in labeled paper envelop, place the envelope in fume hood air stream, and allow fibers to dry overnight. Alternatively, residual THF may be removed by placing the envelope with fibers in a vacuum oven (fitted an appropriate vapor trap) set at 40°C and maintaining a vacuum for at least 1 hour.

10. The fibers are removed from the Kimwipes™ and weighed on an analytical balance. Record the fiber weight as W_f (grams).

11. Calculate the concentration of the resin solution (see step 8) and record concentration as C_o(μg/μL). This concentration will be useful in the analysis of HPLC data.

$$C_o = \frac{(W_o - W_f)}{0.050} \ \mu g/\mu L \qquad 5.5.1(a)$$

12. Mix resin solution (from step 11) on vortex mixer and immediately filter about 4 mL of the resin sample solution through a 0.2 μm Teflon™ membrane filter into a dry, clean glass vial. Immediately cap the vial to prevent contamination and solvent loss. This solution will be used for HPLC analysis. The remaining (unfiltered) solution in the flask can be used to determine soluble resin content and insoluble content (steps 16 and 17).

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13. The extractable resin content and fiber content, not corrected for the presence of volatiles and insoluble components in the prepreg resin and remaining on the fibers, are calculated -

$$\text{wt - \% extractable resin} = 100\% \times \frac{(W_o - W_f)}{W_o} \qquad 5.5.1(b)$$

$$\text{wt - \% fiber} = 100\% - \text{wt \% extractable resin} \qquad 5.5.1(c)$$

14. Glass fibers may be placed in a muffle furnace and heated at 650 to 800°C to remove nonextractable surface material. After cooling to room temperature, the fibers are reweighed and their weights are recorded as W_f'.

15. The amount of nonextractable fiber surface material in glass fiber prepregs is calculated -

$$\text{wt - \% nonextractables} = 100\% \times \frac{(W_f - W_f')}{W_o} \qquad 5.5.1(d)$$

16. Insoluble content. The extractable or THF-soluble resin content may also be determined by filtering the solution prepared in step 8 through a 0.2 μm Teflon™ membrane filter. Using a volumetric pipet, an aliquot (e.g., 10 mL) of the filtered solution is transferred to a pre-weighed aluminum pan (weight W_A) which is then placed into a fume hood to evaporate the solvent. A stream of filtered air or nitrogen can be directed over the surface of the pan to accelerate evaporation. After 9 mL or more of the solvent is removed leaving an oily residue of resin, the pan can be placed in a vacuum oven and heated at about 50°C for several hours to remove residual solvent. After cooling to room temperature, the pan is reweighed (W_A') and the resin content is calculated -

$$\text{wt - \% soluble resin} = 100\% \times (W_A' - W_A) \times \frac{5}{W_o} \qquad 5.5.1(e)$$

Differences in the weight-percent resin determined using Equations 5.6.1(b) and 5.6.1(e) may be attributed to the presence of volatiles and insoluble (nonfibrous) components in the prepreg.

17. Insoluble content. The amount of insoluble components may be determined by the following procedure. An aliquot of the solution from step 12 can be centrifuged to precipitate the insoluble components. The precipitate is washed at least three times with the solvent, dried, and weighed.

5.5.2 Procedure for HPLC/HPSEC analysis of glass, aramid, and graphite fiber prepregs. Mix resin solution (prepared in Section 5.6.1, step 12) on a vortex mixer and immediately filter about 4 mL of the resin sample solution through a 0.2 μm Teflon™ membrane filter into a dry, clean glass vial.

Immediately cap the vial to prevent contamination and solvent loss. The sample is now ready for HPLC analysis.

If the HPLC analysis is not run immediately, the sample solution should be kept in a cool, dark location. If care is taken during storage, the THF solution will remain stable and may be analyzed weeks after its preparation with no apparent effect on the HPLC analysis.

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5.5.2.1 *Reverse phase HPLC analysis.* The epoxy resin prepreg analysis can be run using any of a number of commercially available HPLC instruments. An integrator/recorder or state-of-the-art HPLC data analysis system is recommended for data acquisition, plotting, and reporting. HPLC operating conditions were selected for simplicity and compatibility with most commercial HPLC equipment.

HPLC System: Waters Associates model-244 instrument with M6000A solvent delivery systems, M720 system controller, 710B WISP auto-injection system, M440 UV detector, and M730 data module. Similar systems available from other manufacturers may also be used.

Solvents: Acetonitrile (distilled-in-glass) and reagent grade water prepared from distilled water using a Millipore Milli-Q2 (Millipore Corp., Bedford, MA) or equivalent water purification system. Purging the solvents with helium is recommended.

Column: Waters Associates μBondapak C18. (Similar columns available from other manufacturers may also be used).

Injection Volume: 10 mL

Flow Rate: 2.0 mL/min

Mobile Phase (solvent program):

Time	% Acetonitrile	% Water	Curve
0	45	55	*
12 min	100	0	7
16 min	100	0	*
20 min	45	55	6

Detector: UV 254nm Run Time: 20 minutes

5.5.2.2 *Size Exclusion Chromatography (SEC) analysis.* The SEC analysis of the prepreg resins can be run using HPLC instrumentation as described above.

Solvent: THF (distilled-in-glass) A helium purge should be maintained on THF for optimum results.

Columns: IBM SEC type A and type C, 5 micron (columns from other manufacturers, such as the Waters μStyragel 1000, 500, 100, 100 Å, are also acceptable).

Injection Volume: 10 μL Flow Rate: 1 mL/min
Detector: UV 254nm Run Time: 15 minutes

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Calculations: Integrated peak areas are converted to area percentages (% area).

5.5.3 **Procedure for Fourier transform infrared spectroscopy (FTIR).** Several droplets of the resin/THF solution (prepared in Section 5.5.1, step 12) are placed on the surface of a polished salt plate (preferably KBr). The sample is analyzed as soon as the THF has evaporated. A Perkin Elmer model 1550 or 1700 FTIR spectrometer with model 7500 computer or an equivalent instrument is used to scan and record the spectrum (500 to 4000 cm) of the salt plate with and without the sample on its surface. The analysis should be conducted with the salt plate and sample in a purged, dry nitrogen atmosphere at room temperature. Depending upon the sample, 100 to 200 scans of the spectrum may be required to optimize spectral resolution. It also may be necessary to deposit more or less sample on the salt plate. The spectrum of the sample obtained by subtracting that of the salt plate is plotted, reported, and stored on a computer disk.

5.5.4 **Procedure for differential scanning calorimetry (DSC).** This test can be performed using a DuPont Instruments 9900 Thermal Analyzer/Controller and model 912 DSC accessory or an equivalent instrument.

Specimen:	Prepreg (10 to 30 mg) in an aluminum sample pan
Reference:	Empty sample pan
Heating Rate:	10°C/min
Temperature Range:	Room temperature to 350°C
Atmosphere:	Dry nitrogen gas purge
Data Handling:	Data is stored on a computer disk and a plot of heat flow dH/dt (μW/sec) vs temperature (°C) is produced.
Heat of Reaction:	The calibration routine and integration program provided with the thermal analyzer is used to calculate heats of reaction ΔH of thermoset prepreg resins.
Glass Transition:	A cooling device attached to the DSC cell may be needed to facilitate glass transition temperature T_g measurements of thermoset prepreg resins; i.e., it is often necessary to initiate temperature scans at -50°C or lower since such resins typically have T_g values below room temperature. The thermal analyzer may have a software routine to assist in determining T_g values.

5.5.5 **Procedure for dynamic mechanical analysis (DMA).** A single ply of prepreg is cut into a 1.1 cm x 1.7 cm strip and the strip is mounted in a DuPont model 982 or 983 DMA accessory. A DuPont 9900 or 1090 controller is used to run the test and plot the results. Equivalent instruments may also be used.

Heating Rate:	5°C/min
Temperature Range:	Room temperature to 350°C
Atmosphere:	Dry, nitrogen gas purge
Data Handling:	Data is stored on a computer disk and a plot of storage modulus and tan δ is plotted as a function of temperature.
Glass Transition:	The temperature of the damping peak maximum is assigned as the T_g value.
Gelation:	Gelation occurs when the Young's modulus starts increasing rapidly (several orders of magnitude) over a narrow temperature range. Gelation temperature depends upon heating rate and mechanical frequency. Therefore, both heating rate and frequency should be included when DMA gelation temperatures are reported.

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Gelation Time: In the isothermal mode, the time to gelation is determined by rapidly heating a sample to the desired temperature, holding the temperature constant and monitoring the change in Young's modulus with time. Gelation time is defined as the time it takes for the modulus to start rapidly increasing (several orders of magnitude).

5.5.6 **Procedure for rheological characterization.** A Rheometrics Dynamic Spectrometer (RDS) or equivalent system is used for this test. Samples are prepared by cutting three 25-mm diameter circles from a single ply of prepreg. The three plies are stacked and placed between the rheometer's parallel plates.

Heating Rate:	2°C/min
Temperature Range:	Room temperature to the onset of gelation (for thermosets)
Atmosphere:	Air or a blanket of nitrogen gas
Geometry:	25-mm diameter parallel plate
Gap:	Typically 0.8 mm, but may be adjusted according to sample characteristics.
Data Reporting:	Shear moduli (storage and loss) and complex viscosity are plotted as a function of temperature.

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Chapter 6. LAMINA AND LAMINATE CHARACTERIZATION

Page

6.1 INTRODUCTION. 6-2

6.2 NON-DESTRUCTIVE EVALUATION. 6-2

6.3 PHYSICAL ANALYSIS TECHNIQUES. 6-3
6.3.1 Thermal analysis. 6-3
6.3.2 Rheological analysis. 6-3
6.3.3 Fiber volume. 6-3
6.3.4 Void volume. 6-3
6.3.5 Density. 6-4
6.3.6 Dimensional stability. 6-4
6.3.7 Moisture weight gain. 6-4

6.4 ENVIRONMENTAL EXPOSURE TESTING. 6-4

6.5 FLAMMABILITY. 6-5

6.6 LAMINA, LAMINATE, AND FABRIC MECHANICAL PROPERTY TESTS. . . . 6-5
6.6.1 General. 6-5
6.6.1.1 Test method acceptance criteria. 6-6
6.6.2 Tensile tests. 6-6
6.6.2.1 General considerations. 6-6
6.6.2.2 0° Unidirectional laminae. 6-7
6.6.2.3 90° Unidirectional laminae. 6-7
6.6.2.4 Bidirectional and fabric laminates. 6-7
6.6.3 Compression tests. 6-8
6.6.4 Flexure tests. 6-8
6.6.5 Shear tests. 6-8
6.6.6 Fatigue. 6-10
6.6.7 Creep. 6-10
6.6.8 Damage tolerance and laminate testing. 6-11

6.7 FILAMENT WINDING MECHANICAL PROPERTY TESTS. 6-12

6.8 BRAIDING MECHANICAL PROPERTY TESTS. 6-12

6.9 THICK-SECTION COMPOSITES MECHANICAL PROPERTY TESTS 6-14
6.9.1 Introduction. 6-14

6.10 THERMAL AND MOISTURE ABSORPTION PROPERTIES. 6-14

6.11 ELECTRICAL PROPERTIES. 6-14

REFERENCES 6-15

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6.1 INTRODUCTION. The use of composite materials continues to increase as new performance, reliability, and durability requirements drive hardware designs to higher levels of structural efficiency. Additionally, government requirements are becoming more stringent to ensure proper levels of structural integrity are maintained. These design drivers, among others, have resulted in a growing recognition that certification or qualification of aerospace structure requires an extensive combination of analysis, testing, and documentation.

Further, because of the large number of design variables inherent to composite structure, analytic models are even more necessary than for metallic structure to ensure completeness of the hardware qualification process. Inherent in all structural analysis models are material, physical, and mechanical property characterization data. Ideally, these analytic models would permit analysts to predict full-scale structural response (e.g. stability, deflections, strength, life) directly from a generic (lamina) material database. In truth, test data is required at design development (element, subcomponent, component) and full-scale article test levels as well as the generic (coupon) levels of evaluation.

The purpose of Chapter 6 is to provide guidelines on testing procedures for characterization of physical and mechanical lamina (ply) properties. While current procedures emphasize development of a lamina-level database, this does not preclude higher-level testing.

A laminate is a product made by bonding together two or more layers of material or materials, and a lamina is a single ply or layer in a laminate. The material forming each layer typically consists of a carbon, glass, or organic (polymeric) fiber reinforcement embedded in a thermoplastic or thermosetting resin matrix. While retaining their identities in the composite, the constituents combine to provide specific characteristics and properties.

Many techniques are used to characterize the chemical, physical, and mechanical properties of composite materials. The purpose of this chapter is to provide information on techniques that may be used to analyze and evaluate these properties. The test methods discussed in each section may not be appropriate for all types of composite materials. Currently, more studies are being conducted to investigate how variations in reinforcement and resin chemistry and morphology may affect the physical properties and long term performance of composites. Where possible, the limitations of existing test methods are discussed.

6.2 NON-DESTRUCTIVE EVALUATION. A variety of non-destructive testing (NDT) techniques are available for detecting both surface and interior flaws in composites. Visual inspection and liquid penetrant methods can be used for identifying surface defects, while more sophisticated techniques are required for detecting internal flaws (i.e. voids, delamination, foreign inclusions, disbonds). These techniques include ultrasonics, radiography, thermography, acoustic emission, X-ray and eddy-current testing. The basic principles and procedures for these methods are covered in the MIL-HDBK-728 series, while more specific information on the theory and interpretation of data can be found in the following:

MIL-HDBK-731 Thermography
MIL-HDBK-732 Acoustic Emission
MIL-HDBK-733 Radiography
MIL-HDBK-787 Ultrasonic

These documents do not discuss the recent advances in NDT techniques, which is currently an active field of research and development.

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6.3 PHYSICAL ANALYSIS TECHNIQUES.

6.3.1 Thermal analysis. The thermal analytical techniques described in chapter 4, section 4.3.1 may also be used to evaluate composite materials. Information obtained from thermal analysis includes the glass transition temperature, crystalline melt temperature, expansion/contraction properties, thermal stability, and extent of cure for thermosets.

The glass transition temperature (T_g) is the temperature at which a material passes from a glassy state to a rubbery state, defining the upper use temperature region for a given composite material. Below T_g large scale molecular motions do not occur and the material possesses the properties desirable for structural components (i.e. strength, stiffness), above T_g large scale molecular motion is possible and the material properties are quite different. The changes in properties are much more evident in thermoplastic materials than in the crosslinked thermosets. The standard approach for measuring T_g is by thermal analytical techniques. These include differential thermal analysis (DTA), differential scanning calorimetry (DSC), thermal mechanical analysis (TMA), and dynamic mechanical analysis (DMA). The order of sensitivity for the analysis of composites is, DSC least sensitive, then DTA, TMA, and DMA.

The crystalline melt temperature (T_m) of semi-crystalline thermoplastic composites can be obtained from DSC or DTA experiments. In addition, an estimate of the degree of crystallinity can be made. This becomes an important parameter since the properties of semi-crystalline thermoplastic composites may be dependent upon the degree of crystallinity of the matrix resin. The heating required for processing prepregs into composite structures may have an affect on the degree of crystallinity as well as the crystal structure.

The coefficient of thermal expansion (CTE) can be obtained from TMA. The CTE is obtained from the slope of the linear expansion versus temperature curve. Thermoset composites typically exhibit different linear expansion regions above and below T_g. ASTM Test Method E 831-86 "Linear Thermal Expansion of Solid Materials by Thermomechanical Analysis" describes the technique (Reference 6.3.1).

The extent of cure for thermosetting composites can be evaluated by thermal analysis. Dynamic DSC or DTA can be used to detect residual heat of reaction, indicating incomplete cure. TMA and DMA may also be used to obtain relative information on the extent of cure for thermosets.

6.3.2 Rheological analysis.

6.3.3 Fiber volume. The fiber volume of composites may be obtained from ASTM Test Method D 3171-76 "Fiber Content of Resin-Matrix Composites by Matrix Digestion" (Reference 6.3.3(a)). Values in terms of volume require determining the density of the composite sample, which may be obtained by a liquid displacement technique described in ASTM Test Method D 792-86 (Reference 6.3.3(b)). Another method based on known values of areal weight and fiber density has recently been published (Reference 6.3.3 (c)). Elastostatic measurements (Reference 6.3.3(d)) and ultrasonic methods (Reference 6.3.3(e)) have also been used for determining fiber volume.

6.3.4 Void volume. The void volume of composite materials is not easily determined. Ultrasonic NDT methods can estimate void size and give qualitative information, however quantitative analysis is not currently possible. Other methods have been tried with varying degrees of success. These include, sonic methods, radiography, electrical properties, microwave techniques, and infrared/thermal NDT methods (References 6.3.4(a) - 6.3.4(d)). Methods such as ASTM Test Method D 2734-70 "Void

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Content of Reinforced Plastics", require accurate determinations of the specific gravity of the resin and fibers in order to calculate the void content (Reference 6.3.4(e)).

6.3.5 Density. The density or specific gravity of composite samples may be determined from the following test methods (References 6.3.3(b) and 6.3.5):

ASTM D 792-86 "Specific Gravity and Density of Plastics by Displacement".

ASTM D 1505-85 "Density of Plastics by the Density-Gradient Technique".

6.3.6 Dimensional stability. Dimensional changes in composite materials are typically a function of temperature and/or moisture. Changes in length or volume of a sample can be detected by mechanical, optical or electrical transducer and recorded as a function of temperature or time. Several techniques for measuring linear expansion, such as dial gauges, micrometers, telescopes, linear variable differential transformers, interferometers, and X-ray diffraction patterns, have been used. As discussed in Section 6.3.1, TMA is a good technique for determining linear thermal expansion. Standard test methods for obtaining thermal expansion values are (References 6.3.1, 6.3.6(a) and (b)):

ASTM D 864-52 "Coefficient of Cubical Thermal Expansion of Plastics".

ASTM D 696-79 "Coefficient of Linear Thermal Expansion of Plastics".

ASTM E 831-86 "Linear Thermal Expansion of Solid Materials by Thermo-mechanical Analysis".

6.3.7 Moisture weight gain.

6.4 ENVIRONMENTAL EXPOSURE TESTING. The evaluation of environmental effects on composite materials has been an area of investigation for a number of years (References 6.4 (a) - 6.4(g)). Both field exposure and in the laboratory accelerated exposure methods have been used. Typically, hygrothermal effects are being measured, and they are expressed as a change in the mechanical and/or physical properties of the material. However, the determination of changes in the chemical properties of the resin and/or fiber material should also be evaluated, as well as exposure to other environments (References 6.4(h) - 6.4(l)).

The type of environment used for exposure testing should be realistic with respect to the materials intended application. Although not intended for materials testing, MIL-STD-810E provides some information on environmental testing and possible effects of various exposure conditions on military equipment performance (Reference 6.4(m)). ASTM standards which may be applicable include (References 6.4(n) - (q)) :

ASTM D 543-87 "Resistance of Plastics to Chemical Reagents", provides guidance for determining changes in materials upon exposure to standard reagents.

ASTM D 756-78(83) "Determination of Weight and Shape Changes of Plastics Under Accelerated Service Conditions", seven procedures for exposing plastics to combinations of extreme humidity and temperature that will accelerate the material changes taking place in a sheltered space subjected to humidity and temperature variations.

ASTM D 4364-84 "Performing Accelerated Outdoor Weathering of Plastics Using Concentrated Natural Sunlight", provides a procedure for performing accelerated outdoor exposure testing of plastics using a Fresnel-reflector accelerated outdoor weathering test machine.

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ASTM D 1435-85 "Outdoor Weathering of Plastics", defines test conditions for the exposure of plastic materials to weathering.

Physical effects of exposure may include changes in surface properties, electrical properties, dimensions and weight. Visual inspection of the material can give qualitative information on color and/or gloss changes, crazing/cracking and fiber blooming. Weight and dimensional changes will yield information on the type of interaction between the material and the environment (fluid absorption, corrosion, erosion, chemical reaction). Chemical property changes may result from thermo-oxidative, photo-oxidative, or chemical degradation modes. Mechanical property changes are usually observed as changes in tensile, compressive, shear and flexural strength, as well as decreased resistance to crack growth.

The mechanical test methods used for determining the effects of exposure must be sensitive to the type of deterioration process being investigated. Changes in the resin/fiber interface or matrix material would require tensile and compression testing in transverse direction, and in-plane shear. Section 6.6 on Lamina, Laminate, and Fabric Mechanical Test Methods provides information on the types of standard testing techniques currently in use. In addition, the application of a static and/or dynamic stress during exposure should be considered. This type of testing may require special equipment or test fixtures which will maintain the desired load during environmental exposure. Ideally, the validity of laboratory tests would need to be established by comparison with field tests and in service inspection data.

6.5 FLAMMABILITY. The flammability characteristics of organic polymer matrix and fiber materials is an important consideration when designing composite structures (Reference 6.5(a)). Along with a loss of mechanical properties, these materials may release toxic gases and dense smoke during pyrolysis. The flammability properties can be altered by the addition of flame retardant additives (Reference 6.5(b)). The additives may or may not be incorporated into the backbone of the polymer matrix, depending upon the resin formulation. Test methods used for evaluating plastic materials are listed below. These methods were not developed for testing composites (References 6.5(c) - (g)) .

ASTM D 568-77 "Rate of Burning and/or Extent and Time of Burning of Flexible Plastics in a Vertical Position".

ASTM D 635-81 "Rate of Burning and/or Extent and Time of Burning of Self- Supporting Plastics in a Horizontal Position".

ASTM D 2863-87 "Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of Plastics (Oxygen Index)".

ASTM D 2843-88 "Density of Smoke from the Burning or Decomposition of Plastics".

ASTM F 814-84B "Specific Optical Density of Smoke Generated by Solid Materials for Aerospace Applications".

6.6 LAMINA, LAMINATE, AND FABRIC MECHANICAL PROPERTY TESTS.

6.6.1 General. Section 6.6 contains test methods for determining material property data for composite materials. The purpose of this section is to provide for uniformity in the use of standard test methods and, ultimately, to provide for combining the experimental data. The reader is referred to Chapter 8 for reporting of data to MIL-HDBK-17. The test methods are representative of procedures

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used in the composite materials industry and were selected after review of user material specifications. Specific standards are included to allow the user to perform tests consistent with industry practice; however, inclusion of these standards should not be considered an endorsement of any standard or organization by MIL-HDBK-17.

These discussions reflect the current dynamic state of test methods development for composite materials. The intent is to inform the reader of the availability of test standards, and to alert the reader to the limitations of these test methods. New test methods, or modifications to a test standard that require a dissimilar test fixture or deviate from the recommended specimen configuration, are not included in MIL-HDBK-17 unless there is a general consensus that mention of the new technique or modification will lessen confusion. By restricting the discussion to test methods which are standards, or to tests which are being evaluated for standardization, MIL-HDBK-17 seeks to avoid an endorsement of a technique which may be popular, but which has not undergone the careful scientific scrutiny that insures a thorough and comprehensive evaluation of the precision, accuracy, and validation of the test.

It should be noted that procedures which may enhance the preparation of the test specimen may be given. These statements are deliberately general in terms of applicability and are based upon the experience of the composite materials industry. The reader should further investigate any enhancements of a standard before proceeding with a test program.

6.6.1.1 *Test method acceptance criteria.* The test method shall have undergone a rigorous review of its precision and bias by an independent association that may include representatives from material suppliers, end-users, academia, or government. This review shall include interlaboratory (round-robin) testing conducted under the sponsorship of a standards-making organization, and it shall also be included in a referenceable, open-literature publication. The round-robin test data shall be made available to the MIL-HDBK-17 Coordination Group for review, validation, and acceptability of the test method to MIL-HDBK-17.

If there is no available standard test method, then the MIL-HDBK-17 Coordination Group may select "common practice" test methods for inclusion in the Handbook. A "common practice" test method is one that is widely used within industry, academia, or government, but variations exist in test specimen or test fixture design. However, there are no discernible technical differences between the various test methods, i.e., the test methods determine the same material property with similar degrees of precision and bias.

A "common practice" test method shall also have begun the process towards acceptance by a standards-making organization as either a standard or a guideline for practice to independently determine its precision and bias. "Common practice" test methods are clearly identified in MIL-HDBK-17.

6.6.2 Tensile tests.

6.6.2.1 *General considerations.* Tensile testing of laminates will generally be accomplished in accordance with ASTM Test Method D 3039 (Reference 6.6.2.1). This test employs a straight-sided specimen with bonded end tabs for gripping, and is applicable to 0° and 90° unidirectional constructions as well as bidirectional and fabric laminates. Details for specific orientations are discussed in Sections 6.6.2.2, 6.6.2.3, and 6.6.2.4. The test yields tensile ultimate strength, tensile modulus, and the major Poisson's ratio.

Experience within the aerospace industry has indicated that the critical factors in obtaining consistent tensile test results are tab design, tab application, fiber alignment, and specimen machining

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quality. With respect to machining, it is important that methods be used that will not introduce cracks, as these could cause premature failure due to stress concentration. If rough cutting is performed, a secondary operation should be used that will remove any edge damage. Variables that can affect the degree of damage and effectiveness of damage removal are cutting tool design, cutting/grinding tool speed, feed rate, amount of material removed per pass, and use of liquid coolants. In general, liquid coolants are recommended to prevent thermal damage, but may not be necessary when removing small amounts of material per pass. Any coolant used shall not adversely affect the composite being machined. Tool design is particularly important when machining aramid reinforced composites (which are often difficult to cut cleanly). Although specific machining procedures depend on coupon configuration and material, it is generally advantageous to remove small amounts of material per pass and use a slow feed rate and relatively high tool speed, thereby keeping surface temperatures below the intended service temperature of the composite.

The purpose of tabbing is to introduce uniform loading into the specimen. The degree to which this is realized depends upon a number of factors including tab bevel angle, and the mechanical properties of the tab material relative to the laminate being tested. In general, the tab material should be less stiff than the test laminate.

Surface preparation prior to tab bonding may include light sanding or grit blasting to promote adhesion. Care must be taken not to abrade any part of the laminate gage section (any area not covered by the tab), as this damage could cause premature failure.

Thermal considerations are also important. If the coefficient of thermal expansion of the tab material is significantly different from that of the laminate, bond failure could occur either during bond cycle cool-down, or during elevated temperature testing. In addition, bond cycles that will advance the cure state of the laminate cannot be used, as this will invalidate the data. For this reason, many companies use 250°F (120°C) curing adhesives when tabbing 350°F (180°C) cured laminates. This is of less concern when postcured laminates are being tested.

6.6.2.2 *0° Unidirectional laminae.* When preparing 0° specimens, care must be taken to insure that the fibers are aligned parallel to the coupon load axis. A misalignment of only 0.5° has been shown to affect ultimate strength by as much as 5%. To accommodate capacities of standard composite test machines, the width of 0° unidirectional specimens is usually specified as 0.500 inches (12.7 mm).

6.6.2.3 *90° Unidirectional laminae.* As specified in ASTM Test Method D 3039, the 90° tensile specimen is wider and shorter than the 0° coupon. Since this test is totally matrix dependent, the handling and precise alignment of the specimen are critical. Special alignment fixtures are often used in conjunction with hydraulic or pneumatic test grips to prevent introduction of out-of-plane forces.

As an alternate, a sandwich beam specimen in accordance with ASTM Test Method C 393 may be used for 90° testing (Reference 6.6.2.3). This specimen is less susceptible to handling damage, but panel warpage is a problem.

6.6.2.4 *Bidirectional and fabric laminates.* The specimen for these laminates is wider than the 0° unidirectional coupon. For fabrics it is mandatory to orient the warp and fill for each ply in the same direction, and to report the direction tested. A modified ASTM D 3039 dog-bone shaped specimen is frequently utilized for testing fabrics. As larger tow, fewer ends per inch fabrics have become more common, it has become practical to increase the gage width to 1.000 inch (25.4 mm) to reduce scatter of test results.

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6.6.3 Compression tests. Compression test methods for composite materials generally fall into three categories: unsupported coupon, supported coupon, and sandwich beam. ASTM Test Method D 3410, "Compressive Properties of Unidirectional or Cross-ply Fiber-Resin Composites", describes the specimens and fixtures for the unsupported coupon and sandwich beam tests (Reference 6.6.3(a)). The unsupported coupon has a recommended gage length of 0.500 inch (12.7 mm) and can be placed in one of the two compression fixtures given in ASTM D 3410. The two fixtures have split collet-type grips at both ends but they differ in the way load is introduced and in the gripping arrangement. The sandwich beam is a rectangular bonded beam with a 6-ply, all 0° composite test skin. The skin is bonded to a high density core. It should be noted that the sandwich beam specimen is generally expensive to test and unsuitable for environmental testing. There are also questions concerning the influence of honeycomb on the test results.

ASTM Test Method D 695, "Compressive Properties of Rigid Plastics", was not designed for composite materials, but nonetheless it has been modified by the aerospace industry for use with composites (Reference 6.6.3(b)). References will often be made to a "modified ASTM D 695 test", but each company has developed its own modification and there is no one accepted test fixture, gage length, or test procedure. In general, a modified ASTM D 695 test supports the face of the specimen and uses a gage section ranging from 0.1 to 0.2 inches (2.5 to 5.1 mm) in length. The supported specimen face of this test method, in addition to the smaller gage section, generally results in higher compressive strength and moduli values than ASTM D 3410. These higher values make it the method of choice for many designers.

There is considerable debate as to whether a modified ASTM D 695 test results in a true compressive failure. The constrained specimen may not be representative of a compressive failure in composite materials. ASTM D 3410 may produce a more accurate mechanistic compressive failure in composites, but the design values are generally lower and may not be representative of a compressive failure mode in an aircraft structure.

Regardless of choice, ASTM D 3410 or modified ASTM D 695, it has become more practical to incorporate 1.000 inch (25.4 mm) gage width laminate specimens for fabrics with low end count per inch weaves (such as 6000 filament graphite tow fabrics).

6.6.4 Flexure tests. There is not a recommended test method for determining the flexural properties of composite laminates. Even though there are approved flexure test methods, there is some debate as to the validity of the results.

Within the aerospace industry, flexure testing is primarily used for quality control. ASTM Test Method D 790, "Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials", was originally written for plastics but has since been modified and approved for composites (Reference 6.6.4). In some cases, ASTM Test Method C 393, "Flexure Test of Flat Sandwich Constructions", has been adapted for use with composite laminates (Reference 6.6.2.3).

6.6.5 Shear tests. The use of continuous fiber reinforced composites as high performance structural materials in modern aerospace vehicles necessitates the accurate measurement of the material response to mechanical and thermal loads. Most of the current composite material shear test methods were originally developed for metals, wood, and adhesives. However, coupling effects, nonlinear behavior of the matrix or the fiber-matrix interface, and the presence of normal stresses combine to make the present shear test methods of questionable value for composite materials (References 6.6.5(a) - (d)).

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The shear test methods most widely used in the aerospace industry to measure the interlaminar and in-plane shear moduli and shear strengths of composite materials are subject to the above-mentioned limitations. For instance, in the Short Beam Shear test (ASTM Test Method D 2344), the shear stress distribution depends upon the beam length-to-thickness ratio and eccentricities of the rollers applying the loads (Reference 6.6.5(a),(e)). Stinchcomb et al. (Reference 6.6.5(b)) found that a good quality specimen failed in microbuckling or a combination of shear and microbuckling and that a shear failure only occurred in poor quality specimens. The Short Beam Shear test should not be utilized for determining the interlaminar shear strength. Short beam shear is useful for quality control and perhaps for selecting candidate materials, but it does not provide an accurate measurement of interlaminar shear strength.

Similar problems exist for the Rail Shear test (ASTM Guide D 4255), which is one of the most analyzed methods for determining the in-plane shear modulus of composite materials (Reference 6.6.5(f)). Stress concentrations at the corners and significant normal stresses are produced in the test section depending upon the method of load introduction, stiffness of rails, and properties of the composite (Reference 6.6.5(c)).

The torsional shear of a thin-walled circular tube (ASTM Test Method E 143) is considered the most desirable method to obtain both the shear strength and the shear modulus of a material (Reference 6.6.5(g)). Though from the applied mechanics viewpoint this is the ideal shear test, the torsional tube presents many problems to the experimentalist. Cost of fabricating tubular specimens can be prohibitive unless the manufacturer is involved in filament winding. Preparation of a tube is time-intensive and requires more material than a flat specimen. Composite tubes can also be extremely fragile and difficult to handle. Further, the specimen must be mounted concentrically in the test apparatus to prevent the introduction of bending moments, and the tube must be free to move axially to avoid introducing axial forces. Buckling of the tube must also be prevented. This requires special test equipment which can be expensive to manufacture. Yet, for all its limitations, when shear properties from this test are available, these values are the standard by which results from other shear tests are judged.

The ±45 Off-Axis Tensile Shear Test (ASTM Practice D 3518) consists of loading a ±45-degree symmetric laminate uniaxially in tension (Reference 6.6.5(h)). This test has the advantage of being economical with material and time and it involves a simple test procedure (Reference 6.6.5(d)). Researchers have reported good correlation with unidirectional laminates between the results of the ±45-degree shear test and other shear test methods (References 6.6.5(d,i-k)), including the torsion tube. However, like all other shear test methods, this shear test is not without its problems. The stress/strain response tends toward ductile characteristics as a result of interply effects (Reference 6.6.5(j)). The stiffness is considered reliable up to 1.3% strain, but at higher stress levels the stiffness is underestimated due to edge effects (Reference 6.6.5(d)). Strength is also governed by these edge effects and is considered a conservative, lower-bound design value. Note that ±45 off-axis shear test results with fabric materials become more difficult to interpret due to the nature of the product form, mechanical testing, etc.

However, for determining the initial in-plane shear modulus, the ±45 Off-Axis Tensile Shear Test (ASTM Practice D 3518) may provide a value which reflects the actual stress state in a laminated structure. The specimen's ductile stress-strain response resulting from interply effects is indicative of the fact that a "pure" shear stress does not exist in the specimen, yet it may mimic the actual stress state that may occur in a laminate and may be representative of the interaction of one lamina upon adjoining laminae. The resulting value is an "effective" shear modulus, and may be a more realistic determination for the designer.

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6.6.6 Fatigue. Static testing of unidirectional composite coupons is useful for material characterization, comparison of materials, and for prediction of application laminate properties through the use of lamination plate theory. In the area of fatigue, however, no generalized methodology has yet been devised to predict laminate behavior from unidirectional coupon data. Hence, the development of fatigue design values becomes a unique problem for each application lay-up. Many studies have been undertaken, and much has been written concerning life prediction for specific laminates under cyclic loading spectra. Even at this level, empirical methods have been favored due to the inadequacy of results obtained from cumulative damage models, fracture mechanics analyses, and other theoretical approaches (References 6.6.6(a) and (b)).

ASTM Test Method D 3479, "Tension - Tension Fatigue of Oriented Fiber, Resin Matrix Composites", is a generalized coupon testing method (Reference 6.6.6(c)). However, because composite fatigue is so application dependent, it is important that the laminates represent the application and that the laminates testing account for the service load spectra and environmental conditions. Currently this is accomplished in composite hardware programs through a "building block" test approach involving coupon, element, and component specimens, all representative of full-scale structural details.

It is important to note that, for the majority of current aircraft composite structure, fatigue capability does not become a limiting factor if all static strength concerns have been thoroughly and successfully addressed. Exceptions to this are high-cycle components such as those found in helicopter dynamic systems.

6.6.7 Creep. Creep is defined as the change in a property over time when subjected to a constant forcing function. Here we are interested in the mechanical aspects of creep and monitor the change in measured strain with time when the specimen is subjected to a constant load. This change in strain must be indicative of an effective change in stiffness of the test specimen. This change may be (or may not be) recoverable upon unloading.

Creep should be considered if the end use involves high stress in the matrix-dominated direction, high temperature, or exposure to a harsh chemical environment - in other words, if there is a chance of matrix softening. In composites with a thermoplastic matrix, concern for creep is important, particularly if the service temperature is near or above the glass transition temperature. In a thermoset matrix, creep is expected to be small due to cross-linking. In general, creep testing does not provide primary design data. Designs should be checked for creep deformation if the working load involves major shearing action, e.g., short beam bending, etc. In composites, large shear stresses can be generated near a structural discontinuity; however, creep can be beneficial in some of these instances in relaxing the stress and avoiding catastrophic failure.

In fiber-reinforced plastics (composites), one can assume that creep will be more important when the composite is loaded in a matrix dominated manner than in a fiber dominated manner. For instance, the creep of a unidirectional specimen tensile loaded in the fiber direction is expected to be small and hence only of secondary importance. However, loading a specimen in a matrix dominated manner is not as straightforward as one would expect. Testing a unidirectional specimen in a transverse tensile manner one would think must essentially load the matrix, and this is not so. There are several explanations. One of the explanations is that loading a transverse specimen puts the matrix in a bidirectional state of stress (tensile) because the fibers prevent the matrix from laterally contracting (i.e., Poisson's effect) and thus the amount of creep response is restricted. Another argument for low creep response in transverse specimens is that the specimens are weak and the strains are small, so the change in strain would also be small. Another way to load the matrix is in shear and the creep response should be large. The most convenient way to load the matrix in shear is to load a [±45]

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specimen in tension. Although there is some argument that this test does not produce pure intralaminar shear, it at least produces some shearing and can be thought analogous to loading a unidirectional laminate in shear. Experience has shown that the resulting creep is significant. Other loadings that would be interesting to examine with respect to creep response would be compression of unidirectional specimens in the fiber direction and three-point bend loading of unidirectional specimens (in both of these methods shear plays a role).

Common experimental procedure is to apply a dead weight tensile load to a [±45] specimen at 35 MPa, 70 MPa, or 105 MPa (5, 10, or 15 ksi) and monitor the strain as a function of time. The strain reading at the first application of the full load is designated as the strain at zero time. Subsequent measurements are timed from that zero time reading. Readings are taken at 1, 2, 3, 10, 20, 30, 60, 100, and 200 minutes, and then as convenience dictates. Strain as a function of time is plotted on semi-log graph paper and the test continued for at least 30,000 minutes (or 3 weeks). Testing should be done at controlled (constant) humidity and temperature conditions (References 6.6.7(a) and (b)). Generally, specimens are 25 mm wide, 150 mm long and ~1 - 1.5 mm thick (1.0 in. wide, 6 in. long, and ~0.04 - 0.06 in. thick). These dimensions are open to question; there is some evidence that wider samples will creep less than narrow samples.

6.6.8 Damage tolerance and laminate testing. The resistance of composite materials to damage has gained broad interest in the aerospace community. To work toward characterizing the damage tolerance of these materials, several tests have been developed. These include laminate tests such as compression after impact and various tests for determining strain energy release rate.

For compression after impact testing, the most widely used configuration is that discussed in NASA Publications 1092 and 1141 (References 6.6.8(a) and (b)). Among the factors to be considered in using these methods are the actual energy of the falling tup in impact, the rigidity of the holding fixture, specimen preparation techniques, and the materials used and actual configuration of the compression test fixture. Other compression after impact configurations are also in use, and no industry wide standard has been accepted.

Strain energy release rate, G_c, is currently the subject of considerable research. The most common method for determining G_c, the mixed mode strain energy release rate, is the edge delamination test. This test involves measuring the load and strain at onset of the first delamination of a specified lay-up when tested in tension. For determination of G_c, basic laminate properties must be known. Detailed procedures outlining lay-up, test method, and calculations are given in NASA Publications 1092 and 1142 (References 6.6.8(a) and (b)). Further discussion may be found in Reference 6.6.8(c).

Mode I strain energy release rate, G_{Ic}, is usually determined by using a hinged double cantilever beam specimen. The two halves of a pre-cracked specimen are pulled apart in tension so that the crack propagates between laminate layers in the center of the specimen. Load and crack-opening displacement are monitored throughout the test. Details of the specimen configuration and test procedure are given in Reference 6.6.8(a).

Factors that may effect the test results include resin content and specimen geometry. There are indications that an optimal resin content range may exist outside of which G_{Ic} values will be lower. The crack length over which the data is reduced is another variable which requires further standardization.

Mode II strain energy release rate, G_{IIc}, may be determined by using an end-notched flexure specimen. This approach is discussed in Reference 6.6.8(d). The test is designed so that a precrack is allowed

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to propagate through the middle plane of the specimen during Mode II loading. Issues of test specimen geometry, loading rates, lay-up, and other factors have not yet been standardized.

ASTM Committee D-30 is currently working on test procedures for these properties. The approaches currently being considered are: Mode I - the double cantilever beam test; Mode II - the end-notch flexure test; and mixed mode - the edge delamination and the cracked-lap shear tests.

6.7 FILAMENT WINDING MECHANICAL PROPERTY TESTS. The mechanical behavior of filament wound structures is typically different from the behavior of flat laminated structures. Some noted differences result from the type of cure, resin void content, microcracking, and free edge construction. However, filament wound structures require the same mechanical property data for design and analysis as used for general laminated structures. The majority of filament wound structures are used in the rocket motorcase community and, consequently, most of the test specimens are in the form of cylinders or bottles that more closely simulate the geometry of the structures to be designed and analyzed.

In November 1983, the Joint Army, Navy, NASA, and Air Force (JANNAF) Interagency Propulsion Committee charted by the Department of Defense (DoD) formed the Composite Motorcase Subcommittee (CMCS) (Reference 6.7(a)). The CMCS is concerned with the application of composites materials in the construction of rocket motorcases for strategic and tactical missiles, space propulsion systems, and cartridge cases for gun propulsion. The CMCS consists of four working panels: (1) Testing and Inspection (T&I), (2) Design and Analysis (D&A), (3) Processing, and (4) Materials. The T&I Panel surveyed industry on test methods which resulted in seventeen different tension tests, seventeen different compression tests, and sixteen different shear tests that were being used to obtain mechanical property data. The T&I and D&A Panels joined to evaluate the test methods via a JANNAF Workshop (Reference 6.7(b)). A panel of experts in filament wound composites was selected and tasked to make recommendations for test methods. A joint T&I and D&A Member JANNAF Workshop was held in April 1986 to discuss the panel of experts' recommendations and to have an industry selection of JANNAF Interim Standard Test Methods to be used for the determination of uniaxial material properties for the design and analysis of filament wound structures. Table 6.7 contains the test methods selected and the output parameters obtained from each test. An element of orientation is included to identify directions of the output parameters.

The JANNAF CMCS is presently engaged in Round Robin Testing (RRT) of the Interim Standard Test Methods. The RRT effort includes industry and government laboratories. Until the results of the RRT are known, JANNAF cannot recommend detailed test methods for uniaxial material characteristics of filament wound structures. The composite motorcase community also has a vital interest in characterizing the mechanical behavior of filament wound structures subjected to biaxial and triaxial states of stress. The JANNAF efforts are being coordinated with MIL-HDBK-17, ASTM D-30 Committee, SACMA, IDA, and the DoD Standardization Program for Composites Technology (CMPS).

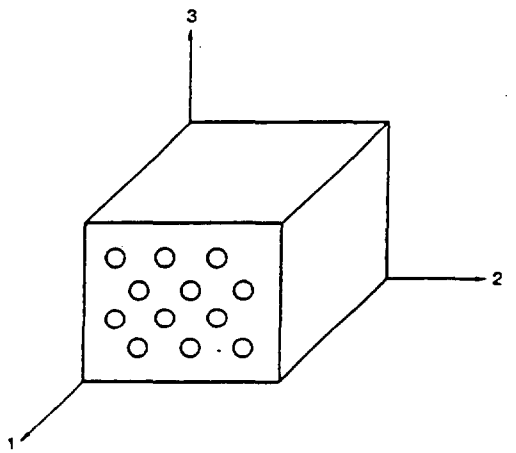
6.8 BRAIDING MECHANICAL PROPERTY TESTS. The goal of testing braided specimens is to produce mechanical property data which mimics the performance of that section of the braided structure in a representative manner. Often this is difficult to achieve due to the local contour or details which are not accurately formed into the test specimens. Therefore certain assumptions as to the applicability of the specimens to the final part must be made prior to the start of the allowables testing.

Most specimens assume pristine manufacturing was used to make the coupon and that this is representative of the final part. This initial coupon may or may not mimic the final product depending upon care and reproducibility in the design of the part. Test specimens for tension, compression,

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TABLE 6.7 *JANNAF interim standard test methods.*

0° Tension:	Pressurized NOL Ring, Pressurized Tube (90° Wind)	E_{11}, ν_{12}, σ_{11}, ϵ_{11}
90° Tension:	Tube (90° Wind)	E_{22}, ν_{21}, σ_{22}, ϵ_{22}
0° Compression:	Flat Laminate (0°)	E_{11}, ν_{12}, σ_{11}, ϵ_{11}
90° Compression:	Tube (90° Wind)	E_{22}, ν_{21}, σ_{22}, ϵ_{22}
In-Plane Shear:	Torsion Tube (90° Wind)	G_{12}, τ_{12}, γ_{12}
Transverse Shear:	Iosipescu	G_{23}, τ_{23}, γ_{23}



shear, pin bearing, interlaminar shear and interlaminar tension are described in other sections of this handbook. In some cases, specimen geometry may be more accurately represented by the actual shape of the zone -- for example a tapered coupon (preferably symmetric) which contains the essential geometry of the actual tapered braid. In general, the specific mechanical properties of interest are torsional stiffness and strength, shear stiffness and strength, tension, compression and flexural moduli and strengths, bearing strength for bolted assemblies, and adhesive bond strength where required. In the event that the braided parts could be damaged or degraded by the environment, a series of tests to provide a measure of the damage tolerance and environmental effects on braided materials must be included. Damage criteria must include manufacturing defects which are not detectable non-destructive evaluations.

Several candidate configurations for these types of tests are listed in NASA CR 1092 - Revised, 1982 and should be considered as part of a design allowables program based upon intended use and environment.

Fatigue and creep tests should be considered whenever the part is a dynamic component. The highly complex courses followed by the braided yarns through the matrix material cause fiber load paths which are difficult to predict. The chance that the matrix is loaded beyond the level predicted by laminated design and experience are near certain. Therefore, a lower confidence in understanding how a braided part behaves will require a rigorous fatigue and creep test plan owing to the complex

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interweave of the braided yarns in the part. Tests designed to determine fatigue or creep effects are probably part specific and should be conducted on a braided specimen which is identical to the finished product. Scaling effects and the use of similitude in test specimen design should be avoided whenever practical. Otherwise, the results may not mimic the critical defect which results in a failure of the part prior to the predicted value.

In general, it is accurate to state that for both two- and three-dimensional braided composites, the criticality of defects is the single most important factor in the design of the part and the corresponding application of the allowables which are used. This is due to the nature of complex, net-shape braided preforms as well as physical limitations in the ability of the yarns to fully cover an area of rapid change in contour. The good news is that net-shape braided parts do provide a means to build these parts without intensive hand labor suitable to production environments and fully capable of doing the intended job. This is true provided that the designer or analyst has made the correct choices in the type of specimens used for determining braided allowables.

6.9 THICK-SECTION COMPOSITES MECHANICAL PROPERTY TESTS.

6.9.1 Introduction.

6.10 THERMAL AND MOISTURE ABSORPTION PROPERTIES.

6.11 ELECTRICAL PROPERTIES.

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Chapter 7. STRUCTURAL ELEMENT CHARACTERIZATION

Page

7.1 INTRODUCTION. 7-2

7.2 MECHANICALLY-FASTENED JOINTS 7-2
7.2.1 Definitions. 7-2
7.2.2 Failure modes. 7-2
7.2.3 Design requirements. 7-3
7.2.4 Material bearing strength. 7-5
7.2.4.1 Significance 7-5
7.2.4.2 Specimen testing and design 7-5
7.2.4.3 Replication requirements 7-5
7.2.4.4 Test conditions 7-8
7.2.4.5 Bearing strength calculations 7-8
7.2.5 Bearing strength of joints. 7-8
7.2.5.1 Rationale 7-8
7.2.5.2 Specimen design and testing 7-9
7.2.5.3 Test matrices 7-13
7.2.5.4 Data reduction 7-14
7.2.5.5 Effects of gaps/shimming 7-14
7.2.6 Notch tension/compression strength 7-17
7.2.7 Bearing/by-pass strength. 7-17
7.2.8 Shear-out strength. 7-17
7.2.9 Fastener pull-thru strength. 7-17

7.3 BONDED JOINTS. 7-17

7.4 OTHER TOPICS. 7-17

REFERENCES 7-18

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7.1 INTRODUCTION. Testing and analysis of composite joints and other elements are essential for maintaining the structural integrity of composite structures and ensuring their reliability. Any joint in a composite structure is a potential failure site. Without proper design, a joint can act as a damage initiation point, which can lead to a loss in structural stability and eventual failure of a component. Two types of joints are in common use, namely (1) mechanically-fastened joints and (2) adhesively-bonded joints. These guidelines define the recommended approach for testing bolted joints to determine design properties against various failure modes. A detailed analysis of the stress distribution around a fastener hole is not presented here but is available in Volume 3. Discussion on both theoretical and empirical approaches to the stress analysis of bolted joints in composite materials can be found in Reference 7.1.

In addition to the recommended approaches to obtain bolted joints design properties for the different failure modes, a straightforward test method to determine material bearing strength is provided in Section 7.2.4. The bearing property measured by this test is the upper bound value that can be achieved in a realistic structural joint. The test is useful for qualification purposes or for material-to-material comparisons.

7.2 MECHANICALLY-FASTENED JOINTS.

7.2.1 Definitions. The following definitions are relevant to this section.

Bearing Area -- The diameter of the hole multiplied by the thickness of the specimen.

Bearing Load -- A compressive load on an interface.

Bearing Strain -- The ratio of the deformation of the bearing hole in the direction of the applied force, to the pin diameter.

Bearing Strength -- The maximum bearing stress which can be sustained.

Bearing Stress -- The applied load divided by the bearing area.

Edge Distance Ratio -- The distance from the center of the bearing hole to the edge of the specimen in the direction of the applied load, divided by the diameter of the hole.

Proportional Limit Bearing Stress -- The bearing stress value corresponding to the deviation from linearity of the bearing stress versus hole elongation curve.

Ultimate Bearing Strength -- The bearing stress corresponding to total failure of the test specimen.

Yield Bearing Strength -- The bearing stress at the intersection of the bearing load-deformation curve with the tangent modulus drawn from the 4% elongation value of the original hole diameter.

7.2.2 Failure modes. An important consideration in joint testing and analysis is the selection of the type of test method with due attention to the failure mode which is likely to result with a specific joint design in a particular composite system. A brief discussion on various failure modes is provided in this section.

The occurrence of a particular failure mode is dependent primarily on joint geometry and laminate lay-up. Composite bolted joints may fail in various modes as shown in Figure 7.2.2. The likelihood of a particular failure mode is influenced by bolt diameter (D), laminate width (w), edge distance (e),

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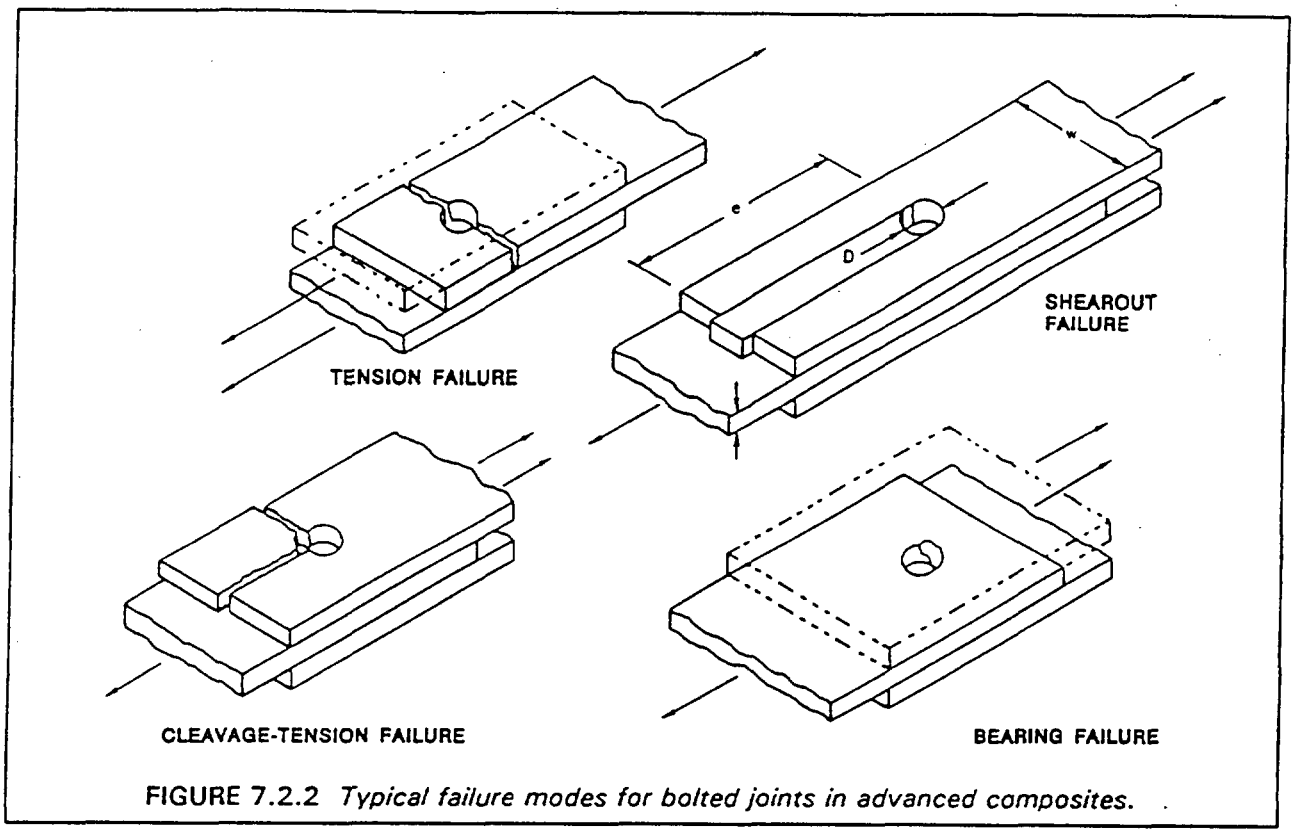


FIGURE 7.2.2 *Typical failure modes for bolted joints in advanced composites.*

and thickness (t). The type of fastener used can also influence the occurrence of a particular failure mode.

Net section tension/compression failures occur when the bolt diameter is a sufficiently large fraction of the strip width. This fraction is about one-quarter or more for near-isotropic lay-ups in graphite/epoxy systems. It is characterized by failure of the plies in the primary load direction.

Cleavage failures occur because of the proximity of the end of the specimen. A cleavage failure can be triggered from a net-section tension failure. This type of failure often initiates at the end of the specimen rather than adjacent to the fastener.

In some instances the bolt head may be pulled out through the laminate after the bolt is bent and deformed. This mode is frequently associated with countersunk fasteners and is highly dependent on the particular fastener used.

Finally, it is important to note that for any given geometry, the failure mode may vary as a function of lay-up and stacking sequence.

7.2.3 Design requirements. In order to design against the different failure modes and the interactions between them, the capability of the composite has to be determined by test for:

- Bearing
- Notch/Net Tension/Compression
- Bearing/By-Pass
- Shear-Out.

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These are described in Sections 7.2.5 to 7.2.8. The amount of testing will vary among manufacturers and regulating agencies depending on the confidence assigned to the analysis capability of each company. The philosophy of MIL-HDBK-17 is to provide guidance as to amount of testing that would be typical, but not necessarily the minimum.

The bearing, net tension/compression, and bearing/by-pass failure mode criticality is best illustrated by a plot shown in Figure 7.2.3. This figure, which is typically used by airframe designers, encompasses the five failure possibilities as a function of bolt load and strain in the joining members. This plot is usually determined by tests which are described in Section 7.2.5 to 7.2.7. At zero bearing (no bolt load), the failure is in net tension or compression (points A and E). Open-hole or filled-hole coupons described in Section 7.2.6 are used to determine this property. The line between A and C represents the reduction of net tension strength due to the bearing load. Similarly the line from E to C^1 represents the effect of bolt load on net compression strength. Points C and C^1 are the strengths of a single fastener joint where all the load is reacted by the bolt. Section 7.2.5 describes the tests required to establish this design point for different joint variables. In practice, joints C and C^1 are not much different so that a tension bearing test is usually sufficient.

Plots such as Figure 7.2.3 may be different for each distinct laminate, fastener type, and environmental condition, but many application ranges may be covered by one plot. The shape of the curves could also change depending on the percentage of 0°, 90° or ± 45° direction plies in the laminate. The intent of the sections that follow is to provide guidance on how to establish by test the critical points of Figure 7.2.3. The number of laminates to be tested is governed by analysis capability and degree of confidence in extrapolation.

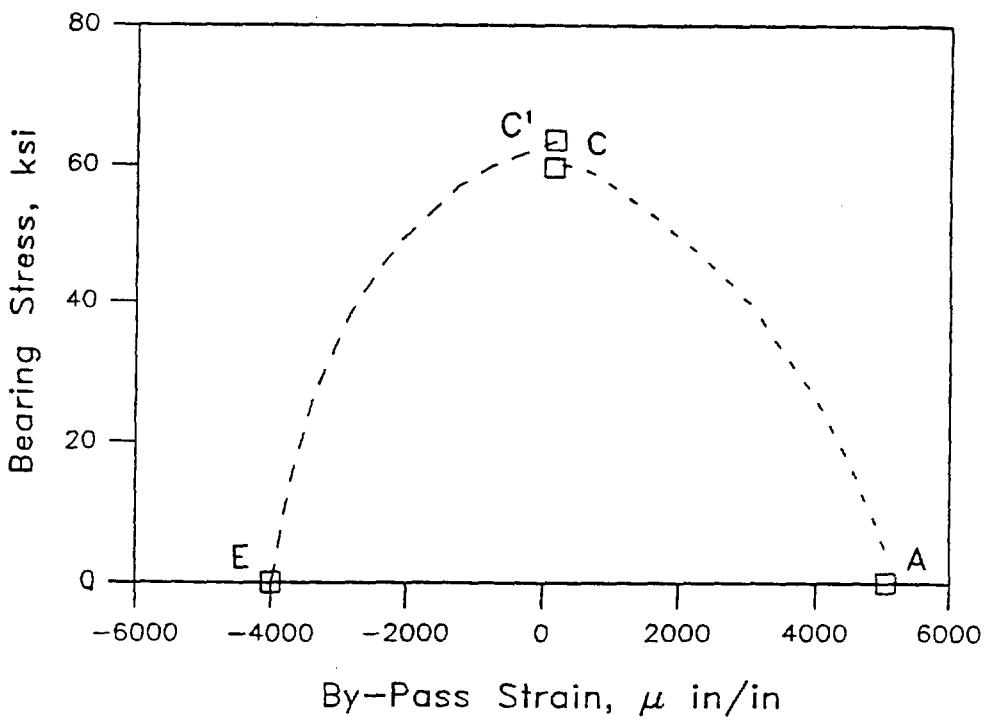


FIGURE 7.2.3 *Bearing/by-pass interaction.*

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The shear-out mode of failure is usually avoided in design by providing sufficient edge distance between the holes or the free edge and balanced laminate configuration. However, in certain rework situations shear-out critical joints cannot be avoided. In those situations, a test program must be undertaken to establish allowables (see Section 7.2.8).

Net tension/compression and shear-out strengths are a strong function of laminate configuration, joint geometry, and hole size, but only marginally dependent on fastener type, joint configuration, or environment. On the other hand, bearing and fastener pull-through strengths are greatly influenced by the type of fastener used and its characteristics such as clamp effects, bolt stiffness, head and tail areas, countersinking, etc. The laminate lay-up and stacking sequence do not affect bearing or pull-through strengths significantly unless extreme lay-ups are used. The extreme being defined as a laminate having highly concentrated plies or plies in only two directions.

7.2.4 Material bearing strength.

7.2.4.1 *Significance*. The proposed test introduces the bearing load in a double shear configuration. In actual applications, load transfer in a single shear configuration is more commonplace, resulting in larger stress concentrations in the thickness direction, and lowering the realizable bearing strength; these single-shear tests are also discussed in Section 7.2.5. In other words, the bearing strength values measured in this test cannot be applied to single shear joints.

Only a tensile loading condition is proposed for evaluating bearing failures; under compression, the larger edge distance ($e >> 3D$) should only influence the bearing stress at failure minimally unless a shear-out mode of failure is possible (e.g., a laminate with a large percent of 0° plies).

Both final bearing failure and bearing deformation are needed for material characterization purposes. Therefore, it is recommended that the bearing stress variation as a function of hole deformation be documented, and that bearing stress values corresponding to the proportional limit, yield bearing (see Section 7.2.1), and ultimate failure be recorded.

In summary, bearing strength, as measured by the test in this section, is considered a material property for relative evaluation and design. In realistic structural joints, factors like geometry, fastener type, and load eccentricity will significantly influence the realizable fraction of the bearing strength measured in the proposed test. Bearing strength tests more appropriate in design of joints are discussed in Section 7.2.5.

7.2.4.2 *Specimen testing and design*. Using standard test equipment, the specimen shall be tested in the double shear arrangement shown in Figure 7.2.4.2. Test specimens shall conform to the dimensions and tolerances shown. Recommended b/D and e/D values are 6 and 3, respectively, and D = 1/4 inch (6.4 mm). A $[+45/0/-45/90]_{ns}$ lay-up, where n is selected so that the total thickness is approximately 0.125 inch (3.2 mm) is also recommended. The bearing hole shall be located as shown in Figure 7.2.4.3. The hole shall be drilled undersized and reamed to size. The bolt shall be made of hardened steel (Rockwell Hardness of C60 - C64) and shall be 0.249 (+0.0000/-0.0010). Thus, a close tolerance fit between the specimen and pin is required since a loose fit will tend to give lower results. Prior to assembly, the specimen, bolt, and adjacent areas should be cleaned of all foreign matter and contamination, especially lubricants.

7.2.4.3 *Replication requirements*. Five replicates shall be tested as a minimum. If three tests are conducted on one specimen, as shown in Figure 7.2.4.3, a total of six tests on two specimens will suffice.

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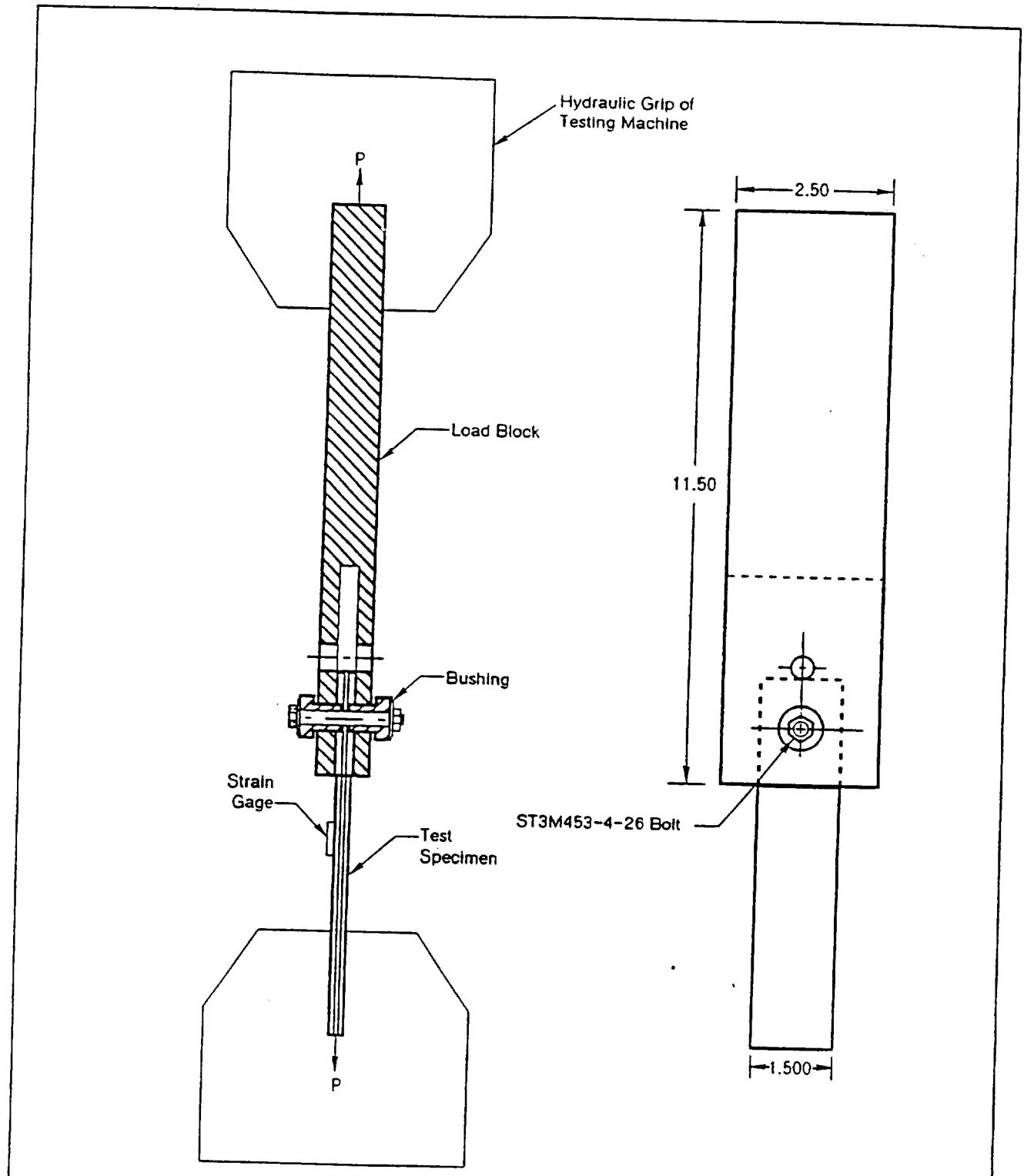
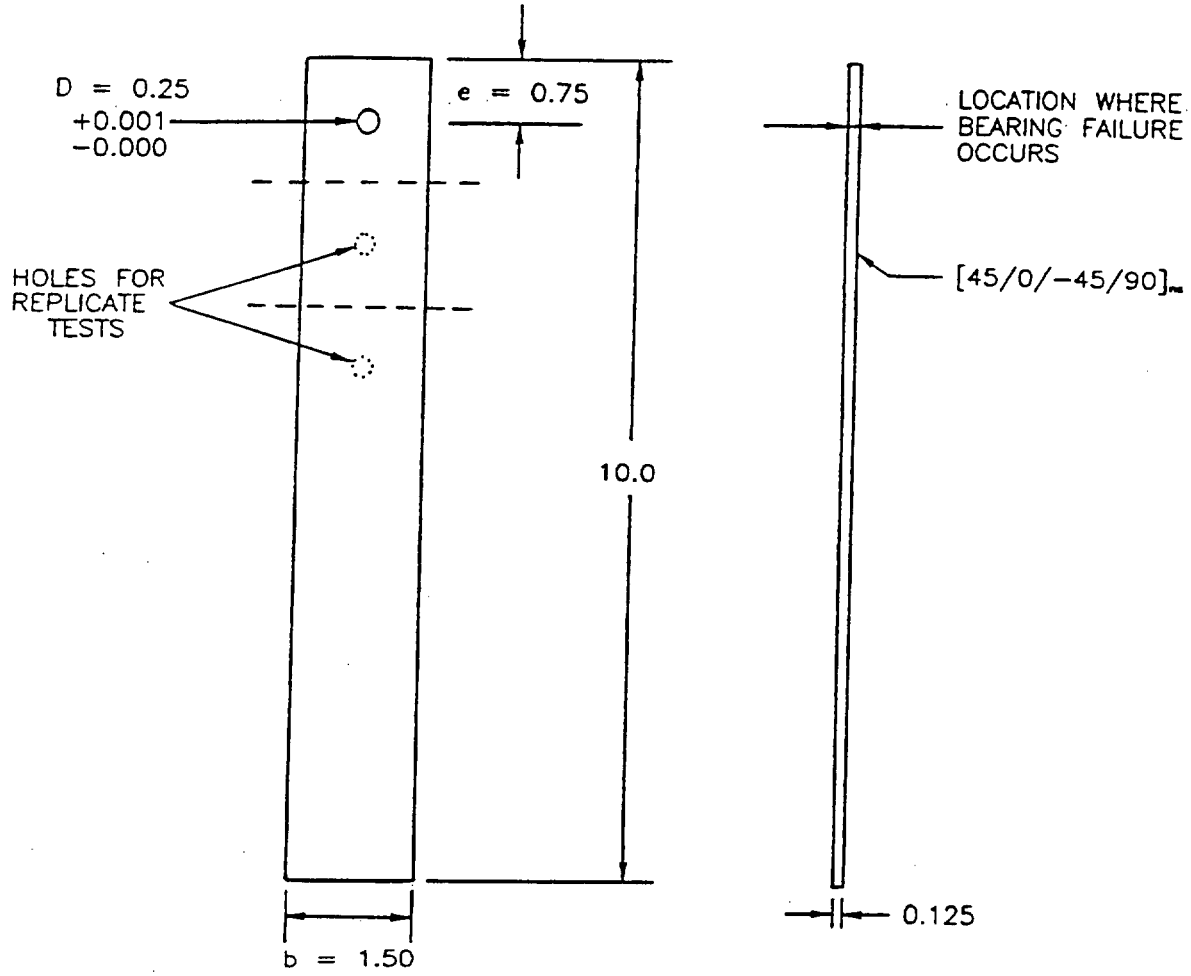


FIGURE 7.2.4.2 *Test arrangement for material bearing strength measurement.*

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ALL DIMENSIONS IN INCHES

FIGURE 7.2.4.3 *Test specimen geometry.*

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7.2.4.4 *Test conditions*. Tests shall be conducted under as-fabricated or ambient dry or room temperature dry (RTD) conditions, and one hot, wet conditions. The hot, wet test shall be conducted on specimens after they are preconditioned to equilibrium level moisture contents (see Section 2.2.2.2), at temperatures that are dependent on the wet glass transition temperature for the material wet T_g). The recommended test temperatures for the hot, wet test is T_g - 50F° (T_g - 28C°).

Measurement of specimen dimensions, test procedures, etc., shall follow the recommendations of MIL-STD-1312-B/Test Methods, "Fasteners Test Methods, X. Mechanically Attached Composite Shear Joint", (Reference 7.2.4.4). The determination of yield bearing strength is shown for a typical curve in Figure 7.2.4.4.

7.2.4.5 *Bearing strength calculations*. Bearing strength should be calculated using the following equation:

$$F^{br} = P/tD \qquad 7.2.4.5$$

where

F^{br}	=	bearing strength, psi (Pa)
P	=	bearing load, lb_f (N)
D	=	bearing hole diameter, in. (m)
t	=	specimen thickness, in. (m)

Superscripts bry and bru are commonly used to differentiate between yield and ultimate bearing strengths.

Bearing strength values are reported in MIL-HDBK-17 as typical or average values. Therefore, bearing strength values which are available for each specific condition should be analyzed to produce typical property values as described in Chapter 8.

7.2.5 Bearing strength of joints.

7.2.5.1 *Rationale*. This section describes single shear test specimens and the associated test matrices required to obtain bearing strength of single-lap joints. The resulting test data can be applicable either for the selection and screening of fasteners and the design of bolted joints or both. If the actual joint configuration is double shear, the test specimen and procedure of Section 7.2.4 would be more appropriate.

Bearing strength is a function of joint geometry and stiffnesses of the members and the fastener. It should be noted that for a 0/±45/90 family of laminates with 20-40% of 0° plies and 40-60% of ±45° plies the bearing strength is essentially constant. In addition, fastener characteristics such as clamp-up force, and head and tail configuration have a significant effect. However, for a specific laminate family, a specific fastener, and equal thickness lamina joining members, the parameter with the greatest influence is t/D. This was recognized by the aircraft designers and all the bearing data for metals is presented in MIL-HDBK-5 (Reference 7.2.5.1) in terms of the t/D parameter, Figure 7.2.5. The slope of this nondimensional curve is the bearing strength which decreases with increased t/D until for sufficiently thick lamina members shear failure occurs in the bolt. The data generated using the recommended test specimens, procedures, and test matrices will produce equivalent data for composite joints.

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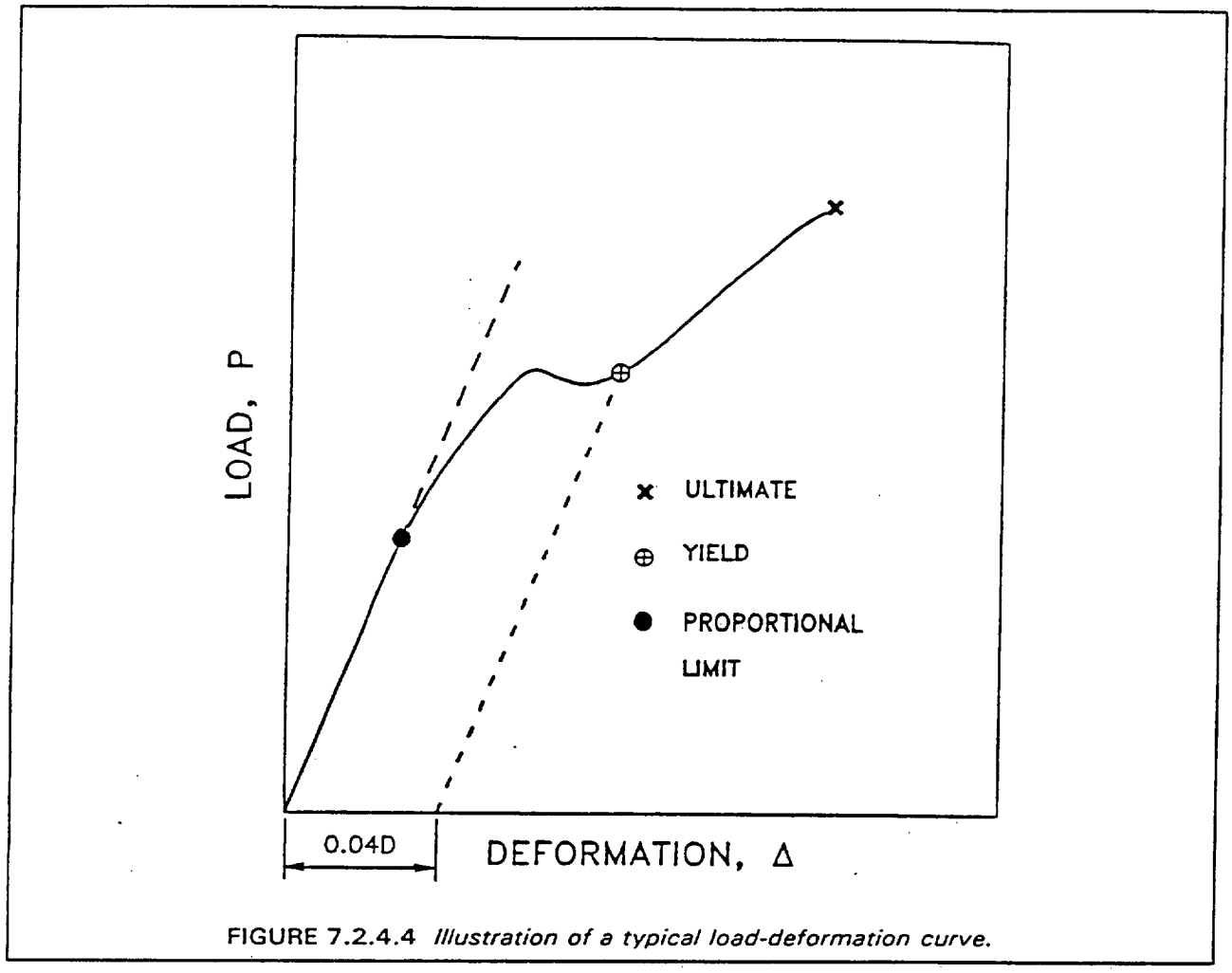


FIGURE 7.2.4.4 *Illustration of a typical load-deformation curve.*

In the design process there may be instances where the joint configuration may not correspond to the test configurations recommended here, i.e., unequal joining members, gaps, solid shims, fuel sealing provisions. These effects on bearing strengths should be evaluated by modifying the specimen geometry as needed. The test procedures, presented here are still applicable.

7.2.5.2 *Specimen design and testing.* Recommended composite-to-composite and composite-to-metal bearing specimen geometries are shown in Figures 7.2.5.2(a) and (b). Both are single lap geometries. Although it is generally more difficult to test than the double lap configuration, the single lap test configuration is more representative of most critical aircraft bolted joint applications. The single lap induces both bending and shear loads on the fastener, while the double lap induces mostly shear loads.

The joint configurations shown in Figures 7.2.5.2(a) and (b) may be used to generate both design and fastener screening data. In some cases, the fastener supplier and/or the requester may want to evaluate fastener behavior in a single fastener configuration. The recommended single fastener joint configuration is shown in Figure 7.2.5.2(c). This is the same specimen specified in MIL-STD-1312-X (Reference 7.2.5.2). It should be recognized that this joint configuration may be subject to high bending due to the load eccentricity transmitted through the bolt. The bending can be reduced by increasing the stiffness of the two laps, through increased thickness, or material stiffness. It should

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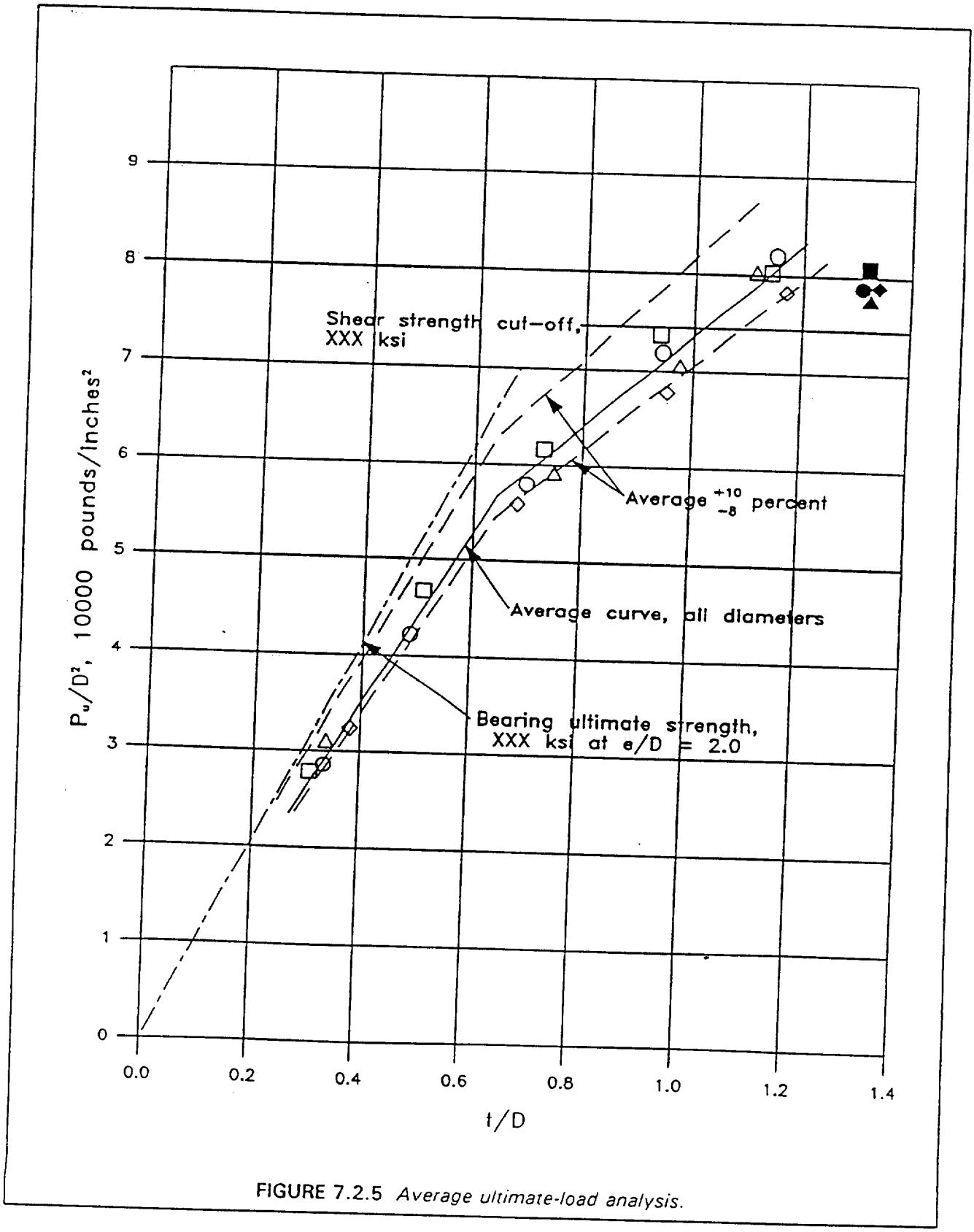


FIGURE 7.2.5 *Average ultimate-load analysis.*

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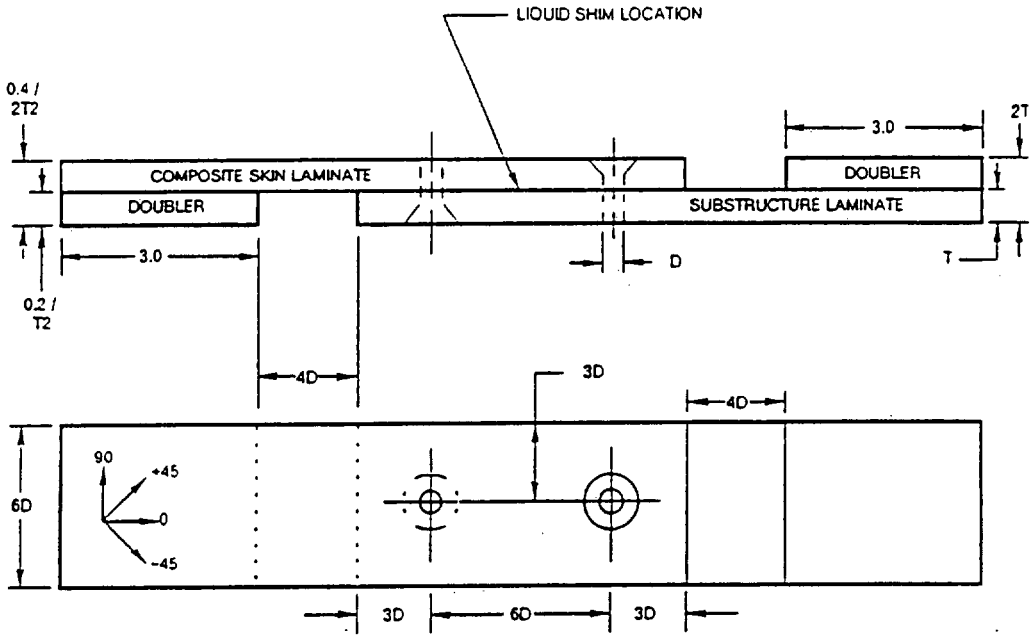


FIGURE 7.2.5.2(a) *Composite-to-composite joint configuration.*

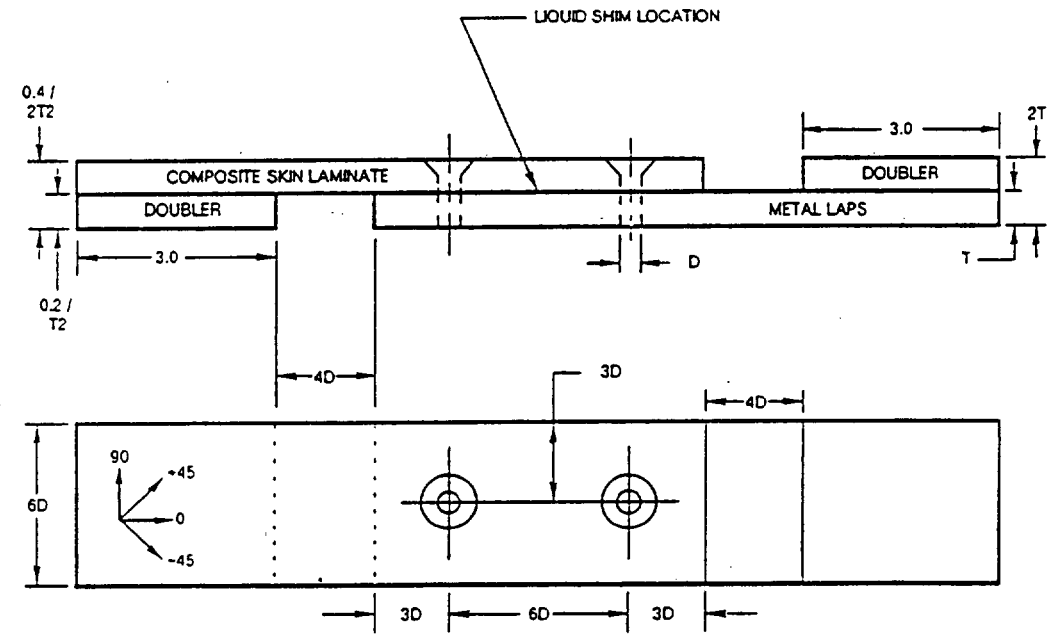


FIGURE 7.2.5.2(b) *Composite-to-metal joint configuration.*

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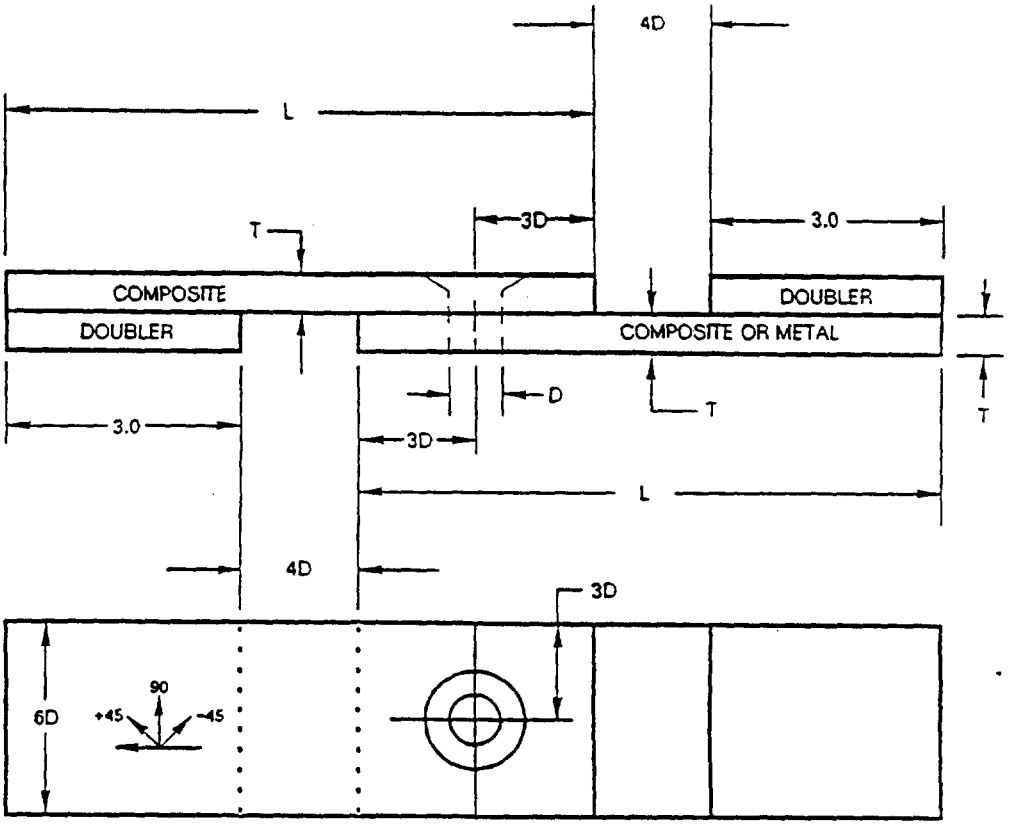


FIGURE 7.2.5.2(c) *Single fastener lap joint.*

also be noted that the single fastener joint is generally not representative of multi-fastener joint applications (e.g., joint rotation, deflection). Therefore, it should be used mostly for fastener screening purposes.

When tested, the specimen geometries shown in Figures 7.2.5.2(a) and (b) are intended to result in composite bearing failures (as opposed to tension, or cleavage failures). Fastener pull-thru's and fastener failures, though not acceptable as a measure of composite bearing strength, do provide a measure of joint strength for a particular fastener type.

The metal tongue and doubler dimensions (Figures 7.2.5.2(a) and (b)) are specified to align the load path along the interface between the two laps. The doublers and the tongue shall be bonded to the composite bearing specimens prior to loading, or prior to specimen conditioning for environmental hot/wet tests. Fiberglass/epoxy can be substituted for metal doublers.

Tests shall be conducted according to the test matrices shown in Tables 7.2.5.3(a) through 7.2.5.3(c) and MIL-STD-1312-X test procedures (Sections 1-4.1.3 and Section 5). Hot/wet tests shall be conducted after specimens have been preconditioned. Test data requirements are specified in MIL-STD-1312-X, Section 6.

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7.2.5.3 *Test matrices.* This section describes the recommended test matrices for composite-to-composite and composite-to-metal bolted joints, for both static and fatigue testing.

For composite-to-composite bolted joints the recommended test matrix for single shear bearing strength testing is given in Table 7.2.5.3(a) and the associated test specimen configuration is given in Figure 7.2.5.2(a). The test data generated from the full test matrix of Table 7.2.5.3(a) will be sufficient to design composite-to-composite mechanical joints against bearing failure for one material and one fastener type. For other fasteners, the tests with note (1) should be sufficient to provide correction factors which would be applicable to all other not-tested conditions. These are labeled as fastener supplier tests or screening tests. For screening tests, in addition to t2 thickness, a third thickness specimen (t3) is shown so that sufficient test data would be generated to construct Figure 7.2.5.

Table 7.2.5.3(a) *Composite-to-composite mechanically fastened joint test matrix for bearing strength.*

GEOMETRY	SKIN MEMBER THICKNESS (IN)	LAY-UP	BOLT DIAMETER (IN)	ENVIRONMENT (TEMP/% MOIST)	NUMBER OF TESTS
COMPOSITE TO COMPOSITE	0.2	25/50/25	0.25 D2	RT/ambient RT/ambient	10 (1,2) 5 (1)
	0.2	50/40/10	0.25 D2	RT/ambient RT/ambient	5 5
	t2	25/50/25	0.25 D2	RT/ambient RT/ambient	5 (1) 5 (1)
	t2	50/40/10	0.25 D2	RT/ambient RT/ambient	5 5
	t3	25/50/25	0.25 D2	RT/ambient RT/ambient	5 (1) only 5 (1) only
COMPOSITE TO COMPOSITE	0.2	25/50/25	0.25 D2	hot/wet hot/wet	5 5
	0.2	50/40/10	0.25 D2	hot/wet hot/wet	5 5
	t2	25/50/25	0.25 D2	hot/wet hot/wet	5 5
	t2	50/40/10	0.25 D2	hot/wet hot/wet	5 5

Notes:

(1) Supplier fastener screening tests
(2) Add 5 replications with 0.03 ± 0.003 liquid shim gap between members (optional)
(3) Single shear configuration per Figure 7.2.5.1

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In the design test matrix two different composite lay-ups are shown at each thickness. The lay-up varies from quasi-isotropic $(45/0\text{-}45/90)_{ns}$ to an orthotropic lay-up of 50% 0° plies in the load direction $(45/0/\text{-}45/90/0_2/45/0/\text{-}45/0)_{ns}$. One other thickness (t2) and bolt diameter (D2) are left unspecified; their choice should be dependent on the application. Two environments should be tested, room temperature as received and hot/wet. The selection of hot/wet temperature and moisture content should be guided by Section 2.2.2 of Volume 1.

The baseline 0.2 inch thick quasi-isotropic lay-up with the 0.25 inch bolt diameter could be used to evaluate the effect of a 0.03 inch thick or thicker liquid shim gap between the two members (option Note (2); also see Section 7.2.5.1). The test specimen configuration in Figure 7.2.5.2(a) shows the location of the liquid shim. A metal spacer can be used instead of the liquid shim if the spacer is unbonded to the composite.

For composite-to-metal bolted joints, the recommended test matrix for single shear bearing strength testing is given in Table 7.2.5.3(b) and the associated test specimen configuration is given in Figure 7.2.5.2(b). The general comments from the composite-to-composite bolted joints section also apply to the composite-to-metal bolted joints since the test matrices are the same. The composite-to-composite configuration is more critical than the composite-to-metal joint with respect to the design of the fastener tail; therefore, the composite-to-composite test specimen is more useful for the evaluation of fasteners by the fastener supplier. Because of the above reason, note (1) in Table 7.2.5.3(b) has been designated as tests required for a different fastener. The metal fixture shown in Figure 7.2.5.2(b) is designed so that the effective load line passes through the joint interface.

The recommended fatigue matrix is given in Table 7.2.5.3(c) and is based on using the test specimen configuration of Figure 7.2.5.2(a). Constant amplitude fatigue is suggested with a stress ratio of R = -0.2 (compressive load is 20 percent of tensile load). Frequency of loading should be selected so as to avoid excessive heating at the joint area of the specimen. For current material systems this translates to 5 Hz. This test matrix should be repeated for each fastener under consideration. The fifteen specimen per test will allow three replicates at five stress levels. A load level of half the static strength is a good starting point. All tests should be conducted at room temperature/ambient environment.

The specimens with specified thickness and bolt diameter (t = 0.2 in. and D = 0.25 in.) have been sized to fail in bearing. Specimens should be selected based on assuring bearing failure and avoiding bolt shearing, or net tension failures either in composite or metal members.

7.2.5.4 *Data reduction*. The procedures of data reduction described for the pure bearing test of Section 7.2.4 apply to the bearing tests in this section. In addition, the failure mode must be adequately described.

7.2.5.5 *Effects of gaps/shimming*. Although in most composite applications the use of bonded joints appears more weight-efficient, bolted joints still predominate due to their higher joint reliability and the need to disassemble some joints. In the assembly of composite structure, gaps between mating surfaces will occur and the disposition of these gaps is required prior to clamp-up of the fastener. Closing excessive unshimmed gaps when installing fasteners can cause delaminations in the composite structure, however, residual gaps of any size may reduce joint performance.

Test data show that the strength of bolted composite joints depends partially on bolt diameter, composite thickness, shimmed gap thickness, and the type of shimming material used. Examples of strength reduction curves are shown in Figures 7.2.5.4 and 7.2.5.5 for the diameter to thickness ratio and shimmed gap effects for a single shear composite joint in a multiple bolt splice. These are not

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Table 7.2.5.3(b) *Composite-to-metal mechanically fastened joint test matrix for bearing strength.*

GEOMETRY	SKIN MEMBER THICKNESS (IN)	LAY-UP	BOLT DIAMETER (IN)	ENVIRONMENT (TEMP/% MOIST)	NUMBER OF TESTS
COMPOSITE TO METAL	0.2	25/50/25	0.25 D2	RT/ambient RT/ambient	10 (1,2) 5 (1)
	0.2	50/40/10	0.25 D2	RT/ambient RT/ambient	5 5
	t2	25/50/25	0.25 D2	RT/ambient RT/ambient	5 (1) 5 (1)
	t2	50/40/10	0.25 D2	RT/ambient RT/ambient	5 5
COMPOSITE TO COMPOSITE	0.2	25/50/25	0.25 D2	hot/wet hot/wet	5 5
	0.2	50/40/10	0.25 D2	hot/wet hot/wet	5 5
	t2	25/50/25	0.25 D2	hot/wet hot/wet	5 5
	t2	50/40/10	0.25 D2	hot/wet hot/wet	5 5

Notes:
(1) Alternate fastener tests
(2) Add 5 replications with 0.03 ± 0.003 liquid shim gap required between members (optional)
(3) Single shear configuration per Figure 7.2.5.2

Table 7.2.5.3(c) *Mechanically fastened joint fatigue test matrix for bearing fatigue.*

GEOMETRY	THICKNESS (IN)	LAY-UP	TOTAL NUMBER OF TESTS
COMPOSITE TO COMPOSITE	t1	25/50/25	15 (1)
	t1	50/40/10	15
COMPOSITE TO COMPOSITE	t1	25/50/25	15 (1)
	t1	50/40/10	15

Notes:
(1) Supplier fastener screening tests
(2) Constant amplitude fatigue (R = -0.2) to 4% hole elngation measured across a single hole
(3) Single shear configuration; same as static tests

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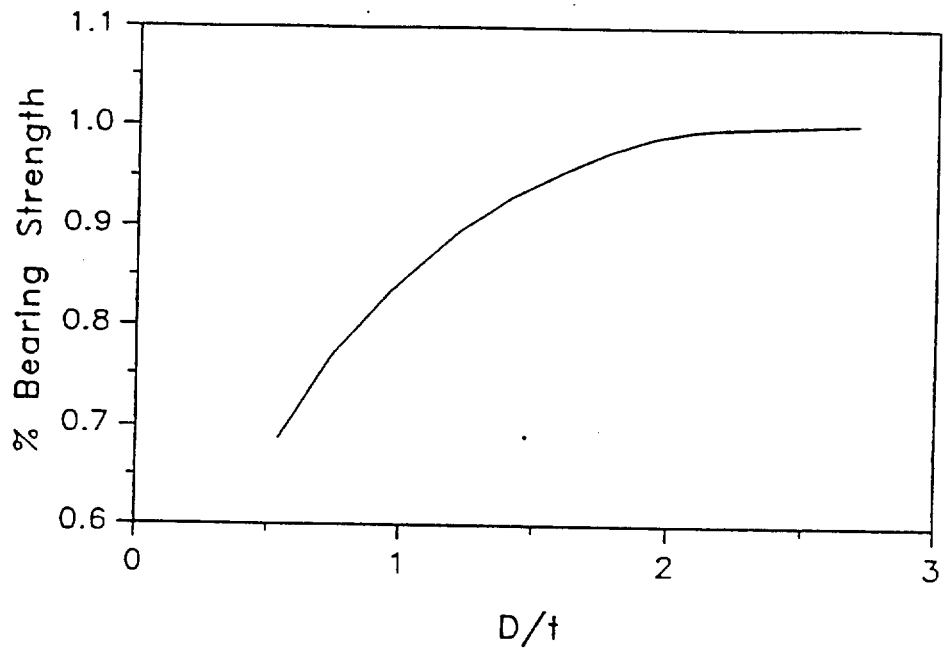


FIGURE 7.2.5.4 *Bearing strength reduction vs D/t.*

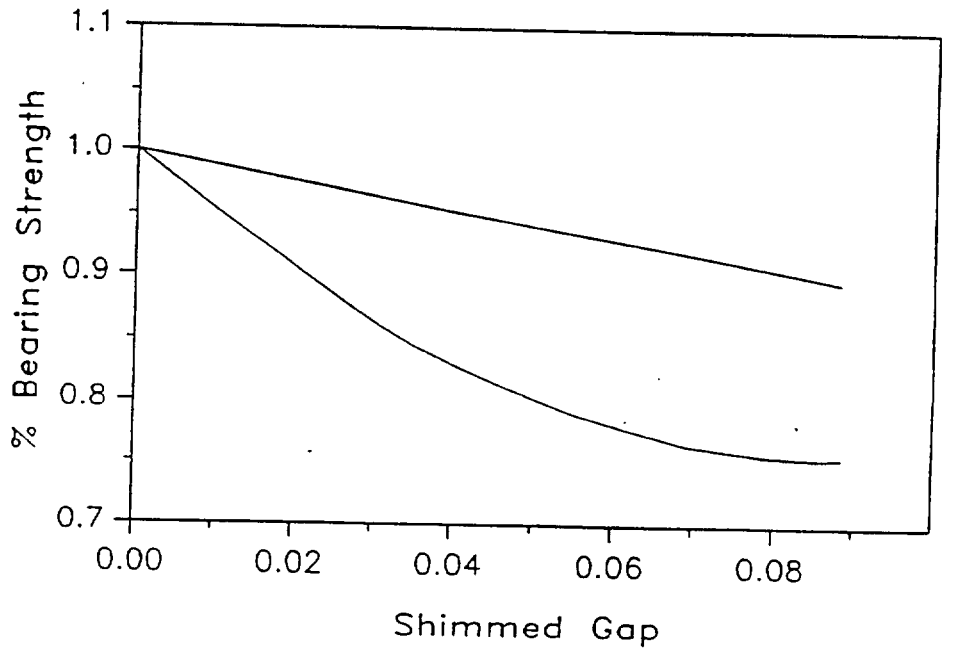


FIGURE 7.2.5.5 *Bearing strength reduction vs shimmed gap.*

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generic curves and generation of similar data would be required for specific user application. The reduction factors are then used to reduce the nominal allowable bearing stress. The nominal bearing allowable for a particular material system would be obtained using tests with configurations minimizing bolt bending to obtain uniformity of stress through the thickness (a large diameter to thickness ratio clevis or multi-fastener test) and using all pertinent statistical and environmental knockdowns. Joint strength reduction factors are greater for joints using liquid shims for filling the gaps than joints using metal or composite solid shims.

7.2.6 Notch tension/compression strength.

7.2.7 Bearing/by-pass strength.

7.2.8 Shear-out strength.

7.2.9 Fastener pull-thru strength.

7.3 BONDED JOINTS.

7.4 OTHER TOPICS.

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Chapter 8. ANALYSIS AND PRESENTATION OF DATA

Page

8.1 GENERAL. 8-5
8.1.1 Introduction. 8-5
8.1.2 Data documentation requirements. 8-5
8.1.3 Normalization. 8-5
8.1.4 Symbols. 8-8

8.2 MATERIAL AND SPECIMEN REQUIREMENTS. 8-10
8.2.1 Statistically-based material properties. 8-10
8.2.2 Typical properties. 8-10

8.3 DETERMINATION OF PROPERTIES. 8-11
8.3.1 Introduction. 8-11
8.3.2 Mechanical properties. 8-11
8.3.2.1 Ultimate stress and strain. 8-11
8.3.2.2 Transverse tensile properties. 8-11
8.3.2.3 Elastic constants and Poisson's ratio. 8-11
8.3.3 Chemical properties. 8-11
8.3.4 Physical properties. 8-11
8.3.4.1 Density. 8-11
8.3.4.2 Composition. 8-11
8.3.4.3 Water absorption. 8-11
8.3.4.4 Glass transition temperature. 8-11
8.3.5 Thermal properties. 8-12
8.3.6 Electrical properties. 8-12
8.3.7 Typical stress-strain curves. 8-12
8.3.7.1 Fitting equations. 8-13

8.4 PRESENTATION OF DATA. 8-17
8.4.1 Properties and definitions. 8-17
8.4.1.1 Sign convention. 8-17
8.4.2 Organization of data in handbook. 8-17
8.4.2.1 Fiber properties. 8-17
8.4.2.2 Matrix properties. 8-17
8.4.2.3 Composite properties. 8-17
8.4.3 Sample tables. 8-18
8.4.4 Sample graphs. 8-22

8.5 CALCULATION OF STATISTICALLY-BASED MATERIAL PROPERTIES. . . 8-23
8.5.1 Introduction. 8-23
8.5.1.1 Statistical terms. 8-23
8.5.1.2 Sample size requirements. 8-27
8.5.2 Definition of computational procedures. 8-27
8.5.3 Detecting outliers and characterizing the population. 8-29
8.5.3.1 Detecting outliers. 8-29
8.5.3.2 The k-sample Anderson-Darling test. 8-30

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:30Z
Check the source to verify that this is the current version before use.

Page

8.5.3.3 Equality of variance test. 8-30
8.5.3.4 Investigating the form of the distribution. 8-30
8.5.4 Normal analysis procedure in the presence of batch-to-batch variation. 8-31
8.5.5 Normal analysis procedure for a single population. 8-33
8.5.5.1 Computational procedure. 8-33
8.5.5.2 Lognormal procedure. 8-34
8.5.6 Two-parameter Weibull analysis procedure for a single population. . 8-34
8.5.6.1 Estimating the shape and scale parameters of a Weibull distribution. 8-34
8.5.6.2 Computational procedure. 8-37
8.5.7 Nonparametric analysis procedure for a single population. 8-38
8.5.7.1 Sample size greater than 28. 8-38
8.5.7.2 The Hanson-Koopmans method. 8-38
8.5.8 Linear statistical models. 8-39

8.6 STATISTICAL PROPERTIES. 8-43
8.6.1 Introduction. 8-43
8.6.2 Outlier detection procedure. 8-43
8.6.2.1 The maximum normed residual method. 8-43
8.6.3 Subpopulation compatibility tests. 8-44
8.6.3.1 The k-sample Anderson-Darling test. 8-44
8.6.3.2 Test for equality of variances. 8-45
8.6.4 Goodness-of-fit tests. 8-46
8.6.4.1 Anderson-Darling test for normality. 8-46
8.6.4.2 Anderson-Darling test for lognormality. 8-47
8.6.4.3 Anderson-Darling test for Weibullness. 8-48
8.6.4.4 Testing for normality in a regression setting. 8-48
8.6.5 Exploratory Data Analysis. 8-49
8.6.5.1 The quantile box plot. 8-49
8.6.5.2 The Informative Quantile function. 8-50
8.6.6 General linear statistical models. 8-53
8.6.6.1 Example: Simple linear regression problems. 8-55
8.6.6.2 Example: One-way fixed effects analysis of variance model. . 8-56
8.6.7 Analysis of variance procedures. 8-59
8.6.7.1 Computational procedures. 8-60
8.6.7.1.1 Computations based on individual measurements. . 8-60
8.6.7.1.2 Computations based on summary statistics. 8-60
8.6.7.2 ANOVA Table. 8-61
8.6.8 Methods for developing A-basis values. 8-61
8.6.8.1 Weibull model. 8-62
8.6.8.2 Normal model. 8-62
8.6.8.3 Lognormal model. 8-62
8.6.8.4 Nonparametric method. 8-62
8.6.8.5 Hanson-Koopmans method. 8-62
8.6.8.6 ANOVA method. 8-62
8.6.8.7 Linear regression. 8-63

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T13:30Z
Check the source to verify that this is the currentversion before use.

Page

8.7 SAMPLES OF COMPUTATIONAL PROCEDURES. 8-64
8.7.1 Problem 1 - Outlier detection and multiple sample tests. 8-64
8.7.2 Problem 2 - Outlier detection and ANOVA method. 8-70
8.7.3 Problem 3 - Weibull distribution. 8-73
8.7.4 Problem 4 - Normal distribution. 8-76
8.7.5 Problem 5 - Lognormal distribution. 8-78
8.7.6 Problem 6 - Nonparametric method. 8-80
8.7.7 Problem 7 - Insufficient data. 8-81
8.7.8 Problem 8 - Regression analysis. 8-82

8.8 CALCULATIONS FOR STATISTICAL TABLES. 8-86
8.8.1 One-sided tolerance factors, k_B, for the normal distribution, 0.95 confidence. 8-86
8.8.2 One-sided tolerance factors, V, for the Weibull distribution, 0.95 confidence. 8-86
8.8.3 Ranks, r, of observations for determining B-values for an unknown distribution. 8-87
8.8.4 0.20 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. 8-87
8.8.5 0.95 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. 8-88
8.8.6 One-sided tolerance factors, T, for the one-way random effects ANOVA, 0.95 confidence. 8-89
8.8.7 Critical values for the k-sample Anderson-Darling test at the 0.05 significance level. 8-90
8.8.8 0.95 Quantiles of the chi-squared distribution associated with γ degrees of freedom. 8-90
8.8.9 Critical values for the MNR outlier detection procedure (α = 0.05). . 8-90
8.8.10 Nonparametric B-basis material properties for small samples. 8-91
8.8.11 One-sided tolerance limit factors, V_A, for the Weibull distribution, 0.95 confidence. 8-91
8.8.12 One-sided tolerance limit factors, k_A, for the normal distribution, 0.95 confidence. 8-91
8.8.13 Ranks, r, of observations for determining A-basis values for an unknown distribution. 8-91
8.8.14 The Hanson-Koopmans method exponents for A-basis values. . . . 8-91
8.8.15 0.1225 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. 8-91
8.8.16 One-sided tolerance factors for A-basis calculations, T_A, for the one-way random effects ANOVA, 0.95 confidence. 8-92

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:30Z
Check the source to verify that this is the current version before use.

Page

STATISTICAL TABLES 8-93
8.8.1 One-sided tolerance factors, k_B, for the normal distribution, 0.95 confidence. 8-93
8.8.2 One-sided tolerance factors, V, for the Weibull distribution, 0.95 confidence. 8-95
8.8.3 Ranks, r, of observations for determining B-values for an unknown distribution. 8-97
8.8.4 0.20 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. 8-98
8.8.5 0.95 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. 8-100
8.8.6 One-sided tolerance factors, T, for the one-way random effects ANOVA, 0.95 confidence. 8-102
8.8.7 Critical values for the k-sample Anderson-Darling test at the 0.05 significance level. 8-108
8.8.8 0.95 Quantiles of the chi-squared distribution associated with γ degrees of freedom. 8-109
8.8.9 Critical values for the MNR outlier detection procedure (α = 0.05). 8-110
8.8.10 Nonparametric B-basis material properties for small samples. . . . 8-111
8.8.11 One-sided tolerance limit factors, V_A, for the Weibull distribution, 0.95 confidence. 8-112
8.8.12 One-sided tolerance limit factors, k_A, for the normal distribution, 0.95 confidence. 8-114
8.8.13 Ranks, r, of observations for determining A-basis values for an unknown distribution. 8-116
8.8.14 The Hanson-Koopmans method exponents for A-basis values. . . 8-117
8.8.15 0.1225 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. 8-118
8.8.16 One-sided tolerance factors for A-basis calculations, T_A, for the one-way random effects ANOVA, 0.95 confidence. 8-120

REFERENCES 8-126

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8.1 GENERAL. This section of Chapter 8 covers general information applicable to the sections that follow. Information specific to individual properties is to be found in the pertinent sections.

8.1.1 Introduction. The mechanical properties in MIL-HDBK-17 are used in the design of aerospace structures and elements. Thus it is exceedingly important that the values presented in MIL-HDBK-17 reflect as accurately as possible the actual properties of the materials covered.

The statistical procedures used in determining the materials properties values presented in this handbook are described in Sections 8.5 through 8.8. A software package which performs the statistical techniques presented in these sections is available from the MIL-HDBK-17 Secretariat upon request. Section 8.2 presents the definitions for typical and B-basis values and defines the minimum number of batches and specimens to obtain a B-basis property value. Guidelines for the computation of individual mechanical properties are presented in Section 8.3. Section 8.4 describes the procedures and formats for presenting the material property data in Volume 2.

8.1.2 Data documentation requirements. The purpose of this section is to outline data documentation requirements necessary to establish the validity of a manufacturer's material system physical, chemical, and mechanical property database. These requirements are necessary to establish justification for the inclusion of data in MIL-HDBK-17. The essence of documentation requirements is complete traceability and control of the database development process from material production, through procurement, fabrication, machining, gaging, environmental conditioning, testing, data acquisition, data normalization, and final statistical interpretation.

The information in Table 8.1.2 should be documented in any material property determination. Note that many of the items are marked as needed for submission of data to the Secretariat. All other information should be traceable and available to the Secretariat for validation of outliers. This list is based on the information necessary for lamina/laminate level mechanical property testing. The information required for other tests or material levels is similar. For instance, prepreg property testing would require the prepreg, reinforcement, and matrix material analysis information, as well as appropriate information on specimen preparation and testing procedure. For testing on laminae made from prepreg, redundant information on the matrix and reinforcement would not be required.

8.1.3 Normalization. The values of fiber-dominated strength and stiffness properties (i.e., 0° tension, 0° compression) are dependent upon the volume fraction of fiber present in the laminate. In typical applications, the strength and stiffness have been found to vary linearly with percent fiber. Properties other than 0° tension and 0° compression strengths and stiffnesses may also be dependent on fiber volume fraction, but not in a well-defined way. For example, interlaminar shear properties are frequently found to remain fairly constant over a given fiber fraction range, but become lower when the fiber fraction falls above or below this range. Normalization is not considered appropriate for such properties.

In order to compare the properties of laminates of different fiber contents, it has become common practice to report values normalized to a given fiber volume fraction. This is accomplished by determining the fiber volume fraction in the cured test laminates by an appropriate method, and then multiplying the raw property values by the factor: (nominal fiber volume fraction)/(determined fiber volume fraction).

It may be shown that, for fiber/fabric of a given areal weight, the per ply thickness depends entirely on the fiber volume fraction. Hence, normalization may also be performed on a cured ply thickness basis. Although the true relationship is of the form, cured ply thickness = K/fiber fraction (where K is a constant), the departure from linearity in the 0.45 to 0.65 fiber volume fraction range is less than 0.1%. The normal thickness is chosen to correspond to a given fiber volume fraction or to the

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TABLE 8.1.2 *Documentation requirements*, continued on next page.

Material identification • material identification ○ material procurement specification
Prepreg analysis • ply manufacturer • date of manufacturer • material lot number • commercial designation • material form • reinforcement areal weight • resin content ○ moisture content ○ high performance liquid chromatography results ○ infrared spectroscopy results ○ differential scanning calorimetry results ○ rheological dynamic spectroscopy results
Reinforcement analysis • specification ○ precursor type • commercial designation • manufacturer • date of manufacturer • lot number • surface treatment identification • specific gravity, measured • nominal filament count • fiber areal weight • twist
Matrix material analysis • specification • material chemical family • commercial designation • manufacturer • lot number ○ gel time and test conditions • specific gravity, measured ○ dynamic mechanical analysis results for neat resin ○ high performance liquid chromatography results ○ infrared spectroscopy results ○ differential scanning calorimetry results

• Required for submission to Secretariat
○ Recommended documentation in any material property determination

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TABLE 8.1.2 *Documentation requirements*, concluded.

Processing information • process specification • panel manufacturer • date of manufacture ○ panel identification number • process cure cycle
Lamina analysis • form (panel, tube, etc.) • ply count • layup code • dimensions ○ NDT conclusions • fiber volume • resin content (weight or volume) • density ○ glass transition temperature (wet and dry, batch basis) and test method Specimen preparation ○ specimen lay-out data and numbering system • specimen drawing ○ specimen acceptance criteria (machining specifications, etc.)
Mechanical testing • test matrix performed • test procedure (including deviations from standard procedures) • specimen dimensions for each specimen • environmental conditioning (drying) procedures • moisture absorption data to equilibrium • fastener type and torque-up conditions (if any) • test temperature and moisture content at failure, environment • continuous stress-strain data to failure • failure mode identification and location • all non-normalized (raw) data
Data analysis † normalized data and procedures used ○ statistical procedures used ○ statistically-reduced data and parameters (by batch and pooled) ○ statistical hypothesis checks

• Required for submission to Secretariat
○ Recommended documentation in any material property determination
† Optional for submission to Secretariat

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expected thickness under given process conditions.

The thickness method may be particularly useful for fiber reinforcements such as aramid, where direct, repeatable methods for determining fiber volume of cured laminates are not well established or easily applied on a routine basis. The thickness method has the added advantage of allowing normalization of each individual test specimen, rather than applying a common fraction to all specimens from a given panel (based on panel fiber volume fraction). The latter approach might be less accurate since the thickness (and hence fiber fraction) will vary throughout the panel. The per ply thickness approach is not considered appropriate for test panels wound from rovings, however, since the per ply thickness is influenced greatly by the wind spacing and roving bandwidth, and may not consistently correspond to a fiber volume fraction.

The recommended data practices are as follows:

1. For 0° tension and 0° compression strengths and stiffnesses, normalized values shall be reported. Normalization may be by fiber volume fraction or thickness as appropriate for the material. The method of normalization shall be stated, and the normalizing factor shall be within the material process specification range. Raw data before normalization shall also be reported.

8.1.4 Symbols. The symbols which are used in Chapter 8, particularly those that are used in Sections 8.5 through 8.8, and not commonly used throughout this handbook are listed below, each with its definition and the section in which it is first used. These are primarily used to denote statistical variables.

SYMBOL	DEFINITION	SECTION
A_i	population	8.6.3.1
ADK	k-sample Anderson-Darling statistic	8.6.3.1
B	B-basis value	8.5.1.1
C.V.	critical value	8.6.2.1
e	error, residual	8.6.4.4
EV	equality of variances test statistic	8.6.3.2
F	F statistic	8.5.8
$F_{0.2}$	0.20 quantile of the F-distribution	8.5.4
IQ	informative quantile function	8.6.5.2
k	number of batches	8.5.4
k_n	Hanson-Koopmans coefficient	8.5.7.2
k^*	see Equation 8.5.4(i)	8.5.4
k_B	one-sided tolerance limit factor	8.5.5.1
MNR	maximum normed residual test statistic	8.6.2.1
N	total number of observations in several batches	8.5.4
n	number of observations in a sample	8.5.1.1
n′	effective sample size	8.5.4
n^*	see Equation 8.5.4(d)	8.5.4
n_i	number of observations in sample i	8.6.6.2
OSL	observed significance level	8.5.3.4
Q	quantile function	8.6.5.1

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SYMBOL	DEFINITION	SECTION
$\hat{Q}$	quantile function estimate	8.6.5.1
$q_{0.10}$	tenth percentile of the underlying population distribution	8.5.7.1
R	ratio of between-batch to within-batch variances	8.5.4
$\hat{R}$	see Equation 8.5.4(e)	8.5.4
r	rank of observation	8.5.7.1
r_i	normed residual	8.6.2.1
RME	relative magnitude of error	8.8
S	sample standard deviation	8.5.1.1
S^2	sample variance	8.5.1.1
S_b^2	sample between-batch variance	8.5.4
S_e^2	sample within-batch variance	8.5.4
S_y	estimated variance of errors from the regression line	8.5.8
T	tolerance limit factor	8.5.4
$t_{\gamma,0.95}$	0.95 quantile of the central t-distribution with degrees of freedom γ	8.5.4
$t_{\gamma,0.95}(\delta)$	0.95 quantile of the non-central t-distribution with non-centrality parameter δ and degrees of freedom γ	8.5.4
TIQ	truncated informative quantile function	8.6.5.2
V	one-sided tolerance limit factor for the Weibull distribution	8.5.6.2
$\bar{X}$	sample mean, overall mean	8.5.1.1
x_i	observation i in a sample	8.5.1.1
x_{ij}	jth observation in sample i	8.6.7
$x_{(r)}$	rth observation, sorted in ascending order, observation of rank r	8.5.7.1
α	(1) significance level	8.8.9
	(2) true intercept of regression equation	8.6.6.1
	(3) scale parameter of Weibull distribution	8.5.6.1
$\hat{\alpha}$	estimate of scale parameter of Weibull distribution	8.5.6.1
β	(1) true slope of regression equation	8.6.6.1
	(2) shape parameter of Weibull distribution	8.5.6.1
$\hat{\beta}$	estimate of shape parameter of Weibull distribution	8.5.6.1
γ	degrees of freedom	8.5.4
δ	noncentrality parameter	8.5.4
ϵ	error	8.6.6.1
μ	population mean	8.5.1.1
σ	population standard deviation	8.6.6.1
σ^2	population variance	8.6.6.1
σ_b^2	population between-batch variance	8.5.4
σ_e^2	population within-batch variance	8.5.4

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8.2 MATERIAL AND SPECIMEN REQUIREMENTS.

8.2.1 Statistically-based material properties. The statistically-based properties included in MIL-HDBK-17 are B-basis values. A B-value, as defined in Section 1.7, is the value above which at least 90 percent of the population of values is expected to fall, with a confidence of 95 percent. In order to obtain a B-value by using a nonparametric method as described in Section 8.5.7.1, twenty-nine measurements are required. For the purposes of obtaining a reasonable evaluation of material variation, properties as presented in this handbook are based on a minimum of six measurements from at least five batches of a material. This criterion provides for a minimum of thirty measurements. Fewer measurements can be evaluated according to the statistical methods described in Section 8.5.7.2; these data will be included in the handbook but B-basis values will not be reported for them. B-values are included for all ultimate strength and strain-to-failure properties.

8.2.2 Typical properties. Typical, sample mean, values are presented for all properties in MIL-HDBK-17. Statistical assurance is associated only with the ultimate strength and strain-to-failure.

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8.3 DETERMINATION OF PROPERTIES.

8.3.1 Introduction. This section outlines the procedures for calculating material property values from the experimental records of the individual specimens. The statistical methods described in Sections 8.5 through 8.8 and the statistical definitions presented in Section 1.7 are used to determine the necessary values.

8.3.2 Mechanical properties.

8.3.2.1 *Ultimate stress and strain.* The ultimate stress and strain-to-failure will receive the full statistical treatment described in Section 8.5.2 including outlier detection, data pooling testing, and determination of distribution.

8.3.2.2 *Transverse tensile properties.* A B-basis value will not be reported for transverse strength in tension for unidirectional tape materials but the remaining statistical parameters will be tabulated.

8.3.2.3 *Elastic constants and Poisson's ratio.* The elastic constants, Young's modulus or shear modulus, shall be collected from each of the experiments, with a minimum of six specimens from each of five batches, for each property. For each experiment, the modulus will be calculated from the slope of the load-deformation curve within the linear portion of the curve or, if the load-deformation curve is nonlinear, the modulus will be calculated from the slope of a secant line between load-deformation points at 0.001 in/in (m/m) and 0.003 in/in (m/m) strain. The results will be examined for batch-to-batch variation and pooled if possible. Minimum and maximum values will be reported in Volume 2. If stress-strain data are provided for each experiment, the data can be fit by algebraic functions as described in Section 8.3.7. Poisson's ratio values will be determined for the same range of data as the modulus values. Typical values of Poisson's ratios determined from the thirty experiments for tensile and compressive loading conditions will be included in the data summary for a given material.

8.3.3 Chemical properties.

8.3.4 Physical properties. The following requirements have been established for determining physical properties from experimental results and reporting those properties in MIL-HDBK-17.

8.3.4.1 *Density.* The density will be reported for each batch used in the experiments to determine any of the mechanical properties. Typical density values will be part of the material description presented with each set of data.

8.3.4.2 *Composition.*

8.3.4.3 *Water absorption.* The equilibrium moisture content for the relative humidity values from Tables 2.4(a) and (b) will be determined from a minimum of three specimens. This value will be presented in the data summary for each material. If additional information is available for moisture content as a function of temperature and relative humidity, those values will be presented graphically.

Values for the moisture expansion coefficient will be treated in the same way as those for the thermal expansion coefficient.

8.3.4.4 *Glass transition temperature.* The glass transition temperature will be determined from a minimum of three measurements per dry and wet test condition. This information will be a basis for determining the maximum use temperature.

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8.3.5 Thermal properties. The coefficient of linear thermal expansion will be determined from a minimum of five measurements at each condition. Room temperature values will be listed in the data summary and additional values of the thermal expansion coefficient will be presented graphically as a function of temperature.

The specific heat and the thermal conductivity will be presented in a similar fashion. When data are adequate to present curves showing specific heat, thermal conductivity, and mean coefficient of thermal expansion over a range of temperatures, graphical presentation is used in addition to tabular presentation. A smooth curve is drawn through the plotted points to depict the overall trend of the data. The smooth curves for specific heat, thermal conductivity, and thermal expansion are then shown in a single figure as shown in Figure 8.3.5. The reference temperature for thermal expansion should be shown on the figure. In Figure 8.3.5, the reference temperature of 70°F (21°C) indicates that the mean coefficient of expansion between 70°F (21°C) and the indicated temperature is plotted.

8.3.6 Electrical properties.

8.3.7 Typical stress-strain curves. Typical stress-strain curves will be determined from a minimum of two stress-strain curves from each batch. It is highly preferable to have stress-strain records from all of the experiments completed to evaluate a given mechanical property. The best fit of the stress-strain data for each experiment to one of seven algebraic functions will be obtained as described in References 8.3.7(a) and (b). The seven functions evaluated for best fit are:

1) linear
2) parabolic
3) inverse parabolic
4) Ramberg-Osgood exponential

Figure 8.3.5 *Thermal properties as a function of temperature.*

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4) Ramberg-Osgood exponential
5) bilinear
6) parabolic - linear
7) parabolic - exponential

The equations used for each function are presented in the Section 8.3.7.1. The average error of fit is defined by the product of the root mean square (RMS) stress error and the RMS percent stress error. By using the product of these two values, sensitivity to errors at high stress values and the initial portion of the stress-strain curve is provided. The average error of fit is determined for each function. The function with the smallest average error of fit is used to describe each set of data.

An average stress-strain curve is determined from these best-fit functions. The minimum stress-to-failure for all sets of test data is divided into a hundred increments. At each increment, the average stress based on each best-fit function is determined. The average stresses are again fit to the algebraic functions to obtain an average stress-strain curve. The function with the smallest average error of fit and the constants for this function will be reported with the average stress-strain curve in Volume 2.

All of the ultimate stress and strain-to-failure values, not just those used in determining the typical stress-strain curve, will be included as a scatter plot at the top of the averaged stress-strain curve. Example stress-strain curves which include the scatter plot are shown in Figures 8.3.7(a) and (b).

8.3.7.1 *Fitting equations.* The fitting equations for the stress-strain curves based on References 8.3.7(a) and (b) are presented below. Also included with each type of curve are the equations for the secant modulus and tangent modulus curves. These curves will be presented in MIL-HDBK-17 as discussed in Section 8.4.4. Each of the functions are based on the terms stress (s) and strain (e). The secant modulus functions are calculated as the secant modulus between the current value of strain and zero strain.

$$E_s = \frac{s(e) - s(0)}{e - 0} = \frac{s(e)}{e} \qquad 8.3.7.1(a)$$

The equation for the tangent modulus at any value of strain is:

$$E_t = \frac{ds}{de} \qquad 8.3.7.1(b)$$

Linear:

$$s = C_1 e \qquad 8.3.7.1(c)$$

$$E_t = E_s = C_1 \qquad 8.3.7.1(d)$$

Parabolic:

$$s = C_1 e + C_2 e^2 \qquad 8.3.7.1(e)$$

$$E_t = C_1 + 2C_2 e \qquad 8.3.7.1(f)$$

$$E_s = C_1 + C_2 e \qquad 8.3.7.1(g)$$

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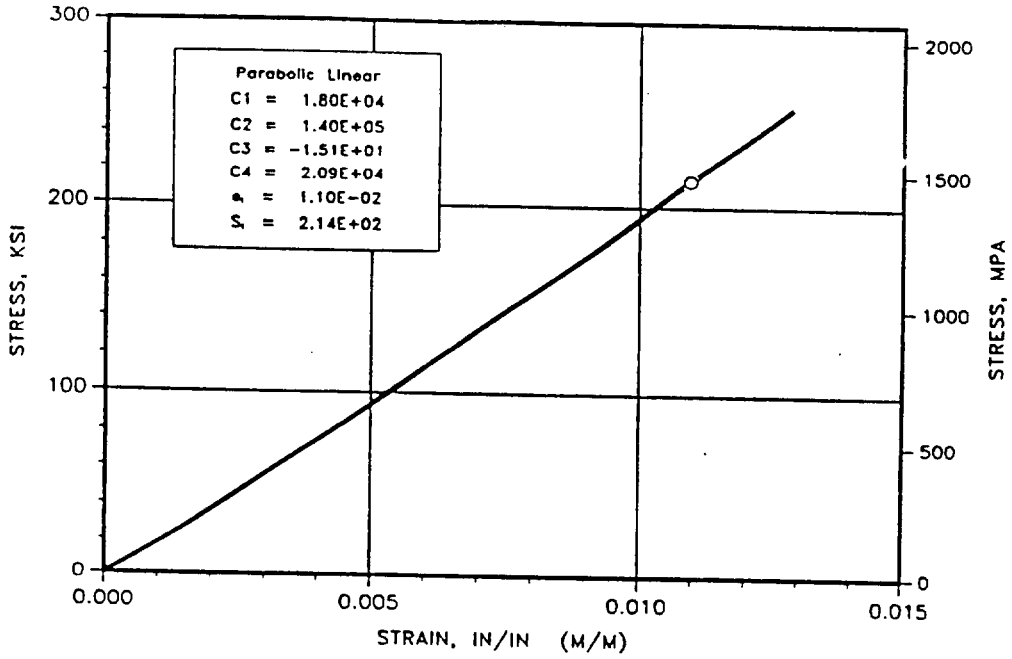


Figure 8.3.7(a) *Tensile stress-strain curve for AS4/3501-6 carbon/epoxy.*

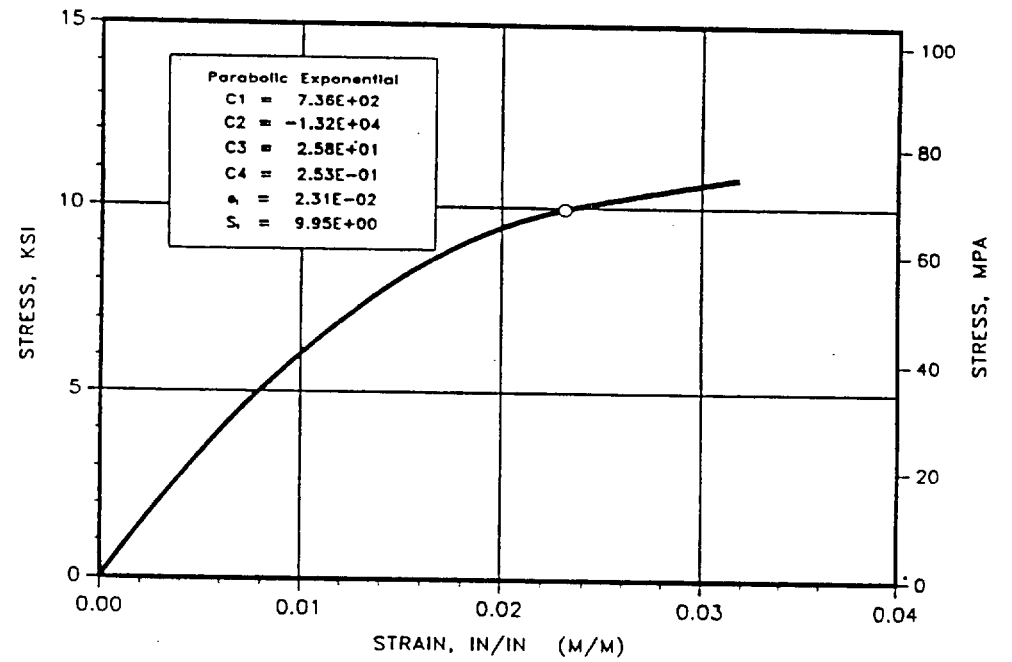


Figure 8.3.7(b) *Shear stress-strain curve for AS4/3501-6 carbon/epoxy.*

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$$E_t = \pm (C_2^2 + 4C_3\varepsilon)^{-1/2} \qquad 8.3.7.1(i)$$

$$E_s = \frac{C_2 \pm (C_2^2 + 4C_3\varepsilon)^{1/2}}{2C_3\varepsilon} \qquad 8.3.7.1(j)$$

The ± symbol has the same sign as the constant C_3.

Ramberg-Osgood Exponential:

$$\varepsilon = \frac{s}{C_2} + 0.002\left[\frac{s}{C_1}\right]^n, \qquad n = C_3 \qquad 8.3.7.1(k)$$

Both secant modulus and tangent modulus values for the Ramberg-Osgood exponential function are found numerically.

The remaining functions have two fitted portions of the curves. The intersection of these two portions (e_i, s_i) is found as part of the fitting procedure.

Bilinear:

Below (e_i,s_i)

$$s = C_1\varepsilon \qquad 8.3.7.1(c)$$

$$E_t = E_s = C_1 \qquad 8.3.7.1(d)$$

Above (e_i,s_i)

$$s = C_2 + C_3\varepsilon \qquad 8.3.7.1(l)$$

$$E_t = C_3 \qquad 8.3.7.1(m)$$

$$E_s = \frac{C_2}{\varepsilon} + C_3 \qquad 8.3.7.1(n)$$

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$$E_s = C_1 + C_2\theta \qquad 8.3.7.1(f)$$

$$s = C_3 + C_4\theta \qquad 8.3.7.1(o)$$

$$E_t = C_4 \qquad 8.3.7.1(p)$$

$$E_s = \frac{C_3}{\theta} + C_4 \qquad 8.3.7.1(q)$$

Parabolic - Exponential:
Below (e_i, s_i)

$$s = C_1 + C_2\theta^2 \qquad 8.3.7.1(e)$$

$$E_t = C_1 + 2C_2\theta \qquad 8.3.7.1(f)$$

$$E_s = C_1 + C_2\theta \qquad 8.3.7.1(g)$$

Above (e_i, s_i)

$$s = C_3\theta^n, \quad n = C_4 \qquad 8.3.7.1(r)$$

$$E_t = nC_3\theta^{n-1} \qquad 8.3.7.1(s)$$

$$e_s = C_3\theta^{n-1} \qquad 8.3.7.1(t)$$

In all cases, the type of curve and the values of the constants will be shown on the typical stress-strain curve figures. When there are two regions in a stress-strain curve, the value of the strain and stress and the intersection of the two regions is also shown on the figure.

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8.4 **PRESENTATION OF DATA.** All data are presented in Volume 2 of the handbook, MIL-HDBK-17-2. This section describes how the data are presented and organized in that volume.

8.4.1 **Properties and definitions.** The properties and their definitions are found in the appropriate chapters of Volume 1. Fiber properties and methods for obtaining them are discussed in Chapter 3. Resin properties are presented in Chapter 4. Methods for characterizing prepreg materials are discussed in Chapter 5 and properties and definitions for laminae and laminates are presented in Chapter 6. The statistical methods used in determining these properties are discussed in this chapter. Material system codes and laminate orientation codes are defined in Chapter 1 of Volume 2.

8.4.1.1 *Sign convention.* All compressive values, represented by a superscript c, are reported as positive numbers. Thus, a positive compression strength indicates failure due to a load applied in the opposite direction of a positive tensile failure.

8.4.2 **Organization of data in handbook.** The data in Volume 2 is divided into chapters of fiber properties, resin properties, and composite properties organized by fiber and then resin.

8.4.2.1 *Fiber properties.* Chapter 2 in Volume 2 provides data for fiber properties. Sections are included for different types of fiber, e.g., glass fibers and carbon fibers. In each section, the general characteristics of the type of fiber are given, as well as an index of suppliers, designations, and abbreviations. For each specific fiber, data are organized in the following manner. The X's in the subsection number are determined by the type of fiber and the specific fiber described.

2.X.X.1	Supplier and product data
2.X.X.2	Chemical and physical properties
2.X.X.2.1	Typical range of chemical constituents
2.X.X.2.2	Expected bound in physical properties
2.X.X.3	Thermal-mechanical properties
2.X.X.3.1	Stress-strain curves
2.X.X.3.2	Environmental effects

8.4.2.2 *Matrix properties.* Matrix or resin properties are included in Chapter 3 which is divided into sections according to the type of resin. For example, Section 3.2 gives data for epoxies and Section 3.3 provides data for polyester resins. The subsections for each specific resin are the same as those in Chapter 2 given above.

8.4.2.3 *Composite properties.* The remaining chapters of Volume 2 provide data for prepreg, lamina, laminate, and joint properties. There are individual chapters for each family of composites based on fiber type. For example, Chapter 4 describes glass fiber composites. Within each chapter, there is an index of suppliers, designations, and abbreviations. Sections are included based on the resin type used with the fiber described in the chapter, e.g., Section 4.3 provides properties for epoxy-glass composites.

Properties are organized in the following manner for each specific composite:

X.X.X.1	Supplier and product data
X.X.X.2	Prepreg chemical and physical properties
X.X.X.2.1	Physical description
X.X.X.2.2	Resin content
X.X.X.2.3	Fiber content
X.X.X.2.4	Volatiles content

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X.X.X.2.6	Inorganic fillers and additives content
X.X.X.2.7	Areal weight
X.X.X.2.8	Tack and drape
X.X.X.2.9	Resin flow
X.X.X.2.10	Gel time
X.X.X.3	Lamina chemical properties
X.X.X.4	Lamina physical properties
X.X.X.5	Lamina mechanical properties
X.X.X.5.1	Data summaries
X.X.X.5.2	Typical stress-strain curves
X.X.X.6	Thermal properties
X.X.X.7	Electrical properties
X.X.X.8	Laminate thermal-mechanical properties
X.X.X.8.1	Index of properties by lay-up
X.X.X.8.2	Strength properties
	a. Lay-up No. 1
	b. Lay-up No. 2
X.X.X.8.3	Thermal properties
X.X.X.8.4	Electrical properties
X.X.X.9	Joint thermal-mechanical properties
X.X.X.9.1	Index of properties by joint and composite system
X.X.X.9.2	Bearing strength
	a. System No.1
	b. System No. 2

8.4.3 Sample tables. Sample tables for mechanical property data presentation in Volume 2 are given in Tables 8.4.3(a) through (c). Table 8.4.3(a) is a summary page giving information about the material system and the mean strength and modulus for each test and environmental condition. Table 8.4.3(b) shows the format for data which are normalized (see Section 8.1.3) and Table 8.4.3(c) is the format for data which are not normalized. In these tables, a number of values are given for several properties. For ultimate stress and strain, the following values are reported:

Typical
Maximum
Minimum
Standard deviation
B-basis
Type of distribution
Distribution constants, C_1 and C_2
Number of specimens
Number of batches
Test method
Source code

The B-basis value is determined for the type of distribution which has been found to fit the data as described in Sections 8.5.2 and 8.5.5 - 7. The type of distribution which was used is listed as:

Weibull = Two-parameter Weibull distribution
Normal = Normal distribution
Lognormal = Lognormal distribution
Nonpara. = Nonparametric method
ANOVA = Analysis of variance method

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TABLE 8.4.3(a) *Format for lamina/laminate property summary table.*

MATERIAL:					*Family Material Orientation* Summary
PREPREG:					
FIBER:			RESIN:		
FIBER DENSITY:	lb/in^3		RESIN DENSITY:	lb/in^3	
FIBER AREAL WEIGHT:	g/m^2 (lb/ft^2)		COMP. DENSITY:	lb/in^3	
NOMINAL PLY THICKNESS:	in.				
T_g(dry):	°F	T_g(wet): °F	T_g METHOD:		
PROCESSING:					

MECHANICAL PROPERTY SUMMARY

		RTD	CTD	RTW	ETW
F_1^{tu}	(ksi)				
E_1^t	(Msi)				
F_2^{tu}	(ksi)				
E_2^t	(Msi)				
F_1^{cu}	(ksi)				
E_1^c	(Msi)				
F_2^{cu}	(ksi)				
E_2^c	(Msi)				
F^{su}	(ksi)				
G^s	(Msi)				
F^{lsu}	(ksi)				
G^{ls}	(Msi)				

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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TABLE 8.4.3(b) *Format for normalized lamina/laminate property table.*

MATERIAL:						*Family Material Orientation Class of data*
SPECIMEN LAY-UP:			COMP. DENSITY:			
RESIN CONTENT:	wt%		VOID CONTENT:	%		
FIBER VOLUME:	%		PLY THICKNESS:	in.		
NORMALIZED BY:						
Temperature(°F) Moisture Content(%) Equilibrium at T, RH Source Code						
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{cu} (ksi)	Mean Minimum Maximum C.V. (%) B-value Distribution C_1 C_2 No. Specimens No. Batches Test Method						
e_1^{cu} (μin/in)	Mean Minimum Maximum C.V. (%) B-value Distribution C_1 C_2 No. Specimens No. Batches Test Method						
E_1^c (Msi)	Mean Minimum Maximum C.V. (%) No. Specimens No. Batches Test Method						
ν_{12}^c	Mean No. Specimens No..Batches Test Method						

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TABLE 8.4.3(c) *Format for as-measured lamina/laminate property table.*

MATERIAL:						*Family* *Material* *Orientation* *Class of data*
SPECIMEN LAY-UP:			COMP. DENSITY:			
RESIN CONTENT: wt%			VOID CONTENT: %			
FIBER VOLUME: %			PLY THICKNESS: in.			
NORMALIZED BY:						
Temperature(°F) Moisture Content(%) Equilibrium at T, RH Source Code						
	Mean Minimum Maximum C.V. (%)					
F_1^{cu} (ksi)	B-value Distribution C_1 C_2					
	No. Specimens No. Batches Test Method					
	Mean Minimum Maximum C.V. (%)					
e_1^{cu} (μin/in)	B-value Distribution C_1 C_2					
	No. Specimens No. Batches Test Method					
	Mean Minimum Maximum C.V. (%)					
E_1^c (Msi)	No. Specimens No. Batches Test Method					
	Mean					
ν_{12}^c	No. Specimens No. Batches Test Method					

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The constants C_1 and C_2 are the mean and standard deviation for the normal distribution, the mean and standard deviation of the log-data for the lognormal distribution, and α and β for the two-parameter Weibull distribution.

Typical, minimum, and maximum values are give for the elastic constants. Typical values only are provided for Poisson's ratio and any other properties shown. For each set of properties, the test method, the number of specimens, the number of batches, and a source code are given.

8.4.4 Sample graphs. Sample graphs for the stress-strain curves are given in Section 8.3.7. Figure 8.4.4 shows a graph of the tangent modulus and the secant modulus as a function of strain for the data of Figure 8.3.7(a).

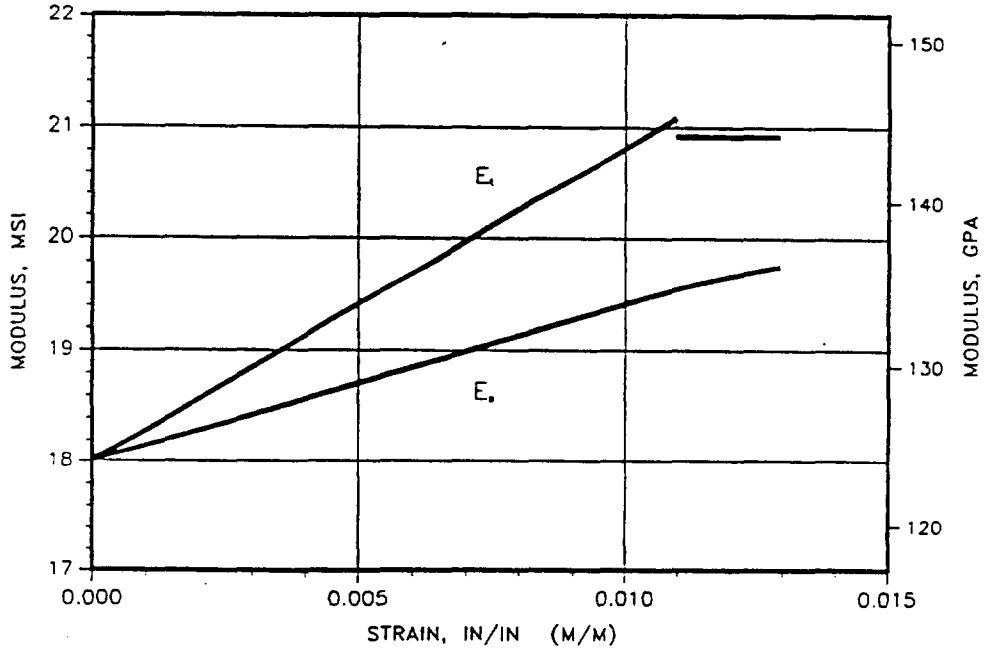


Figure 8.4.4 *Modulus curves for AS4/3501-6, see Figure 8.3.7(a).*

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8.5 CALCULATION OF STATISTICALLY-BASED MATERIAL PROPERTIES.

8.5.1 **Introduction.** Section 8.5 contains computational methods for obtaining B-basis values from composite material failure data. In Section 8.5.1.1, definitions are provided for a number of statistical terms which are used frequently in these guidelines. A discussion of the data requirements for publishing a B-basis value in MIL-HDBK-17 is presented in Section 8.5.1.2.

A step-by-step procedure for selecting the appropriate computational method is outlined in Section 8.5.2 with a flowchart showing the steps which must be taken. Section 8.5.3 contains procedures which evaluate several different statistical models and determine which, if any, of the models adequately describes the data. These procedures include methods for detecting outliers, for testing the compatibility of several batches of data, and for investigating the form of the underlying population from which a sample is drawn.

There are various methods for computing B-basis values. The assumption of a statistical model for the data determines which of these methods should be used. Sections 8.5.4 through 8.5.8 contain methods for computing B-basis values under the various models. Statistical tools used in these methods, as well as supplemental methods and tables, are presented in Section 8.6.

Finally, Section 8.7 contains examples illustrating the computational methods. Illustrative sets of data are used to demonstrate the steps that must be followed and the calculations which must be performed in calculating B-basis values.

8.5.1.1 *Statistical terms.* Proper use of the following statistical terms and equations will do much to alleviate misunderstanding in the presentation of data analyses. Definitions for other statistical terms are included in Section 1.7.

Population - The set of measurements about which inferences are to be made or the totality of possible measurements which might be obtained in a given testing situation. For example, "all possible ultimate tensile strength measurements for Composite Material A, conditioned at 95% relative humidity and room temperature". In order to make inferences about a population, it is often necessary to make assumptions about its distributional form. The assumed distributional form may also be referred to as the population.

Sample - The collection of measurements or observations taken from a specified population.

A-basis value - A statistically-based material property; a 95% lower confidence bound on the first percentile of a specified population of measurements. Also a 95% lower tolerance bound for the upper 99% of a specified population.

B-basis value - A statistically-based material property; a 95% lower confidence bound on the tenth percentile of a specified population of measurements. Also a 95% lower tolerance bound for the upper 90% of a specified population.

Compatible - Descriptive term referring to different batches which may be treated as coming from the same population.

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Location parameters and statistics:

Population mean - The average of all potential measurements in a given population weighted by their relative frequencies in the population. The population mean is the limit of the sample mean as the sample size increases.

Sample mean - The average of all observations in a sample and an estimate of the population mean. If the notation $X_1, X_2, \ldots, X_n$ is used to denote the n observations in a sample, then the sample mean is defined by:

$$\bar{X} = \frac{X_1 + X_2 + \ldots + X_n}{n} \qquad 8.5.1.1(a)$$

or

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} \qquad 8.5.1.1(b)$$

Sample median - After ordering the observations in a sample from least to greatest, the sample median is the value of the middle-most observation if the sample size is odd and the average of the two middle-most observations if the sample size is even. If the population is symmetric about its mean, the sample median is also a satisfactory estimator of the population mean.

Dispersion statistics:

Sample variance - The sum of the squared deviations from the sample mean, divided by n - 1, where n denotes the sample size. The sample variance is defined by:

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n - 1} \qquad 8.5.1.1(c)$$

or

$$S^2 = \frac{n \sum_{i=1}^{n} (X_i)^2 - \left(\sum_{i=1}^{n} X_i \right)^2}{n(n - 1)} \qquad 8.5.1.1(d)$$

Sample standard deviation - The square root of the sample variance. The sample standard deviation is denoted by S.

Probability distribution terms:

Probability distribution - A formula which gives the probability that a value will fall within prescribed limits. When the word *distribution* is used in this handbook, it should be interpreted to mean *probability distribution*.

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Normal Distribution - A two parameter (μ,σ) family of probability distributions for which the probability that an observation will fall between a and b is given by the area under the curve

$$f(x) = \frac{1}{\mu\sqrt{2\pi}} \exp\left[-(x-\mu)^2/2\,\sigma^2\right] \qquad 8.5.1.1(e)$$

between a and b. A normal distribution with parameters (μ,σ) has population mean μ and variance σ^2.

Lognormal Distribution - A probability distribution for which the probability that an observation selected at random from this population falls between a and b $(0 < a < b < \infty)$ is given by the area under the normal distribution between ℓog a and ℓog b.

Two-Parameter Weibull Distribution - A probability distribution for which the probability that a randomly selected observation from this population lies between a and b $(0 < a < b < \infty)$ is given by

$$\exp\left[-(a/\alpha)^{\beta}\right] - \exp\left[-(b/\alpha)^{\beta}\right] \qquad 8.5.1.1(f)$$

where α is called the scale parameter and β is called the shape parameter.

Probability function terms:

Cumulative Distribution Function - A function, usually denoted by F(x), which gives the probability that a random variable lies between any prescribed pair of numbers, that is

$$P\{a < x \leq b\} = F(b) - F(a) \qquad 8.5.1.1(g)$$

Such functions are non-decreasing and satisfy

$$\lim_{x \to -\infty} F(x) = 0 \quad \text{and} \quad \lim_{x \to \infty} F(x) = 1 \qquad 8.5.1.1(h)$$

The cumulative distribution function is related to the probability density function by

$$f(x) = \frac{d}{dx} F(x) \qquad 8.5.1.1(i)$$

provided that F(x) is differentiable.

F-distribution - A probability distribution which is employed in the analysis of variance, regression analysis, and tests for equality of variance. Tables of this distribution are readily available.

Probability Density Function - A function $f(x) \geq 0$ for all x with

$$\int_{-\infty}^{\infty} f(x)\,dx = 1 \qquad 8.5.1.1(k)$$

The probability density function determines the cumulative distribution function F(x) by

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$$F(x) = \int_{-\infty}^{x} f(t)\,dt \qquad 8.5.1.1(l)$$

Note that the limits $(-\infty, \infty)$ may be conventional; for example, the exponential distribution satisfies the definition by defining its probability density function as

$$f(x) = \begin{cases} 0 & x \le 0 \\ e^{-x} & x > 0 \end{cases} \qquad 8.5.1.1(m)$$

The probability density function is used to calculate probabilities as follows:

$$P\{a \le x \le b\} = \int_{a}^{b} f(x)\,dx \qquad 8.5.1.1(n)$$

Error and Variability:

Fixed Effect - A systematic shift in a measured quantity due to a particular level change of a treatment or condition. The change in level for a treatment or condition is often under the control of the experimenter. A measured quantity could be compressive strength or tensile modulus. A treatment or condition could be test temperature, fabricator, etc. For a fixed effect, the shift in the measured quantities is to be interpreted as a consistent change not only in the context of the observed data but also with respect to future data under the same treatment or condition.

Random Effect - A shift in a measured quantity due to a particular level change of an external, usually uncontrollable, factor. The level of this factor is regarded as a random draw from an infinite population. The specific level of a random effect is never under the control of the experimenter however it may remain fixed within a limited subgroup of observed data. A measured quantity could be compressive strength or tensile modulus. An external factor could be batch production leading to batch-to-batch differences. Fabricator-to-fabricator differences may be considered a random effect if the number of fabricators involved are considered to be a small random sample of all present and future fabricators. For a random effect, the shift in the measured quantities is viewed as a random variable having mean zero and a non-zero variance. Within a subgroup experiencing a fixed level of an external factor, the measured quantities are correlated (shifting as a cluster around a population average with the magnitude of the shift depending on the level of the factor). Therefore, to obtain the most independent information concerning the population of response values, it is better to have more subgroups than to have more measurements per subgroup.

Random Error - That part of the data variation which is due to unknown or uncontrolled external factors and which affects each observation independently and unpredictably. It is the residual error in a model under analysis (i.e., the variability remaining after the variability due to fixed and random effects has been removed). Random error is a special case of random effects. In both cases, the level of the random effect or error is uncontrollable but random errors vary independently from measurement to measurement (i.e., there are no random error shifts shared in common by several measurements). An important example of random error is the specimen-to-specimen variability occurring within a subgroup experiencing constant levels of treatment, condition, and other external factors (fixed and random effects).

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Material Variability - A source of variability due to the spatial and consistency variations of the material itself and due to variations in its processing (e.g., the inherent microstructure, defect population, cross-link density, etc.). Components of material variability can be any combination of fixed effects, random effects, and random error.

8.5.1.2 *Sample size requirements.* The sample size requirements for publishing a B-basis value in MIL-HDBK-17 depend upon the status of material and process specifications for the subject composite material system. If no standard specifications have been developed for the material system, a minimum of five batches of material and six specimens per batch are required for each fabricator that wishes to publish a B-basis value in MIL-HDBK-17.

In the event that standard material and process specifications have been developed for a composite material system, several fabricators may jointly develop data to publish a single B-basis value in MIL-HDBK-17. In this case, a minimum of three fabricators, three batches of material per fabricator, and six specimens per batch are required.

The sample specimens should be obtained from randomly selected areas of randomly selected panels of material fabricated to the material specification for which the B-basis value is desired. The number of specimens from each batch should be as nearly equal as possible with the largest batch size being no more than one and a half times the smallest batch size.

It should be noted that the magnitude of a B-basis value is a function of the amount of data obtained, the number of batches represented, and the uniformity of the batches produced. In general, B-basis values will tend to increase if the number of batches is increased or the number of specimens per batch is increased. B-basis values also tend to increase as population variability is reduced.

If the minimum data requirements specified above cannot be met, these guidelines may be used to calculate "preliminary" B-basis values. These data will be included as interim or screening data in MIL-HDBK-17 but the B-basis values will not be reported in the handbook until the minimum data requirements are met.

8.5.2 **Definition of computational procedures.** The procedure used to determine a B-basis value depends on the characteristics of the data. The step-by-step procedure for selecting the appropriate computational method is illustrated by the flowchart in Figure 8.5.2. Details for the specific computational methods are provided in later sections.

If the sample is made up of data from several different batches, each of the batches should be screened for outliers by the method described in Section 8.5.3.1. One should then determine whether or not the data from the different batches should be analyzed as one sample by employing the k-sample Anderson-Darling test as specified in Section 8.5.3.2. If the k-sample test does not reject the hypothesis that the batches are from the same population, the batches should be combined and the data should be analyzed as a single sample.

If the hypothesis that the batches are from the same population is rejected, indicating significant batch-to-batch variability, then the B-basis value should be computed by the ANOVA method of Section 8.5.4. A diagnostic test for the equality of variance assumption is discussed in Section 8.5.3.3.

If the sample represents a single batch of material, or several batches which are to be analyzed as a single sample, the single sample of data should be screened for outliers as discussed in Section 8.5.3.1. Note that if only a single batch is represented, then the B-basis value should only be

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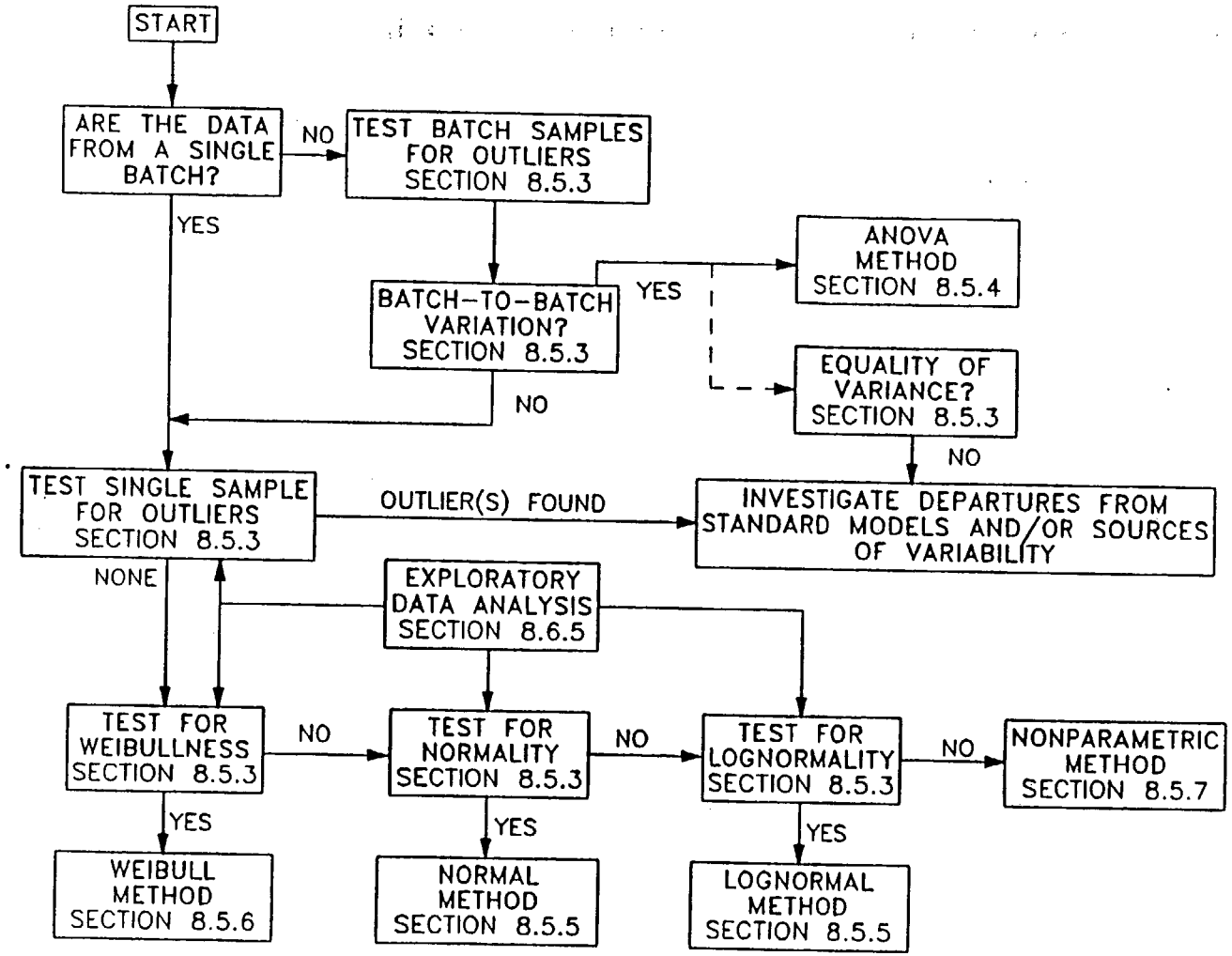


FIGURE 8.5.2 *Flowchart illustrating computational procedures for B-basis material property values.*

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considered a "preliminary" value. One should then determine which, if any, distributional form will be assumed in calculating the B-basis value. The use of goodness-of-fit tests for the two-parameter Weibull, normal, and lognormal distributions is described in Section 8.5.3.4. These tests establish the degree to which the underlying population may be fitted by one of these distributions. The two-parameter Weibull distribution is tested first. If it does not adequately fit the data, then the normal distribution is tested. If neither of these provide an adequate fit, then the lognormal distribution is tested. Exploratory data analysis (EDA) techniques, described in Section 8.6.5, can provide graphical illustrations of the distribution of the sample in support of the goodness-of-fit tests.

If one of the distributions mentioned above provides a good fit to the data, the B-basis value is computed by the corresponding two-parameter Weibull, normal, or lognormal method as described in Sections 8.5.5 or 8.5.6. If none of the distributions adequately fit the data, and the sample contains 29 or more observations, the nonparametric method described in Section 8.5.7.1 is used to compute the B-basis value. If none of the distributions fit the data adequately, and the sample size is less than 29, the Hanson-Koopmans method, described in Section 8.5.7.2, may be employed. This method, however, does not produce an approved B-basis value due to the small sample size. These nonparametric procedures should only be used when batch-to-batch variation is small.

All values, both B-basis and mean values, will be rounded to three significant figures by the usual rule. In other words, values with a fourth figure less than five will be rounded down. Values with a fourth figure greater than or equal to five will be rounded up.

8.5.3 Detecting outliers and characterizing the population. Before a B-basis value is computed, the data must be screened for erroneous values and the structure of the population from which the data were sampled must be investigated in order to determine the appropriate method. The methods discussed in this section address these two issues. These methods include the maximum normed residual outlier test (Section 8.5.3.1), the k-sample Anderson-Darling test (Section 8.5.3.2), an equality of variance test (Section 8.5.3.3), and goodness-of-fit tests for the two-parameter Weibull, normal, and lognormal distributions (Section 8.5.3.4).

8.5.3.1 *Detecting outliers.* An observation is said to be an outlier if it is an observation that has been recorded in error. For example, an erroneous observation could be the result of clerical error or the incorrect setting of environmental conditions during testing. Since outliers may have a substantial influence on the statistical analysis of the data, it is mandatory that the data be screened for outliers prior to the calculation of a B-basis value.

Initially, the data is subjected to the statistical outlier procedure discussed in Section 8.6.2, using a significance level of 0.05, to identify potential outliers. The test screens for outliers in a single sample. When different batches are represented, the data from each batch should be tested separately. Since the outlier detection procedure may miss erroneous observations, the sample should also be visually inspected for observations suspected of being outliers.

If the statistical outlier procedure or the visual inspection identifies an observation as a potential outlier, the observation will be investigated to ascertain if a cause for the outlying value can be ascribed. If it can be corrected, as is often the case for clerical errors, this should be done. If it is determined to be an erroneous observation and it cannot be corrected, as might be the case with improper setting of environmental conditions during testing, then it is discarded. If no cause can be determined for the outlying value, it must be retained in the data set.

When errors in data collection and recording are discovered, it should be determined whether similar errors occurred for other data points that were not identified as potential outliers and these values

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should also be corrected or discarded. If any observations are corrected or discarded, both the statistical outlier procedure and the visual inspection should be repeated.

8.5.3.2 *The k-sample Anderson-Darling test.* The k-sample Anderson-Darling test is used to test the hypothesis that the mechanical property data from different batches are independent random samples from the same population. The procedure for performing this test is presented in Section 8.6.3.1. If the test statistic is less than the critical value from Equation 8.6.3.1(j), then the batches should be treated as a single sample and one should proceed to the goodness-of-fit tests for investigating the form of the distribution discussed in Section 8.5.3.4.

If the test statistic is greater than or equal to the critical value, it is concluded that the batches are not identically distributed and the test for equality of variance should be performed. This test is discussed in the next section.

8.5.3.3 *Equality of variance test.* The ANOVA method (Section 8.5.4) is derived under the assumption that the variances within each batch are equal. The recommended procedure for testing this assumption is presented in Section 8.6.3.2. If the test does reject the hypothesis that the variances are equal, it is recommended that an investigation of the reason for the non-equal variances be carried out. This may reveal problems in the generation of the data or in the fabrication of the material.

8.5.3.4 *Investigating the form of the distribution.* The method employed in calculating the B-basis value for a single sample depends on the distributional form which is assumed for the data. The most frequently used parametric procedures involve the two-parameter Weibull, normal, or lognormal distributions. Section 8.6.4 contains procedures for performing a goodness-of-fit test for each of these distributions. These goodness-of-fit tests yield an observed significance level (OSL) for each of the distributions listed above. The OSL from each of the tests measures the probability of observing an Anderson-Darling statistic as extreme as the value calculated assuming that the given distribution is the correct one.

As shown in Figure 8.5.2, the test for the Weibull model (Section 8.6.4.3) is to be performed first. If the Weibull model cannot be demonstrated to adequately fit the data, then the normal and lognormal (Section 8.6.4.2) tests are performed in succession. This preference for the Weibull distribution is based on several factors:

1) Theoretical considerations suggest that the Weibull distribution is appropriate for the strength distribution of brittle materials such as composite fibers. For an elementary account of this result, see Reference 8.5.3.4(a).

2) The "Chain-of-Bundles" model for the strength of two- and three-dimensional unidirectional composites suggests that the Weibull model is appropriate for the strength distribution of such composites. This result is stated in References 8.5.3.4(b) and (c).

3) Empirical support for the Weibull model is provided by the model's good fit to data.

To determine which method should be used in calculating the B-basis value, the goodness-of-fit tests should be employed in the following manner. The two-parameter Weibull distribution is tested first. If the OSL is larger than 0.05, the Weibull distribution method of Section 8.5.6 should be used to compute the B-basis value. If the OSL is less than or equal to 0.05, the hypothesis that the data is a sample from a normal distribution is tested next. If this test for normality gives an OSL larger than 0.05, the normal distribution method of Section 8.5.5 should be used to compute the B-basis value.

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If the normal OSL is less than or equal to 0.05, the lognormal distribution is tested next. If this results in an OSL greater than 0.05, the lognormal distribution method of Section 8.5.5 should be used to compute the B-basis value. If this OSL is less than or equal to 0.05, it is concluded that the data is not a sample from any of these families of parametric distributions, and the nonparametric procedure of Section 8.5.7.1 should be used to compute the B-basis value, provided that the sample size is greater than or equal to 29. If the sample size is less than 29, then the Hanson-Koopmans method, described in Section 8.5.7.2, may be employed.

The exploratory data analysis (EDA) techniques of Section 8.6.5 may be used to graphically display the distribution of measurements in the sample and to provide graphical evidence for the conclusions obtained through the goodness-of-fit tests.

8.5.4 Normal analysis procedure in the presence of batch-to-batch variation. When failure data from different batches are to be analyzed, it is possible that the variation from batch to batch will be significant. This will prevent pooling the data and using the methods of Section 8.5.5, 8.5.6 or 8.5.7 to compute B-basis values from a single sample. This section contains a method, referred to as the ANOVA method, for computing B-basis values when there is significant batch-to-batch variation. It is a modification of a method suggested in Reference 8.5.4(a). The original paper on this topic (Reference 8.5.4(b)) is of some historical interest but should not be used for computation.

The method is based on the one-way analysis of variance (ANOVA) random effects model discussed in Section 8.6.7. The assumptions are that

(1) the data from each batch are normally distributed,
(2) the within-batch variance is the same from batch to batch, and
(3) the batch means are normally distributed.

There is no test available for the first assumption. Simulation studies, however, suggest that moderate violation of this assumption does not have an adverse effect on the properties of the ANOVA model. The second assumption should be validated by performing the test described in Section 8.6.3.2. This test is currently recommended as a diagnostic, since extensive simulation suggests that violation of this assumption will likely result in conservatism, although non-conservatism can arise in some situations. There is no useful test for the third assumption unless data from many (20 or more) batches are available.

In the analysis, all batches are treated the same, and no distinction is made between batches from different fabricators. For this reason, the sample size requirements in Section 8.5.2 require that data from at least three fabricators be included in the sample, with at least three batches from each fabricator. This attempts to insure that any variability among fabricators, while not explicitly estimated, will be included in the analysis via its contribution to the batch-to-batch variability.

In what follows, X_{ij} denotes the jth data value in the ith batch, n_i denotes the sample size of batch i, k is the number of batches, and N is the total number of data values ($N = n_1 + n_2 + \ldots + n_k$). If the entire set of data values is not available, means, standard deviations, and sample sizes for the batches are sufficient to perform the calculations.

Computational Procedure. The first step in computing the B-basis value is to compute estimates of the between-batch and within-batch variances (S_b^2 and S_e^2, respectively). Computation of the

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intermediate values MSB and MSE, is discussed in Section 8.6.7.2. The within-batch variance (σ_e^2) is estimated by

$$S_e^2 = MSE \qquad 8.5.4(a)$$

The between-batch variance (σ_b^2) is estimated by

$$S_b^2 = (MSB - MSE)/n' \qquad 8.5.4(b)$$

where

$$n' = \frac{N - n^*}{k - 1} \qquad 8.5.4(c)$$

and

$$n^* = \sum_{i=1}^{k} \frac{n_i^2}{N} \qquad 8.5.4(d)$$

If (MSB-MSE)/n' is negative, then set S_b^2 is equal to 0. For more information on estimation procedures for the one-way random effects model, see Reference 8.5.4(c).

The quantity n' may be thought of as the "effective sample size". When the number of specimens per batch is the same, n' is equal to the common sample size, n.

The ratio of the between-batch to within-batch variances ($R = \sigma_b^2/\sigma_e^2$) must also be estimated. An approximate upper 80% confidence bound for this ratio is given by the quantity $\hat{R}$, which is computed as

$$\hat{R} = [(MSB/MSE)/F_{0.2} - 1]/n' \qquad 8.5.4(e)$$

where $F_{0.2}$ is the 0.20 quantile of an F distribution with k - 1 numerator degrees of freedom and N - k denominator degrees of freedom (Reference 8.5.4(d)). If $\hat{R}$ is negative, then set $\hat{R}$ equal to zero. Table 8.8.4 contains the $F_{0.2}$ values required in Equation 8.5.4(e).

The formula for the B-basis value is then

$$B = \bar{X} - T\sqrt{S_b^2 + S_e^2} \qquad 8.5.4(f)$$

where $\bar{X}$ is the overall mean (average of all N measurements) defined in Equations 8.6.7.1.1(a) and 8.6.7.1.2(a), and T is a tolerance limit factor defined below and tabulated in Tables 8.8.6. The tolerance limit factor T is defined in statistical terms as

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$$T = \sqrt{\frac{n^*\hat{R}+1}{N(\hat{R}+1)}}\; t_{\gamma,0.95}(\delta) \qquad 8.5.4(g)$$

where $t_{\gamma,0.95}(\delta)$ is the 0.95 quantile of the non-central t distribution with noncentrality parameter

$$\delta = 1.282\sqrt{N(\hat{R}+1)/(n^*\hat{R}+1)} \qquad 8.5.4(h)$$

and degrees of freedom approximated by

$$\gamma = \frac{(\hat{R}+1)^2}{\frac{(\hat{R}+\frac{1}{n^*})^2}{k^*-1} + \frac{(n^*-\frac{1}{n^*})^2}{k^*(n^*-1)}} \qquad 8.5.4(i)$$

where $k^* = N/n^*$. T may be computed from the following approximate formula:

$$T = 1.282 + 1.282\,(t_{\gamma,0.95}/\delta) + 1.80\,(1/\gamma) - 1.85\,(1/\gamma^2) + 0.567(\delta/\gamma) + 5.24\,(\delta/\gamma^2) - 1.08(\delta^2/\gamma^2) + 0.0166(\delta^3/\gamma^2) + 7.79(1/\gamma^4) \qquad 8.5.4(j)$$

(Note: $t_{\gamma,0.95}$ is the 0.95 quantile of the central t-distribution with γ degrees of freedom.) Tolerance limit factors are tabulated in Table 8.8.6 for various values of n^*, k^* and $\hat{R}$.

Equations 8.5.4(e) through 8.5.4(i) were developed under the assumption of equal sample sizes. The performance of the method is not affected substantially by unequal sample sizes as long as the largest batch size is no more than one and one half times the smallest batch size. If sample sizes are equal, then $k^* = k$ and $n^* = n$.

8.5.5 **Normal analysis procedure for a single population.** This procedure should be used when a B-basis value is to be computed from a single sample of data that is assumed to be normally distributed. See Section 8.5.1.1 for a definition of the normal distribution. Section 8.5.5.1 contains the computational procedure and Section 8.5.5.2 describes how this procedure is applied to lognormal data. Further information on this procedure may be found in Reference 8.5.5.

8.5.5.1 *Computational procedure.* In order to compute B-basis values for a normal population, it is first necessary to obtain estimates of the population mean and standard deviation. The sample mean and standard deviation, $\bar{X}$ and S, are computed from the sample of available measurements using the equations presented in Section 8.5.1.1.

The B-basis value is then calculated by use of the following formula:

$$B = \bar{X} - k_B S \qquad 8.5.5.1$$

where

$\bar{X}$ = the sample mean based on n observations
S = the sample standard deviation
k_B = the one-sided tolerance-limit factor from Table 8.8.1.

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A numerical approximation to the k_B values is given in Equation 8.8.1.

8.5.5.2 *Lognormal procedure.* If the sample is assumed to follow a lognormal distribution, then the computational procedure presented in Section 8.5.5.1 is used to calculate the B-basis value. However, the calculations are performed using the logarithms of the data rather than the original observations. Either the natural or the common logarithm may be used. The computed B-basis value must then be transformed back to the original units by applying the inverse of the log transformation which was used. See Section 8.5.1.1 for definition of the lognormal distribution.

If, for example, the natural logarithm is used to transform the data, the B-basis value would be calculated as follows:

$$B = \exp\ (\bar{X}_L - k_B S_L) \qquad 8.5.5.2(a)$$

where

$$\bar{X}_L = \frac{\sum_{i=1}^{n} \ln(X_i)}{n} \qquad 8.5.5.2(b)$$

$$S_L = \sqrt{\frac{\sum_{i=1}^{n} [\ln(X_i) - \bar{X}_L]^2}{n-1}} \qquad 8.5.5.2(c)$$

and k_B is the one-sided tolerance-limit factor from Table 8.8.1.

8.5.6 **Two-parameter Weibull analysis procedure for a single population.** This procedure should be used when a B-basis value is to be computed from a single sample of data which is assumed to follow a two-parameter Weibull distribution. See Section 8.5.1.1 for a definition of the two-parameter Weibull distribution.

In order to compute a B-basis value for a two-parameter Weibull population, it is first necessary to obtain estimates of the population shape and scale parameters. Section 8.5.6.1 contains a step-by-step procedure for calculating maximum likelihood estimates of these parameters. The computational procedure for calculating B-basis values using these estimates and the tolerance limit factors in Table 8.8.2 is outlined in Section 8.5.6.2. For further information on these procedures, see Reference 8.5.6.

8.5.6.1 *Estimating the shape and scale parameters of a Weibull distribution.* In order to calculate the B-basis value for a random sample from the two-parameter Weibull distribution, estimates of the shape and scale parameters are needed. The procedure described below is the maximum likelihood method, and the estimates obtained are known as the maximum likelihood estimates. The shape parameter estimate is denoted by $\hat{\beta}$, and the scale parameter by $\hat{\alpha}$. The estimates are obtained by solving the pair of likelihood equations given below for $\hat{\beta}$ and $\hat{\alpha}$.

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$$\hat{\alpha}\,\hat{\beta}\,n - \frac{\hat{\beta}}{\hat{\alpha}^{\hat{\beta}-1}} \sum_{i=1}^{n} X_i^{\hat{\beta}} = 0 \qquad 8.5.6.1(a)$$

$$\frac{n}{\hat{\beta}} - n \ln \hat{\alpha} + \sum_{i=1}^{n} \ln X_i - \sum_{i=1}^{n} \left[\frac{X_i}{\hat{\alpha}}\right]^{\hat{\beta}} (\ln X_i - \ln \hat{\alpha}) = 0 \qquad 8.5.6.1(b)$$

Equation 8.5.6.1(a) can be rewritten as

$$\hat{\alpha} = \left(\frac{\sum_{i=1}^{n} X_i^{\hat{\beta}}}{n}\right)^{1/\hat{\beta}} \qquad 8.5.6.1(c)$$

By substituting Equation 8.5.6.1(c) into Equation 8.5.6.1(b), the following equation is obtained.

$$\frac{n}{\hat{\beta}} + \sum_{i=1}^{n} \ln X_i - \frac{n}{\sum_{i=1}^{n} X_i^{\hat{\beta}}} \sum_{i=1}^{n} X_i^{\hat{\beta}} \ln X_i = 0 \qquad 8.5.6.1(d)$$

Since everything in Equation 8.5.6.1(d) is known except $\hat{\beta}$, this equation can be solved numerically, and the solution of $\hat{\beta}$ that is obtained is substituted into Equation 8.5.6.1(c) to obtain $\hat{\alpha}$, the estimate of α.

Figure 8.5.6 shows FORTRAN source code for three routines which compute the estimates of $\hat{\alpha}$ and $\hat{\beta}$ by the method described above. WBLEST is a subroutine which returns the estimates of the parameters, $\hat{\beta}$ and $\hat{\alpha}$. FNALPH is a function which calculates the estimate of the scale parameter, $\hat{\alpha}$. GFUNCT is a function which evaluates Equation 8.5.6.1(d). Arguments to WBLEST are

X	=	a vector of length NOBS containing the data (input),
NOBS	=	the number of data values, N (input),
BETA	=	estimate of the shape parameter (output),
ALPHA	=	estimate of the scale parameter (output).

The algorithm by which the FORTRAN code computes the estimates is described in the following paragraph.

Equation 8.5.6.1(d) is a monotonically increasing continuous function of $\hat{\beta}$. Designate the left-hand side of Equation 8.5.6.1(d) divided by n as $G(\hat{\beta})$ and obtain a solution for $\hat{\beta}$ by the following iterative procedure. Let S_y denote the standard deviation of $Y_1,\ldots,Y_n$ where $Y_1 = \ell n(X_i)$ for $i = 1,\ldots,n$. Calculate $I = 1.28/S_y$ as an initial guess at the solution and calculate G(I). If G(I) > 0, then find the smallest positive integer k such that $G(I/2^k) < 0$ and let $L = I/2^k$ and $H = I/2^{k-1}$. If G(I) < 0, then find the smallest positive integer k such that $G(2^k I) > 0$ and let $L = 2^{k-1} I$ and $H = 2^k I$. In

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```
C-------------------------------------------------------------------
      SUBROUTINE WBLEST(X,NOBS,ALPHA,BETA)
C
C  COMPUTE MLES FOR SHAPE PARAMETER (BETA) AND SCALE PATAMETER
C  (ALPHA) BY SOLVING THE EQUATION G(BETA) = 0, WHERE G IS
C  A MONOTONICALLY INCREASING FUCTION OF BETA.
C  THE INITIAL ESITMATE IS: RI=(1.28)/(STD. DEV. OF LOG(X)'S)
C  AND THE TOLERANCE IS   :  2*RI/(10**6).
C
      DIMENSION X(NOBS)
C
      RN = FLOAT(NOBS)
      SUMY = 0.0
      SUMYSQ = 0.0
      DO  2 I = 1, NOBS
         Y = ALOG(X(I))
         SUMY = SUMY + Y
         SUMYSQ = SUMYSQ + (Y**2)
    2 CONTINUE
      YSTD = SQRT((SUMYSQ - (SUMY**2)/RN)/(RN - 1.0))
      XGM = EXP(SUMY/RN)
      RI = 1.28/YSTD
      TOL = 2.0*.000001*RI
      BETAH = RI
      GFM = GFUNCT(X,NOBS,BETAH,XGM)
C
C  IF G(BETAH) .GE.  0, DIVIDE THE INTIAL ESTIMATE BY 2 UNTIL
C  THE ROOT IS BRACKETED BY BETAL AND BETAH.
C
      IF(GFM .GE. 0.0) THEN
           DO 3 J = 1, 20
                BETAH = BETAM
                BETAM = BETAM/2.0
                GFM = GFUNCT(X,NOBS,BETAM,XGM)
                IF (GFM .LE. 0.0) GO TO 4
    3      CONTINUE
           STOP 'GFM NEVER LE 0'
    4      CONTINUE
           BETAL = BETAM
      ENDIF
C
C    IF G(BETAM) .LT.  0, MULTIPY THE THE INITIAL ESTIMATE BY 2
C    UNTIL THE ROOT IS BRACKETED BY BETAL AND BETAH
C
      IF(GFM .LT. 0.0) THEN
           DO 7 J = 1, 20
                BETAL=BETAM
                BETAM=BETAM*2.0
                GFM=GFUNCT(X,NOBS,BETAM,XGM)
                IF(GFM .GE. 0.0) GO TO 8
    7      CONTINUE
           STOP 'GFM NEVER GE 0'
    8      CONTINUE
           BETAH = BEHAM
      ENDIF
C
C  SOLVE THE EQUATION G(BETA) = 0 FOR BETA BY BISECTING THE
C  INTERVAL (BETAL,BETAH) UNTIL THE TOLERANCE IS MET
C
   10 CONTINUE
      BETAM = (BETAL + BETAH) / 2.0
      GFM = GFUNCT(X,NOBS,BETAM,XGM)
      IF(GFM .GE. 0.0) THEN
           BETAH = BETAM
      ENDIF
      IF(GFM .LT. 0.0) THEN
           BETAH = BETAM
      ENDIF
      IF((BETAH - BETAL) .GT. TOL) GO TO 10
C
      BETA = (BETAL + BATAH) / 2.0
      ALPHA = FWALPH(X,NOBS,BETA,XGM)
      RETURN
      END
C-------------------------------------------------------------------
```

FIGURE 8.5.6 *FORTRAN routines for calculating two-parameter Weibull shape and scale parameter estimates*, continued on next page

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```
C-------------------------------------------------------------
      FUNCTION FNALPH(X,NOBS,BETA,XGM)
C
C  COMPUTE MLE FOR TWO-PARAMETER WEIBULL SCALE PARAMETER (ALPHA)
C  XGM IS THE GEOMETRIC MEAN OF THE X'S
C
      DIMENSION X(NOBS)
      RN = FLOAT(NOBS)
C
      SUMZ = 0.0
      DO 20 I = 1, NOBS
        SUMZ = SUMZ + (X(I)/XGM)**BETA
   20 CONTINUE
C
      FNALPH = XGM*(SUMZ/RN)**(1./BETA)
C
      RETURN
      END
C-------------------------------------------------------------



C-------------------------------------------------------------
      FUNCTION GFUNCT(X,NOBS,BETA,XGM)
C
C  COMPUTE G FUCTION USED IN ESTIMATING THE TWO-PARAMETER WEIBULL
C  SHAPE PARAMETER (BETA).
C  XGC IS THE GEOMETRIC MEAN OF THE X'S USED IN ESTIMATING ALPHA.
C
      DIMENSION X(NOBS)
      RN = FLOAT(NOBS)
C
      ALPHA = FNALPH(X,NOBS,BETA,XGM)
      SUMYZ = 0.0
      DO 10 I = 1, NOBS
          SUMYZ = SUMYZ + ALOG(X(I))*((X(I)/ALPHA)**BETA - 1.)
   10 CONTINUE
C
      GFUNCT = (SUMYZ/RN) - 1.0/BETA
C
      RETURN
      END
C-------------------------------------------------------------
```

FIGURE 8.5.6 *FORTRAN routines for calculating two-parameter Weibull shape and scale parameter estimates*, concluded.

either case, the interval (L,H) contains the solution to $G(\hat{\beta}) = 0$. Now calculate G(M) where M = (L + H)/2. If G(M) = 0, then the solution is $\hat{\beta} = M$. If G(M) > 0, then let H = M. If G(M) < 0 then let L = M. The new interval (L,H) still contains the solution to $G(\hat{\beta}) = 0$ but is only half as long as the old interval. Calculate a new M-value and begin the process of interval halving again. The process is repeated until H-L < $2l/10^6$. The solution to $G(\hat{\beta}) = 0$ is then taken to be M = (L + H)/2. The solution is in error by at most $l/10^6$.

8.5.6.2 *Computational procedure.* For given parameter estimates $\hat{\alpha}$ and $\hat{\beta}$, the computation of a B-basis value is carried out by

$$B = \hat{Q} \exp\left\{\frac{-V}{\hat{\beta}\sqrt{n}}\right\} \qquad 8.5.6.2(a)$$

where

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$$\hat{Q} = \hat{\alpha}\ (0.10536)^{1/\hat{\beta}} \qquad 8.5.6.2(b)$$

and V is the value in Table 8.8.2 corresponding to a sample of size n. A numerical approximation to the V values is given in Equation 8.8.2(b).

8.5.7 Nonparametric analysis procedure for a single population. This procedure should be used when a material property value is to be computed from failure data and the form of the distribution of the population is unknown. The distribution should be considered unknown if tests indicate significant departure from the two-parameter Weibull, normal, and lognormal distributions.

8.5.7.1 *Sample size greater than 28.* First, order the sample so that the observations are in increasing order. Then, using Table 8.8.3, determine the r value corresponding to the sample size n. For sample sizes lying between tabulated values, use the r value associated with the largest tabulated sample size which is smaller than the actual n. The B-basis value is the rth lowest observation in the ordered sample. For example, in a sample of size n = 30, the lowest (r = 1) observation is the B-basis value. As another example, in a sample of size n = 600, the 48th ordered observation (r = 48) is the B-basis value. For sample sizes less than 29, a B-basis value acceptable for MIL-HDBK-17 cannot be calculated with this procedure; the Hanson-Koopmans procedure can provide a B-basis value for these small sample sizes.

The value of r for any sample size n can be computed directly by using the following relationships. If $q_{0.10}$ is the 0.10 quantile of the underlying population distribution, then the number of observations in the sample less than $q_{0.10}$ is a binomial random variable with n trials and probability of success p = 0.10. Letting $X_{(r)}$ denote the rth ordered sample value, the statement $X_{(r)} \leq q_{0.10}$ is true if and only if there are r or more values in the sample which are less than or equal to $q_{0.10}$. Using this relationship, the B-basis value is the rth ordered sample value, $X_{(r)}$, where $r \geq 1$ is the largest integer solution to

$$\sum_{w=r}^{n} \binom{n}{w} (0.10)^{w}\ (0.90)^{n-w} \geq 0.95 \qquad 8.5.7.1(a)$$

and

$$\binom{n}{w} = \frac{n!}{w!(n-w)!} \qquad 8.5.7.1(b)$$

Table 8.8.3 was generated from this equation. A numerical approximation to the tabulated r values as a function of n is also given in Section 8.8. Further information on this procedure may be found in Reference 8.5.5.

8.5.7.2 *The Hanson-Koopmans method.* Although it can be demonstrated that for $n < 29$ there is no useful nonparametric method, the following procedure, due to D. L. Hanson and L. H. Koopmans (Reference 8.5.7.2), may be a useful method for obtaining a B-basis value for sample sizes not exceeding 28. This procedure requires the assumption that the observations are a random sample from a probability density function with a log-concave cumulative distribution function. That is, if the probability function is f(x), and the cumulative distribution function is F(x), then

$$h(x) = -\log F(x) \qquad 8.5.7.2(a)$$

is a concave function of x, for $x \geq 0$. This assumption is satisfied by a large class of probability distributions, however it is not, strictly speaking, nonparametric.

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There is substantial empirical evidence that suggests the composite strength data that will be encountered by users of this handbook satisfies the assumptions underlying the Hanson-Koopmans method and, consequently, this procedure can be recommended for use when n is less than 29. However, in view of the required assumption, this is not an unconditional recommendation.

The B-basis value is calculated as follows. Order the observations from smallest to largest. Let $X_{(1)}$ be the smallest observation, $X_{(2)}$ the second smallest, and so on. In general, we denote the r-th smallest value by $X_{(r)}$. Then the B-basis value is given by

$$B = X_{(r)}\left[\frac{X_{(1)}}{X_{(r)}}\right]^k \qquad 8.5.7.2(b)$$

where r and k depend on n and are determined from Table 8.8.10. This equation for the B-basis value should not be employed if $X_{(r)} = X_{(1)}$.

Example. Assume that 25 observations have been obtained. Order these in increasing value. From the table, for n = 25, r = 11; therefore, we record the first and eleventh values. For purposes of illustration, let the smallest value be 36.64 and the eleventh value be 47.50. Then the B-basis value is

$$B = 47.50\left(\frac{36.64}{47.50}\right)^{1.087} = 35.8$$

8.5.8 Linear statistical models. Let Y be the material property being studied and let $X^{(1)}, X^{(2)}, \ldots, X^{(k)}$ be variables, whose potential value as linear predictors of Y is under investigation. Specifically, we are assuming a (linear) model of the form

$$Y = \beta_0 + \beta_1 X^{(1)} + \beta_2 X^{(2)} + \cdots + \beta_k X^{(k)} + \epsilon \qquad 8.5.8(a)$$

where $\beta_0, \beta_1, \beta_2, \ldots, \beta_k$ may be either statistical parameters to be estimated or random variables.

The predictor variables $X^{(1)}, X^{(2)}, \ldots, X^{(k)}$ are assumed to be fixed (not random) and their values will sometimes be selected by the experimenter as part of the experimental design. If $\beta_0, \beta_1, \beta_2, \ldots, \beta_k$ are statistical parameters, then the model in Equation 8.5.8(a) is called a **fixed effects model**. If $\beta_0, \beta_1, \beta_2, \ldots, \beta_k$ are random variables, then the model is called a **random effects model** or **components of variance model**. If some of the β's are random variables and some are statistical parameters, then the model is known as a **mixed model**. The "error", ϵ, is assumed to be a random variable with mean zero. In the remainder of this section, only the fixed effects model is discussed.

In particular, the statement that Equation 8.5.8(a) is a linear model refers to linearity in $\beta_0, \beta_1, \beta_2, \ldots, \beta_k$ and not necessarily in $X^{(1)}, X^{(2)}, \ldots, X^{(k)}$. Thus

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_k X^k + \epsilon \qquad 8.5.8(b)$$

is a linear model (by letting $X^{(i)} = X^i$) and also

$$Y = \beta_0 + \beta_1 e^{-X} + \beta_2 e^{-2X} + \cdots + \beta_k e^{-kX} + \epsilon \qquad 8.5.8(c)$$

is a linear model (by letting $X^{(i)} = e^{-iX}$).

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Note that the assumption that $X^{(1)}, X^{(2)}, \ldots, X^{(k)}$ are known constants applies to Equations 8.5.8(b) and 8.5.8(c) as well. For example in Equation 8.5.8(b), if X is a known constant, then $X^2, X^3, \ldots, X^k$ are also known quantities.

The discussion here is intended to be superficial. For more complete treatments of this very extensive subject, the reader should consult References 8.5.8(a) - 8.5.8(d).

The remainder of this section deals exclusively with the special case k = 1. This special case is called the simple linear regression model. More general situations are discussed in Section 8.6.6. Assume again that β_0, β_1 are fixed unknown parameters and not random variables. Thus we are considering the model

$$Y = \beta_0 + \beta_1 X + e \qquad 8.5.8(d)$$

Assume that the experimenter chooses *n* X values $X_1, X_2, \ldots, X_n$ which need not be distinct, and observes that the corresponding Y values; thus the data consists of the *n* pairs

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$$

where the X's have been chosen by the experimenter and the Y's are the observed "responses" at each X.

In order for the statistical analysis to be valid we must have $n \geq 3$ and at least two distinct X values. Let $\hat{\beta}_0$ and $\hat{\beta}_1$ denote estimators of β_0 and β_1 respectively. Then for any X (which need not be one of the experimental values $X_1, X_2, \ldots, X_n$), a predicted or fitted value denoted by $\hat{Y}$ is obtained, that is

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X \qquad 8.5.8(e)$$

It is customary to estimate β_0 and β_1 using the principle of least squares, which may be defined as follows. Let β_0^* and β_1^* be any estimators of β_0 and β_1 respectively. Let

$$Q(\beta_0^*, \beta_1^*) = \sum_{i=1}^{n} (Y_i - \hat{Y}_i^*)^2 \qquad 8.5.8(f)$$

where $\hat{Y}_i^* = \beta_0^* + \beta_1^* X_i$.

The least square estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are the values of β_0^* and β_1^*, which minimize $Q(\beta_0^*, \beta_1^*)$. They are given by

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \qquad 8.5.8(g)$$

and

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$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(X_i-\bar{X})(Y_i-\bar{Y})}{\sum_{j=1}^{n}(X_j-\bar{X})^2} \qquad 8.5.8(h)$$

where

$$\bar{Y} = \sum_{i=1}^{n} Y_i/n \qquad 8.5.8(i)$$

and

$$\bar{X} = \sum_{i=1}^{n} X_i/n \qquad 8.5.8(j)$$

It is sometimes more convenient to calculate $\hat{\beta}_1$ by the following equivalent formula,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} X_iY_i - n\overline{XY}}{\sum_{i=1}^{n} X_i^2 - n\bar{X}^2} \qquad 8.5.8(k)$$

Statistical significance (at level α) of a regression means that there is evidence that $\beta \neq 0$ (with a probability of $\leq \alpha$ of reaching this conclusion when $\beta = 0$). If $\beta \neq 0$, then X is of value as a linear predictor of Y. In order for the usual test of significance to be valid, the following additional assumptions are required. The Y 's are independent normally distributed random variables with common variance σ^2 and means $\beta_0 + \beta_1 X_i$, $i = 1, 2, \ldots, n$.

To test whether the regression is significant at level α, let

$$S_y^2 = \frac{\sum_{i=1}^{n}(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2}{n-2} \qquad 8.5.8(l)$$

and define

$$SSE = \sum_{i=1}^{n}(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 \qquad 8.5.8(m)$$

$$SST = \sum_{i=1}^{n}(Y_i - \bar{Y})^2 \qquad 8.5.8(n)$$

and

$$SSR = SST - SSE \qquad 8.5.8(o)$$

Then define

$$F = \frac{SSR}{S_y^2} \qquad 8.5.8(p)$$

which has the F-distribution with 1 and n-2 degrees of freedom. The regression is considered significant if the value in Equation 8.5.8(p) exceeds the $1-\alpha$ quantile of the F-distribution with $\nu_1 = 1$ and $\nu_2 = n - 2$ degrees of freedom. Table 8.8.5 provides these values for $\alpha = 0.05$.

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Since the mean of the material property variable Y varies linearly with X, adjustments need to be made in computing a B-basis value. The meaning of such B-values is a straightforward extension from the single sample case. Namely, for a given X , the B-value should, with 95% confidence, fall below the tenth percentile of the distribution of Y-values at the value X . For A-basis values, exchange 99% for 90% above.

A B-basis value, in the case of simple linear regression, can be determined as follows. For $X = X_0$, compute B as

$$B = (\hat{\beta}_0 + \hat{\beta}_1 X_0) - k_B S_y \qquad 8.5.8(q)$$

where S_y is the square root of S_y^2 in 8.5.8(l),

$$k_B = \left[\frac{1+\Delta}{n}\right]^{\frac{1}{2}} t_{\gamma,0.95}(\delta) \qquad 8.5.8(r)$$

and $t_{\gamma,0.95}(\delta)$ is the 95th percentile of the non-central t-distribution with $\gamma = n-2$ degrees of freedom and noncentrality parameter

$$\delta = \frac{1.282}{\left[\frac{1+\Delta}{n}\right]^{\frac{1}{2}}} \qquad 8.5.8(s)$$

with

$$\Delta = \frac{n(X_0 - \bar{X})}{\sum_{i=1}^{n} (X_i - \bar{X})^2} \qquad 8.5.8(t)$$

The following approximation to k_B may be used when n is greater than or equal to 10 and $0 \le \Delta \le 10$:

$$k_B = 1.282 + \exp\left[0.595 - 0.508\,\ell n(n) + \frac{4.62}{n} + \left(0.486 - \frac{0.986}{n}\right)\ell n(1.82 + \Delta)\right] \qquad 8.5.8(u)$$

Under the stated conditions, this approximation has a relative error not exceeding 1%. A numerical illustration is given in Problem 8, Section 8.7.8.

To adapt Equation 8.5.8(q) to A-basis values, replace 1.282 in Equation 8.5.8(s) with 2.326. For A-basis values, k_A can be approximated by

$$k_B = 2.326 + \exp\left[0.659 - 0.514\,\ell n(n) + \frac{6.58}{n} + \left(0.481 - \frac{1.42}{n}\right)\ell n(3.71 + \Delta)\right] \qquad 8.5.8(v)$$

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8.6 STATISTICAL PROPERTIES.

8.6.1 Introduction. This section includes a number of statistical tools for use in the analysis of data for MIL-HDBK-17. These tools are intended to supplement the basic procedures described in Section 8.5.

Various statistical techniques are employed in the analysis of material-property data. This section presents brief descriptions of procedures that are used most frequently in this application. More detailed descriptions of these and other statistical techniques and tables can be found in a number of workbooks and texts, some of which are given as references.

When procedures other than those described below are employed in the preparation of data proposals, they should be described adequately in the proposal to allow a proper evaluation of their validity and relevance.

8.6.2 Outlier detection procedure. An observation is said to be an outlier if it is an observation that has been recorded in error. Since outliers may have a substantial influence on the statistical analysis of a set of data, it is often desirable that a data set be screened for outliers prior to data analysis. The maximum normed residual method for detecting potential outliers is described below. For more information on this test procedure, see References 8.6.2(a) and 8.6.2(b).

8.6.2.1 *The maximum normed residual method.* The maximum normed residual (MNR) test is a screening procedure for identifying outliers in a single set of data. It involves an examination of the residuals (signed distances from the mean divided by the standard deviation) to determine whether or not they are unusually large. The test assumes that the non-erroneous data values are independently and identically normally distributed. Also, the outlier procedure searches for one outlier at a time and, therefore, the significance level pertains to a single decision.

The first step is to compute the normed residuals and the MNR statistic. If $X_1, X_2, \ldots X_n$ denote the data values in the sample of size n, the normed residuals are defined as:

$$r_i = \frac{X_i - \bar{X}}{S}, \quad i = 1,2,\ldots,n \qquad 8.6.2.1(a)$$

where $\bar{X}$ and S are the sample mean and sample standard deviation, respectively, defined in Section 8.5.1.1. The MNR statistic is the maximum of the absolute values of the normed residuals:

$$MNR = \max_i \{|r_i|\} \qquad 8.6.2.1(b)$$

Next, the MNR statistic is compared to the critical value for the sample size n from Table 8.8.9. The critical values for this test are computed from the following formula:

$$C.V. = \frac{n-1}{\sqrt{n}} \sqrt{\frac{t^2}{n-2+t^2}} \qquad 8.6.2.1(c)$$

where t is the $[1 - \alpha/(2n)]$ quantile of the t distribution with n - 2 degrees of freedom and α is the significance level of the test. Critical values for sample sizes 3 through 200 and a significance level

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of α = 0.05 are tabulated in Table 8.8.9. Values for other sample sizes or other significance levels may be computed using the above formula.

If MNR is smaller than the critical value, then no outliers are detected in the sample. If MNR is larger than the critical value, then the data value associated with the largest absolute normed residual is declared to be a potential outlier.

If an outlier is detected, the procedure is repeated on the reduced set of data (eliminating the potential outlier). This process is repeated until no potential outliers are detected. Note that in the second iteration, the mean, standard deviation, and critical value are computed using a sample size of n - 1. In the third iteration, the sample size will be n - 2, and so on.

8.6.3 Subpopulation compatibility tests. This section contains tests for determining the compatibility of two or more populations. The k-sample Anderson-Darling test in Section 8.6.3.1 is used to test the hypothesis that the populations from which two or more independent samples were taken are identically distributed. The equality of variance test in Section 8.6.3.2 is used to test the hypothesis that the variances of the populations from which two or more independent random samples were taken are equal.

8.6.3.1 *The k-sample Anderson-Darling test.* The k-sample Anderson-Darling statistic tests the hypothesis that the populations from which two or more independent random samples were drawn are identical. The test can be applied to determine whether two or more products differ with regard to their strength distributions. The test is a nonparametric statistical procedure and only requires the assumption of independent random samples from their respective populations.

Consider the populations $A_1, A_2, ..., A_k$. Let $X_{11}, X_{12}, ..., X_{1n_1}$, denote a sample of n_1 data points from population A_1, let $X_{21}, X_{22}, ..., X_{2n_2}$, denote a sample of the n_2 data points from population A_2, and so forth. Furthermore, let $N = n_1 + n_2 + ... + n_k$ represent the total number of data points in the combined samples.

Let L denote the total number of distinct data points in the combined samples and $Z_{(1)}, Z_{(2)}, ..., Z_{(l)}$ denote the distinct values in the combined data set ordered from least to greatest. The k-sample Anderson-Darling statistic is defined by

$$ADK = \frac{N-1}{N^2(k-1)} \sum_{i=1}^{k} \left[\frac{1}{n_i} \sum_{j=1}^{L} h_j \frac{(NF_{ij} - n_i H_j)^2}{H_j(N-H_j) - Nh_j/4} \right] \qquad 8.6.3.1(a)$$

where

h_j = the number of values in the combined samples equal to $Z_{(j)}$

H_j = the number of values in the combined samples less than $Z_{(j)}$ plus one half the number of values in the combined samples equal to $Z_{(j)}$, and

F_{ij} = the number of values in sample corresponding to population A_i which are less than $Z_{(j)}$ plus one half the number of values in the sample corresponding to population A_i which are equal to $Z_{(j)}$.

Under the hypothesis of no difference in the populations, the mean and variance of ADK are approximately 1 and

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$$\sigma_N^2 = \mathrm{Var(ADK)} = \frac{aN^3 + bN^2 + cN + d}{(N-1)(N-2)(N-3)(k-1)^2} \qquad 8.6.3.1(b)$$

with

$$a = (4g-6)(k-1) + (10-6g)S \qquad 8.6.3.1(c)$$

$$b = (2g-4)k^2 + 8Tk + (2g-14T-4)S - 8T + 4g - 6 \qquad 8.6.3.1(d)$$

$$c = (6T+2g-2)k^2 + (4T-4g+6)k + (2T-6)S + 4T \qquad 8.6.3.1(e)$$

$$d = (2T+6)k^2 - 4Tk \qquad 8.6.3.1(f)$$

where

$$S = \sum_{i=1}^{k} \frac{1}{n_i} \qquad 8.6.3.1(g)$$

$$T = \sum_{i=1}^{N-1} \frac{1}{i} \qquad 8.6.3.1(h)$$

and

$$g = \sum_{i=1}^{N-2} \sum_{j=i+1}^{N-1} \frac{1}{(N-i)j} \qquad 8.6.3.1(i)$$

If

$$\mathrm{ADK} \geq 1 + \sigma_N \left[1.645 + \frac{0.678}{(k-1)^{0.5}} - \frac{0.362}{k-1} \right] \qquad 8.6.3.1(j)$$

one may conclude (with a five percent risk of being in error) that the samples were drawn from different populations. Otherwise, the hypothesis that the samples were selected from identical populations is not rejected. Table 8.8.7 contains the critical values (Equation 8.6.3.1(j)) for the case of equal sample sizes. For more information on the k-sample Anderson-Darling test, see Reference 8.6.3.1.

8.6.3.2 *Test for equality of variances.* This section describes a test suggested by Lehmann (Reference 8.6.3.2) for determining whether two or more estimates of variance differ significantly. With k independent random samples, the sample variance for the ith sample is denoted by S_i^2 and has $\nu_i = n_i - 1$ degrees of freedom, where n_i is the size of the ith sample.

Letting $Z_i = \ell n\,(S_i^2)$, the test statistic is computed as

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$$EV = \frac{1}{2}\sum_{i=1}^{l} \gamma_i Z_i^2 - \frac{1}{2}(N-k)\,\bar{Z}^2 \qquad 8.6.3.2(a)$$

where

$$\bar{Z} = \sum_{i=1}^{k} \gamma_i Z_i/(N-k) \qquad 8.6.3.2(b)$$

is a weighted average of the Z_i values, and N is the total sample size. This statistic is compared to the 0.95th quantile of a chi-square distribution with k - 1 degrees of freedom, values of which are tabulated in Table 8.8.8. (A numerical approximation to the tabulated values is given in Section 8.8.8.) If the test statistic is greater than or equal to the tabulated value, then the variances are declared to be significantly different. If the statistic is less than the tabulated value, then the hypothesis of equality of variance is not rejected.

8.6.4 **Goodness-of-fit tests.** The computational procedure selected to establish B-basis values by statistical techniques is dependent upon the underlying distribution of measurements. The most frequently used parametric procedures involve the use of the normal, lognormal, or two-parameter Weibull distributions. This section contains methods that are used to establish how well a population may be fitted by one of these distributions.

Four goodness-of-fit test procedures are described below. The purpose of each is to indicate whether a distributional assumption should be rejected. The methods presented are based on the Anderson-Darling goodness-of-fit test statistic, which is sensitive to discrepancies in the tail regions. The tests of fit to the normal, lognormal, and Weibull distributions for a single sample are presented in Sections 8.6.4.1, 8.6.4.2 and 8.6.4.3, respectively. A test for normality in a regression setting is discussed in Section 8.6.4.4.

An observed significance level (OSL) may be computed for each test. The OSL is the probability of obtaining a value of the test statistic at least as large as that obtained if the hypothesis that the data are actually from the distribution being tested is true. If the OSL is less than or equal to 0.05, the hypothesis is rejected (with at most a five percent risk of being in error) and one proceeds as if the data are not from the distribution being tested.

In what follows, unless otherwise noted, the sample size is denoted by n, the sample observations by $X_1, \ldots, X_n$, and the sample observations ordered from least to greatest by $X_{(1)}, \ldots, X_{(n)}$.

8.6.4.1 *Anderson-Darling test for normality:* The Anderson-Darling test for normality is used to test the hypothesis that the population from which a given sample of data was drawn is normally distributed. See Section 8.5.1.1 for a definition of the normal distribution.

The test compares the cumulative normal distribution function that fits the data best with the cumulative distribution function of the observed data. Let

$$Z_{(i)} = (X_{(i)} - \bar{X})/S \qquad i = 1,\ldots,n \qquad 8.6.4.1(a)$$

where $X_{(i)}$ is the ith smallest sample observation, $\bar{X}$ is the sample average, and S is the sample standard deviation. Equations for computing these sample statistics are presented in Section 8.5.1.1.

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The Anderson-Darling test statistic is

$$AD = \sum_{i=1}^{n} \frac{1-2i}{n} \{ \ell n [F_o(Z_{(i)})] + \ell n [1-F_o(Z_{(n+1-i)})] \} - n \qquad 8.6.4.1(b)$$

where F_o is the standard normal distribution function. The standard normal distribution function $F_o(x)$ is the area to the left of the value x under the standard normal density curve. (See Equation 8.5.1.1(e).)

The observed significance level (OSL) is calculated as

$$OSL = 1/\{1+\exp[-0.48 + 0.78 \ell n (AD^*) + 4.58 (AD^*)]\} \qquad 8.6.4.1(c)$$

where

$$AD^* = (1 + 4/n - 25/n^2) AD \qquad 8.6.4.1(d)$$

The OSL is a measure of the goodness-of-fit of a normal distribution to the data. Specifically, the OSL measures the probability of observing an Anderson-Darling statistic at least as extreme as the value calculated if a normal distribution is in fact the underlying distribution.

If OSL $\leq$ 0.05, one may conclude (at five percent risk of being in error) that the population from which the sample was drawn is not normally distributed. Otherwise, the hypothesis that the population is normally distributed is not rejected. For further information on this test procedure, see Reference 8.5.6.

8.6.4.2 *Anderson-Darling test for lognormality.* The lognormal distribution is a positively skewed distribution which is simply related to the normal distribution. If something is lognormally distributed, then its logarithm is normally distributed. The natural (base e) logarithm is used in MIL-HDBK-17.

In order to test the goodness-of-fit of the lognormal distribution, take the log of the data and perform the Anderson-Darling test for normality from Section 8.6.4.1. Using the natural logarithm, let

$$Z_{(i)} = \frac{\ell n (X_{(i)}) - \bar{X}_L}{S_L} \qquad 8.6.4.2(a)$$

where $X_{(i)}$ is the ith smallest sample observation and X_L and S_L are the mean and standard deviation of the $\ell n (X_i)$ values defined by

$$\bar{X}_L = \frac{\sum_{i=1}^{n} \ell n (X_i)}{n} \qquad 8.6.4.2(b)$$

$$S_L = \left\{ \frac{\sum_{i=1}^{n} [\ell n (X_i) - \bar{X}_L]^2}{n-1} \right\}^{\frac{1}{2}} \qquad 8.6.4.2(c)$$

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The Anderson-Darling statistic is computed using equation 8.6.4.1(b) and the observed significance level (OSL) is computed as in Equation 8.6.4.1(c).

8.6.4.3 *Anderson-Darling test for Weibullness.* The Anderson-Darling test for two-parameter Weibullness is used to test the hypothesis that the population from which a given sample of data was drawn has a two-parameter Weibull distribution. See Section 8.5.1.1 for a definition of the two-parameter Weibull distribution.

The test compares the cumulative Weibull distribution function that fits the data best with the cumulative distribution function of the data. The first step is to compute the parameter estimates $\hat{\alpha}$ and $\hat{\beta}$. A procedure for computing these estimates is given in Section 8.5.6.1. Then let

$$Z_{(i)} = [X_{(i)} / \hat{\alpha}\,]^{\hat{\beta}} \qquad i = 1, \ldots, n \qquad 8.6.4.3(a)$$

The Anderson-Darling test statistic is then

$$AD = \sum_{i=1}^{n} \frac{1 - 2i}{n} \{ \ln [1 - \exp(-Z_{(i)})] - Z_{(n+1-i)} \} - n \qquad 8.6.4.3(b)$$

and the observed significance level is calculated as

$$OSL = 1 / \{1 + \exp [-0.10 + 1.24 \ln (AD^*) + 4.48 (AD^*)]\} \qquad 8.6.4.3(c)$$

where

$$AD^* = (1 + 0.2/\sqrt{n})\, AD \qquad 8.6.4.3(d)$$

The OSL is a measure of the goodness-of-fit of the two-parameter Weibull distribution to the data. Specifically, the OSL measures the probability of observing an Anderson-Darling statistic at least as extreme as the value calculated if the two-parameter Weibull distribution is in fact the underlying distribution.

8.6.4.4 *Testing for normality in a regression setting.* This section contains a test for normality of the dependent variable in a regression setting. The test is an Anderson-Darling test for normality performed on the residuals

$$e_i = y_i - (a + b x_i) \qquad i = 1, \ldots, n \qquad 8.6.4.4(a)$$

from the regression (see Section 8.6.6) assuming equality of variance of the residuals over the range of the independent variable. Letting

$$Z_{(i)} = e_{(i)} / S_y \qquad i = 1, \ldots, n \qquad 8.6.4.4(b)$$

where $e_{(1)}, \ldots, e_{(n)}$ are the ordered residuals from smallest to largest and S_y is defined in Section 8.6.6. The Anderson-Darling test statistic is computed using Equation 8.6.4.1(b), and the observed significance level (OSL) is computed as in Equation 8.6.4.1(c).

If OSL $\leq$ 0.05, conclude that the dependent variable is not normally distributed. Otherwise, the assumption that the dependent variable is normally distributed is not rejected. The justification for this procedure may be found in Reference 8.6.4.5.

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8.6.5 Exploratory Data Analysis. Exploratory Data Analysis (EDA) techniques can be used to help familiarize the analyst with a set of data. The advantage of such techniques is that they are simple and visual and often point out the problems in the data which can be corrected, or at least noted before more involved techniques are undertaken. The disadvantage of these techniques is that they are only exploratory in nature and yield no quantitative measure upon which conclusions can be based. Due to the relative unfamiliarity with the materials being analyzed, the use of EDA as both an educational tool and a data screening technique can be invaluable; however, due to the subjectivity involved with the use of these techniques, it is essential to confirm any and all conclusions based on EDA with other statistical procedures. Two EDA techniques are described below; the Quantile Box Plot and the Informative Quantile Functions. Those readers desiring a more complete treatment of the subject are referred to Reference 8.6.5.

8.6.5.1 *The quantile box plot.* The quantile box plot provides a graphical summary of the sample values. This procedure depicts the symmetry, tail sizes, and median value of the sample as well as indicating the possible existence of outliers and inhomogeneous data.

Let F(x) be the underlying distribution function. The uth quantile of F(x), q_u, is the solution to the equation $F(q_u) = u$. The quantile function, Q(u), is defined by

$$Q(u) = F^{-1}(u) \qquad 0 < u < 1 \qquad 8.6.5.1(a)$$

(see Figure 8.6.5.1(a)). Letting $X_{(1)} \leq X_{(2)} \leq \ldots \leq X_{(n)}$ denote the ordered measurements for a sample of size n, Q(u) is estimated by the piecewise linear function

$$\hat{Q}(u) = \left(nu - j + \frac{1}{2}\right) X_{(j+1)} + \left(j + \frac{1}{2} - nu\right) X_{(j)} \qquad 8.6.5.1(b)$$

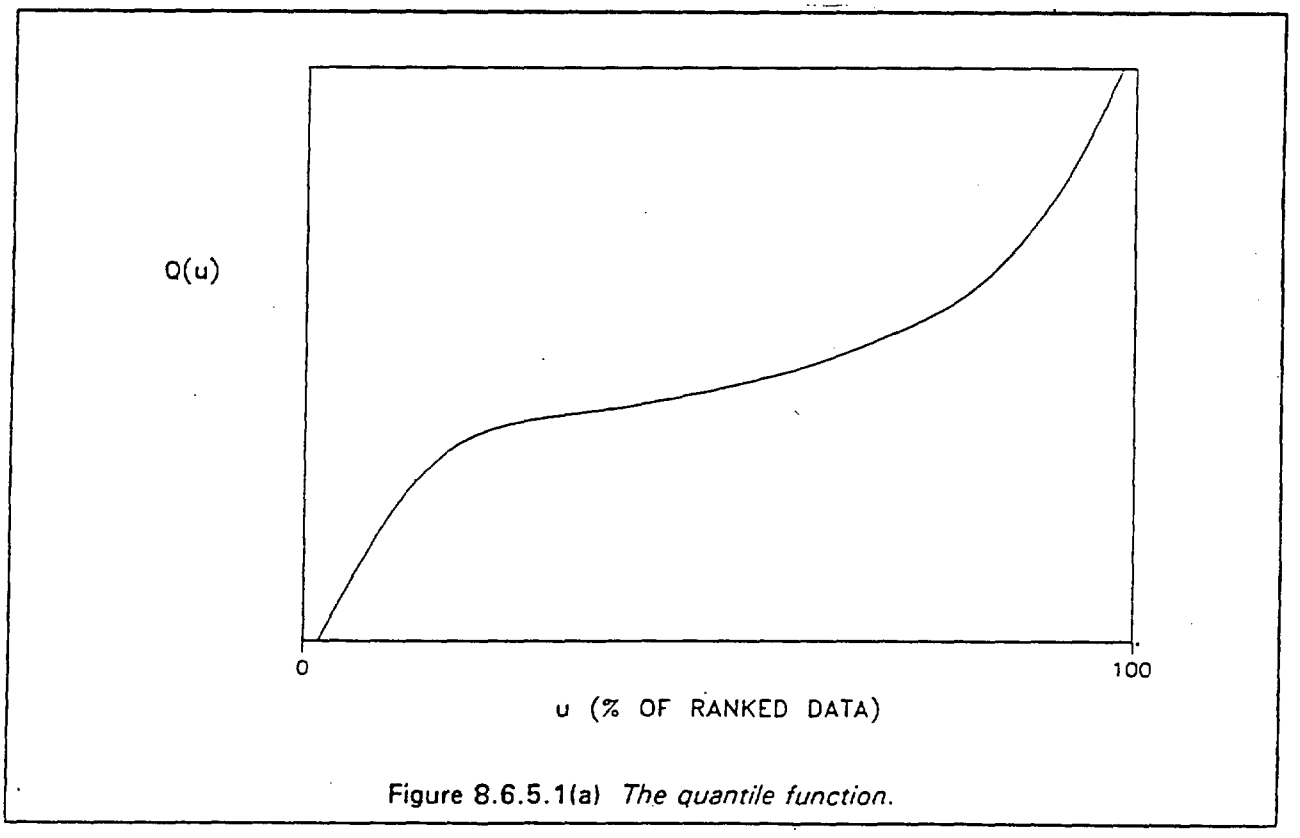


Figure 8.6.5.1(a) *The quantile function.*

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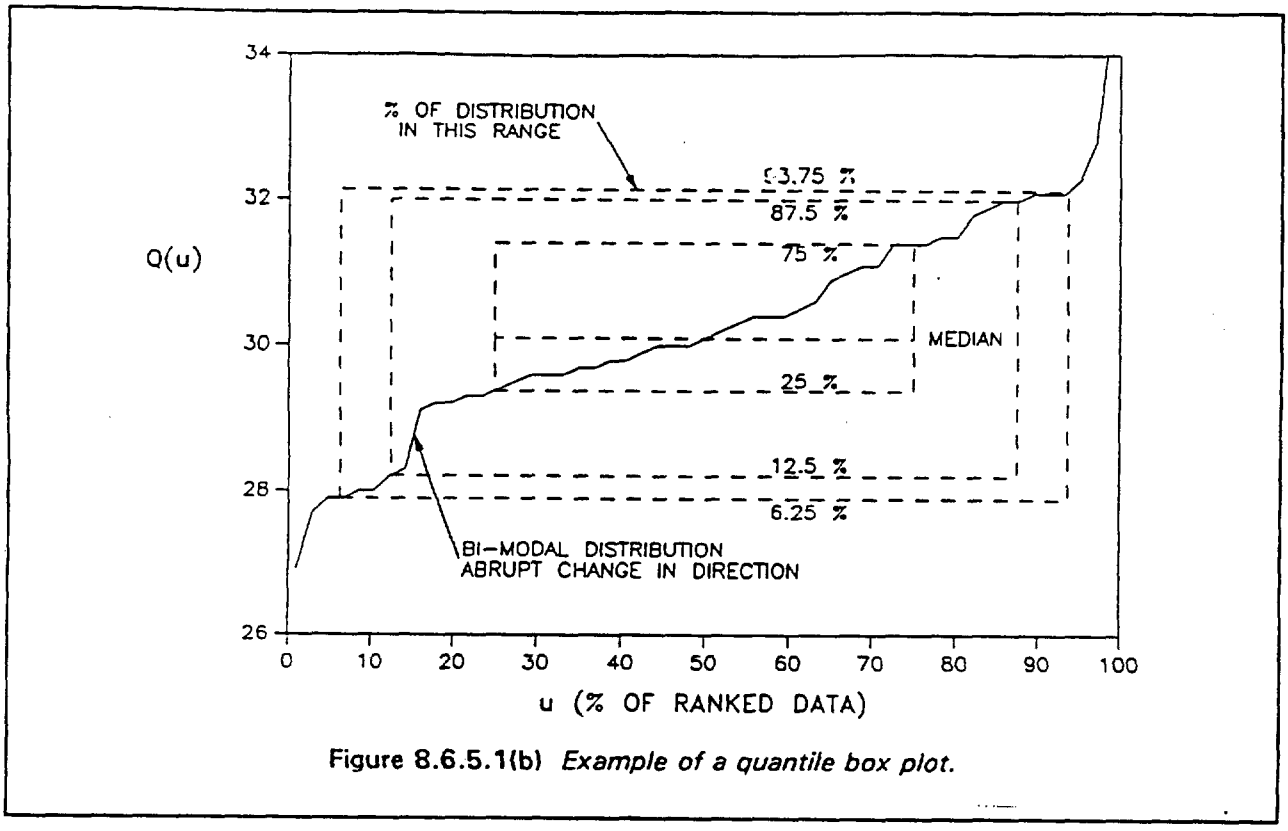


Figure 8.6.5.1(b) *Example of a quantile box plot.*

for

$$\frac{2j-1}{2n} \le u < \frac{2j+1}{2n} \qquad 8.6.5.1(c)$$

The quantile box plot is a graph of the above function, with three rectangles (boxes) drawn to represent various quantiles and a line segment indicating the sample median. The innermost box has boundaries of 0.25 and 0.75 on the u axis, and boundaries of Q(0.25) and Q(0.75) on the Q(u) axis. The other two boxes are drawn in a similar manner with boundaries of 0.125 to 0.875 and 0.0625 to 0.9375, respectively, on the u axis. The line segment indicating the sample median is drawn from coordinates (0.25, Q(0.5)) to (0.75, Q(0.5)).

Figure 8.6.5.1(b) is an example of a quantile box plot. The boxes enable the analyst to get a feeling for the symmetry and tail sizes of the underlying distribution. Flat spots in Q(u) indicate modal values. Sharp rises in Q(u) for u in the vicinity of 0 or 1 indicate the possible presence of outliers in the data. Sharp rises in Q(u) within the boxes indicate the possible existence of two (or more) populations or gaps in the data. A thorough treatment of the use of the Quantile Box plot can be found in Reference 8.6.5.1.

8.6.5.2 *The Informative Quantile function.* Techniques for obtaining B-basis values for unimodal data can be divided into two main categories: techniques for specific parametric families, and nonparametric techniques. The Informative Quantile (IQ) function may be used as an aid in identifying a parametric model which provides a satisfactory fit to the data. Parametric techniques have been most thoroughly discussed for the normal, lognormal, and two-parameter Weibull parametric families;

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thus only these techniques will be considered here. Henceforth in this section, any reference to the Weibull parametric family should be interpreted as a reference to the two-parameter Weibull parametric family.

The IQ function was developed to identify which univariate location-scale parametric distribution best describes an ordered group of data. A univariate location-scale parametric distribution is one whose distribution function F(x) can be expressed as

$$F(x) = F_o[(x - a)/b] \qquad 8.6.5.2(a)$$

where a and b are the location and scale parameters respectively, and $F_o(x)$ is the "standard" distribution with a = 0 and b = 1. The IQ function identifies the standard distributional form and is thus independent of the values of the location and scale parameters.

The Weibull and lognormal parametric families are not location-scale parametric families. However, these distributions are simply related to two location-scale families: the normal and the extreme value families.

The IQ function is defined as

$$IQ(u) = \frac{Q(u) - Q(0.5)}{2[Q(0.75) - Q(0.25)]} \qquad 8.6.5.2(a)$$

where Q(u) is the quantile function defined in Equation 8.6.5.1(a). In order to determine whether the data can be adequately modeled by either the normal or extreme value distribution, a plot of the truncated IQ function, defined by

$$TIQ(u) \begin{cases} -1 & \text{if } IQ(u) \le -1 \\ IQ(u) & \text{if } -1 < IQ(u) \le 1 \\ 1 & \text{if } IQ(u) > 1 \end{cases} \qquad 8.6.5.2(b)$$

is compared to the graph of the exact TIQ plots for these distributions (see Figures 8.6.5.2(a) and (b)). Though the TIQ plots for the data will be considerably less smooth than the exact TIQ plots, they may be compared for general shape and tail behavior.

In order to determine the adequacy of either the lognormal or the Weibull distribution, use the natural logarithms of the data to define the quantile function. Thus, Equation 8.6.5.1(b) becomes

$$\hat{Q}(u) = (nu - j + \frac{1}{2})\ \ln(X_{(j+1)}) + (j + \frac{1}{2} - nu)\ \ln(X_{(j)}) \qquad 8.6.5.2(c)$$

for

$$\frac{2j - 1}{2n} \le u < \frac{2j + 1}{2n} \qquad 8.6.5.2(d)$$

The IQ and TIQ functions in Equations 8.6.5.2(a) and 8.6.5.2(b) are defined using this quantile function.

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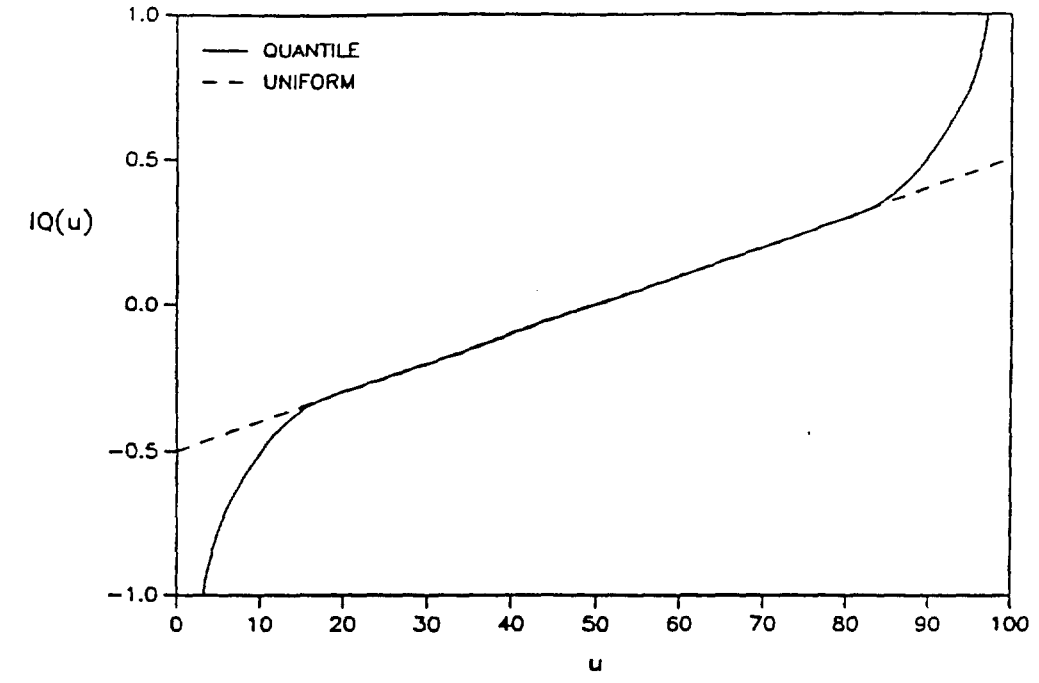


Figure 8.6.5.2(a) *TIQ plot of the normal distribution parametric family.*

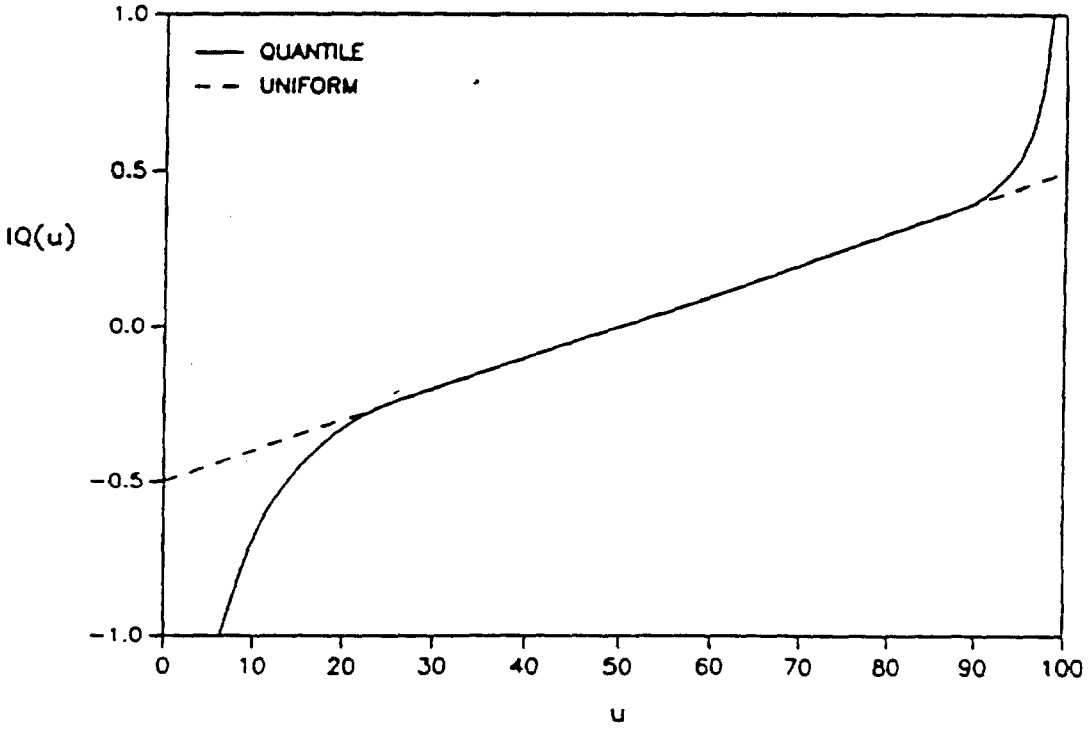


Figure 8.6.5.2(b) *TIQ plot of the extreme value distribution parametric family.*

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Thus, to determine whether the data can be adequately modeled by the normal distribution, compare the TIQ plot for the original data to the TIQ plot for the normal distribution in Figure 8.6.5.2(a). To determine whether the data can be adequately modeled by the lognormal distribution, compare the TIQ plot for the log data to the TIQ plot for the normal distribution in Figure 8.6.5.2(a). The adequacy of the two-parameter Weibull distribution is determined by comparing the TIQ plot for the log data to the exact TIQ plot for the extreme value distribution in Figure 8.6.5.2(b). For further information concerning the quantile function and the informative quantile function, the reader is referred to References 8.6.5.2(a) and 8.6.5.2(b).

8.6.6 General linear statistical models. General linear statistical models will be described in this section. Let

$$Y_i=\beta_1 X_{i1}+\beta_2 X_{i2}+\cdots+\beta_k X_{ik}+\epsilon_i, \quad i=1,2,\ldots,n \qquad 8.6.6(a)$$

where $\beta_1, \beta_2, \ldots, \beta_k$ are assumed to be fixed unknown statistical parameters. The predictor variables $X_{i1}, X_{i2}, \ldots, X_{ik}$ $i=1,2,\ldots,n$ are known constants, usually chosen by the experimenter. The errors $\epsilon_1, \epsilon_2, \ldots, \epsilon_n$ are assumed to be independent random variables with mean zero and common variance σ^2.

In most applications it is assumed that the first predictor variable X_{i1} is constant, equal to 1 for $i = 1, \ldots, n$. In that case, β_1 represents the intercept term of the model and the model is said to be a model with intercept. The remainder of this section deals only with such models. For a more complete discussion of this subject, the reader is directed to References 8.5.8(a) through 8.5.8(d).

It is frequently advantageous to write the system of equations 8.6.6(a) in matrix notation, which simplifies the various formulae that are needed. Thus Equation 8.6.6(a) is rewritten as

$$Y=X\beta+\epsilon \qquad 8.6.6(b)$$

where

$$Y=\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad X=\begin{bmatrix} X_{11} & X_{12} & \cdots & X_{1k} \\ X_{21} & X_{22} & \cdots & X_{2k} \\ \vdots & \vdots & & \vdots \\ X_{n1} & X_{n2} & \cdots & X_{nk} \end{bmatrix} \qquad 8.6.6(c)$$

$$\beta=\begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix} \quad \epsilon=\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

In order to proceed further, it is required that $n \geq k$.

If X is of rank k, then the least squares estimates are found by minimizing

$$\epsilon^T\epsilon=(Y-X\beta)^T(Y-X\beta) \qquad 8.6.6(d)$$

with respect to β and the solution in given by

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$$\hat{\beta} = (X^T X)^{-1} X^T Y \qquad 8.6.6(e)$$

where A^T denotes the transpose of the matrix A.

Then let

$$SST = (Y - \bar{Y})^T (Y - \bar{Y}) \qquad 8.8.6(e)$$

where

$$(Y - \bar{Y}) = \begin{bmatrix} Y_1 - \bar{Y} \\ Y_2 - \bar{Y} \\ \vdots \\ Y_k - \bar{Y} \end{bmatrix}$$

and let

$$SSE = (Y - \bar{X}\hat{\beta})^T (Y - \bar{X}\hat{\beta}) \qquad 8.8.6(f)$$

Then

$$SSR = SST - SSE \qquad 8.8.6(g)$$

and

$$F = \frac{SSR/(k-1)}{SSE/(n-k)} \qquad 8.8.6(h)$$

has the F-distribution with k-1 and n-k degrees of freedom.

To test the significance of the regression one compares the value in Equation 8.6.6(h) with the $1-\alpha$ quantile of the F-distribution with γ_1 = k-1 and γ_2 = n-k degrees of freedom. If the value in Equation 8.6.6(j) exceeds this quantile, the regression is judged significant at significance level α, i.e., the risk is 100α% that the regression is judged significant when in fact it is not. Table 8.8.5 produces these quantile values for the customary significance level of $\alpha = 0.05$.

Since the mean of the material property Y_i varies linearly with the predictor variables $X_{i2}, ..., X_{ik}$, one has to define the A- and B-values conditional on these predictors. For example, for a particular set of predictor values $X_{02}, ..., X_{0k}$, a B-value will, with 95% confidence, fall below 90% of all Y-values that can be observed under that particular set of predictor values. For A-basis values, exchange 99% for 90% above.

A B-basis value can now be determined as follows. For any vector $X_0^T = (1, X_{02}, ..., X_{0k})$, which need not be any of the experimental values, compute B as follows:

$$B = X_0\hat{\beta} - k_B\sqrt{\frac{SSE}{(n-k)}} \qquad 8.6.6(i)$$

where

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$$k_B = \sqrt{X_0(X^TX)^{-1}X_0^T}\, t_{\gamma,0.95}(\delta) \qquad 8.6.6(j)$$

$$\delta = \frac{1.282}{\sqrt{X_0(X^TX)^{-1}X_0^T}} \qquad 8.6.6(k)$$

and $\gamma = n - k$ where $t_{\gamma,0.95}(\delta)$ is the 95th percentile of the non-central t-distribution with n - k degrees of freedom and non-centrality parameter δ.

To adapt Equation 8.6.6(i) to A-basis values, replace 1.282 by 2.326 in Equation 8.6.6(k). The approximation in Equations 8.5.8(u) and (v) are not generally applicable here and the reader is directed to the algorithm in Reference 8.6.6(a) for evaluating the non-central t-distribution.

8.6.6.1 *Example: Simple linear regression problems.* In order to present the simple linear regression model of Equation 8.5.8(d) in matrix notation Equation 8.6.6(a), set k = 2 and let β_1 be the intercept and β_2 the slope of the regression line. The matrices are as follows.

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad X = \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix} \quad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \quad e = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$$

Thus, $X\beta + \epsilon$ is the n-element column vector whose rows are

$$Y_i = \beta_1 + \beta_2 X_i + e_i, \quad i = 1, 2, \ldots, n \qquad 8.6.6.1(a)$$

which is the model for the simple linear regression described in Section 8.5.8. (It is customary when using the matrices to label the parameters $\beta_1, \beta_2, \ldots, \beta_k$, which differs slightly from the notation used in Section 8.5.8. There the intercept, denoted here β_1, was denoted β_0 and the slope, denoted here β_2, was denoted β_1. The present notation describes the parameters as elements of the β vector, the subscripts denoting their position in the vector.)

Thus

$$X^TX = \begin{bmatrix} n & \sum_{i=1}^{n} X_i \\ \sum_{i=1}^{n} X_i & \sum_{i=1}^{n} X_i^2 \end{bmatrix} \qquad 8.6.6.1(b)$$

so that,

$$(X^TX)^{-1} = \frac{1}{n\sum_{i=1}^{n} X_i^2 - \left(\sum_{i=1}^{n} X_i\right)^2} \begin{bmatrix} \sum_{i=1}^{n} X_i^2 & -\sum_{i=1}^{n} X_i \\ -\sum_{i=1}^{n} X_i & n \end{bmatrix} \qquad 8.6.6.1(c)$$

Equations 8.6.6.1(a), 8.6.6.1(b) and 8.6.6.1(c) can easily be obtained by performing the indicated matrix multiplications. Consequently, it can be verified that $\hat{\beta}^T = (\hat{\beta}_1, \hat{\beta}_2)$, where

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$$\hat{\beta}_1 = \bar{Y} - \hat{\beta}_2 \bar{X} \qquad 8.6.6.1(d)$$

and

$$\hat{\beta}_2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{j=1}^{n} (X_j - \bar{X})^2} \qquad 8.6.6.1(e)$$

as was previously obtained in Equations 8.5.8(g) and 8.5.8(h).

8.6.6.2 *Example: One-way fixed effects analysis of variance model.* This section is concluded with a discussion of the one-way fixed effects analysis of variance model using matrix notation. The model is given by

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}, \qquad \begin{matrix} i = 1, 2, \ldots, r \\ j = 1, 2, \ldots, m \end{matrix} \qquad 8.6.6.2(a)$$

where each of the parameters τ_i, i = 1, 2, . . ., r, can be chosen to denote the difference between the mean of the ith group of data elements and the grand mean, averaged over all r groups. (See Equation 8.6.6.2(d).) This model can be written in the form of Equation 8.6.6(b) as follows: write the matrix X by

$$X = \begin{bmatrix} 1 & 1 & 0 & \ldots & 0 \\ 1 & 1 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 1 & 0 & \ldots & 0 \\ 1 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 0 & 0 & \ldots & 1 \end{bmatrix} \qquad 8.6.6.2(b)$$

where X has rm rows and r + 1 columns. Every entry in the first column is a "1". The first m elements of the second column are "1" and the rest are zero. The third column has all elements zero except for the second group of m elements, and so on. Also, the matrices Y, β, and ϵ can be written

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$$Y = \begin{bmatrix} Y_{11} \\ \vdots \\ Y_{1m} \\ Y_{21} \\ \vdots \\ Y_{2m} \\ \vdots \\ \vdots \\ Y_{rm} \end{bmatrix} \quad \beta = \begin{bmatrix} \mu \\ \tau_1 \\ \tau_2 \\ \vdots \\ \tau_r \end{bmatrix} \quad \epsilon = \begin{bmatrix} e_{11} \\ \vdots \\ e_{1m} \\ e_{21} \\ \vdots \\ e_{2m} \\ \vdots \\ \vdots \\ e_{rm} \end{bmatrix} \qquad 8.6.6.2(c)$$

This exhibits the principal distinction between an analysis of variance model and a regression model. For analysis of variance models, the X matrix consists of "zeros" and "ones," which, except for the first column, serve as indicators for the various treatments, blocks, and so on. Note further that the matrix X in Equation 8.6.6.2(b) is not of full rank. This can be resolved in this specific example by introducing the restriction

$$\sum_{i=1}^{n} \tau_i = 0 \qquad 8.6.6.2(d)$$

so that the parameter τ_i describes the mean departure of the ith group from the grand mean of all groups. The linear restriction of Equation 8.6.6.2(d) permits the reduction of parameters by one; since, for example, a parameter, $\tau_r = \tau_1 - \tau_2 - \ldots - \tau_{r-1}$ can be written. Thus X can be replaced by X^*, where

$$X^* = \begin{bmatrix} 1 & 1 & 0 & \ldots & 0 \\ 1 & 1 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 1 & 0 & \ldots & 0 \\ 1 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & -1 & -1 & \ldots & -1 \end{bmatrix} \qquad 8.6.6.2(e)$$

where X^* has rm rows and r columns. Also X^* has (full) rank r. The least square estimates are given $(X^{*T}X^*)^{-1}X^{*T}Y$ (see Equation 8.6.6(d)). By direct calculation, one obtains

$$(X^{*T}X^*) = \begin{bmatrix} rm & 0 & 0 & \ldots & 0 \\ 0 & 2m & m & \ldots & m \\ 0 & m & 2m & \ldots & m \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & m & m & \ldots & 2m \end{bmatrix} \qquad 8.6.6.2(f)$$

and therefore,

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$$(X^{*T}X^{*})^{-1} = \begin{bmatrix} \frac{1}{rm} & 0 & 0 & \cdots & 0 \\ 0 & \frac{r-1}{rm} & -\frac{1}{rm} & \cdots & -\frac{1}{rm} \\ 0 & -\frac{1}{rm} & \frac{r-1}{rm} & \cdots & -\frac{1}{rm} \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & -\frac{1}{rm} & -\frac{1}{rm} & \cdots & \frac{r-1}{rm} \end{bmatrix} \qquad 8.6.6.2(g)$$

Let $T_i = \sum_{j=1}^{m} Y_{ij}$. Then

$$X^{*}Y = \begin{pmatrix} \sum_{i=1}^{r} T_i \\ T_1 - T_r \\ T_2 - T_r \\ \vdots \\ T_2 - T_r \end{pmatrix} \qquad 8.6.6.2(h)$$

Thus

$$\hat{\mu} = \bar{Y} = \sum_{j=1}^{m}\sum_{i=1}^{r} Y_{ij}/mr \qquad 8.6.6.2(i)$$

and

$$\hat{\tau}_i = \sum_{j=1}^{n} Y_{ij}/n - \hat{\mu}, \qquad i = 1, 2, \ldots, k \qquad 8.6.6.2(j)$$

As before (Section 8.5.8), define

$$SSE = \sum_{ij} (Y_{ij} - \hat{\mu} - \hat{\tau}_i)^2 \qquad 8.6.6.2(k)$$

$$S_y^2 = \frac{SSE}{mr - r} \qquad 8.6.6.2(l)$$

and

$$SST = \sum_{ij} (Y_{ij} - \bar{Y})^2 \qquad 8.6.6.2(m)$$

Then, defining

$$SSR = SST - SSE \qquad 8.6.6.2(n)$$

a test of the hypothesis, $\tau_i = 0$, $i = 1, 2, \ldots, k$ may be carried out by defining

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$$F = \frac{SSR/(k-1)}{S_y^2} \qquad 8.6.6.2(o)$$

which has the F-distribution with r - 1 and mr - r degrees of freedom. Thus, the treatments $\{\tau_i\}$ would be considered significant at level α if Equation 8.6.6.2(o) exceeds the 1-α quantile for γ_1 = r - 1 and γ_2 = mr - r degrees of freedom. Table 8.8.5 provides these values for α = 0.05. A B-basis value is given by

$$B = (\hat{\mu} + \hat{\tau}_i) - k_B S_y, \quad i = 1, 2, ..., r \qquad 8.6.6.2(p)$$

when the ith treatment applies, where

$$k_B = [X_0(X^TX)^{-1}X_0^T]^{\frac{1}{2}}\, t_{\gamma,0.95}(\delta) \qquad 8.6.6.2(q)$$

By direct calculation, using Equation 8.6.6.2(g), it follows that

$$k_B = \frac{1}{\sqrt{m}}\, t_{\gamma,0.95}(\delta) \qquad 8.6.6.2(r)$$

where γ = rm - r (the number of degrees of freedom) and δ, the non-centrality parameter is given by

$$\delta = 1.282\sqrt{m} \qquad 8.6.6.2(s)$$

8.6.7 Analysis of variance procedures. This section contains a discussion of the standard one-way analysis of variance (ANOVA) procedures for analyzing multiple samples of data. Two different models are referred to: the fixed effects model and the random effects model. The fixed effects model is appropriate when inferences are to be made only to the populations from which the actual samples were drawn. The random effects model is appropriate when the samples are viewed as representing a random selection from a population. In the random effects model, inferences are made to the whole collection of populations, not just those from which samples were actually drawn. Although the two models involve different assumptions and lead to different interpretations of the data, the computational formulas for computing sums of squares are the same.

The basic model can be written as

$$X_{ij} = \mu + b_i + e_{ij}, \quad \begin{matrix} i = 1, ..., k \\ j = 1, ..., n_i \end{matrix} \qquad 8.6.7(a)$$

where k is the number of different samples, n_i is the number of observations in the ith sample, and X_{ij} represents the jth observation in the ith sample.

The ANOVA models represent each observation as the sum of three components; μ is the overall average of the population, b_i is the effect attributed to the ith sample, and e_{ij} is a random error term which represents unexplainable sources of variation. The error terms, e_{ij}, are assumed to be independently distributed normal random variables with a mean of zero and a variance of σ_e^2 (the within-sample variance). In the fixed effects model, the b_i values are assumed to be population parameters, taking on a specific value for each sample. In the random effects model, the b_i are assumed to be independent random variables following a normal distribution with a mean of zero and a variance of σ_b^2 (the between-sample variance). See Reference 8.6.7 for more information on analysis of variance procedures.

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8.6.7.1 *Computational procedures.* The analysis of data based on either of the above models involves computing sums of squares which measure variability between samples and within samples. Two procedures for computing these quantities are presented. The first assumes that all of the original observations are available. The second assumes that only the batch means, standard deviations and sample sizes are available. Both of these procedures yield the same results for the computed sums of squares.

8.6.7.1.1 *Computations based on individual measurements.* When all of the observations in a sample are available, the first step is to compute averages. The overall average is computed as

$$\bar{X} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} X_{ij}/N \qquad 8.6.7.1.1(a)$$

$$N = \sum_{i=1}^{k} n_i \qquad 8.6.7.1.1(b)$$

where is the total sample size. The sample averages are computed as

$$\bar{X}_i = \sum_{j=1}^{n_i} X_{ij} / n_i \quad \text{for} \quad i = 1, \ldots, k \qquad 8.6.7.1.1(c)$$

From these quantities, the required sums of squares can be computed. The between-sample sum of squares is computed as

$$SSB = \sum_{i=1}^{k} n_i \bar{X}_i^2 - N\bar{X}^2 \qquad 8.6.7.1.1(d)$$

and the total sum of squares is computed as

$$SST = \sum_{i=1}^{k} \sum_{j=1}^{n_i} X_{ij}^2 - N\bar{X}^2 \qquad 8.6.7.1.1(e)$$

The within-sample, or error, sum of squares is computed by subtraction

$$SSE = SST - SSB \qquad 8.6.7.1.1(f)$$

8.6.7.1.2 *Computations based on summary statistics.* It is often the case that only summary statistics are available for each sample. If these summary statistics contain the sample averages $\bar{X}_i$, the standard deviations of the data from each sample (S_i) and the sample sizes (n_i), the sums of squares can be computed as follows. First, compute the overall mean,

$$\bar{X} = \sum_{i=1}^{k} n_i \bar{X}_i / N \qquad 8.6.7.1.2(a)$$

The between-sample sum of squares is computed as in the previous section

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$$SSB = \sum_{i=1}^{k} n_i \bar{X}_i^2 - N\bar{X}^2 \qquad 8.6.7.1.2(b)$$

where

$$N = \sum_{i=1}^{k} n_i \qquad 8.6.7.1.2(c)$$

Each of the sample standard deviations has $n_i - 1$ degrees of freedom. The sum of squares within samples can be computed by taking a sum of the squared standard deviations weighted by their degrees of freedom:

$$SSE = \sum_{i=1}^{k} (n_i - 1) S_i^2 \qquad 8.6.7.1.2(d)$$

The total sum of squares, SST, is the sum of SSB and SSE.

8.6.7.2 *ANOVA Table.* Once the sum of squares computations have been completed, an ANOVA table is constructed to display the information about sources of variation in the data. A typical ANOVA table which is used for both the fixed effects and random effects models is shown below. The first column identifies the source of variation. The degrees of freedom and the computed sums of squares are listed in the second and third columns, respectively. The fourth column contains mean squares which are defined as the sum of squares divided by its degrees of freedom. The final column contains an F statistic which is equal to the ratio of the mean squares. This statistic is used to test the hypothesis that there is significant sample-to-sample variation. The statistic is compared to the upper 0.95th quantile of an F distribution with k - 1 numerator degrees of freedom and N - k denominator degrees of freedom. Table 8.8.5 contains these critical F values. If the computed statistic is greater than the tabulated F value, this indicates that there is statistically significant sample-to-sample variation. If the computed statistic is less than the tabulated value, then the variation between samples is not statistically significant.

Source	Degrees of Freedom	Sum of Squares	Mean Squares	F Test
Samples	k-1	SSB	MSB = SSB/(k-1)	F = MSB/MSE
Error	N-k	SSE	MSE = SSE/(N-k)	
Total	N-1	SST		

8.6.8 Methods for developing A-basis values. The A-basis value for a particular distribution is defined as a 95% lower confidence limit on the first percentile of the distribution. However, accurate parametric estimation of the first percentile is extremely dependent on the model being correct. Since the probability model being assumed is approximate, it will happen quite often that such estimates will be inaccurate. This problem can be overcome by using nonparametric estimates (which

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require 300 or more specimens) or, to some extent, by the Hanson-Koopmans method (which can be quite conservative for small samples).

Consequently, the procedures outline herein for determining A-basis values should be regarded as providing a safety level for the B-basis values, but not as 95% confidence limits on the first percentile.

8.6.8.1 *Weibull model.* Proceed as in Section 8.5.6, except replace the value V with the corresponding value V_A from Table 8.8.11. These factors are appropriate for both two-parameter and three-parameter Weibull models.

8.6.8.2 *Normal model.* Proceed as in Section 8.5.5, except replace the value k_B with the corresponding value from Table 8.8.12.

8.6.8.3 *Lognormal model.* Proceed as in Section 8.5.5.2, except replace the value k_B with the corresponding value from Table 8.8.12.

8.6.8.4 *Nonparametric method.* Proceed as in Section 8.5.7.1, except replace MIL-HDBK-17 Table 8.8.3 with Table 8.8.13. If $n < 299$ (which will nearly always be the case), then use the Hanson-Koopmans method below.

8.6.8.5 *Hanson-Koopmans method.* Find the value k_A corresponding to the sample size n in Table 8.8.14, then proceed as follows. If x_n denotes the largest sample value, and x_1 denotes the smallest sample value, then the A-basis value is:

$$A = x_{(n)} \left[\frac{x_{(1)}}{x_{(n)}} \right]^{k_A} \qquad 8.6.8.5$$

8.6.8.6 *ANOVA method.* To obtain A-basis values by the ANOVA method, perform the calculations as in Section 8.5.4 with the following changes:

$$A = x_n \left(\frac{x_n}{x_1} \right)^k \qquad 8.6.8(a)$$

1) Replace $F_{0.2}$ with $F_{0.1225}$ in Equation 8.5.4(e) and in the paragraph which follows this equation.

2) Replace Table 8.8.4 with Table 8.8.15.

3) Replace Table 8.8.6 with Table 8.8.16.

4) In Equation 8.5.4(h), replace 1.282 with 2.326.

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5) Replace Equation 8.5.4(j) with Equation 8.6.8.6.

$$T = 2.326 + 2.326\,(t_{\tau,0.95}/\delta) + 11.091\,(1/\tau) - 2.0818\,(1/\tau^2) + 0.079011\,(\delta/\tau) + 0.20642\,(\delta/\tau^2) - 0.0040166\,(\delta^2/\tau^2) - 2.3836 \times 10^{-5}\,(\delta^3/\tau^2) + 21.930\,(1/\tau^4) \qquad 8.6.8.6(b)$$

(Note: $t_{\gamma,0.95}$ is the 0.95 quantile of the central t-distribution with γ degrees of freedom.) Tolerance limit factors are tabulated in Table 8.8.6 for various values of n^*, k^* and $\hat{R}$.

This approximation is always within 7.5% of the tabulated values, and usually within 5%. The exact solution of the tolerance limit factor can be calculated based on the algorithm of the non-central t-distribution in Reference 8.6.8.6.

6) To determine $F_{0.1225}$ use Equation 8.8.16(a) with $z = -1.1625$.

7) When there is one numerator degree of freedom or one denominator degree of freedom, Equation 8.6.8(a) cannot be used. the corresponding equations necessary for A-basis calculations are given below:

For $\tau_1 = 1$

$$F_{0.1225} = \left[0.1541391 + 0.03945031/\tau_2 + 0.00543174/\tau_2^2 - 0.00585461/\tau_2^3 - 0.00165541/\tau_2^4 \right]^2 \qquad 8.6.8.6(c)$$

For $\tau_2 = 1$

$$F_{0.1225} = \left[0.1541391 + 0.03945031/\tau_1 + 0.00543174/\tau_2^1 - 0.00585461/\tau_1^3 - 0.00165541/\tau_1^4 \right]^{-2} \qquad 8.6.8.6(d)$$

8.6.8.7 *Linear regression.* The method for calculating A-basis values for linear regression are noted at the end of Section 8.6.6 and Equation 8.5.8(v).

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8.7 **SAMPLES OF COMPUTATIONAL PROCEDURES.** It is appropriate at this point to review the computational procedures described in Sections 8.5.3 through 8.5.8. To do so, hypothetical sets of data have been created which represent mechanical property measurements from both tensile and compression tests. The data sets for Problems 1 through 8, which follow, are reported in Table 8.7.

In progressing through the sample problems, the flowchart in Figure 8.5.2 has been followed, and appropriate references to specific sections of these guidelines have been made.

8.7.1 **Problem 1 - Outlier detection and multiple sample tests.** The data set for this problem consists of tensile strength measurements from nine batches of material. This problem illustrates the outlier detection procedure, the k-sample Anderson-Darling test, the test for normality of multiple samples, and the homogeneity of variance test.

Problem 1 - Step 1. The first step is to screen the data for outliers using the MNR procedure as described in Sections 8.5.3.1 and 8.6.2. The screening procedure is performed separately on each batch. The relevant calculations for the seventh batch, with a sample mean of 72.52 and a sample standard deviation of 8.08, are shown in the table below.

X_i	$\lvert r_i \rvert = \left\lvert \frac{X_i - \bar{X}}{s} \right\rvert = \left\lvert \frac{X_i - 72.52}{8.08} \right\rvert$
65.4	0.88
69.4	0.39
69.9	0.32
71.5	0.13
86.4	1.72

The MNR statistic is the largest absolute residual, or 1.72. Since this is greater than the n = 5 critical value of 1.715 from Table 8.8.9, the observation 86.4 is identified as a potential outlier in this batch.

Similar calculations for the remaining batches identify no additional potential outliers in this set of data. Visual inspection of the data also does not identify any additional potential outliers.

No cause could be determined for the potential outlier, 86.4, and it is, therefore, retained in the data set.

Problem 1 - Step 2. The k-sample Anderson-Darling test described in Section 8.6.3.1 will be employed next to determine whether or not the data from the nine batches should be combined. The first step is to order the pooled sample. Table 8.7.1 lists the 47 sorted, distinct values in the column labeled $Z_{(j)}$. The remaining columns show the $h_{(j)}$, H_j, and F_{1j} values used in calculating the terms in the statistic arising from the first batch (i = 1). The column labeled f_{1j} shows the number of times that $Z_{(j)}$ is represented in the first batch and is used in calculating F_{1j}. From these numbers, it follows that

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TABLE 8.7 *Sample data sets for Section 8.7.*, continued on next page.

Problem 1		Problem 2		Problem 3	
Batch	Data	Batch	Data	Batch	Data
1	65.4	1	61.3	1	67.4
1	63.6	1	68.5	1	61.8
1	63.4	1	62.5	1	72.6
1	61.3	1	66.0	1	56.0
1	61.0	1	66.6	1	69.0
1	61.6	1	64.8	1	75.0
1	65.4	1	69.5	1	65.4
2	64.0	2	66.5	2	59.4
2	64.8	2	64.7	2	66.2
2	66.4	2	64.9	2	74.8
2	61.2	2	65.2	2	75.7
2	64.7	2	70.3	2	63.9
3	68.4	3	66.0	3	70.5
3	68.8	3	72.7	3	61.7
3	66.8	3	67.1	3	66.6
3	63.6	3	67.7	3	66.1
3	69.3	3	65.7	3	72.7
4	66.8	4	61.9	4	65.5
4	67.9	4	68.0	4	61.7
4	68.9	4	63.3	4	60.6
4	65.6	4	74.6	4	75.0
4	64.6	4	66.2	4	64.4
4	71.1	4	68.2	4	75.3
4	64.1	4	69.1	4	71.7
5	74.6	5	68.9	5	51.6
5	74.2	5	65.0	5	64.3
5	62.7	5	70.9	5	74.2
5	73.3	5	65.4	5	75.4
5	68.6	5	66.5	5	60.5
5	74.0	5	64.9	5	47.1
6	67.4	6	75.8	6	73.5
6	68.0	6	75.2	6	77.0
6	74.6	6	71.5	6	66.7
6	74.9	6	69.6	6	66.4
6	65.5	6	66.1	6	65.0
7	69.4	7	72.8	7	64.0
7	71.5	7	75.0	7	66.2
7	69.9	7	66.3	7	64.3
7	66.4	7	69.5	7	60.4
7	65.4	7	71.9	7	72.4
8	64.7	8	71.9	8	71.0
8	70.0	8	71.0	8	66.9
8	70.2	8	69.5	8	76.8
8	79.2	8	69.5	8	64.4
8	76.8	8	72.6	8	70.0
8	78.5	8	74.6	8	72.1
9	70.7	9	68.7	9	73.9
9	80.6	9	76.3	9	78.5
9	68.3	9	76.6	9	68.0
9	87.0	9	109.9	9	75.0
9	81.1	9	66.2	9	79.5
9	79.7	9	72.4	9	60.6
9	82.3	9	72.8	9	71.4

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TABLE 8.7 *Sample data sets for Section 8.7*, concluded.

Problem 4		Problem 5		Problem 6		Problem 7	
Batch	Data	Batch	Data	Batch	Data	Batch	Data
1	30.4	1	30.4	1	72.3	1	63.9
1	29.6	1	29.0	1	65.3	1	76.2
1	29.3	1	34.8	1	74.3	1	76.9
1	27.9	1	30.5	1	68.5	1	67.9
1	29.8	1	27.3	1	65.7	1	80.4
1	30.5	1	27.9	1	64.3	2	62.2
1	29.4	1	29.4	1	72.8	2	79.6
2	31.5	2	29.8	2	74.5	2	62.4
2	31.1	2	32.4	2	66.3	2	76.6
2	27.9	2	33.2	2	74.3	2	80.5
2	34.5	2	29.9	2	69.3	3	72.2
2	29.5	2	30.5	2	64.9	3	61.3
3	29.6	3	31.1	3	75.3	3	74.9
3	29.8	3	32.0	3	68.5	3	81.6
3	29.1	3	31.0	3	75.2	3	55.9
3	30.0	3	29.3	3	74.5	4	58.2
3	30.4	3	29.9	3	69.6	4	77.0
4	32.1	4	32.2	4	65.6	4	87.5
4	29.7	4	29.9	4	74.6	4	63.0
4	26.9	4	29.1	4	76.5	4	77.4
4	30.0	4	29.9	4	72.0	5	66.2
4	31.4	4	32.7	4	69.2	5	75.8
4	28.3	4	31.6	4	74.3	5	62.6
4	31.9	4	31.6	4	78.7	5	78.0
5	30.0	5	27.5	5	68.1	5	78.4
5	31.5	5	29.3	5	72.1		
5	32.0	5	29.1	5	66.6		
5	30.2	5	29.5	5	75.0		
5	30.6	5	28.7	5	67.2	Problem 8	
5	31.1	5	29.5	5	73.4	x	y
6	29.7	6	29.9	6	66.4	0.112	32.0
6	29.3	6	28.9	6	72.6	0.112	32.0
6	30.4	6	29.2	6	67.5	0.116	30.0
6	32.1	6	29.1	6	76.3	0.116	32.0
6	32.8	6	31.1	6	65.1	0.119	28.0
7	29.9	7	31.1	7	73.3	0.119	30.4
7	30.9	7	30.4	7	72.6	0.121	26.8
7	32.1	7	33.4	7	67.8	0.122	28.0
7	31.4	7	30.0	7	69.4	0.124	29.2
7	32.0	7	27.3	7	67.3	0.124	28.4
8	29.2	8	31.3	8	75.2	0.126	27.3
8	31.0	8	30.2	8	72.8	0.126	28.0
8	31.8	8	30.2	8	67.8	0.127	29.2
8	30.3	8	29.1	8	76.2	0.128	26.0
8	28.2	8	33.0	8	72.8	0.129	27.6
8	28.0	8	34.4	8	68.7	0.133	26.8
9	30.1	9	28.4	9	72.3	0.135	26.8
9	29.2	9	30.1	9	76.0	0.135	24.0
9	27.7	9	30.5	9	65.8	0.135	25.2
9	28.0	9	29.4	9	69.9	0.137	24.8
9	31.4	9	28.7	9	67.8		
9	29.6	9	32.3	9	74.6		
9	32.3	9	29.6	9	71.3		

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TABLE 8.7.1 *Illustration of k-sample Anderson-Darling statistic calculations for the first batch (i = 1).*

j	$Z_{(j)}$	h_j	H_j	f_{1j}	F_{1j}
1	61.0	1	0.5	1	0.5
2	61.2	1	1.5	0	1.0
3	61.3	1	2.5	1	1.5
4	61.6	1	3.5	1	2.5
5	62.7	1	4.5	0	3.0
6	63.4	1	5.5	1	3.5
7	63.6	2	7.0	1	4.5
8	64.0	1	8.5	0	5.0
9	64.1	1	9.5	0	5.0
10	64.6	1	10.5	0	5.0
11	64.7	2	12.0	0	5.0
12	64.8	1	13.5	0	5.0
13	65.4	3	15.5	2	6.0
14	65.5	1	17.5	0	7.0
15	65.6	1	18.5	0	7.0
16	66.4	1	19.5	0	7.0
17	66.8	2	21.0	0	7.0
18	67.4	1	22.5	0	7.0
19	67.9	1	23.5	0	7.0
20	68.0	1	24.5	0	7.0
21	68.5	1	25.5	0	7.0
22	68.4	1	26.5	0	7.0
23	68.6	1	27.5	0	7.0
24	68.8	1	28.5	0	7.0
25	68.9	1	29.5	0	7.0
26	69.3	1	30.5	0	7.0
27	69.4	1	31.5	0	7.0
28	69.9	1	32.5	0	7.0
29	70.0	1	33.5	0	7.0
30	70.2	1	34.5	0	7.0
31	70.7	1	35.5	0	7.0
32	71.1	1	36.5	0	7.0
33	71.5	1	37.5	0	7.0
34	73.3	1	38.5	0	7.0
35	74.0	1	39.5	0	7.0
36	74.2	1	40.5	0	7.0
37	74.6	2	42.0	0	7.0
38	74.9	1	43.5	0	7.0
39	76.8	1	44.5	0	7.0
40	78.5	1	45.5	0	7.0
41	79.2	1	46.5	0	7.0
42	79.9	1	47.5	0	7.0
43	80.6	1	48.5	0	7.0
44	81.1	1	49.5	0	7.0
45	82.3	1	50.5	0	7.0
46	86.4	1	51.5	0	7.0
47	87.0	1	52.5	0	7.0

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$$\frac{1}{n_i}\sum_{j=1}^{L} h_j \frac{(NF_{ij} - n_i H_j)^2}{H_j(N - H_j) - Nh_j/4} = \frac{1}{7}\sum_{j=1}^{47} h_j \frac{(53F_{ij} - 7H_j)^2}{H_j(53 - H_j) - 53h_j/4} = 363.33 \qquad 8.6.8.6(d)$$

When these calculations are repeated for the remaining eight batches, the k-sample Anderson-Darling statistic is computed as

$$\begin{aligned} ADK &= \frac{N-1}{N^2(k-1)}\sum_{i=1}^{k}\left[\frac{1}{n_i}\sum_{j=1}^{L}\frac{(NF_{ij} - n_i H_j)^2}{h_j(N - H_j) - Nh_j/4}\right] \\ &= \frac{52}{53^2(9-1)}(363.33 + \ldots + 324.11) \\ &= 2.67 \end{aligned}$$

The computed value of the statistic is compared to the critical value from Equation 8.6.3.1(j), which is 1.44. Since the computed value of 2.67 is greater than the critical value of 1.44, the hypothesis that the populations from which these groups were drawn are identical is rejected.

Problem 1 - Step 3. The equality of variance test described in Section 8.6.3.2 is used to determine if the within-batch variances are significantly different. Relevant sample statistics for the nine batches are given in the table below.

Batch	n_i	S_i^2	Y_i	$z_i = \ell n\ (S_i^2)$
1	7	3.46	6	1.24
2	5	3.61	4	1.28
3	5	5.34	4	1.68
4	7	6.25	6	1.83
5	6	22.37	5	3.11
6	5	19.01	4	2.95
7	5	65.29	4	4.18
8	6	33.76	5	3.52
9	7	44.09	6	3.79

The test statistic is then calculated as follows.

$$\begin{aligned} EV &= \frac{1}{2}\sum_{i=1}^{k}\gamma_i Z_i^2 - \frac{1}{2}(n-k)\bar{Z}^2 \\ &= \frac{1}{2}\sum_{i=1}^{9}\gamma_i Z_i - \frac{1}{2}(53-9)(2.61) \\ &= 24.63 \end{aligned}$$

The critical chi-square value from Table 8.8.8 corresponding to k - 1 = 8 degrees of freedom is 15.51. Since the test statistic is greater than the critical value, the hypothesis that the variances

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are equal is rejected. Since the equality of variance test is a diagnostic test, a B-basis value can be calculated using the ANOVA method. However, a nonconservative B-basis value can result is some instances. Potential problems with consistency in fabrication or processing of the different batches should be examined.

Problem 1 - Step 4. The ANOVA method of Section 8.5.4 is used to calculate a B-basis value. (See Problem 2 for the details of the computations using the ANOVA method.) The B-basis value is calculated as

$$B = \bar{X} - T\sqrt{S_b^2 + S_e^2} = 69.78 - 2.077\sqrt{21.93 + 22.18} = 55.99$$

For presentation in MIL-HDBK-17, this B-basis value would be rounded to 56.0.

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8.7.2 **Problem 2 - Outlier detection and ANOVA method.** The data set for this problem consists of tensile strength measurements from nine batches of material. This problem illustrates the outlier detection procedure and the calculation of B-basis values by the ANOVA method.

Problem 2 - Step 1. The first step is to screen the data for outliers using the MNR procedure described in Sections 8.5.3.1 and 8.6.2. The screening procedure is performed separately on each batch. (See Problem 1 for details of the outlier detection computations.)

A single observation, the value 109.9 in the ninth batch, is identified as a potential outlier. Assume that the potential outlier was the result of a clerical error, and its value should have been 69.6. The remaining calculations assume that this correction has been made. Upon repeating the entire outlier screening procedure, no potential outliers or extreme outliers are identified.

Problem 2 - Step 2. The k-sample Anderson-Darling test statistic for the data is ADK = 1.81. (See Problem 1 for a detailed computation of the k-sample statistic.) Since 1.81 is greater than the critical value of 1.44, conclude that the batches are not from the same population.

Problem 2 - Step 3. The test for equality of variances between the batches gives a calculated statistic of EV = 5.60, which is less than the critical value of 15.51 from Table 8.8.8. (See Problem 1 for a detailed computation of this test statistic.) Thus, one should proceed as if the data are normally distributed and the variances are equal, and use the ANOVA method of Section 8.5.4 to calculate a B-basis value.

Problem 2 - Step 4. Summary statistics for the data are given in the table below. Preliminary ANOVA calculations covered in Section 8.6.7.1.2 are:

$$\bar{X} = \sum_{i=1}^{k} n_i \bar{X}_i / N = [7(65.60) + \ldots + 7(71.80)]/53 = 68.84$$

$$SSB = \sum_{i=1}^{k} n_i \bar{X}_i^2 - N\bar{X}^2 = [7(65.60)^2 + \ldots + 7(71.80)^2 - 53(68.84)^2 = 317.255$$

$$SSE = \sum_{i=1}^{k} (n_i - 1) S_i^2 = [6(2.99)^2 + \ldots + 6(3.88)^2 = 460.680$$

$$MSB = SSB/(k-1) = 317.255/(9-1) = 39.66$$

$$MSE = SSE/(N-k) = 460.680/(53-9) = 10.47$$

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Batch	n_i	$\bar{X}_i$	S_i
1	7	65.60	2.99
2	5	66.32	2.33
3	5	67.84	2.84
4	7	67.33	4.17
5	6	66.93	2.45
6	5	71.64	4.03
7	5	71.10	3.33
8	6	71.52	1.96
9	7	71.80	3.88

Preliminary calculations covered in Section 8.5.4 are:

$$n^* = \sum_{i=1}^{k} n_i^2/N = (7^2 + \ldots + 7^2)/53 = 6.0189$$

$$n' = (N-n^*)/(k-1) = (53-6.0189)/(9-1) = 5.8726$$

$$k^* = N/n^* = 53/6.0189 = 8.8056$$

$$S_e^2 = MSE = 10.47$$

$$S_b^2 = (MSB - MSE)/n' = (39.66 - 10.47)/5.8726 = 4.97$$

and

$$F_{0.2,k-1,N-k} = F_{0.2,8,44} = 0.56 \qquad \text{(from Table 8.8.4)}$$

The upper bound for the ratio of between-batch to within-batch variability is

$$\hat{R} = [(MSB/MSE)/F_{0.2} - 1]/n' = [(39.66/10.47)/0.56 - 1]/5.8726 = 0.98$$

The tolerance limit factor T should be obtained from the approximation in Section 8.5.4 rather than from Table 8.8.4 because $\hat{R}$, k^* and n^* are not integers. The noncentrality parameter is calculated as

$$\delta = 1.282\sqrt{\frac{N(\hat{R}+1)}{n^*\hat{R}+1}} = 1.282\sqrt{\frac{53\,(0.98+1)}{(6.0189)\,(0.98)+1}} = 5.15$$

The degrees of freedom parameter is calculated as

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$$\gamma = \frac{(\hat{R}+1)^2}{\frac{(\hat{R}+\frac{1}{n^*})^2}{k^*-1} + \frac{(\frac{n^*-1}{n^*})^2}{k^*(n^*-1)}}$$

$$\gamma = \frac{(0.98+1)^2}{\frac{\left(0.98+\frac{1}{6.0189}\right)^2}{8.8056-1} + \frac{\left(\frac{6.0189-1}{6.0189}\right)^2}{8.8056(6.1089-1)}}$$

$$= 21.40$$

and

$$t_{\gamma,0.95} = t_{21.4,0.95} = 1.72$$

The tolerance limit factor is calculated as follows:

$$T = 1.282 + 1.282\frac{t_{\gamma,0.95}}{\delta} + \frac{1.80}{\gamma} - \frac{1.85}{\gamma^2} + 0.567\frac{\delta}{\gamma} + 5.24\frac{\delta}{\gamma^2} - 1.08\frac{\delta^2}{\gamma^2} + 0.0166\frac{\delta}{\gamma^2} + \frac{7.79}{\gamma^4}$$

$$= 1.282 + 1.282\frac{1.72}{5.15} + \frac{1.80}{21.4} - \frac{1.85}{21.4^2} + 0.567\frac{5.15}{21.4} + 5.24\frac{5.15}{21.4^2} - \frac{5.15^2}{21.4^2} + 0.0166\frac{5.15^3}{21.4^2} + \frac{7.79}{21.4^4}$$

$$= 1.94$$

Thus, a B-basis value is calculated as

$$B = \bar{X} - T\sqrt{S_b^2 + S_e^2} = 68.84 - 1.94\sqrt{4.97 + 10.47} = 61.22$$

For presentation in MIL-HDBK-17, this B-basis value would be rounded to 61.2.

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8.7.3 Problem 3 - Weibull distribution. The data set for this problem consists of tensile strength measurements from nine batches of material. This problem illustrates the two-parameter Weibull goodness-of-fit test and the calculation of B-basis values by the Weibull method.

Problem 3 - Step 1. There are no detected outliers in this set of data. (See Problem 1 for details of the outlier detection computations.)

Problem 3 - Step 2. The k-sample Anderson-Darling test statistic is ADK = 0.99. (See Problem 1 for details of the computation of the k-sample statistic.) Since this is less than the critical value of 1.44, conclude that the data from the batches may be combined into a single sample.

Problem 3 - Step 3. The maximum normed residual (MNR) test is performed on the pooled data. No potential outliers are detected in the pooled data. (see Problem 1 - Step 1 for details of the outlier detection procedure.)

Problem 3 - Step 4. In order to perform the two-parameter Weibull goodness-of-fit test described in Section 8.6.4.3, it is necessary to compute estimates of the scale and shape parameters, $\hat{\alpha}$ and $\hat{\beta}$. A procedure for doing this is described in Section 8.5.6.1. The geometric mean of the data is computed as

$$\bar{X}_G = \exp\left[\frac{1}{N}\sum_{i=1}^{N} \ln(X_i)\right] = \exp\left[\frac{1}{53}\sum_{i=1}^{53} \ln(X_i)\right] = 67.501$$

For a given value of $\hat{\beta}$, $\hat{\alpha}$ is calculated as

$$\hat{\alpha} = \bar{X}_G\left[\frac{1}{N}\sum_{i=1}^{N}\left(\frac{X_i}{\bar{X}_G}\right)^{\hat{\beta}}\right]^{\frac{1}{\hat{\beta}}}$$

$$\alpha = (67.501)\left[\frac{1}{53}\sum_{i=1}^{53}\left(\frac{X_i}{67.501}\right)^{\hat{\beta}}\right]^{\frac{1}{\hat{\beta}}}$$

In order to calculate $\hat{\beta}$, define the function $G(\hat{\beta})$ by

$$G(\hat{\beta}) = \frac{1}{N}\sum_{i=1}^{N} \ln(X_i)\left(\left[\frac{X_i}{\hat{\alpha}}\right]^{\hat{\beta}} - 1\right) - \frac{1}{\hat{\beta}}$$

$$= \frac{1}{53}\sum_{i=1}^{53} \ln(X_i)\left(\left[\frac{X_i}{\hat{\alpha}}\right]^{\hat{\beta}} - 1\right) - \frac{1}{\hat{\beta}}$$

where $\hat{\alpha}$ is calculated as above. The estimate, $\hat{\beta}$, is the solution to the Equation $G(\hat{\beta}) = 0$. An iterative technique for solving this equation is given in Section 8.5.6.2, and begins by setting

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$$\hat{\beta} = \frac{1.28}{S_y} = \frac{1.28}{0.1044} = 12.2605$$

The solution is $\hat{\beta} = 12.27$, which in turn gives $\hat{\alpha} = 70.77$.

The first five ordered observations are listed below with the transformations necessary to compute the goodness-of-fit test statistic.

X_i	$Z_{(i)} = \left(\frac{X_{(i)}}{\hat{\alpha}}\right)^{\hat{\beta}} = \left(\frac{X_{(i)}}{70.77}\right)^{12.27}$
47.1	0.0068
51.6	0.0207
56.0	0.0566
59.4	0.1166
60.4	0.1431
⋮	⋮

The Anderson-Darling goodness-of-fit statistic and observed significance level are calculated according to Section 8.6.4.3 as follows.

$$\begin{aligned} AD &= \sum_{i=1}^{N} \frac{1-2i}{N} \{\ln[1-\exp(-Z_{(i)})] - Z_{(N+1-i)}\} - N \\ &= \sum_{i=1}^{53} \frac{1-2i}{53} \{\ln[1-\exp(-Z_{(i)})] - Z_{(54-i)}\} - 53 \\ &= 0.535 \end{aligned}$$

$$AD^* = (1 + 0.2/\sqrt{N})\,AD = (1 + 0.2/\sqrt{53})\,0.535 = 0.5497$$

$$\begin{aligned} OSL &= \frac{1}{1 + \exp(-0.10 + 1.24 \ln AD^* + 4.48 AD^*)} \\ &= \frac{1}{1 + \exp[-0.10 + 1.24 \ln(0.550) + 4.48(0.550)]} \\ &= 0.165 \end{aligned}$$

Since the Weibull goodness-of-fit test yields an OSL value greater than 0.05, there is insufficient evidence to contradict the assumption that the data follow a two-parameter Weibull distribution. Hence, the two parameter Weibull method in Section 8.5.6.2 should be used to compute the B-basis value.

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Problem 3 - Step 5. The parameter estimates $\hat{\alpha}$ and $\hat{\beta}$ calculated in the previous step are used to compute the B-basis value for the sample as described in Section 8.5.6. The quantities necessary to compute the B-basis value are:

$$V = 4.670 \text{ (from Table 8.8.2)}$$

$$\hat{\alpha} = 70.77$$

$$\hat{\beta} = 12.27$$

$$\hat{Q} = \hat{\alpha}\,(0.10536)^{1/\hat{\beta}} = (70.77)(0.10536)^{1/12.27} = 58.91$$

The B-basis value is calculated as

$$B = \hat{Q} \exp\{-V/(\hat{\beta}\sqrt{N})\} = 58.91 \exp\{-4.670/[(12.27)\sqrt{53}]\} = 55.91$$

For presentation in MIL-HDBK-17, this B-basis value would be rounded to 55.9.

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8.7.4 **Problem 4 - Normal distribution.** The data set for this problem consists of compression test measurements from nine batches of material. This problem illustrates the normal goodness-of-fit test and the calculation of B-basis values by the normal method.

Problem 4 - Step 1. There are no detected outliers in this set of data. (See Problem 1 for details of the outlier detection calculations.)

Problem 4 - Step 2. The k-sample Anderson-Darling test statistic is ADK = 1.05 (see Problem 1 for a detailed computation of the k-sample statistic). Since this is less than the critical value of 1.44, conclude that the data from the batches may be combined and treated as a single sample. The next step is to investigate the form of the distribution.

Problem 4 - Step 3. The maximum normed residual (MNR) test is performed on the pooled data. No potential outliers are detected in the pooled data. (see Problem 1 - Step 1 for details of the outlier detection procedure.)

Problem 4 - Step 4. The Weibull goodness-of-fit test yields an observed significance level of 0.025. (See Problem 3 for details of the computation for the Weibull goodness-of-fit test.) Since this is less than 0.05, the normal goodness-of-fit test described in Section 8.6.4.1 is performed.

Problem 4 - Step 5. The average and standard deviation of the sample are 30.25 and 1.495, respectively. The first five ordered observations are listed below with the Z-values and the values of the standard normal distribution necessary for calculation of the normal Anderson-Darling statistic.

$X_{(i)}$	$Z_{(i)} = \frac{X_{(i)} - \bar{X}}{S} = \frac{X_{(i)} - 30.25}{1.495}$	$F_o(Z_{(i)})$
26.9	-2.25	0.0122
27.7	-1.71	0.0436
27.9	-1.58	0.0571
27.9	-1.58	0.0571
28.0	-1.51	0.0655
⋮	⋮	⋮

$$AD = \sum_{i=1}^{N} \frac{1-2i}{N} \{\ln[F_o(Z_{(i)})] + \ln[1 - F_o(Z_{(N+1-i)})]\} - N$$

$$= \sum_{i=1}^{53} \frac{1-2i}{53} \{\ln[F_o(-Z_{(i)})] + \ln[1 - F_o(Z_{(54-i)})\} - 53$$

$$= 0.365$$

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$$AD^* = \left[1 + \frac{4}{N} - \frac{25}{N^2}\right]AD = \left[1 + \frac{4}{53} - \frac{25}{53^2}\right](0.363) = 0.3872$$

$$OSL = \frac{1}{1 + \exp(-0.48 + 0.78 \ln AD^* + 4.58 AD^*)}$$

$$= \frac{1}{1 + \exp[0.48 + 0.78 \ln(0.387) + 4.58(0.387)]}$$

$$= 0.365$$

Since the normal goodness-of-fit test yields an OSL value (0.365) greater than 0.05, there is insufficient evidence to contradict the assumption that the data are normally distributed. Hence, the normal method in Section 8.5.5 is used to compute a B-basis value.

Problem 4 - Step 6. From Table 8.8.1, the one-sided tolerance limit factor, k_B, is 1.634. The B-basis value for a normally distributed sample is computed as

$$B = \bar{X} - k_B S = 30.25 - (1.634)(1.495) = 27.81$$

For presentation in MIL-HDBK-17, this B-basis value would be rounded to 27.8.

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8.7.5 **Problem 5 - Lognormal distribution.** The data set for this problem consists of compression test measurements from nine batches of material. This problem illustrates the lognormal goodness-of-fit test and the calculation of B-basis values by the lognormal method.

Problem 5 - Step 1. There are no detected outliers in this set of data. (See Problem 1 for details of the outlier detection calculations.)

Problem 5 - Step 2. The k-sample Anderson-Darling test statistic is ADK = 1.24. (See Problem 1 for details of the computation of the k-sample statistic.) Since this is less than the critical value of 1.44, conclude that the data from the
batches may be combined into a single sample.

Problem 5 - Step 3. The maximum normed residual (MNR) test is performed on the pooled data. No potential outliers are detected in the pooled data. (see Problem 1 - Step 1 for details of the outlier detection procedure.)

Problem 5 - Step 4. The observed significance levels (OSL) for the two-parameter Weibull and the normal goodness-of-fit tests are given below:

Distribution	OSL
Two-parameter Weibull	0.000
Normal	0.034

(See Problems 3 and 4 for details of the computations for these tests.) Since the OSLs are both less than 0.05, neither of the distributions adequately describe the data. Thus, the lognormal goodness-of-fit test is performed.

Problem 5 - Step 5. In order to perform the lognormal goodness-of-fit test described in Section 8.6.4.2, the natural logarithms of the data are used. The average and standard deviation of the transformed data are

$$\bar{X}_L = 3.41$$
$$S_L = 0.0548$$

The first five ordered observations are listed below with the transformations necessary to compute the goodness-of-fit statistic. The goodness-of-fit statistic and observed significance level are calculated as:

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$$AD = \sum_{i=1}^{N} \frac{1-2i}{N} \{\ln[F_o(Z_{(i)})] + \ln[1 - F_o(Z_{(N+1-i)})]\} - N$$
$$= \sum_{i=1}^{53} \frac{1-2i}{53} \{\ln[F_o(-Z_{(i)})] + \ln[1 - F_o(Z_{(54-i)})\} - 53$$
$$= 0.650$$

$$AD^* = \left[1 + \frac{4}{N} - \frac{25}{N^2}\right] AD = \left[1 + \frac{4}{53} - \frac{25}{53^2}\right] (0.650) = 0.6933$$

$$OSL = \frac{1}{1 + \exp(-0.48 + 0.78 \ln AD^* + 4.58 AD^*)}$$
$$= \frac{1}{1 + \exp[0.48 + 0.78 \ln(0.693) + 4.58(0.693)]}$$
$$= 0.083$$

$X_{(i)}$	$\ln(X_{(i)})$	$Z_{(i)} = \frac{\ln(X_{(i)}) - \bar{X}_L}{S_L} = \frac{\ln(X_{(i)}) - 3.41}{0.0548}$	$F_o[Z_{(i)}]$
27.3	3.31	-1.82	0.0344
27.3	3.31	-1.82	0.0344
27.5	3.31	-1.82	0.0344
27.9	3.33	-1.45	0.0735
28.4	3.35	-1.09	0.1379
⋮	⋮	⋮	⋮

Since the lognormal goodness-of-fit test results in an OSL value greater than 0.05, there is insufficient evidence to contradict the assumption that the data are lognormally distributed. Hence, the lognormal method in Section 8.5.5.2 is used to compute a B-basis value.

Problem 5 - Step 6. The B-basis value for lognormally distributed data is computed as

$$B = \exp(\bar{X}_L - k_B S_L) = \exp(3.41 - 1.634(0.0548)) = 27.68$$

For presentation in MIL-HDBK-17, this B-basis value would be rounded to 27.7.

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8.7.6 **Problem 6 - Nonparametric method.** The data set for this problem consists of tensile strength measurements for nine batches of material. This problem illustrates the calculation of B-basis values by the nonparametric method.

Problem 6 - Step 1. There are no detected outliers in this set of data. (See Problem 1 for details of the outlier detection calculations.)

Problem 6 - Step 2. The k-sample Anderson-Darling test statistic is ADK = 0.89. (See Problem 1 for details of the computation of the k-sample statistic.) Since this is less than the critical value of 1.44, conclude that the data from the batches may be combined into a single sample.

Problem 6 - Step 3. The maximum normed residual (MNR) test is performed on the pooled data. No potential outliers are detected in the pooled data. (see Problem 1 - Step 1 for details of the outlier detection procedure.)

Problem 6 - Step 4. The results of the goodness-of-fit tests for the three distributions are:

Distribution	OSL
Two-parameter Weibull	0.013
Normal	0.008
Lognormal	0.007

(See problems 3, 4, and 5 for details of the computations for each of these tests.)

Since all of the observed significance levels are less than 0.05, it is concluded that the data do not follow any of the three distributions. Thus, the nonparametric method described in Section 8.5.7 must be used to calculate the B-basis value.

Problem 6 - Step 5. The first step in computing a B-basis value by the nonparametric method is to order the data values from smallest to largest. The five smallest values are 64.3, 64.9, 65.1, 65.3, and 65.6. The next step is to obtain the appropriate rank from Table 8.8.3 corresponding to the sample of size n. With an n of 53, the rank of the observation to be used as a B-basis value is r = 2. Thus, the second observation, or 64.9, is the B-basis value for this sample. For presentation in MIL-HDBK-17, this B-basis value would be 64.9.

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8.7.7 Problem 7 - Insufficient data. The data set for this problem consists of tensile strength measurements for five batches of material. This problem illustrates the situation where none of the standard distributions adequately fit the data, and there is insufficient data to perform the nonparametric method.

Problem 7 - Step 1. There are no detected outliers in this set of data. (See Problem 2 for details of the outlier detection calculations.)

Problem 7 - Step 2. The k-sample Anderson-Darling test statistic is ADK = 0.59. (See Problem 1 for details of the computation of the k-sample statistic.) Since this is less than the critical value of 1.63, conclude that the data from the batches may be combined into a single sample.

Problem 7 - Step 3. The maximum normed residual (MNR) test is performed on the pooled data. No potential outliers are detected in the pooled data. (see Problem 1 - Step 1 for details of the outlier detection procedure.)

Problem 7 - Step 4. The results of the goodness-of-fit tests for the three distributions are:

Distribution	OSL
Two-parameter Weibull	0.031
Normal	0.011
Lognormal	0.006

(See problems 3, 4, and 5 for details of the computations for each of these tests.)

Since all of the observed significance levels are less than 0.05, it is concluded that the data do not follow any of the three distributions. The nonparametric method cannot be used, however, because there are only 25 data values in the sample. The Hanson-Koopmans method should be used to calculate a B-basis value for these data.

Problem 7 - Step 5. Following the procedure described in Section 8.5.7.2, a B-basis value can be estimated. For n = 25, from Table 8.8.10 it is determined that r = 11 and k = 1.087. After ranking the data in ascending order, the first and eleventh values are found.

$$X_{(1)} = 55.9 \qquad X_{(11)} = 72.2$$

$$B = X_{(r)}\left(\frac{X_{(1)}}{X_{(r)}}\right)^{k}$$

$$= 72.2\left(\frac{55.9}{72.2}\right)^{1.087}$$

$$= 54.7$$

These data can be included in MIL-HDBK-17 as interim data, but the B-value would not be reported in the handbook.

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8.7.8 **Problem 8 - Regression analysis.** The data set for this problem consists of compression test measurements and corresponding specimen thickness measurements. This problem illustrates the regression analysis procedures presented in Sections 8.5.8 and 8.6.6.

Problem 8 - Step 1. In this example, x represents the thickness and y the compressive yield strength determined from a group of compression tests. From the data in Table 8.7, the following quantities may be calculated:

$n = 20$ $\quad (\Sigma x)^2 = 6.23$

$\Sigma x = 2.496$ $\quad (\Sigma y)^2 = 316406.25$

$\Sigma y = 562.5$ $\quad (\Sigma x)(\Sigma y) = 1404$

$\Sigma x^2 = 0.312638$ $\quad \Sigma xy = 69.8974$

$\Sigma y^2 = 15923.85$

$$S_{xx} = \Sigma x^2 - (\Sigma x)^2/n = 0.312638 - (6.23)/20 = 0.0001137$$
$$S_{xy} = \Sigma xy - (\Sigma x)(\Sigma y)/n = 69.8974 - (2.946)(562.5)/20 = -0.3026$$
$$S_{yy} = \Sigma y^2 - (\Sigma y)^2/n = 15923.85 - (316406.25)/20 = 103.5$$

The slope of regression line is:

$$b = \frac{S_{xy}}{S_{xx}} = \frac{-0.3026}{0.001137} = -266$$

The y-intercept of the regression line is:

$$a = \frac{\Sigma y - b\Sigma x}{n} = \frac{562.5}{20} - \frac{(-266)(2.496)}{20} = 28.1 - (-33.2) = 61.3$$

Thus, the final equation of the least squares regression line is:

$$y^* = a + bx = 61.3 - 266x$$

Using this equation, the values of y^* in the table below are computed for the values of x in the data set. The root mean square error is computed as follows:

$$S_y = \sqrt{\frac{\Sigma(y - y^*)^2}{n - 2}} = \sqrt{\frac{23.18}{18}} = 1.13$$

or R^2 is computed as follows:

$$S_y = \left[\frac{S_{yy} - b^2 S_{xx}}{n - 2}\right]^{0.5} = \left[\frac{103.5 - (-266)^2(0.001137)}{18}\right]^{0.5} = 1.13$$

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x	y^*	$e = y - y^*$
0.112	31.5	0.5
0.112	31.5	0.5
0.116	30.5	-0.4
0.116	30.5	1.6
0.119	29.7	-1.6
0.119	29.7	0.8
0.121	29.1	-2.3
0.122	28.9	-0.8
0.124	28.5	0.9
0.124	28.5	0.1
0.126	27.8	-0.5
0.126	27.8	0.2
0.127	27.5	1.7
0.128	27.3	-1.3
0.129	27.0	0.6
0.133	25.9	0.9
0.135	25.4	1.4
0.135	25.4	-1.4
0.135	25.4	-0.2
0.137	24.9	-0.1

$$R^2 = \frac{b^2 S_{xx}}{S_{yy}} = \frac{(-266)^2(0.001137)}{103.5} = 0.78$$

Thus, 78% of the variability in the y data about its average is explained by the linear relationship between y and x.

Problem 8 - Step 2. One of the assumptions made in linear regression analysis is that the residuals are normally distributed about the regression line. The validity of this assumption may be checked by performing a normal goodness-of-fit test on the residuals as discussed in Section 8.6.4.5. Note that the $Z_{(i)}$ values used in the Anderson-Darling statistic are defined as $Z_{(i)} = e_{(i)}/S_Y$, where $e_{(i)}$ is the ith ordered residual and S_y is the root-mean-square error from the regression. The first five ordered residuals and the preliminary goodness-of-fit calculations are shown in the following table.

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$e_{(i)}$	$Z_{(i)} = \frac{e_{(i)}}{S_y} = \frac{e_{(i)}}{1.13}$
-2.3	-2.04
-1.6	-1.42
-1.4	-1.24
-1.3	-1.15
-0.8	-0.71

The normal goodness-of-fit test statistic is 0.2391 with an OSL of 0.562. (See Problem 4 for details of the computation for the normal goodness-of-fit test.) Since the OSL is greater than 0.05, there is insufficient evidence to contradict the assumption that the residuals are normally distributed.

Problem 8 - Step 3. There are multiple y observations for several of the x values. Thus, it is possible to construct an analysis of variance table to test the adequacy of the regression as discussed in Section 8.6.6.2. The sums of squares for the three primary lines of the analysis of variance table are calculated as follows:

$$SSR = b^2 S_{xx} = (-266)^2(0.001137) = 80.45$$

$$SST = S_{yy} = 103.5$$

$$SSE = SST - SSR = 103.5 - 80.45 = 23.05$$

The mean squares are calculated as shown below.

$$MSR = SSR = 80.45$$

$$MSE = SSE/(n - 2) = 23.05/18 = 1.28$$

$$F = MSR/MSE = 80.45/1.28 = 62.8$$

The analysis of variance table is shown below.

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Source of Variation	Degrees of Freedom	Sum of Squares, SS	Mean Squares, MS	F_{calc}
Regression	1	80.45	80.45	F = 62.8
Error	18	23.05	1.28	
Total	19	103.50		

The F value of 62.8 with 1 and n - 2 = 18 degrees of freedom is greater than the value of 4.41 from Table 8.8.5 corresponding to 1 and 18 degrees of freedom, so the regression is declared to be significant.

Problem 8 - Step 4. With the linear regression equation from step 1, lower tolerance limits may be calculated at any thickness (x value) by the procedure in Section 8.5.8. Details for computing a B-basis value at x = 0.120 are given below.

The average thickness value in the data set is:

$$\bar{x} = \Sigma x/n = 2.496/20 = 0.125$$

The Δ factor required to compute the tolerance limit factor, k'_B, is:

$$\Delta = \frac{(x_o - \bar{x})^2}{\sum_{i=1}^{n} (X_i - \bar{x})^2/n} = \frac{(0.120 - 0.125)^2}{(0.01137)/20} = 0.44$$

The approximation for the k' factor is:

$$k'_B = 1.282 + \exp\left[0.595 - 0.508\,\ln(n) + \frac{4.62}{n} + \left(0.486 - \frac{0.986}{n}\right)\ln(1.82 + \Delta)\right]$$

$$= 1.282 + \exp\left[0.595 - 0.508\,\ln(20) + \frac{4.62}{20} + \left(0.486 - \frac{0.986}{20}\right)\ln(1.82 + 1.44)\right]$$

$$= 1.99$$

Thus, a B-basis value at x = 0.120 is computed as

$$B = (a + bx_o) - k_B S_y = [61.3\,266\,(0.120)] - 1.99\,(1.13) = 27.13$$

For presentation in MIL-HDBK-17, this value would be rounded to 27.1.

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8.8 CALCULATIONS FOR STATISTICAL TABLES. This section contains a number of tables which are required for the analyses described in Sections 8.5 and 8.6. Tables 8.8.4 through 8.8.7, 8.8.10, 8.8.15, and 8.8.16 were generated specifically for MIL HDBK-17. The remaining tables were adapted from MIL-HDBK-5 (see Reference 8.5.5).

For some of the tabulated values, theoretical derivations and numerical approximations are provided below. The approximations are useful in computer applications when the software required to generate the tabulated values is not available. The accuracy of the approximations is measured by the relative magnitude of error (RME). The RME is defined as

$$\text{RME} = \frac{|\text{approximate value} - \text{actual value}|}{\text{actual value}} \qquad 8.8$$

and measures the percentage error in the approximate value with respect to the actual value.

8.8.1 One-sided tolerance factors, k_B, for the normal distribution, 0.95 confidence. The k_B values in Table 8.8.1 are calculated as $1/\sqrt{n}$ times the 0.95th quantile of the noncentral t-distribution with noncentrality parameter $1.282\sqrt{n}$ and n - 1 degrees of freedom. An approximation to the k_B values in Table 8.8.1 is:

$$k_B = 1.282 + \exp(0.958 - 0.520\,\ell n(n) + 3.19/n) \qquad 8.8.1$$

This approximation is accurate to within 0.2% of the tabulated values for n greater than or equal to 16.

8.8.2 One-sided tolerance factors, V, for the Weibull distribution, 0.95 confidence. The V values in Table 8.8.2 are calculated using the following statistical results. First, define the random variables

$$A_i = \frac{\ell n(X_i) - \ell n(\tilde{\alpha})}{1/\hat{\beta}} \qquad i = 1, \ldots, n \qquad 8.8.2(a)$$

where X_i is a Weibull random variable with unknown shape and scale parameters β and α and $\tilde{\alpha}$ and $\hat{\beta}$ are the maximum likelihood estimators (MLE's) of β and α given by Equations 8.5.6.1(a) and 8.5.6.1(c). For a particular n, the V value is the 0.95th quantile of the conditional distribution of the random variable

$$V = \frac{\sqrt{n}[\ell n(\hat{Q}) - \ell n(Q)]}{1/\hat{\beta}} \qquad 8.8.2(b)$$

given that

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$$A_i = \frac{\ell n(x_i') - \ell n(\hat{\alpha}')}{1/\hat{\beta}'} \qquad 8.8.2(c)$$

where

$$x_i' = -\ell n\left(1 - \frac{i - 0.5}{n + 0.25}\right) \qquad i = 1,\ldots,n \qquad 8.8.2(d)$$

$$\hat{Q} = \hat{\alpha}\,(0.10536)^{1/\beta} \qquad 8.8.2(e)$$

$$Q = \alpha\,(0.10536)^{1/\beta} \qquad 8.8.2(f)$$

and $\hat{\alpha}'$ and $\hat{\beta}'$ are the MLE's of the two-parameter Weibull scale and shape parameters for the sample x_1', ..., x_n'. The conditional distribution of V is determined by the relationship

$$V = \sqrt{n}\,[Z + \ell n(0.10536)] \qquad 8.8.2(g)$$

where the distribution of Z is given in Theorem 4.1.3 on page 150 of reference 8.6.4.1. Numerical integration was used to determine the V values in Table 8.8.2 based on these results.

An approximation to the V values in Table 8.8.2 is:

$$V = 3.803 + \exp\left\{1.79 - 0.516\,\ell n(n) + \frac{5.1}{n}\right\} \qquad 8.8.2(h)$$

This approximation is accurate to within 0.5% of the tabulated values for n greater than or equal to 16.

8.8.3 Ranks, r, of observations for determining B-values for an unknown distribution. For $n > 29$, an approximation to the ranks for B-basis values in Table 8.8.3 is

$$r = \frac{n}{10} - 1.645\sqrt{\frac{9n}{100}} + 0.23 \qquad 8.8.3$$

rounded to the nearest integer. This approximation is exact for all but 12 values of n in the range of the table ($29 \le n \le 10499$). For this small percentage of n values (0.1%), the approximation errs by one rank on the conservative side.

8.8.4 0.20 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. An approximation to the $F_{0.20}$ values in Table 8.8.4 is

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$$F_{0.20} = \exp\left(2\delta\left[1 + \frac{z^2-1}{3} - \frac{4\sigma^2}{3}\right] + 2\sigma z\sqrt{1 + \frac{\sigma^2(z^2-3)}{6}}\right) \qquad 8.8.4(a)$$

where

$$\delta = 0.5(1/(\gamma_2 - 1) - 1/(\gamma_1 - 1)) \qquad 8.8.4(b)$$

$$\sigma^2 = 0.5(1/(\gamma_2 - 1) + 1/(\gamma_1 - 1)) \qquad 8.8.4(c)$$

$z = -0.84$

γ_1 = numerator degrees of freedom

γ_2 = denominator degrees of freedom.

(See Reference 8.8.4.)

Equations 8.8.4(a-d) are not valid when either γ_1 or γ_2 equals one. The following equations are to be used for these special cases:

For $\gamma_1 = 1$

$$F_{0.20} = \left[0.25334720 + \frac{0.06740204}{\gamma_2} + \frac{0.01068163}{\gamma_2^2} - \frac{0.00912432}{\gamma_2^3} - \frac{0.00291922}{\gamma_2^4}\right]^2 \qquad 8.8.4(e)$$

For $\gamma_2 = 1$

$$F_{0.20} = \left[1.28155100 + \frac{0.84658450}{\gamma_1} + \frac{0.57089040}{\gamma_1^2} + \frac{0.25851690}{\gamma_1^3} + \frac{0.05438575}{\gamma_1^4}\right]^{-2} \qquad 8.8.4(f)$$

8.8.5 0.95 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. An approximation to the $F_{0.95}$ values in Table 8.8.5 is

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$$F_{0.95} = \exp\left(2\delta\left[1 + \frac{z^2-1}{3} - \frac{4\sigma^2}{3}\right] + 2\sigma z\sqrt{1 + \frac{\sigma^2(z^2-3)}{6}}\right) \qquad 8.8.5(a)$$

where

$$\delta = 0.5(1/(\gamma_2 - 1) - 1/(\gamma_1 - 1)) \qquad 8.8.5(b)$$

$$\sigma^2 = 0.5(1/(\gamma_2 - 1) + 1/(\gamma_1 - 1)) \qquad 8.8.5(c)$$

$z = 1.645$

γ_1 = numerator degrees of freedom

γ_2 = denominator degrees of freedom.

(See Reference 8.8.4.)

Equations 8.8.5(a-d) are not valid when either γ_1 or γ_2 equals one. The following equations are to be used for these special cases:

For $\gamma_1 = 1$

$$F_{0.95} = \left[1.95996400 + \frac{2.37227200}{\gamma_2} + \frac{2.82250000}{\gamma_2^2} + \frac{2.555585200}{\gamma_2^3} + \frac{1.58953600}{\gamma_2^4}\right]^2 \qquad 8.8.5(e)$$

For $\gamma_2 = 1$

$$F_{0.95} = \left[0.06270671 + \frac{0.01573832}{\gamma_1} + \frac{0.00200073}{\gamma_1^2} - \frac{0.00243852}{\gamma_1^3} - \frac{0.00064811}{\gamma_1^4}\right]^{-2} \qquad 8.8.5(f)$$

8.8.6 One-sided tolerance factors, T, for the one-way random effects ANOVA, 0.95 confidence. To approximate the values in Table 8.8.6, the tolerance limit factor T is defined in as

$$T = \sqrt{\frac{n^*\hat{R}+1}{N(\hat{R}+1)}}\, t_{\gamma,0.95}(\delta) \qquad 8.8.6(a)$$

where $t_{\gamma,0.95}(\delta)$ is the 0.95 quantile of the non-central t-distribution with noncentrality parameter

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$$\delta = 1.282\sqrt{\frac{N(\hat{R}+1)}{n^*\hat{R}+1}} \qquad 8.8.6(b)$$

degrees of freedom approximated by

$$\gamma = \frac{(\hat{R}+1)^2}{\frac{\left(\hat{R}+\frac{1}{n^*}\right)^2}{k^*-1} + \frac{\left(\frac{n^*-1}{n^*}\right)^2}{k^*(n^*-1)}} \qquad 8.8.6(c)$$

Tolerance limit factors may also be computed from the following approximate formula:

$$T = 1.282 + 1.282\frac{T_{\gamma,0.95}}{\delta} + \frac{1.80}{\gamma} - \frac{1.85}{\gamma^2} + 0.567\frac{\delta}{\gamma} + 5.24\frac{\delta}{\gamma^2} - 1.08\frac{\delta^2}{\gamma^2} + 0.0166\frac{\delta}{\gamma^2} + \frac{7.79}{\gamma^4} \qquad 8.8.6(d)$$

This approximation is accurate to within 0.45% of the tabulated values. This approximation may be used for values of n^*, k^* and $\hat{R}$ both within and beyond the range of the table.

8.8.7 **Critical values for the k-sample Anderson-Darling test at the 0.05 significance level.** The k-sample Anderson-Darling test critical values in Table 8.8.7 were calculated using Equation 8.6.3.1(j) for the case of samples of equal size n.

8.8.8 **0.95 Quantiles of the chi-squared distribution associated with γ degrees of freedom.** An approximation to the chi-squared quantiles ($\chi^2_{0.95}$) in Table 8.8.8 is:

$$\chi^2_{0.95} = \gamma\left[1 - \frac{2}{9\gamma} + 1.645\left(\frac{2}{9\gamma}\right)^{\frac{1}{2}}\right]^3 + \frac{9}{100\gamma} \qquad 8.8.8$$

where γ is the degrees of freedom. This approximation is accurate to within 0.2% of the tabulated values. (See Reference 8.8.8.)

8.8.9 **Critical values for the MNR outlier detection procedure (α = 0.05).** The critical values in Table 8.8.9 are computed by the following formula:

$$C.V. = \frac{n-1}{\sqrt{n}}\sqrt{\frac{t^2}{n-2+t^2}} \qquad 8.8.9$$

where t is the $[1 - \alpha/(2n)]$ quantile of the t-distribution with n - 2 degrees of freedom, α is the significance level of the test, and n is the sample size. Numbers in Table 8.8.9 are computed with a significance level of α = 0.05. (See Reference 8.6.2(b))

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8.8.10 **Nonparametric B-basis material properties for small samples.** The values in Table 8.8.10 are based on Reference 8.5.7.2.

8.8.11 **One-sided tolerance limit factors, V_A, for the Weibull distribution, 0.95 confidence.** The V_A values in Table 8.8.11 are calculated as described in Section 8.8.2 (Reference 8.8.11). An approximation to the V_A values is:

$$V \approx 6.649 + \exp\left[2.55 - 0.526\ln(n) + \frac{4.76}{n}\right] \qquad 8.8.12$$

This approximation is accurate within 0.5% of the tabulated values for n greater than or equal to 16.

8.8.12 **One-sided tolerance limit factors, k_A, for the normal distribution, 0.95 confidence.** The k_A values in Table 8.8.12 are calculated as $1/\sqrt{n}$ times the 0.95th quantile of the noncentral t-distribution with noncentrality parameter $2.326\sqrt{n}$ and n - 1 degrees of freedom (Reference 8.8.12). An approximation to the k_B values in Table 8.8.12 is:

$$k_B \approx 2.326 + \exp\left(1.34 - 0.522\ln(n) + \frac{3.87}{n}\right) \qquad 8.8.13$$

This approximation is accurate to within 0.2% of the tabulated values for n greater than or equal to 16.

8.8.13 **Ranks, r, of observations for determining A-basis values for an unknown distribution.** For $n \geq 299$, an approximation to the ranks for A-basis values in Table 8.8.13 is:

$$r \approx \frac{n}{100} - 1.645\sqrt{\frac{99n}{10{,}000}} + 0.29 + \frac{19.1}{n} \qquad 8.8.14$$

For n less than 299, an A-allowable cannot be computed. This approximation is exact for all but 23 values of n in the range of the table ($299 \leq n \leq 11691$). For this small percentage of n values (0.2%), the approximation errs by one rank on the conservative side (Reference 8.8.13).

8.8.14 **The Hanson-Koopmans method exponents for A-basis values.** The values in Table 8.8.14 are based on Reference 8.8.14.

8.8.15 **0.1225 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom.** Table 8.8.15 was generated specifically for MIL-HDBK-17. An approximation to the $F_{0.1225}$ values in Table 8.8.15 is

$$F_{0.1225} = \exp\left(2\delta\left[1 + \frac{z^2-1}{3} - \frac{4\sigma^2}{3}\right] + 2\sigma z\sqrt{1 + \frac{\sigma^2(z^2-3)}{6}}\right) \qquad 8.8.15(a)$$

where

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$$\delta = 0.5\{1/(\gamma_2 - 1) - 1/(\gamma_1 - 1)\} \qquad 8.8.4(b)$$

$$\sigma^2 = 0.5\{1/(\gamma_2 - 1) + 1/(\gamma_1 - 1)\} \qquad 8.8.4(c)$$

$z = -1.1625$

γ_1 = numerator degrees of freedom

γ_2 = denominator degrees of freedom.

(See Reference 8.8.4.)

These equations are not valid when either γ_1 or γ_2 equals one. The following equations are to be used for these special cases:

For $\gamma_1 = 1$

$$F_{0.1225} = \left[0.1541391 + \frac{0.03945031}{\gamma_2} + \frac{0.00543174}{\gamma_2^2} - \frac{0.00585461}{\gamma_2^3} - \frac{0.00165541}{\gamma_2^4}\right]^2 \qquad 8.8.15(b)$$

For $\gamma_2 = 1$

$$F_{0.1225} = \left[0.1541391 + \frac{0.03945031}{\gamma_1} + \frac{0.00543174}{\gamma_1^2} - \frac{0.00585461}{\gamma_1^3} - \frac{0.00165541}{\gamma_1^4}\right]^{-2} \qquad 8.8.15(c)$$

8.8.16 **One-sided tolerance factors for A-basis calculations, T_A, for the one-way random effects ANOVA, 0.95 confidence.** Table 8.8.16 was generated specifically for MIL-HDBK-17 based on the algorithm of the noncentral t-distribution in Reference 8.5.8(a).

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Table 8.8.1 *One-sided tolerance limit factors, k_B, for the normal distribution, 0.95 confidence*, continued on next page.

N = 2 - 137							
n	k	n	k	n	k	n	k
2	20.581	36	1.725	70	1.582	104	1.522
3	6.157	37	1.718	71	1.579	105	1.521
4	4.163	38	1.711	72	1.577	106	1.519
5	3.408	39	1.704	73	1.575	107	1.518
6	3.007	40	1.698	74	1.572	108	1.517
7	2.756	41	1.692	75	1.570	109	1.516
8	2.583	42	1.686	76	1.568	110	1.515
9	2.454	43	1.680	77	1.566	111	1.513
10	2.355	44	1.675	78	1.564	112	1.512
11	2.276	45	1.669	79	1.562	113	1.511
12	2.211	46	1.664	80	1.560	114	1.510
13	2.156	47	1.660	81	1.558	115	1.509
14	2.109	48	1.655	82	1.556	116	1.508
15	2.069	49	1.650	83	1.554	117	1.507
16	2.034	50	1.646	84	1.552	118	1.506
17	2.002	51	1.642	85	1.551	119	1.505
18	1.974	52	1.638	86	1.549	120	1.504
19	1.949	53	1.634	87	1.547	121	1.503
20	1.927	54	1.630	88	1.545	122	1.502
21	1.906	55	1.626	89	1.544	123	1.501
22	1.887	56	1.623	90	1.542	124	1.500
23	1.870	57	1.619	91	1.540	125	1.499
24	1.854	58	1.616	92	1.539	126	1.498
25	1.839	59	1.613	93	1.537	127	1.497
26	1.825	60	1.609	94	1.536	128	1.496
27	1.812	61	1.606	95	1.534	129	1.495
28	1.800	62	1.603	96	1.533	130	1.494
29	1.789	63	1.600	97	1.531	131	1.493
30	1.778	64	1.597	98	1.530	132	1.492
31	1.768	65	1.595	99	1.529	133	1.492
32	1.758	66	1.592	100	1.527	134	1.491
33	1.749	67	1.589	101	1.526	135	1.490
34	1.741	68	1.587	102	1.525	136	1.489
35	1.733	69	1.584	103	1.523	137	1.488

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Table 8.8.1 *One-sided tolerance limit factors, k_B for the normal distribution, 0.95 confidence*, concluded.

n = 138 - ∞							
n	k	n	k	n	k	n	k
138	1.487	172	1.464	230	1.438	400	1.398
139	1.487	173	1.464	235	1.436	425	1.395
140	1.486	174	1.463	240	1.434	450	1.391
141	1.485	175	1.463	345	1.433	475	1.388
142	1.484	176	1.462	250	1.431	500	1.386
143	1.483	177	1.461	255	1.430	525	1.383
144	1.483	178	1.461	260	1.428	550	1.381
145	1.482	179	1.460	265	1.427	575	1.378
146	1.481	180	1.460	270	1.425	600	1.376
147	1.480	181	1.459	275	1.424	625	1.374
148	1.480	182	1.459	280	1.422	650	1.372
149	1.479	183	1.458	285	1.421	675	1.371
150	1.478	184	1.458	290	1.420	700	1.369
151	1.478	185	1.457	295	1.419	725	1.367
152	1.477	186	1.457	300	1.417	750	1.366
153	1.476	187	1.456	305	1.416	775	1.364
154	1.475	188	1.456	310	1.415	800	1.363
155	1.475	189	1.455	315	1.414	825	1.362
156	1.474	190	1.455	320	1.413	850	1.361
157	1.473	191	1.454	325	1.412	875	1.359
158	1.473	192	1.454	330	1.411	900	1.358
159	1.472	193	1.453	335	1.410	925	1.357
160	1.472	194	1.453	340	1.409	950	1.356
161	1.471	195	1.452	345	1.408	975	1.355
162	1.470	196	1.452	350	1.407	1000	1.354
163	1.470	197	1.451	355	1.406	1500	1.340
164	1.469	198	1.451	360	1.405	2000	1.332
165	1.468	199	1.450	365	1.404	3000	1.323
166	1.468	200	1.450	370	1.403	5000	1.313
167	1.467	205	1.448	375	1.402	10000	1.304
168	1.467	210	1.446	380	1.402	∞	1.282
169	1.466	215	1.444	385	1.401		
170	1.465	220	1.442	390	1.400		
171	1.465	225	1.440	395	1.399		

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TABLE 8.8.2 *One-sided tolerance limit factors, V_B for the Weibull distribution, 0.95 confidence,* continued on next page.

n = 10 - 192							
n	V	n	V	n	V	n	V
10	6.711	45	4.764	80	4.477	130	4.309
11	6.477	46	4.751	81	4.471	132	4.305
12	6.286	47	4.738	82	4.466	134	4.301
13	6.127	48	4.725	83	4.462	136	4.296
14	5.992	49	4.713	84	4.457	138	4.292
15	5.875	50	4.702	85	4.452	140	4.288
16	5.774	51	4.691	86	4.448	142	4.284
17	5.684	52	4.680	87	4.443	144	4.280
18	5.605	53	4.670	88	4.439	146	4.277
19	5.533	54	4.659	89	4.435	148	4.273
20	5.469	55	4.650	90	4.431	150	4.269
21	5.412	56	4.640	91	4.427	152	4.266
22	5.359	57	4.631	92	4.423	154	4.262
23	5.310	58	4.622	93	4.419	156	4.259
24	5.265	59	4.631	94	4.415	158	4.256
25	5.224	60	4.605	95	4.411	160	4.253
26	5.186	61	4.597	96	4.407	162	4.249
27	5.150	62	4.589	97	4.404	164	4.246
28	5.117	63	4.582	98	4.400	166	4.243
29	5.086	64	4.574	99	4.396	168	4.240
30	5.057	65	4.567	100	4.393	170	4.237
31	5.030	66	4.560	102	4.386	172	4.234
32	5.003	67	4.553	104	4.380	174	4.232
33	4.979	68	4.546	106	4.373	176	4.229
34	4.956	69	4.539	108	4.367	178	4.226
35	4.934	70	4.533	110	4.361	180	4.224
36	4.913	71	4.527	112	4.355	182	4.221
37	4.893	72	4.521	114	4.349	184	4.218
38	4.875	73	4.515	116	4.344	186	4.216
39	4.857	74	4.509	118	4.339	188	4.213
40	4.840	75	4.503	120	4.334	190	4.211
41	4.823	76	4.498	122	4.328	192	4.208
42	4.807	77	4.492	124	4.323		
43	4.792	78	4.487	126	4.317		
44	4.778	79	4.482	128	4.314		

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TABLE 8.8.2 *One-sided tolerance limit factors, V_B for the Weibull distribution, 0.95 confidence*, concluded.

n = 194 - ∞					
n	V	n	V	n	V
194	4.206	300	4.119	850	3.992
196	4.204	310	4.113	875	3.989
198	4.201	320	4.108	900	3.986
200	4.199	330	4.103	925	3.983
204	4.195	340	4.098	950	3.981
208	4.191	350	4.093	975	3.979
212	4.186	360	4.089	1000	3.976
216	4.182	370	4.085	1100	3.968
220	4.179	380	4.081	1200	3.960
224	4.175	390	4.077	1300	3.954
228	4.171	400	4.073	1400	3.948
232	4.168	425	4.076	1500	3.943
236	4.164	450	4.067	1600	3.939
240	4.161	475	4.060	1700	3.934
244	4.157	500	4.053	1800	3.931
248	4.154	525	4.047	1900	3.927
252	4.151	550	4.041	2000	3.924
256	4.148	575	4.035	3000	3.901
260	4.145	600	4.030	4000	3.887
264	4.142	625	4.025	5000	3.878
268	4.140	650	4.020	6000	3.872
272	4.137	675	4.016	7000	3.866
276	4.134	700	4.012	8000	3.862
280	4.131	725	4.008	9000	3.859
284	4.129	750	4.005	10000	3.856
288	4.126	775	4.001	15000	3.846
292	4.124	800	3.998	20000	3.840
296	4.121	825	3.995	∞	3.803

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TABLE 8.8.3 *Ranks, r, of observations for determining B-values for an unknown distribution.*

n	r	n	r	n	r
28	†	660	54	3901	360
29	1	682	56	4005	370
46	2	704	58	4109	380
61	3	726	60	4213	390
76	4	781	65	4317	400
89	5	836	70	4421	410
103	6	890	75	4525	420
116	7	945	80	4629	430
129	8	999	85	4733	440
142	9	1053	90	4836	450
154	10	1107	95	4940	460
167	11	1161	100	5044	470
179	12	1269	110	5147	480
191	13	1376	120	5251	490
203	14	1483	130	5354	500
215	15	1590	140	5613	525
227	16	1696	150	5871	550
239	17	1803	160	6130	575
251	18	1909	170	6388	600
263	19	2015	180	6645	625
275	20	2120	190	6903	650
298	22	2226	200	7161	675
321	24	2331	210	7418	700
345	26	2437	220	7727	730
368	28	2542	230	8036	760
391	30	2647	240	8344	790
413	32	2752	250	8652	820
436	34	2857	260	8960	850
459	36	2962	270	9268	880
481	38	3066	280	9576	910
504	40	3171	290	9884	940
526	42	3276	300	10191	970
549	44	3380	310	10499	1000[1]
571	46	3484	320		
593	48	3589	330		
615	50	3693	340		
638	52	3797	350		

†B-value does not exist for n < 28.
[1]For n > 10499, use Equation 8.8.3.

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TABLE 8.8.4 *0.20 Quantiles of the F distribution associated with γ_1 and γ_2 degrees of freedom*, continued on next page.

	γ_1 numerator degrees of freedom								
	1	2	3	4	5	6	7	8	9
1	0.106	0.281	0.373	0.425	0.459	0.482	0.500	0.513	0.523
2	0.0833	0.250	0.347	0.405	0.443	0.470	0.489	0.505	0.517
3	0.0765	0.241	0.341	0.403	0.444	0.473	0.495	0.513	0.526
4	0.0733	0.236	0.338	0.403	0.447	0.478	0.502	0.520	0.535
5	0.0714	0.233	0.337	0.404	0.449	0.482	0.507	0.526	0.542
γ_2									
6	0.0701	0.232	0.337	0.404	0.451	0.485	0.511	0.531	0.548
d 7	0.0693	0.230	0.336	0.405	0.453	0.488	0.514	0.535	0.552
e 8	0.0686	0.230	0.336	0.406	0.454	0.490	0.517	0.539	0.556
n 9	0.0681	0.229	0.336	0.406	0.455	0.492	0.520	0.542	0.560
o 10	0.0677	0.228	0.336	0.407	0.456	0.493	0.522	0.544	0.562
m									
i 11	0.0674	0.228	0.336	0.407	0.457	0.495	0.523	0.546	0.565
n 12	0.0671	0.227	0.336	0.407	0.458	0.496	0.525	0.548	0.567
a 13	0.0669	0.227	0.335	0.408	0.459	0.497	0.526	0.550	0.569
t 14	0.0667	0.227	0.335	0.408	0.459	0.498	0.527	0.551	0.570
o 15	0.0665	0.227	0.335	0.408	0.460	0.498	0.528	0.552	0.572
r									
16	0.0664	0.226	0.335	0.408	0.460	0.499	0.529	0.553	0.573
d 17	0.0662	0.226	0.335	0.409	0.461	0.500	0.530	0.554	0.574
e 18	0.0661	0.226	0.335	0.409	0.461	0.500	0.531	0.555	0.575
g 19	0.0660	0.226	0.335	0.409	0.461	0.501	0.532	0.556	0.576
r 20	0.0659	0.226	0.335	0.409	0.462	0.501	0.532	0.557	0.577
e									
e 21	0.0658	0.226	0.335	0.409	0.462	0.502	0.533	0.558	0.578
s 22	0.0658	0.225	0.335	0.409	0.462	0.502	0.533	0.558	0.579
23	0.0657	0.225	0.335	0.409	0.463	0.503	0.534	0.559	0.580
o 24	0.0656	0.225	0.335	0.410	0.463	0.503	0.534	0.559	0.580
f 25	0.0656	0.225	0.335	0.410	0.463	0.503	0.535	0.560	0.581
f 26	0.0655	0.225	0.335	0.410	0.463	0.503	0.535	0.560	0.582
r 27	0.0655	0.225	0.335	0.410	0.463	0.504	0.535	0.561	0.582
e 28	0.0654	0.225	0.335	0.410	0.464	0.504	0.536	0.561	0.583
e 29	0.0654	0.225	0.335	0.410	0.464	0.504	0.536	0.562	0.583
d 30	0.0653	0.225	0.335	0.410	0.464	0.504	0.546	0.562	0.583
o									
m 40	0.0650	0.224	0.335	0.411	0.465	0.506	0.539	0.565	0.587
60	0.0648	0.224	0.335	0.411	0.476	0.508	0.541	0.568	0.590
120	0.0645	0.224	0.335	0.412	0.477	0.510	0.543	0.571	0.594
∞	0.0643	0.223	0.335	0.412	0.478	0.512	0.556	0.574	0.597

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TABLE 8.8.4 *0.20 Quantiles of the F distribution associated with ν_1 and ν_2 degrees of freedom*, concluded

	ν_1 numerator degrees of freedom									
	10	12	15	20	24	30	40	60	120	∞
1	0.531	0.544	0.556	0.569	0.576	0.582	0.589	0.596	0.602	0.607
2	0.527	0.542	0.557	0.573	0.581	0.589	0.597	0.605	0.613	0.621
3	0.537	0.554	0.573	0.590	0.599	0.608	0.618	0.627	0.637	0.645
4	0.547	0.566	0.585	0.605	0.615	0.625	0.635	0.646	0.657	0.663
5	0.555	0.575	0.595	0.617	0.628	0.639	0.650	0.662	0.674	0.685
ν_2										
6	0.561	0.582	0.604	0.627	0.638	0.650	0.663	0.675	0.688	0.699
d 7	0.566	0.588	0.611	0.635	0.647	0.660	0.673	0.686	0.700	0.712
e 8	0.517	0.593	0.617	0.642	0.655	0.668	0.682	0.696	0.710	0.724
n 9	0.574	0.598	0.622	0.648	0.661	0.675	0.689	0.704	0.719	0.735
o 10	0.578	0.601	0.626	0.653	0.667	0.681	0.696	0.712	0.727	0.744
m										
i 11	0.580	0.605	0.630	0.658	0.672	0.687	0.702	0.718	0.735	0.752
n 12	0.583	0.607	0.634	0.662	0.676	0.692	0.707	0.724	0.741	0.757
a 13	0.585	0.610	0.637	0.665	0.680	0.696	0.712	0.729	0.747	0.762
t 14	0.587	0.612	0.639	0.668	0.684	0.700	0.716	0.734	0.752	0.771
o 15	0.588	0.614	0.642	0.671	0.687	0.703	0.720	0.738	0.757	0.776
r										
16	0.590	0.616	0.644	0.674	0.690	0.707	0.724	0.742	0.761	0.781
d 17	0.591	0.618	0.646	0.676	0.693	0.709	0.727	0.746	0.766	0.786
e 18	0.592	0.619	0.648	0.679	0.695	0.712	0.730	0.749	0.769	0.789
g 19	0.593	0.620	0.649	0.681	0.697	0.715	0.733	0.752	0.773	0.793
r 20	0.594	0.622	0.651	0.682	0.699	0.717	0.736	0.755	0.776	0.796
e										
e 21	0.595	0.623	0.652	0.684	0.701	0.719	0.738	0.758	0.779	0.802
s 22	0.596	0.624	0.653	0.686	0.703	0.721	0.740	0.761	0.782	0.805
23	0.597	0.625	0.655	0.687	0.705	0.723	0.742	0.763	0.785	0.806
o 24	0.598	0.626	0.656	0.689	0.706	0.725	0.744	0.765	0.788	0.810
f 25	0.599	0.627	0.657	0.690	0.708	0.726	0.746	0.767	0.790	0.812
f 26	0.599	0.627	0.658	0.691	0.709	0.728	0.748	0.769	0.793	0.816
r 27	0.600	0.628	0.659	0.692	0.710	0.729	0.750	0.771	0.796	0.820
e 28	0.600	0.629	0.660	0.693	0.711	0.731	0.751	0.773	0.797	0.821
e 29	0.601	0.629	0.660	0.694	0.713	0.732	0.753	0.775	0.799	0.823
d 30	0.601	0.630	0.661	0.695	0.714	0.733	0.754	0.776	0.801	0.827
o										
m 40	0.605	0.635	0.667	0.702	0.722	0.743	0.765	0.789	0.816	0.845
60	0.609	0.640	0.673	0.710	0.731	0.753	0.777	0.804	0.834	0.869
120	0.613	0.645	0.680	0.719	0.741	0.765	0.791	0.822	0.857	0.902
∞	0.618	0.651	0.688	0.729	0.753	0.779	0.809	0.843	0.889	0.977

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TABLE 8.8.5 *0.95 Quantiles of the F distribution associated with γ_1 and γ_2 degrees of freedom,* continued on next page.

		γ_1 numerator degrees of freedom								
		1	2	3	4	5	6	7	8	9
	1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54
	2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38
γ_2	3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
	4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
	5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77
d										
e	6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10
n	7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68
o	8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39
m	9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18
i	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02
n										
a	11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90
t	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80
o	13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71
r	14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65
	15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59
d	16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54
e	17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49
g	18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46
r	19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42
e	20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39
e										
s	21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37
	22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34
	23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32
o	24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30
f	25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28
	26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27
f	27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25
r	28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24
e	29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22
e	30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21
d										
o	40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12
m	60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04
	120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96
	∞	3.84	3.00	2.61	2.37	2.21	2.10	2.01	1.94	1.88

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TABLE 8.8.5 *0.95 Quantiles of the F distribution associated with γ_1 and γ_2 degrees of freedom,* Concluded.

		γ_1 numerator degrees of freedom									
		10	12	15	20	24	30	40	60	120	∞
	1	241.88	243.91	245.95	248.01	249.05	250.10	251.14	252.20	253.25	254.31
	2	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.51
γ_2	3	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
	4	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
	5	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
d											
e	6	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
n	7	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
o	8	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
m	9	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
i	10	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
n											
a	11	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
t	12	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
o	13	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
r	14	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
	15	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
d	16	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
e	17	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
g	18	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
r	19	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
e	20	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
e											
s	21	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
	22	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
	23	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
o	24	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
f	25	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
	26	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
f	27	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
r	28	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
e	29	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
e	30	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
d											
o	40	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
m	60	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
	120	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
	∞	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

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TABLE 8.8.6. *One-sided tolerance factors, T, for the one-way random effects ANOVA, 0.95 confidence*, continued on next page.

$\hat{A} = 0.00$

	k*											
	2	3	4	5	6	7	8	9	10	15	20	
1	20.588	6.157	4.163	3.408	3.007	2.756	2.583	2.454	2.355	2.069	1.927	1.282
2	4.559	3.062	2.600	2.363	2.215	2.112	2.035	1.975	1.927	1.778	1.698	1.282
3	3.156	2.480	2.220	2.073	1.977	1.907	1.854	1.813	1.778	1.670	1.609	1.282
4	2.661	2.226	2.039	1.929	1.855	1.801	1.759	1.726	1.698	1.610	1.560	1.282
5	2.404	2.079	1.930	1.840	1.779	1.733	1.698	1.670	1.646	1.570	1.527	1.282
6	2.244	1.981	1.856	1.779	1.726	1.686	1.655	1.630	1.610	1.542	1.504	1.282
7	2.134	1.911	1.802	1.734	1.686	1.651	1.623	1.601	1.582	1.521	1.486	1.282
n* 8	2.053	1.858	1.760	1.699	1.655	1.623	1.598	1.577	1.560	1.504	1.472	1.282
9	1.990	1.815	1.726	1.670	1.630	1.601	1.577	1.558	1.542	1.490	1.460	1.282
10	1.939	1.781	1.699	1.647	1.610	1.582	1.560	1.542	1.527	1.478	1.450	1.282
11	1.897	1.752	1.676	1.627	1.592	1.566	1.545	1.529	1.515	1.468	1.442	1.282
12	1.862	1.727	1.656	1.610	1.577	1.552	1.533	1.517	1.504	1.460	1.434	1.282
13	1.832	1.706	1.639	1.595	1.564	1.541	1.522	1.507	1.494	1.452	1.428	1.282
14	1.807	1.687	1.623	1.582	1.553	1.530	1.512	1.498	1.486	1.446	1.422	1.282
15	1.784	1.671	1.610	1.570	1.542	1.521	1.504	1.490	1.478	1.440	1.417	1.282
∞	1.282	1.282	1.282	1.282	1.282	1.282	1.282	1.282	1.282	1.282	1.282	1.282

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TABLE 8.8.6. *One-sided tolerance factors, T, for the one-way random effects ANOVA, 0.95 confidence*, continued on next page.

$\hat{R} = 0.25$

n* \ k*	2	3	4	5	6	7	8	9	10	15	20	
1	20.588	6.157	4.163	3.408	3.007	2.756	2.583	2.454	2.355	2.069	1.927	1.282
2	5.341	3.288	2.729	2.455	2.288	2.174	2.089	2.024	1.972	1.810	1.724	1.282
3	3.759	2.717	2.373	2.191	2.075	1.992	1.930	1.882	1.842	1.718	1.650	1.282
4	3.196	2.470	2.208	2.063	1.969	1.901	1.849	1.808	1.775	1.668	1.609	1.282
5	2.908	2.331	2.111	1.987	1.905	1.845	1.799	1.763	1.733	1.638	1.584	1.282
6	2.732	2.241	2.047	1.936	1.861	1.807	1.766	1.732	1.705	1.616	1.566	1.282
7	2.614	2.178	2.002	1.899	1.830	1.780	1.741	1.710	1.684	1.601	1.553	1.282
8	2.529	2.131	1.967	1.872	1.807	1.759	1.722	1.692	1.668	1.588	1.543	1.282
9	2.464	2.095	1.941	1.850	1.788	1.742	1.707	1.679	1.655	1.579	1.536	1.282
10	2.414	2.067	1.919	1.832	1.773	1.729	1.695	1.668	1.645	1.571	1.529	1.282
11	2.373	2.043	1.902	1.818	1.761	1.718	1.685	1.659	1.637	1.565	1.524	1.282
12	2.340	2.023	1.887	1.806	1.750	1.709	1.677	1.651	1.629	1.559	1.519	1.282
13	2.312	2.007	1.875	1.796	1.741	1.701	1.670	1.644	1.623	1.555	1.515	1.282
14	2.288	1.993	1.864	1.787	1.734	1.694	1.664	1.639	1.618	1.551	1.512	1.282
15	2.268	1.981	1.855	1.779	1.727	1.688	1.658	1.634	1.614	1.547	1.509	1.282
∞	1.992	1.807	1.721	1.667	1.630	1.601	1.579	1.560	1.545	1.494	1.465	1.282

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TABLE 8.8.6. *One-sided tolerance factors, T, for the one-way random effects ANOVA, 0.95 confidence*, continued on next page.

$\hat{R} = 0.50$

n* \ k*	2	3	4	5	6	7	8	9	10	15	20	
1	20.588	6.157	4.163	3.408	3.007	2.756	2.583	2.454	2.355	2.069	1.927	1.282
2	6.157	3.511	2.855	2.544	2.358	2.232	2.140	2.069	2.012	1.839	1.747	1.282
3	4.434	2.957	2.522	2.302	2.165	2.070	1.999	1.943	1.898	1.759	1.683	1.282
4	3.807	2.716	2.369	2.187	2.071	1.989	1.928	1.879	1.840	1.717	1.649	1.282
5	3.485	2.581	2.280	2.118	2.014	1.940	1.885	1.841	1.805	1.691	1.628	1.282
6	3.289	2.494	2.221	2.073	1.977	1.908	1.856	1.814	1.781	1.673	1.614	1.282
7	3.157	2.434	2.180	2.041	1.950	1.885	1.835	1.796	1.763	1.661	1.603	1.282
8	3.063	2.390	2.149	2.016	1.930	1.867	1.819	1.781	1.750	1.651	1.595	1.282
9	2.992	2.355	2.125	1.998	1.914	1.853	1.807	1.770	1.740	1.643	1.589	1.282
10	2.936	2.328	2.106	1.983	1.901	1.842	1.797	1.761	1.732	1.637	1.584	1.282
11	2.891	2.306	2.091	1.970	1.891	1.833	1.789	1.754	1.725	1.632	1.580	1.282
12	2.855	2.288	2.078	1.960	1.882	1.826	1.782	1.748	1.719	1.628	1.577	1.282
13	2.824	2.273	2.067	1.951	1.875	1.819	1.776	1.742	1.714	1.624	1.574	1.282
14	2.798	2.260	2.058	1.944	1.868	1.814	1.772	1.738	1.710	1.621	1.571	1.282
15	2.776	2.248	2.050	1.937	1.863	1.809	1.767	1.734	1.707	1.618	1.569	1.282
∞	2.484	2.092	1.937	1.846	1.785	1.740	1.706	1.678	1.655	1.579	1.536	1.282

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TABLE 8.8.6. *One-sided tolerance factors, T, for the one-way random effects ANOVA, 0.95 confidence*, continued on next page.

$\hat{R} = 1.00$

n* \ k*	2	3	4	5	6	7	8	9	10	15	20	
1	20.588	6.157	4.163	3.408	3.007	2.756	2.583	2.454	2.355	2.069	1.927	1.282
2	7.682	3.892	3.063	2.688	2.470	2.325	2.220	2.139	2.076	1.883	1.782	1.282
3	5.787	3.376	2.771	2.482	2.309	2.191	2.104	2.037	1.984	1.820	1.732	1.282
4	5.063	3.148	2.635	2.384	2.230	2.125	2.047	1.986	1.938	1.787	1.706	1.282
5	4.683	3.020	2.557	2.326	2.184	2.086	2.013	1.956	1.910	1.767	1.690	1.282
6	4.451	2.937	2.505	2.288	2.153	2.059	1.990	1.935	1.891	1.754	1.680	1.282
7	4.294	2.880	2.469	2.261	2.131	2.041	1.973	1.921	1.878	1.745	1.672	1.282
8	4.181	2.838	2.442	2.241	2.115	2.027	1.961	1.910	1.868	1.737	1.666	1.282
9	4.095	2.805	2.422	2.225	2.102	2.016	1.951	1.901	1.860	1.732	1.662	1.282
10	4.028	2.780	2.405	2.213	2.092	2.007	1.944	1.894	1.854	1.727	1.658	1.282
11	3.975	2.759	2.392	2.203	2.084	2.000	1.938	1.888	1.849	1.724	1.655	1.282
12	3.931	2.742	2.381	2.194	2.077	1.994	1.932	1.884	1.844	1.720	1.653	1.282
13	3.894	2.727	2.371	2.187	2.071	1.989	1.928	1.880	1.841	1.718	1.650	1.282
14	3.863	2.715	2.363	2.181	2.066	1.985	1.924	1.876	1.838	1.716	1.649	1.282
15	3.836	2.704	2.356	2.176	2.062	1.981	1.921	1.873	1.835	1.714	1.647	1.282
∞	3.486	2.559	2.260	2.102	2.001	1.929	1.875	1.832	1.797	1.686	1.624	1.282

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TABLE 8.8.6. *One-sided tolerance factors, T, for the one-way random effects ANOVA, 0.95 confidence*, continued on next page.

$\hat{R} = 2.00$

n \ k	2	3	4	5	6	7	8	9	10	15	20	
1	20.588	6.157	4.163	3.408	3.007	2.756	2.583	2.454	2.355	2.069	1.927	1.282
2	10.092	4.419	3.337	2.875	2.613	2.441	2.319	2.226	2.153	1.936	1.824	1.282
3	8.129	3.979	3.105	2.716	2.491	2.342	2.234	2.152	2.087	1.891	1.789	1.282
4	7.325	3.779	2.995	2.640	2.432	2.293	2.192	2.115	2.054	1.869	1.771	1.282
5	6.890	3.664	2.931	2.595	2.397	2.264	2.167	2.093	2.034	1.855	1.760	1.282
6	6.617	3.591	2.890	2.566	2.374	2.245	2.151	2.079	2.021	1.846	1.753	1.282
7	6.431	3.539	2.860	2.545	2.358	2.231	2.139	2.068	2.012	1.839	1.748	1.282
8	6.296	3.501	2.838	2.529	2.345	2.221	2.130	2.061	2.005	1.835	1.744	1.282
9	6.193	3.471	2.821	2.517	2.336	2.213	2.123	2.054	1.999	1.831	1.741	1.282
10	6.112	3.448	2.808	2.508	2.328	2.207	2.118	2.050	1.995	1.828	1.739	1.282
11	6.047	3.429	2.797	2.500	2.322	2.202	2.113	2.046	1.991	1.825	1.737	1.282
12	5.994	3.413	2.788	2.493	2.317	2.197	2.110	2.042	1.988	1.823	1.735	1.282
13	5.949	3.400	2.780	2.488	2.312	2.194	2.107	2.039	1.986	1.821	1.734	1.282
14	5.911	3.389	2.773	2.483	2.309	2.190	2.104	2.037	1.984	1.820	1.733	1.282
15	5.878	3.379	2.768	2.479	2.305	2.188	2.101	2.035	1.982	1.818	1.731	1.282
∞	5.442	3.246	2.689	2.422	2.260	2.150	2.069	2.006	1.955	1.800	1.717	1.282

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TABLE 8.8.6. *One-sided tolerance factors, T, for the one-way random effects ANOVA, 0.95 confidence*, concluded.

$\hat{R} = 10.00$

n* \ k*	2	3	4	5	6	7	8	9	10	15	20	
1	20.588	6.157	4.163	3.408	3.007	2.756	2.583	2.454	2.355	2.069	1.927	1.282
2	16.565	5.564	3.893	3.237	2.883	2.658	2.501	2.384	2.294	2.029	1.896	1.282
3	15.433	5.382	3.808	3.183	2.843	2.626	2.474	2.361	2.273	2.016	1.886	1.282
4	14.901	5.294	3.766	3.156	2.823	2.610	2.461	2.350	2.263	2.009	1.881	1.282
5	14.592	5.242	3.741	3.139	2.811	2.600	2.453	2.343	2.257	2.005	1.878	1.282
6	14.390	5.207	3.724	3.129	2.803	2.594	2.448	2.338	2.253	2.002	1.875	1.282
7	14.248	5.183	3.713	3.121	2.797	2.590	2.444	2.335	2.250	2.000	1.874	1.282
8	14.142	5.165	3.704	3.116	2.793	2.586	2.441	2.332	2.248	1.999	1.873	1.282
9	14.061	5.151	3.697	3.111	2.790	2.584	2.439	2.330	2.246	1.998	1.872	1.282
10	13.996	5.139	3.692	3.108	2.787	2.581	2.437	2.329	2.245	1.997	1.871	1.282
11	13.943	5.130	3.687	3.105	2.785	2.580	2.436	2.328	2.244	1.996	1.871	1.282
12	13.899	5.123	3.684	3.102	2.783	2.578	2.434	2.327	2.243	1.996	1.870	1.282
13	13.863	5.116	3.680	3.100	2.782	2.577	2.433	2.326	2.242	1.995	1.870	1.282
14	13.831	5.111	3.678	3.099	2.780	2.576	2.432	2.325	2.241	1.995	1.870	1.282
15	13.804	5.106	3.675	3.097	2.779	2.575	2.432	2.324	2.241	1.994	1.869	1.282
∞	13.429	5.040	3.643	3.076	2.763	2.563	2.421	2.315	2.233	1.989	1.865	1.282

$\hat{R} = \infty$

	2	3	4	5	6	7	8	9	10	15	20	
All n*	20.588	6.157	4.163	3.408	3.007	2.756	2.583	2.454	2.355	2.069	1.927	1.282

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TABLE 8.8.7 *Critical values for the k-sample Anderson-Darling test at the 0.05 significance level, (equal sample sizes).*

n* \ k*	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	2.11	1.80	1.65	1.56	1.50	1.46	1.42	1.39	1.37	1.35	1.33	1.32	1.31	1.29
4	2.20	1.86	1.70	1.60	1.54	1.49	1.45	1.42	1.39	1.37	1.36	1.34	1.33	1.31
5	2.25	1.89	1.73	1.63	1.56	1.51	1.47	1.43	1.41	1.39	1.37	1.35	1.34	1.32
6	2.29	1.92	1.74	1.64	1.57	1.52	1.48	1.45	1.42	1.40	1.38	1.36	1.34	1.33
7	2.32	1.94	1.76	1.65	1.58	1.53	1.49	1.45	1.43	1.40	1.38	1.36	1.35	1.34
8	2.34	1.95	1.77	1.66	1.59	1.53	1.49	1.46	1.43	1.41	1.39	1.37	1.35	1.34
9	2.35	1.96	1.78	1.67	1.59	1.54	1.50	1.46	1.43	1.41	1.39	1.37	1.36	1.34
10	2.37	1.97	1.78	1.67	1.60	1.54	1.50	1.47	1.44	1.41	1.39	1.37	1.36	1.35
11	2.38	1.97	1.79	1.68	1.60	1.55	1.50	1.47	1.44	1.42	1.39	1.38	1.36	1.35
12	2.39	1.98	1.79	1.68	1.60	1.55	1.51	1.47	1.44	1.42	1.40	1.38	1.36	1.35
13	2.39	1.98	1.80	1.68	1.61	1.55	1.51	1.47	1.44	1.42	1.40	1.38	1.36	1.35
14	2.40	1.99	1.80	1.69	1.61	1.55	1.51	1.47	1.44	1.42	1.40	1.38	1.37	1.35
15	2.41	1.99	1.80	1.69	1.61	1.55	1.51	1.48	1.45	1.42	1.40	1.38	1.37	1.35
16	2.41	2.00	1.80	1.69	1.61	1.56	1.51	1.48	1.45	1.42	1.40	1.38	1.37	1.35
17	2.42	2.00	1.81	1.69	1.61	1.56	1.51	1.48	1.45	1.42	1.40	1.38	1.37	1.35
18	2.42	2.00	1.81	1.69	1.62	1.56	1.51	1.48	1.45	1.42	1.40	1.39	1.37	1.35
19	2.42	2.00	1.81	1.70	1.62	1.56	1.52	1.48	1.45	1.43	1.40	1.39	1.37	1.36
20	2.43	2.01	1.81	1.70	1.62	1.56	1.52	1.48	1.45	1.43	1.40	1.39	1.37	1.36
∞	2.49	2.05	1.84	1.72	1.64	1.58	1.53	1.50	1.46	1.44	1.42	1.40	1.38	1.37

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TABLE 8.8.8 *0.95 Quantiles of the chi-squared distribution associated with ν degrees of freedom.*

ν	$\chi^2_{0.95}$
1	3.84
2	5.99
3	7.82
4	9.49
5	11.07
6	12.60
7	14.07
8	15.51
9	16.93
10	18.31
11	19.68
12	21.03
13	22.37
14	23.69
15	25.00
16	26.30
17	27.59
18	28.88
19	30.15
20	31.42
21	32.68
22	33.93
23	35.18
24	36.42
25	37.66
26	38.89
27	40.12
28	41.34
29	42.56
30	43.78

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TABLE 8.8.9 *Critical values for the MNR outlier detection procedure, significance level α = 0.05.*

n	CV	n	CV	n	CV	n	CV	n	CV
-	-	41	3.047	81	3.311	121	3.448	161	3.539
-	-	42	3.057	82	3.315	122	3.451	162	3.541
3	1.154	43	3.067	83	3.319	123	3.453	163	3.543
4	1.481	44	3.076	84	3.323	124	3.456	164	3.545
5	1.715	45	3.085	85	3.328	125	3.459	165	3.547
6	1.887	46	3.094	86	3.332	126	3.461	166	3.549
7	2.020	47	3.103	87	3.336	127	3.464	167	3.551
8	2.127	48	3.112	88	3.340	128	3.466	168	3.552
9	2.215	49	3.120	89	3.344	129	3.469	169	3.554
10	2.290	50	3.128	90	3.348	130	3.471	170	3.556
11	2.355	51	3.136	91	3.352	131	3.474	171	3.558
12	2.412	52	3.144	92	3.355	132	3.476	172	3.560
13	2.462	53	3.151	93	3.359	133	3.479	173	3.561
14	2.507	54	3.159	94	3.363	134	3.481	174	3.563
15	2.548	55	3.166	95	3.366	135	3.483	175	3.565
16	2.586	56	3.173	96	3.370	136	3.486	176	3.567
17	2.620	57	3.180	97	3.374	137	3.488	177	3.568
18	2.652	58	3.187	98	3.377	138	3.491	178	3.570
19	2.681	59	3.193	99	3.381	139	3.493	179	3.572
20	2.708	60	3.200	100	3.384	140	3.495	180	3.574
21	2.734	61	3.206	101	3.387	141	3.497	181	3.575
22	2.758	62	3.212	102	3.391	142	3.500	182	3.577
23	2.780	63	3.218	103	3.394	143	3.502	183	3.579
24	2.802	64	3.224	104	3.397	144	3.504	184	3.580
25	2.822	65	3.230	105	3.401	145	3.506	185	3.582
26	2.841	66	3.236	106	3.404	146	3.508	186	3.584
27	2.859	67	3.241	107	3.407	147	3.511	187	3.585
28	2.876	68	3.247	108	3.410	148	3.513	188	3.587
29	2.893	69	3.252	109	3.413	149	3.515	189	3.588
30	2.908	70	3.258	110	3.416	150	3.517	190	3.590
31	2.924	71	3.263	111	3.419	151	3.519	191	3.592
32	2.938	72	3.268	112	3.422	152	3.521	192	3.593
33	2.952	73	3.273	113	3.425	153	3.523	193	3.595
34	2.965	74	3.278	114	3.428	154	3.525	194	3.596
35	2.978	75	3.283	115	3.431	155	3.527	195	3.598
36	2.991	76	3.288	116	3.434	156	3.529	196	3.599
37	3.003	77	3.292	117	3.437	157	3.531	197	3.601
38	3.014	78	3.297	118	3.440	158	3.533	198	3.603
39	3.025	79	3.302	119	3.442	159	3.535	199	3.604
40	3.036	80	3.306	120	3.445	160	3.537	200	3.606

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TABLE 8.8.10 *Nonparametric B-basis material properties for small samples (Reference 8.5.7.2).*

n	r	k
2	2	35.177
3	3	7.859
4	4	4.505
5	4	4.101
6	5	3.064
7	5	2.858
8	6	2.382
9	6	2.253
10	6	2.137
11	7	1.897
12	7	1.814
13	7	1.738
14	8	1.599
15	8	1.540
16	8	1.485
17	8	1.434
18	9	1.354
19	9	1.311
20	10	1.253
21	10	1.218
22	10	1.184
23	11	1.143
24	11	1.114
25	11	1.087
26	11	1.060
27	11	1.035
28	12	1.010

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TABLE 8.8.11 *One-sided tolerance limit factors, V_A, for the Weibull distribution, 0.95 confidence (Reference 8.8.12)*, continued on next page.

N	V_A	N	V_A	N	V_A	N	V_A
10	12.573	44	8.629	78	8.038	124	7.706
11	12.093	45	8.600	79	8.028	126	7.697
12	11.701	46	8.573	80	8.017	128	7.687
13	11.375	47	8.547	81	8.007	130	7.678
14	11.098	48	8.522	82	7.997	132	7.669
15	10.861	49	8.498	83	7.988	134	7.660
16	10.654	50	8.474	84	7.978	136	7.652
17	10.472	51	8.452	85	7.969	138	7.643
18	10.311	52	8.430	86	7.960	140	7.635
19	10.166	53	8.409	87	7.951	142	7.627
20	10.035	54	8.389	88	7.942	144	7.619
21	9.917	55	8.369	89	7.933	146	7.612
22	9.809	56	8.349	90	7.925	148	7.604
23	9.710	57	8.330	91	7.916	150	7.597
24	9.619	58	8.313	92	7.908	152	7.590
25	9.535	59	8.295	93	7.900	154	7.583
26	9.457	60	8.278	94	7.892	156	7.576
27	9.385	61	8.262	95	7.884	158	7.569
28	9.318	62	8.246	96	7.877	160	7.563
29	9.251	63	8.230	97	7.867	162	7.556
30	9.195	64	8.215	98	7.862	164	7.550
31	9.139	65	8.200	99	7.855	166	7.544
32	9.087	66	8.186	100	7.845	168	7.538
33	9.037	67	8.172	102	7.834	170	7.532
34	8.990	68	8.158	104	7.820	172	7.526
35	8.946	69	8.145	106	7.811	174	7.520
36	8.904	70	8.132	108	7.795	176	7.515
37	8.863	71	8.119	110	7.783	178	7.509
38	8.825	72	8.107	112	7.771	180	7.504
39	8.789	73	8.095	114	7.759	182	7.499
40	8.754	74	8.083	116	7.748	184	7.493
41	8.721	75	8.071	118	7.737	186	7.488
42	8.689	76	8.060	120	7.727	188	7.483
43	8.658	77	8.049	122	7.717	190	7.478

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TABLE 8.8.11 *One-sided tolerance limit factors, V_A, for the Weibull distribution, 0.95 confidence (Reference 8.8.12)*, concluded.

N	V_A	N	V_A	N	V_A	N	V_A
192	7.473	268	7.333	475	7.152	1000	6.989
194	7.469	272	7.328	500	7.138	1100	6.972
196	7.454	276	7.322	525	7.126	1200	6.958
198	7.459	280	7.317	550	7.114	1300	6.945
200	7.455	284	7.312	575	7.103	1400	6.934
204	7.446	288	7.307	600	7.093	1500	6.924
208	7.437	292	7.302	625	7.084	1600	6.915
212	7.429	296	7.297	650	7.075	1700	6.907
216	7.421	300	7.292	675	7.066	1800	6.899
220	7.413	310	7.280	700	7.058	1900	6.892
224	7.404	320	7.270	725	7.051	2000	6.886
228	7.397	330	7.259	750	7.044	3000	6.841
232	7.390	340	7.249	775	7.037		
236	7.383	350	7.240	800	7.031		
240	7.376	360	7.229	825	7.025		
244	7.370	370	7.222	850	7.019		
248	7.363	380	7.214	875	7.013		
252	7.357	390	7.206	900	7.008		
256	7.351	400	7.198	925	7.003		
260	7.345	425	7.183	950	6.998		
264	7.339	450	7.167	975	6.993		

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TABLE 8.8.12 *One-sided tolerance limit factors, k_A, for the normal distribution, 0.95 confidence (Reference 8.8.12)*, continued on next page.

n	k_A	n	k_A	n	k_A	n	k_A
2	37.094	36	2.983	70	2.765	104	2.676
3	10.553	37	2.972	71	2.762	105	2.674
4	7.042	38	2.961	72	2.758	106	2.672
5	5.741	39	2.951	73	2.755	107	2.671
6	5.062	40	2.941	74	2.751	108	2.669
7	4.642	41	2.932	75	2.748	109	2.667
8	4.354	42	2.923	76	2.745	110	2.665
9	4.143	43	2.914	77	2.742	111	2.663
10	3.981	44	2.906	78	2.739	112	2.662
11	3.852	45	2.898	79	2.736	113	2.660
12	3.747	46	2.890	80	2.733	114	2.658
13	3.659	47	2.883	81	2.730	115	2.657
14	3.585	48	2.876	82	2.727	116	2.655
15	3.520	49	2.869	83	2.724	117	2.654
16	3.464	50	2.862	84	2.721	118	2.652
17	3.414	51	2.856	85	2.719	119	2.651
18	3.370	52	2.850	86	2.716	120	2.649
19	3.331	53	2.844	87	2.714	121	2.648
20	3.295	54	2.838	88	2.711	122	2.646
21	3.263	55	2.833	89	2.709	123	2.645
22	3.233	56	2.827	90	2.706	124	2.643
23	3.206	57	2.822	91	2.704	125	2.642
24	3.181	58	2.817	92	2.701	126	2.640
25	3.158	59	2.812	93	2.699	127	2.639
26	3.136	60	2.807	94	2.697	128	2.638
27	3.116	61	2.802	95	2.695	129	2.636
28	3.098	62	2.798	96	2.692	130	2.635
29	3.080	63	2.793	97	2.690	131	2.634
30	3.064	64	2.789	98	2.688	132	2.632
31	3.048	65	2.785	99	2.686	133	2.631
32	3.034	66	2.781	100	2.684	134	2.630
33	3.020	67	2.777	101	2.682	135	2.628
34	3.007	68	2.773	102	2.680	136	2.627
35	2.995	69	2.769	103	2.678	137	2.626

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TABLE 8.8.12 *One-sided tolerance limit factors, k_A, for the normal distribution, 0.95 confidence (Reference 8.8.12)*, concluded.

n	k_A	n	k_A	n	k_A	n	k_A
138	2.625	172	2.591	230	2.552	400	2.494
139	2.624	173	2.590	235	2.549	425	2.489
140	2.622	174	2.589	240	2.547	450	2.484
141	2.621	175	2.588	245	2.544	475	2.480
142	2.620	176	2.587	250	2.542	500	2.475
143	2.619	177	2.587	255	2.540	525	2.472
144	2.618	178	2.586	260	2.537	550	2.468
145	2.617	179	2.585	265	2.535	575	2.465
146	2.616	180	2.584	270	2.533	600	2.462
147	2.615	181	2.583	275	2.531	625	2.459
148	2.613	182	2.583	280	2.529	650	2.456
149	2.612	183	2.582	285	2.527	675	2.454
150	2.611	184	2.581	290	2.525	700	2.451
151	2.610	185	2.580	295	2.524	725	2.449
152	2.609	186	2.580	300	2.522	750	2.447
153	2.608	187	2.579	305	2.520	775	2.445
154	2.607	188	2.578	310	2.518	800	2.443
155	2.606	189	2.577	315	2.517	825	2.441
156	2.605	190	2.577	320	2.515	850	2.439
157	2.604	191	2.576	325	2.514	875	2.438
158	2.603	192	2.575	330	2.512	900	2.436
159	2.602	193	2.575	335	2.511	925	2.434
160	2.601	194	2.574	340	2.509	950	2.433
161	2.600	195	2.573	345	2.508	975	2.432
162	2.600	196	2.572	350	2.506	1000	2.430
163	2.599	197	2.572	355	2.505	1500	2.411
164	2.598	198	2.571	360	2.504	2000	2.399
165	2.597	199	2.570	365	2.502	3000	2.385
166	2.596	200	2.570	370	2.501	5000	2.372
167	2.595	205	2.566	375	2.500	10,000	2.358
168	2.594	210	2.563	380	2.499	∞	2.326
169	2.593	215	2.560	385	2.498		
170	2.592	220	2.557	390	2.496		
171	2.592	225	2.555	395	2.495		

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TABLE 8.8.13 *Ranks, r, of observations for determining A-basis values for an unknown distribution* (Reference (8.6.8 (c)).

n	r	n	r	n	r	n	r
≤298	†	3603	27	6657	54	9627	81
299	1	3719	28	6769	55	9736	82
473	2	3834	29	6879	56	9854	83
628	3	3949	30	6990	57	9954	84
773	4	4064	31	7100	58	10063	85
913	5	4179	32	7211	59	10172	86
1049	6	4293	33	7322	60	10281	87
1182	7	4407	34	7432	61	10390	88
1312	8	4521	35	7543	62	10498	89
1441	9	4635	36	7653	63	10607	90
1568	10	4749	37	7763	64	10716	91
1693	11	4862	38	7874	65	10824	92
1818	12	4975	39	7984	66	10933	93
1941	13	5088	40	8094	67	11041	94
2064	14	5201	41	8204	68	11150	95
2185	15	5314	42	8314	69	11258	96
2306	16	5427	43	8423	70	11366	97
2426	17	5539	44	8533	71	11475	98
2546	18	5651	45	8643	72	11583	99
2665	19	5764	46	8753	73	11691	100[1]
2784	20	5876	47	8862	74		
2902	21	5988	48	8972	75		
3020	22	6099	49	9081	76		
3137	23	6211	50	9190	77		
3254	24	6323	51	9300	78		
3371	25	6434	52	9409	79		
3487	26	6545	53	9518	80		

† A-value does not exist for n < 299
[1] For N > 11691, use Equation 8.8.13

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TABLE 8.8.14 *The Hanson-Koopmans method exponents for A-basis values (Reference 8.6.8(b)).*

n	k	n	k	n	k
2	80.00380	38	1.79301	98	1.31553
3	16.91220	39	1.77546	100	1.30806
4	9.49579	40	1.75868	105	1.29036
5	6.89049	41	1.74260	110	1.27392
6	5.57681	42	1.72718	115	1.25859
7	4.78352	43	1.71239	120	1.24425
8	4.25011	44	1.69817	125	1.23080
9	3.86502	45	1.68449	130	1.21814
10	3.57267	46	1.67132	135	1.20620
11	3.34227	47	1.65862	140	1.19491
12	3.15540	48	1.64638	145	1.18421
13	3.00033	49	1.63456	150	1.17406
14	2.86924	50	1.62313	155	1.16440
15	2.75672	52	1.60139	160	1.15519
16	2.65889	54	1.58101	165	1.14640
17	2.57290	56	1.56184	170	1.13801
18	2.49660	58	1.54377	175	1.12997
19	2.42833	60	1.52670	180	1.12226
20	2.36683	62	1.51053	185	1.11486
21	2.31106	64	1.49520	190	1.10776
22	2.26020	66	1.48063	195	1.10092
23	2.21359	68	1.46675	200	1.09434
24	2.17067	70	1.45352	205	1.08799
25	2.13100	72	1.44089	210	1.08187
26	2.09419	74	1.42881	215	1.07595
27	2.05991	76	1.41724	220	1.07024
28	2.02790	78	1.40614	225	1.06471
29	1.99791	80	1.39549	230	1.05935
30	1.96975	82	1.38525	235	1.05417
31	1.94324	84	1.37541	240	1.04914
32	1.91822	86	1.36592	245	1.04426
33	1.89457	88	1.35678	250	1.03952
34	1.87215	90	1.34796	275	1.01773
35	1.85088	92	1.33944	299	1.00000
36	1.83065	94	1.33120		
37	1.81139	96	1.32324		

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TABLE 8.8.15 *0.1225 Quantiles of the F-distribution associated with ν_1 and ν_2 degrees of freedom*, continued on next page.

		ν_1 degrees of freedom								
		1	2	3	4	5	6	7	8	9
ν_2 degrees of freedom	1	0.0380	0.149	0.220	0.262	0.290	0.310	0.324	0.335	0.344
	2	0.0305	0.140	0.218	0.269	0.304	0.329	0.348	0.362	0.374
	3	0.0281	0.137	0.220	0.276	0.314	0.343	0.364	0.381	0.395
	4	0.0270	0.135	0.221	0.280	0.322	0.352	0.376	0.395	0.410
	5	0.0263	0.134	0.222	0.283	0.327	0.359	0.385	0.405	0.421
	6	0.0259	0.134	0.223	0.285	0.331	0.365	0.391	0.412	0.429
	7	0.0256	0.133	0.223	0.287	0.334	0.369	0.396	0.418	0.436
	8	0.0253	0.133	0.223	0.289	0.336	0.372	0.400	0.423	0.442
	9	0.0251	0.133	0.224	0.290	0.338	0.375	0.404	0.427	0.446
	10	0.0250	0.132	0.224	0.291	0.340	0.377	0.407	0.430	0.450
	11	0.0249	0.132	0.224	0.291	0.341	0.379	0.409	0.433	0.453
	12	0.0248	0.132	0.225	0.292	0.342	0.381	0.411	0.436	0.456
	13	0.0247	0.132	0.225	0.293	0.343	0.382	0.413	0.438	0.459
	14	0.0246	0.132	0.225	0.293	0.344	0.384	0.415	0.440	0.461
	15	0.0246	0.132	0.225	0.294	0.345	0.385	0.416	0.442	0.463
	16	0.0245	0.132	0.225	0.294	0.346	0.386	0.417	0.443	0.465
	17	0.0245	0.132	0.225	0.295	0.346	0.387	0.419	0.445	0.466
	18	0.0244	0.132	0.225	0.295	0.347	0.387	0.420	0.446	0.468
	19	0.0244	0.132	0.226	0.295	0.348	0.388	0.421	0.447	0.469
	20	0.0244	0.132	0.226	0.295	0.348	0.389	0.421	0.448	0.470
	21	0.0243	0.132	0.226	0.296	0.348	0.389	0.422	0.449	0.472
	22	0.0243	0.132	0.226	0.296	0.349	0.390	0.423	0.450	0.473
	23	0.0243	0.131	0.226	0.296	0.349	0.391	0.424	0.451	0.473
	24	0.0243	0.131	0.226	0.296	0.350	0.391	0.424	0.452	0.474
	25	0.0243	0.131	0.226	0.296	0.350	0.391	0.425	0.452	0.475
	26	0.0242	0.131	0.226	0.297	0.350	0.392	0.425	0.453	0.476
	27	0.0242	0.131	0.226	0.297	0.350	0.392	0.426	0.453	0.477
	28	0.0242	0.131	0.226	0.297	0.351	0.393	0.426	0.454	0.477
	29	0.0242	0.131	0.226	0.297	0.351	0.393	0.427	0.455	0.478
	30	0.0242	0.131	0.226	0.297	0.351	0.393	0.427	0.455	0.478
	40	0.0241	0.131	0.226	0.298	0.353	0.396	0.430	0.459	0.483
	60	0.0240	0.131	0.227	0.299	0.354	0.398	0.433	0.463	0.487
	120	0.0239	0.131	0.227	0.300	0.356	0.400	0.437	0.467	0.492
	∞	0.0238	0.131	0.227	0.301	0.358	0.403	0.440	0.471	0.497

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Table 8.8.15 *0.1225 Quantiles of the F-distribution associated with ν_1 and ν_2 degrees of freedom*, concluded.

ν_2 denominator degrees of freedom	ν_1 degrees of freedom									
	10	12	15	20	24	30	40	60	120	∞
1	0.351	0.362	0.373	0.385	0.390	0.396	0.402	0.408	0.413	0.419
2	0.383	0.398	0.413	0.428	0.436	0.444	0.452	0.460	0.468	0.476
3	0.406	0.423	0.441	0.459	0.469	0.479	0.488	0.498	0.508	0.518
4	0.422	0.441	0.462	0.482	0.493	0.504	0.515	0.527	0.539	0.551
5	0.434	0.455	0.477	0.500	0.512	0.524	0.537	0.549	0.563	0.576
6	0.444	0.466	0.489	0.514	0.527	0.540	0.554	0.568	0.582	0.597
7	0.451	0.474	0.499	0.525	0.539	0.553	0.568	0.583	0.599	0.615
8	0.457	0.482	0.507	0.535	0.549	0.564	0.580	0.596	0.613	0.630
9	0.462	0.487	0.514	0.543	0.558	0.574	0.590	0.607	0.625	0.643
10	0.467	0.493	0.520	0.550	0.566	0.582	0.599	0.617	0.636	0.655
11	0.470	0.497	0.525	0.556	0.572	0.589	0.607	0.626	0.645	0.665
12	0.473	0.501	0.530	0.562	0.578	0.596	0.614	0.633	0.654	0.675
13	0.476	0.504	0.534	0.566	0.584	0.602	0.621	0.640	0.661	0.683
14	0.479	0.507	0.537	0.571	0.588	0.607	0.626	0.647	0.668	0.691
15	0.481	0.510	0.541	0.574	0.592	0.611	0.631	0.653	0.675	0.699
16	0.483	0.512	0.543	0.578	0.596	0.616	0.636	0.658	0.681	0.705
17	0.485	0.514	0.546	0.581	0.600	0.620	0.640	0.663	0.686	0.712
18	0.486	0.516	0.548	0.584	0.603	0.623	0.644	0.667	0.691	0.717
19	0.488	0.518	0.551	0.587	0.606	0.626	0.648	0.671	0.696	0.723
20	0.489	0.520	0.553	0.589	0.609	0.629	0.652	0.675	0.701	0.728
21	0.491	0.521	0.554	0.591	0.611	0.632	0.655	0.679	0.705	0.733
22	0.492	0.522	0.556	0.593	0.613	0.635	0.658	0.682	0.709	0.737
23	0.493	0.524	0.558	0.595	0.616	0.637	0.660	0.685	0.712	0.742
24	0.494	0.525	0.559	0.597	0.618	0.640	0.663	0.688	0.716	0.746
25	0.495	0.526	0.561	0.599	0.620	0.642	0.665	0.691	0.719	0.750
26	0.495	0.527	0.562	0.600	0.621	0.644	0.668	0.694	0.722	0.753
27	0.496	0.528	0.563	0.602	0.623	0.646	0.670	0.696	0.725	0.757
28	0.497	0.529	0.564	0.603	0.625	0.647	0.672	0.699	0.728	0.760
29	0.498	0.530	0.565	0.605	0.626	0.649	0.674	0.701	0.730	0.763
30	0.498	0.531	0.566	0.606	0.628	0.651	0.676	0.703	0.733	0.766
40	0.503	0.537	0.574	0.615	0.638	0.663	0.690	0.720	0.753	0.791
60	0.509	0.543	0.582	0.626	0.650	0.677	0.706	0.739	0.777	0.823
120	0.514	0.550	0.591	0.637	0.664	0.693	0.725	0.763	0.808	0.868
∞	0.520	0.557	0.600	0.650	0.679	0.711	0.748	0.793	0.852	1.000

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TABLE 8.8.16 *One-sided tolerance factors for A-basis calculations, T_A, for the one-way random effects ANOVA, 0.95 confidence*, continued on next page.

$\hat{R}$ = 0.00

n \ k	2	3	4	5	6	7	8	9	10	15	20	∞
1	37.094	10.553	7.042	5.741	5.062	4.642	4.354	4.143	3.981	3.520	3.295	2.326
2	7.770	5.164	4.386	3.996	3.755	3.589	3.467	3.372	3.297	3.064	2.941	2.326
3	5.339	4.192	3.764	3.528	3.375	3.266	3.183	3.118	3.065	2.898	2.807	2.326
4	4.502	3.776	3.475	3.300	3.184	3.100	3.035	2.984	2.942	2.807	2.733	2.326
5	4.075	3.540	3.302	3.162	3.066	2.996	2.942	2.899	2.863	2.748	2.684	2.326
6	3.812	3.384	3.186	3.067	2.984	2.924	2.877	2.839	2.807	2.706	2.649	2.326
7	3.633	3.273	3.102	2.997	2.924	2.870	2.828	2.794	2.766	2.674	2.622	2.326
8	3.501	3.190	3.037	2.943	2.877	2.838	2.789	2.759	2.733	2.649	2.601	2.326
9	3.400	3.123	2.986	2.899	2.839	2.794	2.759	2.730	2.706	2.629	2.584	2.326
10	3.320	3.070	2.943	2.864	2.808	2.766	2.733	2.706	2.684	2.611	2.570	2.326
11	3.254	3.025	2.908	2.834	2.781	2.742	2.711	2.686	2.665	2.597	2.557	2.326
12	3.199	2.987	2.878	2.808	2.759	2.722	2.693	2.669	2.649	2.584	2.547	2.326
13	3.152	2.954	2.851	2.786	2.739	2.704	2.676	2.654	2.635	2.573	2.538	2.326
14	3.111	2.926	2.828	2.766	2.722	2.688	2.662	2.640	2.623	2.563	2.504	2.326
15	3.076	2.901	2.808	2.749	2.706	2.674	2.649	2.629	2.611	2.555	2.498	2.326
∞	2.326	2.326	2.326	2.326	2.326	2.326	2.326	2.326	2.326	2.326	2.326	2.326

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TABLE 8.8.16 *One-sided tolerance factors for A-basis calculations, T_A, for the one-way random effects ANOVA, 0.95 confidence*, continued on next page.

$\hat{R} = 0.25$

	k											
	2	3	4	5	6	7	8	9	10	15	20	∞
1	37.094	10.553	7.042	5.741	5.062	4.642	4.354	4.143	3.981	3.520	3.295	2.326
2	9.116	5.519	4.575	4.123	3.852	3.669	3.534	3.432	3.350	3.101	2.970	2.326
3	6.328	4.541	3.974	3.681	3.497	3.368	3.273	3.198	3.138	2.951	2.850	2.326
4	5.348	4.121	3.696	3.468	3.321	3.217	3.139	3.078	3.028	2.871	2.785	2.326
5	4.849	3.884	3.533	3.340	3.215	3.125	3.057	3.003	2.959	2.820	2.743	2.326
6	4.546	3.732	3.426	3.255	3.143	3.062	3.000	2.952	2.912	2.785	2.714	2.326
7	4.342	3.625	3.349	3.193	3.090	3.016	2.959	2.914	2.877	2.759	2.693	2.326
8	4.195	3.545	3.291	3.146	3.051	2.981	2.928	2.885	2.851	2.739	2.677	2.326
9	4.084	3.484	3.246	3.110	3.019	2.953	2.903	2.863	2.829	2.723	2.663	2.326
10	3.997	3.435	3.210	3.080	2.994	2.931	2.883	2.844	2.812	2.710	2.652	2.326
11	3.927	3.394	3.180	3.056	2.973	2.912	2.866	2.829	2.798	2.699	2.643	2.326
12	3.869	3.361	3.155	3.035	2.955	2.897	2.852	2.815	2.786	2.690	2.636	2.326
13	3.821	3.333	3.133	3.018	2.940	2.883	2.839	2.804	2.775	2.682	2.629	2.326
14	3.780	3.308	3.115	3.003	2.927	2.872	2.829	2.795	2.766	2.675	2.624	2.326
15	3.744	3.287	3.099	2.990	2.916	2.862	2.820	2.786	2.759	2.669	2.619	2.326
∞	3.263	2.984	2.865	2.793	2.745	2.708	2.680	2.657	2.638	2.576	2.541	2.326

n

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TABLE 8.8.16 *One-sided tolerance factors for A-basis calculations, T_A, for the one-way random effects ANOVA, 0.95 confidence*, continued on next page.

$\hat{R} = 0.50$

n \ k	2	3	4	5	6	7	8	9	10	15	20	∞
1	37.094	10.553	7.042	5.741	5.062	4.642	4.354	4.143	3.981	3.520	3.295	2.326
2	10.553	5.891	4.777	4.262	3.959	3.756	3.609	3.497	3.408	3.141	3.002	2.326
3	7.491	4.930	4.208	3.851	3.632	3.482	3.372	3.286	3.218	3.007	2.895	2.326
4	6.386	4.515	3.945	3.654	3.472	3.346	3.252	3.179	3.120	2.938	2.839	2.326
5	5.821	4.283	3.793	3.537	3.377	3.264	3.180	3.114	3.061	2.894	2.804	2.326
6	5.479	4.134	3.693	3.460	3.313	3.209	3.131	3.070	3.020	2.865	2.780	2.326
7	5.249	4.030	3.622	3.405	3.267	3.169	3.095	3.038	2.991	2.843	2.762	2.326
8	5.084	3.953	3.569	3.364	3.232	3.139	3.069	3.013	2.968	2.826	2.749	2.326
9	4.960	3.894	3.528	3.331	3.205	3.115	3.048	2.994	2.951	2.813	2.738	2.326
10	4.863	3.847	3.495	3.306	3.183	3.096	3.031	2.979	2.937	2.803	2.729	2.326
11	4.785	3.809	3.469	3.284	3.165	3.081	3.017	2.966	2.925	2.794	2.722	2.326
12	4.721	3.777	3.446	3.267	3.151	3.068	3.005	2.955	2.915	2.787	2.716	2.326
13	4.668	3.751	3.427	3.252	3.138	3.057	2.995	2.946	2.907	2.781	2.711	2.326
14	4.623	3.728	3.411	3.239	3.127	3.047	2.987	2.939	2.899	2.775	2.707	2.326
15	4.584	3.708	3.397	3.228	3.117	3.039	2.979	2.932	2.893	2.771	2.703	2.326
∞	4.073	3.437	3.200	3.069	2.982	2.919	2.871	2.833	2.802	2.702	2.646	2.326

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TABLE 8.8.16 *One-sided tolerance factors for A-basis calculations, T_A, for the one-way random effects ANOVA, 0.95 confidence*, continued on next page.

$\hat{R}$ = 1.00

n \ k	2	3	4	5	6	7	8	9	10	15	20	∞
1	37.094	10.553	7.042	5.741	5.062	4.642	4.354	4.143	3.981	3.520	3.295	2.326
2	13.279	6.544	5.124	4.499	4.140	3.904	3.735	3.608	3.507	3.208	3.054	2.326
3	9.874	5.640	4.618	4.144	3.863	3.675	3.539	3.434	3.351	3.101	2.970	2.326
4	8.582	5.243	4.384	3.974	3.729	3.562	3.441	3.347	3.273	3.046	2.926	2.326
5	7.909	5.020	4.248	3.875	3.649	3.495	3.382	3.295	3.225	3.012	2.899	2.326
6	7.497	4.876	4.159	3.809	3.596	3.450	3.343	3.260	3.193	2.989	2.881	2.326
7	7.219	4.776	4.097	3.763	3.558	3.418	3.315	3.235	3.170	2.973	2.867	2.326
8	7.019	4.703	4.050	3.728	3.530	3.394	3.293	3.216	3.153	2.961	2.857	2.326
9	6.868	4.646	4.015	3.701	3.508	3.375	3.277	3.201	3.140	2.951	2.850	2.326
10	6.750	4.602	3.986	3.679	3.490	3.360	3.264	3.189	3.129	2.943	2.843	2.326
11	6.655	4.566	3.963	3.662	3.476	3.348	3.253	3.179	3.120	2.937	2.838	2.326
12	6.578	4.536	3.943	3.647	3.464	3.338	3.244	3.171	3.113	2.931	2.834	2.326
13	6.513	4.510	3.927	3.635	3.454	3.329	3.236	3.164	3.106	2.927	2.830	2.326
14	6.458	4.489	3.913	3.624	3.445	3.321	3.230	3.158	3.101	2.923	2.827	2.326
15	6.411	4.470	3.901	3.615	3.438	3.315	3.224	3.153	3.096	2.919	2.824	2.326
∞	5.793	4.217	3.734	3.487	3.332	3.224	3.144	3.081	3.030	2.871	2.785	2.326

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TABLE 8.8.16 *One-sided tolerance factors for A-basis calculations, T_A, for the one-way random effects ANOVA, 0.95 confidence*, continued on next page.

$\hat{R}$ = 2.00

n \ k	2	3	4	5	6	7	8	9	10	15	20	∞
1	37.094	10.553	7.042	5.741	5.062	4.642	4.354	4.143	3.981	3.520	3.295	2.326
2	17.646	7.464	5.595	4.814	4.379	4.098	3.900	3.751	3.635	3.294	3.121	2.326
3	14.076	6.687	5.189	4.539	4.169	3.927	3.754	3.624	3.521	3.217	3.062	2.326
4	12.622	6.334	4.998	4.407	4.067	3.843	3.682	3.561	3.465	3.179	3.031	2.326
5	11.838	6.134	4.887	4.329	4.007	3.793	3.639	3.523	3.431	3.155	3.013	2.326
6	11.348	6.004	4.814	4.278	3.966	3.760	3.611	3.498	3.408	3.140	3.000	2.326
7	11.013	5.913	4.762	4.242	3.938	3.736	3.590	3.480	3.392	3.128	2.992	2.326
8	10.770	5.846	4.724	4.214	3.917	3.718	3.575	3.466	3.380	3.120	2.985	2.326
9	10.586	5.794	4.695	4.193	3.900	3.705	3.563	3.456	3.370	3.113	2.980	2.326
10	10.441	5.754	4.671	4.177	3.887	3.694	3.554	3.447	3.363	3.108	2.975	2.326
11	10.324	5.720	4.652	4.163	3.876	3.685	3.546	3.440	3.356	3.104	2.972	2.326
12	10.228	5.693	4.636	4.152	3.867	3.677	3.540	3.435	3.351	3.100	2.969	2.326
13	10.148	5.670	4.623	4.142	3.860	3.671	3.534	3.430	3.347	3.097	2.967	2.326
14	10.080	5.650	4.611	4.134	3.853	3.665	3.529	3.426	3.343	3.094	2.965	2.326
15	10.021	5.633	4.601	4.127	3.848	3.661	3.525	3.422	3.340	3.092	2.963	2.326
∞	9.242	5.400	4.465	4.028	3.769	3.595	3.468	3.371	3.294	3.060	2.937	2.326

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TABLE 8.8.16 *One-sided tolerance factors for A-basis calculations, T_A, for the one-way random effects ANOVA, 0.95 confidence*, concluded.

$\hat{R} = 10.0$

n \ k	2	3	4	5	6	7	8	9	10	15	20	∞
1	37.094	10.553	7.042	5.741	5.062	4.642	4.354	4.143	3.981	3.520	3.295	2.326
2	29.578	9.492	6.566	5.443	4.845	4.471	4.212	4.021	3.874	3.451	3.243	2.326
3	27.473	9.167	6.415	5.347	4.775	4.415	4.165	3.981	3.839	3.428	3.225	2.326
4	26.486	9.010	6.341	5.299	4.740	4.387	4.142	3.961	3.821	3.417	3.216	2.326
5	25.914	8.917	6.298	5.271	4.719	4.370	4.128	3.949	3.810	3.410	3.211	2.326
6	25.540	8.856	6.269	5.253	4.705	4.359	4.119	3.941	3.803	3.405	3.207	2.326
7	25.277	8.812	6.248	5.239	4.695	4.352	4.113	3.936	3.798	3.402	3.205	2.326
8	25.082	8.780	6.233	5.229	4.688	4.346	4.108	3.931	3.795	3.399	3.203	2.326
9	24.931	8.755	6.221	5.222	4.682	4.341	4.104	3.928	3.792	3.397	3.201	2.326
10	24.812	8.735	6.211	5.215	4.678	4.337	4.101	3.925	3.789	3.396	3.200	2.326
11	24.714	8.719	6.203	5.210	4.674	4.334	4.098	3.923	3.787	3.395	3.199	2.326
12	24.633	8.705	6.197	5.206	4.671	4.332	4.096	3.921	3.786	3.394	3.198	2.326
13	24.565	8.693	6.191	5.203	4.668	4.330	4.094	3.920	3.784	3.393	3.198	2.326
14	24.507	8.684	6.187	5.200	4.666	4.328	4.093	3.918	3.783	3.392	3.197	2.326
15	24.456	8.675	6.182	5.197	4.664	4.326	4.092	3.917	3.782	3.391	3.197	2.326
∞	23.764	8.557	6.126	5.160	4.637	4.305	4.073	3.901	3.768	3.382	3.189	2.326

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INDEX

Volume 1

A-basis value 8-61
Additives 5-4
Alignment 5-3
Analysis of variance 8-31
  ANOVA 8-59
  Fixed effects 8-56
Anderson-Darling test 8-30, 8-44, 8-90
ANOVA 8-31, 8-62, 8-70, 8-89, 8-92
Area
  Bearing 7-2
  Cross-sectional 3-10
Areal weight 3-10, 5-5

B-basis values 8-10, 8-23
Bearing area 7-2
Bearing strength of joints 7-8
Braiding 6-12

Chemical properties 2-10
Chemical techniques
  Fiber 3-3
  Matrix 4-2
Chromatography 4-4, 5-2, 5-7, 5-8
Compatibility
  Material 2-15
  Statistical 8-44
Compatible 8-23
Composition
  Fiber 3-3
Compression tests 6-8
Compression after impact testing 6-11
Compressive 8-17
Conditioning 2-6
Creep 6-10
Crystalline melt temperature 6-3

Damage tolerance 6-11
Data pooling 2-8
Data documentation 8-5
Density 3-10, 6-4, 8-11
  Fiber 3-8
Differential scanning calorimetry 5-9
Drape 5-5
Dynamic mechanical analysis 5-9

Electrical resistivity
  Fiber 3-8, 3-20
Elemental analysis 3-3, 4-2
Environmental exposure yesting 6-4
Equality of variance 8-45, 8-68
  Test 8-30
Exploratory data analysis 8-49

Fabric 3-10, 6-7
Failure modes 7-2
Fatigue 6-10
Fiber 3-3
Fiber content 5-4, 5-6
Fiber diameter 3-8, 3-17
Fiber properties 8-17
Fiber volume 6-3
Filament Winding 6-12
Fillers 5-4
Flammability 6-5
Functional group 4-2

Gaps 5-3
Gel time 5-5, 5-10
Glass transition temperature 6-3, 8-11
Goodness-of-fit 8-46

Hanson-Koopmans method 8-38, 8-62, 8-81, 8-91

In-plane shear 6-9
Informative quantile function 8-50
Infrared spectroscopy 3-6, 4-2, 5-2, 5-9
Interlaminar 6-9

Joint 8-17

Lamina 6-2, 8-17
  Mechanical properties 2-9
  Physical properties 2-9
Laminate 6-2, 8-17
Linear models 8-39, 8-53
Linear regression 8-40, 8-48, 8-55, 8-63, 8-82
Lognormal distribution 8-30, 8-34, 8-47, 8-62, 8-78

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